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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 6: Electromagnetic Induction

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 73 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. A magnet held vertically, with its north pole down, is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top, then:

(A) the induced current will flow in the anticlockwise direction.

(B) the induced current will flow in the clockwise direction.

(C) no induced current will flow in the solenoid.

(D) the magnet will fall with a constant velocity.

Check Solution → **[2026 • Set 55-2-1]**

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2. The magnetic flux ϕ (in Wb) linked with a coil is related to time t (in s) as $\phi = 5At^2 + Bt - 2C$. The SI units of A and B are respectively:

(A) Wb s^2, Wb s

(B) Wb s^{-1}, Wb

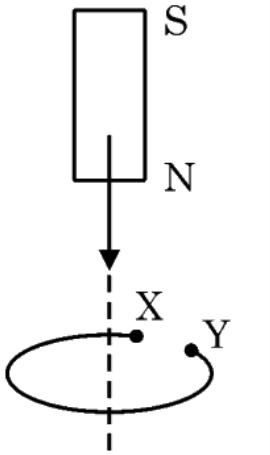
(C) Wb s^{-2}, Wb s^{-1}

(D) Wb s^{-1}, Wb s^{-2}

Check Solution → **[2026 • Set 55-2-1]**

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3. Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct?



(A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(C) Speed of magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

Check Solution → [2026 • Set 55-2-2]

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4. Two identical circular loops *A* and *B* of metal wire are arranged on a table close to each other. Loop *A* is connected to a battery and the current in it increases with time. In response, the loop *B*:

(A) remains stationary

(B) is attracted by loop *A*

(C) is repelled by loop *A*

(D) rotates about its centre of mass which is fixed

Check Solution → [2026 • Set 55-2-3]

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5. A square loop of side 50 cm is placed in a uniform magnetic field of 3·0 *T* acting perpendicular to the plane of the loop. If the loop is rotated through an angle of 90° in 0·3 *s*, the value of emf induced in the loop would be:

(A) 0·25 V

(B) 0·50 V

(C) 0·75 V

(D) 1·0 V

Check Solution → [2026 • Set 55-3-1]

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6. A metallic rod of 1 m length is rotated with a frequency of 40 rev/s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 1 m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of 0·5 T parallel to the axis is present in the region. The value of emf induced between the centre and the metallic ring is close to:

(A) 20 V

(B) 32 V

(C) 40 V

(D) 63 V

Check Solution → [2026 • Set 55-3-2]

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7. A square loop of side L lies in the x-y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, where B_0 is a constant. The magnetic flux through the loop is:

(A) $2\,B_0\,L^2$

(B) $4\,B_0\,L^2$

(C) $3\,B_0\,L^2$

(D) $\sqrt{29}\,B_0\,L^2$

Check Solution → [2026 • Set 55-3-3]

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8. The dimensions of the rate of change of magnetic flux are:

(A) $[MLT^{-3}A^{-1}]$

(B) $[ML^2T^{-3}A^{-1}]$

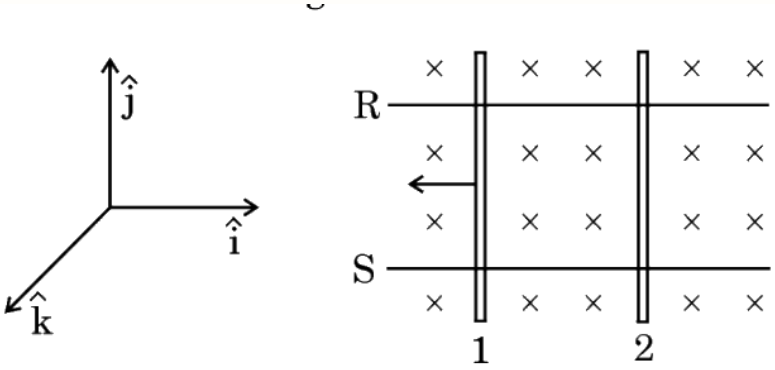
(C) $[ML^2T^{-2}A^{-1}]$

(D) $[ML^2T^{-3}A^{-2}]$

Check Solution → [2026 • Set 55-4-1]

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9. Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along:



(A) $-\hat{i}$

(B) $-\hat{j}$

(C) $\hat{k}$

(D) $\hat{j}$

Check Solution → [2026 • Set 55-4-1]

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10. A closely wound long solenoid of self-inductance L is cut into two identical solenoids. The value of self-inductance of each small solenoid will be:

(A) $\frac{L}{4}$

(B) $\frac{L}{2}$

(C) 2L

(D) 4L

Check Solution → [2026 • Set 55-4-2]

---

11. Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the:

(A) rate at which currents change in the two coils.

(B) relative position and orientation of the coils.

(C) currents in the two coils.

(D) value of voltage induced in one coil due to change in value of current in the other coil.

Check Solution → [2026 • Set 55-4-3]

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12. **Assertion (A):** Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.
Reason (R): Induced emf produced is directly proportional to the magnetic flux.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-5-1]

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13. The magnetic flux through a loop placed in a magnetic field can be changed by changing:

(A) area of the loop only

(B) the value of magnetic field only

(C) orientation of the loop in the magnetic field only

(D) any one or more of the factors given in (A), (B) and (C)

Check Solution → **[2026 • Set 55-5-1]**

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14. A rectangular loop of size 5 cm × 8 cm is lying in x-y plane in a uniform magnetic field $\vec{B} = (2.0\ T)\ \hat{k}$. The total magnetic flux linked with the loop is:

(A) 80 Wb

(B) 16 Wb

(C) 8×10^{-2} Wb

(D) 8×10^{-3} Wb

Check Solution → **[2026 • Set 55-5-2]**

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15. A square loop of side 0.50 m is placed in a uniform magnetic field of 0.4 T perpendicular to the plane of the loop. The loop is rotated through an angle of 60° in 0.2 s. The value of emf induced in the loop will be:

(A) 5 V

(B) 3.5 V

(C) 2.5 V

(D) Zero V

Check Solution → **[2026 • Set 55-5-3]**

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16. **Assertion (A):** It is difficult to move a magnet into a coil of large number of turns when the circuit of the coil is closed.
Reason (R): The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2025 • Set 55-1-1]

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17. A coil has 100 turns, each of area 0.05 m^2 and total resistance 1.5 Ω. It is inserted at an instant in a magnetic field of 90 mT, with its axis parallel to the field. The charge induced in the coil at that instant is:

(A) 3.0 mC

(B) 0.30 *C*

(C) 0.45 *C*

(D) 1.5 *C*

Check Solution → [2025 • Set 55-1-1]

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18. You are required to design an air-filled solenoid of inductance 0.016 *H* having a length 0.81 *m* and radius 0.02 *m*. The number of turns in the solenoid should be:

(A) 2592

(B) 2866

(C) 2976

(D) 3140

Check Solution → [2025 • Set 55-1-1]

---

19. A circular coil of diameter 15 mm having 300 turns is placed in a magnetic field of 30 mT such that the plane of the coil is perpendicular to the direction of magnetic field. The magnetic field is reduced uniformly to zero in 20 ms and again increased uniformly to 30 mT in 40 ms. If the emfs induced in the two time intervals are e_1 and e_2 respectively, then the value of e_1 / e_2 is:

(A) $\frac{1}{2}$

(B) 1

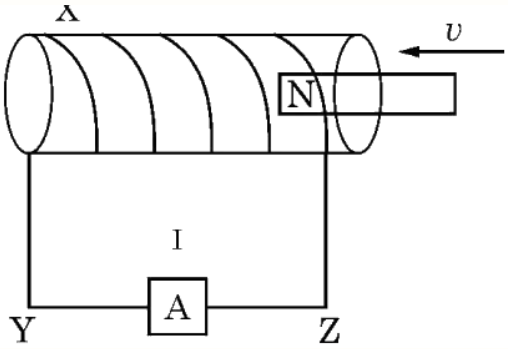
(C) 2

(D) 4

Check Solution → [2025 • Set 55-1-2]

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20. In the figure X is a coil wound over a hollow wooden pipe. A permanent magnet is pushed at a constant speed v from the right into the pipe and it comes out at the left end of the pipe. During the entry and the exit of the magnet, the current in the wire YZ will be from



(A) Y to Z and then Y to Z

(B) Z to Y and then Y to Z

(C) Y to Z and then Z to Y

(D) Z to Y and then Z to Y

Check Solution → [2025 • Set 55-2-1]

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21. Two long solenoids of radii r_1 and r_2 ($r_2 > r_1$) and number of turns per unit length n_1 and n_2, respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is

(A) $\frac{n_1}{n_2}$

(B) $\frac{n_2}{n_1}$

(C) $\frac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\frac{n_2 r_1^2}{n_1 r_2^2}$

Check Solution → [2025 • Set 55-2-2]

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22. A coil of an ac generator, having 100 turns and area 0.1 m^2 each, rotates at half a rotation per second in a magnetic field of 0.02 T. The maximum emf generated in the coil is

(A) 0.31 V

(B) 0.20 V

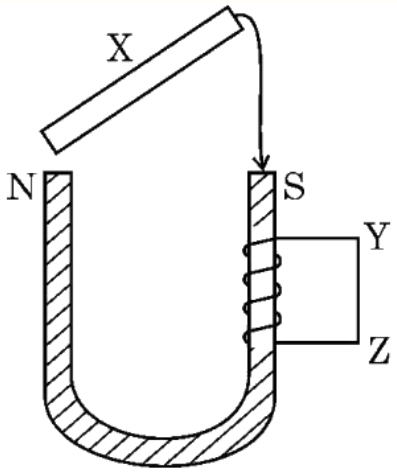
(C) 0.63 V

(D) 0.10 V

Check Solution → [2025 • Set 55-2-2]

---

23. A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet. During fall of the rod, the current in the coil will be



(A) clockwise current

(B) anticlockwise current

(C) alternating current

(D) zero

Check Solution → [2025 • Set 55-2-3]

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24. When current in a coil changes at a steady rate from 8 A to 6 A in 4 ms, an emf of 1.5 V is induced in it. The value of self-inductance of the coil is :

(A) 6 mH

(B) 12 mH

(C) 3 mH

(D) 9 mH

Check Solution → [2025 • Set 55-4-2]

---

25. The magnetic flux linked with a closed coil (in Wb) varies with time t (in s) as $\phi = 5t^2 + 4t - 2$. If the resistance of the circuit is 14 Ω, the magnitude of induced current in the coil at $t = 1$ s will be :

(A) 0.5 A

(B) 1.0 A

(C) 1.5 A

(D) 2.0 A

Check Solution → [2025 • Set 55-4-3]

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26. The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$, where t is in seconds and ϕ is in Wb. The value of emf induced in the coil at $t = 4$ s is:

(A) 32 V

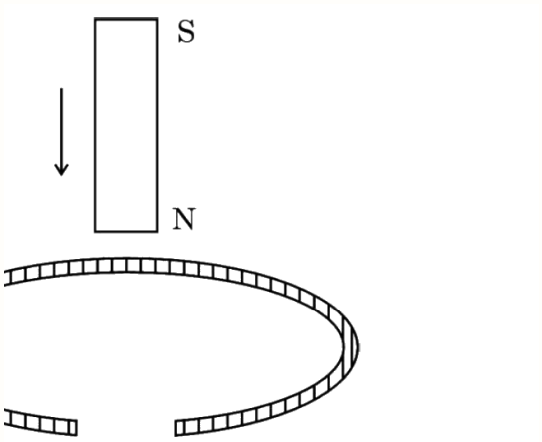
(B) 37 V

(C) 64 V

(D) 69 V

Check Solution → [2025 • Set 55-5-1]

27. A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is:



(A) zero

(B) less than g

(C) g

(D) greater than g

Check Solution → [2025 • Set 55-6-1]

28. A rectangular coil of area A is kept in a uniform magnetic field $\vec{B}$ such that the plane of the coil makes an angle α with $\vec{B}$. The magnetic flux linked with the coil is:

(A) $BA\alpha$

(B) $BA\alpha$

(C) BA

(D) zero

Check Solution → [2025 • Set 55-7-1]

29. A metal rod of length 50 cm is held vertically and moved with a velocity of 10 m/s towards east. The horizontal component of the Earth's magnetic field at the place is 0.4 G. The emf induced across the ends of the rod is:

(A) 0.1 mV

(B) 0.2 mV

(C) 0.8 mV

(D) 1.6 mV

Check Solution → [2025 • Set 55-7-1]

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30. The dimensions of 'self-inductance' are:

(A) $[MLT^{-2}A^{-2}]$

(B) $[ML^{2}T^{-1}A^{-1}]$

(C) $[ML^{-1}T^{-2}A^{-2}]$

(D) $[ML^{2}T^{-2}A^{-2}]$

Check Solution → [2025 • Set 55-7-1]

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31. **Assertion (A):** The mutual inductance between two coils is maximum when the coils are wound on each other.

Reason (R): The flux linkage between two coils is maximum when they are wound on each other.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-1-1]

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32. A conducting circular loop is placed in a uniform magnetic field $B = 50$ mT with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of 1 mm s^{-1}. At the instant the radius of the loop is 4 cm, the induced emf in the loop is:

(A) $\pi\ \mu V$

(B) $2\pi\ \mu V$

(C) $4\pi\ \mu V$

(D) $8\pi\ \mu V$

Check Solution → [2024 • Set 55-1-2]

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33. **Assertion (A):** Lenz's law is a consequence of the law of conservation of energy.
Reason (R): There is no power loss in an ideal inductor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-2-1]

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34. A circular coil of radius 10 cm is placed in a magnetic field $B=(1.0\ i+0.5\ j)$ mT such that the outward unit vector normal to the surface of the coil is $(0.6\ i+0.8\ j)$. The magnetic flux linked with the coil is:

(A) 0.314 μ Wb

(B) 3.14 μ Wb

(C) 31.4 μ Wb

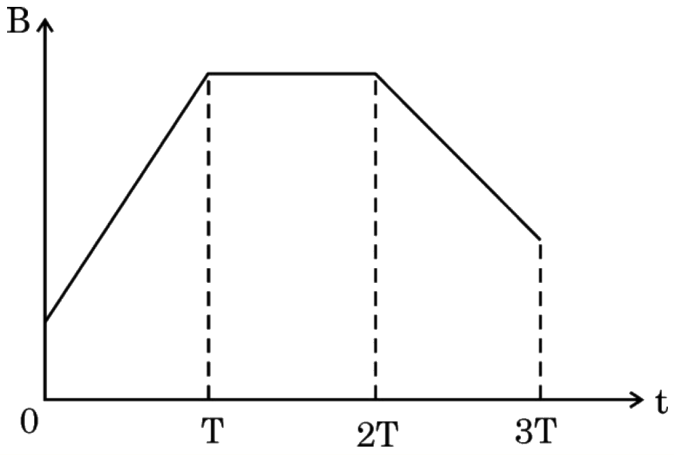
(D) 1.256 μ Wb

Check Solution → [2024 • Set 55-2-1]

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35. A conducting loop is placed in a magnetic field, normal to its plane. The magnitude of the magnetic field varies with time as shown in the figure. If $\varepsilon_1, \varepsilon_2$

and ε_3 are magnitudes of induced emfs during periods 0, T and 2T, then:



(A) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

(B) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

(C) $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

(D) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

Check Solution → [2024 • Set 55-2-2]

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36. The emf induced in a coil rotating in a magnetic field does not depend upon the following:

(A) Area of the coil

(B) Resistance of the coil

(C) Number of turns in the coil

(D) Angular speed of rotation of the coil

Check Solution → [2024 • Set 55-2-3]

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37. The current in a coil of 15 mH increases uniformly from zero to 4 A in 0.004 s. The emf induced in the coil will be:

(A) 22.5 V

(B) 17.5 V

(C) 15.0 V

(D) 12.5 V

Check Solution → [2024 • Set 55-3-1]

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38. Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if:

(A) both l and A are increased

(B) l is decreased and A is increased

(C) l is increased and A is decreased

(D) both l and A are decreased

Check Solution → [2024 • Set 55-3-1]

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39. The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1, the current changes from 4 A to zero in 0.2 s. If the resistance of coil C_2 is 4 Ω, then the charge that flows through it per second will be:

(A) 4.0 C

(B) 1.5 C

(C) 0.05 C

(D) 0.1 C

Check Solution → [2024 • Set 55-3-2]

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40. A coil of N turns is placed in a magnetic field $\vec{B}$ such that $\vec{B}$ is perpendicular to the plane of the coil. $\vec{B}$ changes with time as $B=B_0\cos(\frac{2\pi t}{T})$, where T is time period. The magnitude of emf induced in the coil will be maximum at

(A) $t=\frac{nT}{2}$

(B) $t=\frac{nT}{4}$

(C) $t=\frac{nT}{8}$

(D) $t=nT$

Check Solution → [2024 • Set 55-4-1]

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41. Two coils are placed near each other. When the current in one coil is changed at the rate of 5 A/s, an emf of 2 mV is induced in the other. The mutual inductance of the two coils is:

(A) 0.4 mH

(B) 2.5 mH

(C) 10 mH

(D) 25 H

Check Solution → [2024 • Set 55-5-1]

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42. A coil of area of cross-section 0.5 m^2 is placed in a magnetic field acting normally to its plane. The field varies as $B = 0.5t^2 + 2t$, where B is in tesla and t in seconds. The emf induced in the coil at $t = 1$ s is:

(A) 0.5 V

(B) 1.0 V

(C) 1.5 V

(D) 3.0 V

Check Solution → [2024 • Set 55-5-2]

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43. The mutual inductance of two coils, in a given orientation is 50 mH. If the current in one of the coils changes as $i = 1.0 \sin(100\pi\, t + \frac{\pi}{6})$ A, the peak value of emf (in volt) induced in the other coil will be:

(A) 5

(B) 5π

(C) 0.5π

(D) 0.05π

Check Solution → [2024 • Set 55-5-3]

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44. A circular coil of radius 8.0 cm and 40 turns is rotated about its vertical diameter with an angular speed of 25 rad s^{-1} in a uniform horizontal magnetic field of magnitude 3.0×10^{-2} T. The maximum emf induced in the coil is:

(A) 0.12 V

(B) 0.15 V

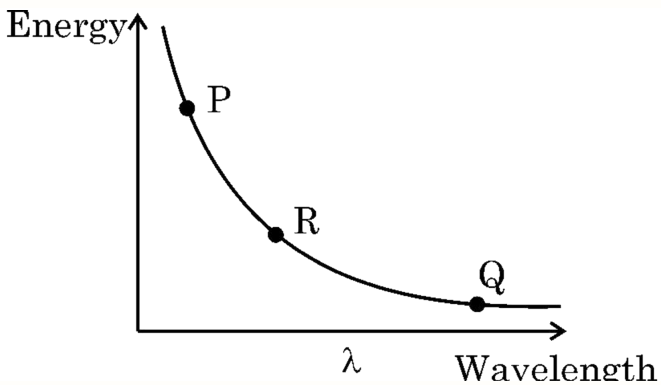
(C) 0.19 V

(D) 0.22 V

Check Solution → [2023 • Set 55-1-1]

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45. Figure shows a rectangular conductor PSRQ in which movable arm PQ has a resistance 'r' and resistance of PSRQ is negligible. The magnitude of emf induced when PQ is moved with a velocity $\vec{V}$ does not depend on:



(A) magnetic field $(\vec{B})$

(B) velocity $(\vec{v})$

(C) resistance (r)

(D) length of PQ

Check Solution → [2023 • Set 55-1-1]

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46. Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current:

(A) increases in A and decreases in B.

(B) decreases in both A and B.

(C) increases in both A and B.

(D) remains same in both A and B.

Check Solution → [2023 • Set 55-1-2]

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47. A bar magnet is dropped in a hollow metallic cylinder along its vertical axis. The acceleration of the falling magnet will be:

(A) zero

(B) equal to g

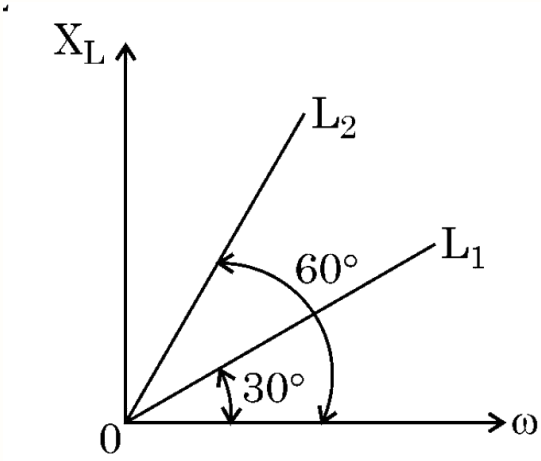
(C) less than g

(D) greater than g

Check Solution → [2023 • Set 55-1-3]

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48. The direction of induced current in the loop abc is:



(A) along abc if I decreases

(B) along acb if I increases

(C) along abc if I is constant

(D) along abc if I increases

Check Solution → [2023 • Set 55-4-1]

49. A square shaped coil of side 10 cm, having 100 turns is placed perpendicular to a magnetic field which is increasing at 1 T/s. The induced emf in the coil is

(A) 0.1 V

(B) 0.5 V

(C) 0.75 V

(D) 1.0 V

Check Solution → [2023 • Set 55-5-1]

50. Laminated iron sheets are used to minimize _____ currents in the core of a transformer.

Check Solution → [2020 • Set 55-1-1]

51. The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become _____ times.

Check Solution → [2020 • Set 55-1-1]

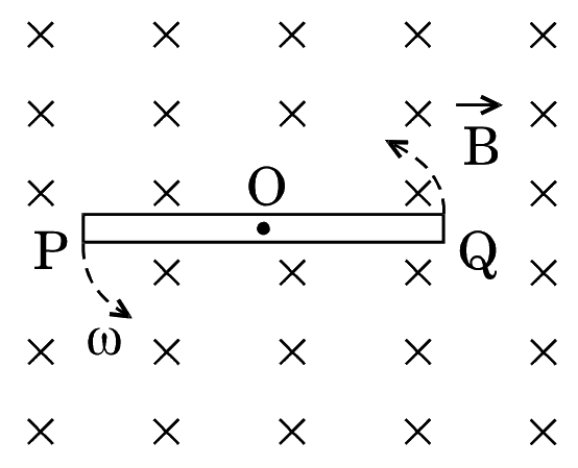
52. A conducting rod of length l is kept parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of emf induced in the conductor?

Check Solution → [2020 • Set 55-2-1]

53. Draw the graph showing variation of the value of the induced emf as a function of rate of change of current flowing through an ideal inductor.

Check Solution → [2020 • Set 55-2-1]

54. A metallic rod PQ of length l is rotated with an angular velocity ω about an axis passing through its mid-point (O) and perpendicular to the plane of the paper, in uniform magnetic field $\vec{B}$, as shown in the figure. What is the potential difference developed between the two ends of the rod, P and Q?

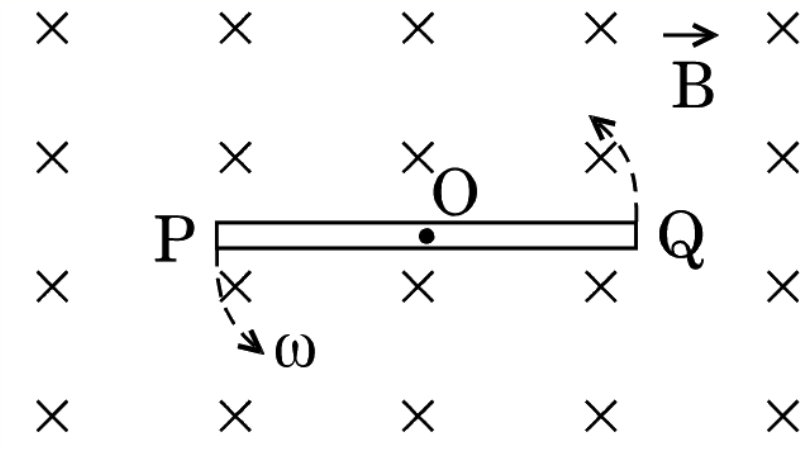


Check Solution →

[2020 • Set 55-2-2]

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55. A metallic rod PQ of length l is rotated with an angular velocity ω in a magnetic field $\vec{B}$ about an axis passing through the mid-point O of the rod and perpendicular to the plane of paper as shown in the figure. What is the potential difference developed between the points O and Q of the rod?



Check Solution →

[2020 • Set 55-2-3]

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56. Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more?

Check Solution → [2020 • Set 55-3-1]

57. In which orientation of the armature coil of an ac generator relative to the magnetic field, will the induced emf be maximum?

Check Solution → [2020 • Set 55-3-3]

58. Plot a graph showing variation of induced e.m.f. with the rate of change of current flowing through a coil.

OR

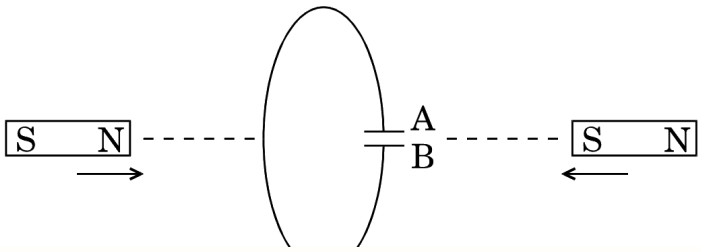
A series combination of an inductor (L), capacitor (C) and a resistor (R) is connected across an ac source of emf of peak value E_0 and angular frequency (ω). Plot a graph to show variation of impedance of the circuit with angular frequency (ω).

Check Solution → [2020 • Set 55-5-1]

59. A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop? Justify.

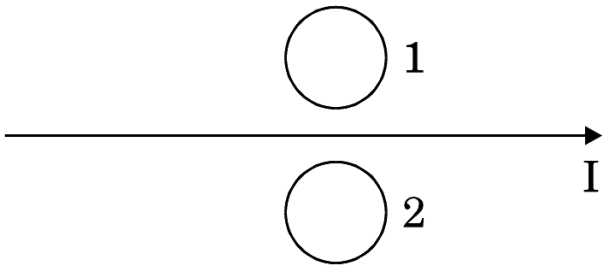
Check Solution → [2017]

60. Predict the polarity of the capacitor in the situation described below:



Check Solution → [2017]

61. What is the direction of induced currents in metal rings 1 and 2 when current I in the wire is increasing steadily?

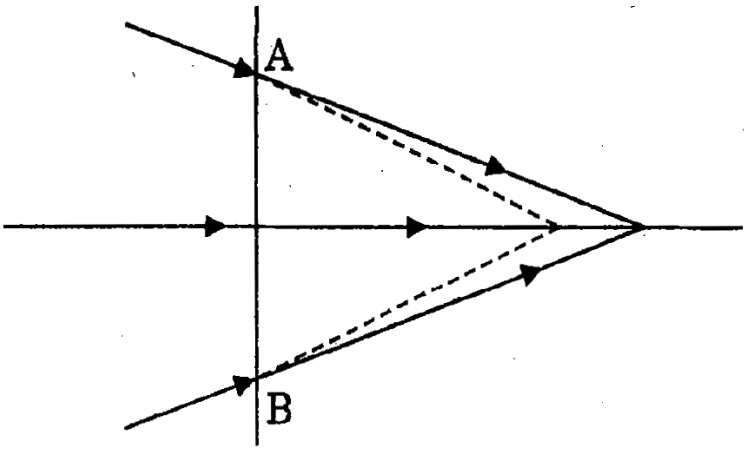


Check Solution → [2017]

62. Two spherical bobs, one metallic and the other of glass, of the same size are allowed to fall freely from the same height above the ground. Which of the two would reach earlier and why ?

Check Solution → [2014]

63. A conducting loop is held below a current carrying wire PQ as shown. Predict the direction of the induced current in the loop when the current in the wire is constantly increasing.



Check Solution → [2014]

64. Give the direction in which the induced current flows in the coil mounted on an insulating stand when a bar magnet is quickly moved along the axis of the coil from one side to the other as shown in the figure.

Check Solution → [2013]

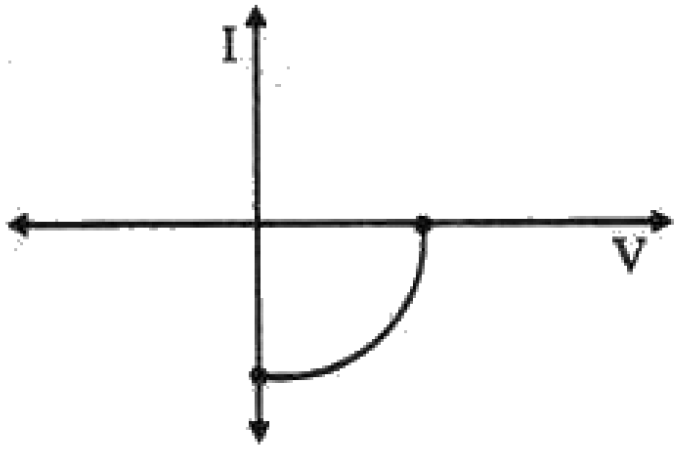
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65. Define self-inductance of a coil. Write its S.I. units.

Check Solution → [2012]

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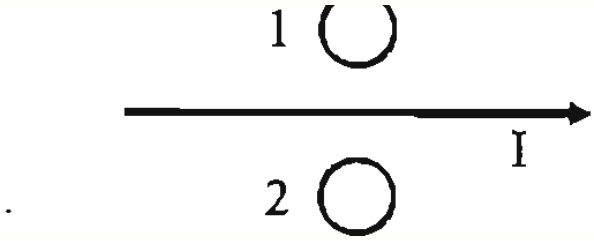
66. A bar magnet is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the directions of induced current in each coil.



Check Solution → [2012]

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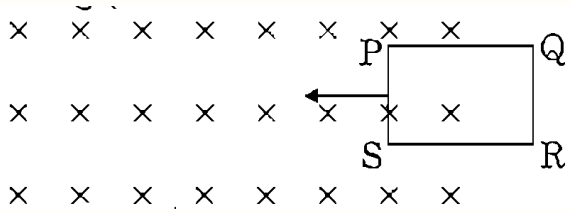
67. Predict the directions of induced currents in metal rings 1 and 2 lying in the same plane where current *I* in the wire is increasing steadily.



Check Solution → [2012]

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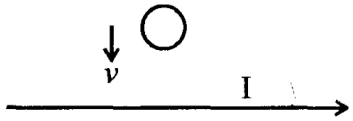
68. A closed loop (PQRS) of wire is moved into a uniform magnetic field at right angles to the plane of the paper as shown in the figure. Predict the direction of the induced current in the loop.



Check Solution → [2012]

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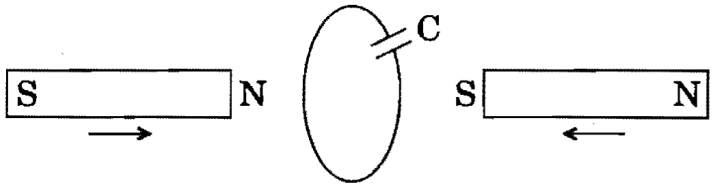
69. Predict the direction of induced current in a metal ring when the ring is moved towards a straight conductor with constant speed v. The conductor is carrying current I in the direction shown in the figure.



Check Solution → [2012]

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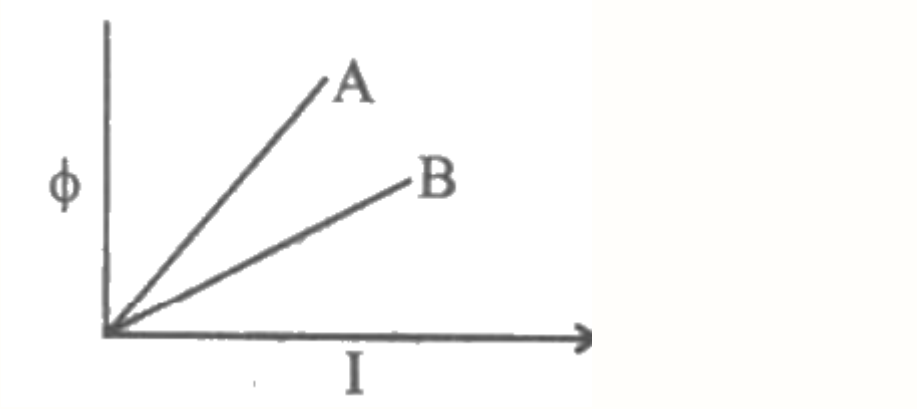
70. Two bar magnets are quickly moved towards a metallic loop connected across a capacitor C as shown in the figure. Predict the polarity of the capacitor.



Check Solution → [2011]

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71. A plot of magnetic flux (ϕ) versus current (I) is shown in the figure for two inductors A and B. Which of the two has larger value of self inductance?



Check Solution → [2010]

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72. When current in a coil changes with time, how is the back emf induced in the coil related to it?

Check Solution → [2008]

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73. Write S.I. unit of magnetic flux. Is it a scalar or a vector quantity?

Check Solution → [2003]

SECTION B

2-Mark Questions

VSA · 21 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

74. A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ω *t*, where *t* represents time elapsed:
(a) Magnetic flux ϕ linked with the coil, and
(b) emf induced in the coil.

Check Solution → **[2026 • Set 55-5-1]**

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75. A square loop of side 10 cm, free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of 0.2 *T*. If it is rotated at the uniform speed of 60 rpm, find the emf induced in the loop.

Check Solution → **[2026 • Set 55-5-2]**

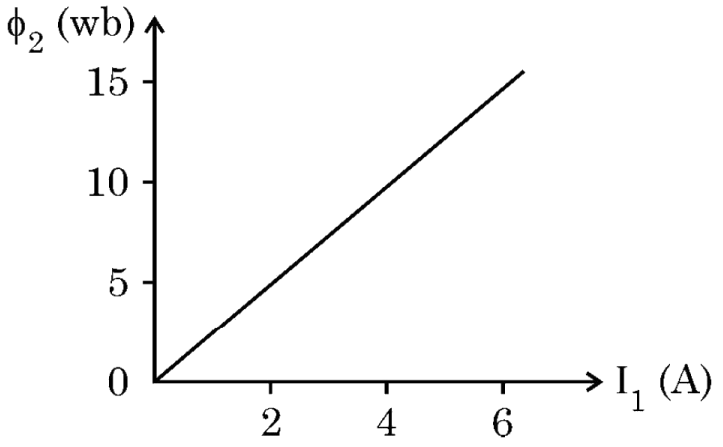
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76. A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position? Explain.

Check Solution → **[2026 • Set 55-5-3]**

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77. Two coils C_1 and C_2 are placed close to each other. The magnetic flux ϕ_2 linked with the coil C_2 varies with the current I_1 flowing in coil C_1, as shown in the figure. Find
(i) the mutual inductance of the arrangement, and
(ii) the rate of change of current $(\frac{dI_1}{dt})$ that will induce an emf of 100 V in coil C_2.



Check Solution → [2023 • Set 55-5-1]

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78. Two identical circular discs, one of copper and another of aluminium, are rotated about their geometrical axes with same angular speed in the same magnetic field acting perpendicular to their planes. Compare the (i) induced emf, and (ii) induced current produced in discs between its centre and edge. Justify your answers.

Check Solution → [2021]

---

79. Two coplanar and concentric coils 1 and 2 have respectively the number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 \gg r_1$). Deduce the expression for mutual inductance of this system.

Check Solution → [2020 • Set 55-3-1]

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80. The energy stored in a solenoid of inductance L is U. The number of turns per unit length of the solenoid is doubled. Keeping the current and all other factors same, find (a) change in inductance of the solenoid, and (b) the final energy stored in the inductor.

Check Solution → [2020 • Set 55-3-2]

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81. (a) Define the SI unit of self-inductance.
(b) The self-inductance of a solenoid is L. If the number of turns per unit length

in it is doubled and the area of cross-section is halved, find the new inductance of the solenoid.

Check Solution → [2020 • Set 55-3-3]

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82. A 0.5 m long solenoid of 10 turns/cm has area of cross-section 1 cm^2. Calculate the voltage induced across its ends if the current in the solenoid is changed from 1 A to 2 A in 0.1 s.

OR

A small flat search coil of area 5 cm^2 with 140 closely wound turns is placed between the poles of a powerful magnet producing magnetic field 0.09 T and then quickly removed out of the field region. Calculate

(a) change of magnetic flux through the coil, and

(b) emf induced in the coil.

Check Solution → [2019 • Set 55-3-1]

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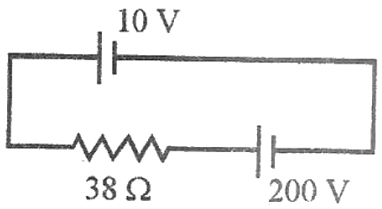
83. State Lenz's Law.

A metallic rod held horizontally along east-west direction, is allowed to fall under gravity. Will there be an emf induced at its ends? Justify your answer.

Check Solution → [2013]

---

84. Figure shows a bar magnet M falling under gravity through an air cored coil C. Plot a graph showing variation of induced e.m.f. (E) with time (t). What does the area enclosed by the E--t curve depict?



Check Solution → [2013]

---

85. Two identical loops, one of copper and the other of aluminium, are rotated with the same angular speed in the same magnetic field. Compare (i) the induced emf and (ii) the current produced in the two coils. Justify your answer.

Check Solution → [2012]

---

86. Define self-inductance of a coil. Show that magnetic energy required to build up the current I in a coil of self-inductance L is given by $\frac{1}{2}LI^2$.

Check Solution → [2012]

---

87. A metallic rod of 'L' length is rotated with angular frequency of 'ω' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius L, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field $\vec{B}$ parallel to the axis is present everywhere. Deduce the expression for the emf between the centre and the metallic ring.

Check Solution → [2012]

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88. Derive the expression for the self inductance of a long solenoid of cross sectional area A and length l, having n turns per unit length.

Check Solution → [2012]

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89. What are eddy currents? Write any two applications of eddy currents.

Check Solution → [2011]

---

90. A current is induced in coil C_1 due to the motion of current carrying coil C_2.
(a) Write any two ways by which a large deflection can be obtained in the galvanometer G.
(b) Suggest an alternative device to demonstrate the induced current in place of a galvanometer.

Check Solution → [2011 • Set 55-1-1]

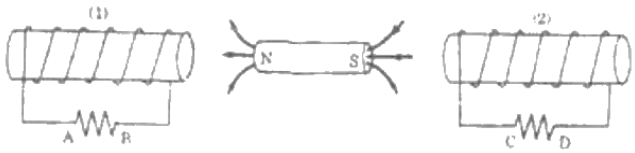
91. Current in a circuit falls steadily from 5.0 A to 0.0 A in 100 ms. If an average e.m.f. of 200 V is induced, calculate the self-inductance of the circuit.

Check Solution → [2011 • Set 55-2-1]

92. A jet plane is traveling west at 450 ms^{-1}. If the horizontal component of earth's magnetic field at that place is 4×10^{-4} Tesla and the angle of dip is 30°, find the emf induced between the ends of wings having a span of 30 m.

Check Solution → [2008]

93. In the figure given below, a bar magnet moving towards the right or left induces an e.m.f. in the coils (1) and (2). Find, giving reason, the directions of the induced currents through the resistors AB and CD when the magnet is moving **(a)** towards the right, and **(b)** towards the left.



Check Solution → [2005]

94. Two circular coils, one of radius r and the other of radius R are placed coaxially with their centres coinciding. For $R \gg r$, obtain an expression for the mutual inductance of the arrangement.

Check Solution → [2004]

SECTION C

3-Mark Questions

SA · 51 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

95. (i) Define mutual inductance of a pair of coils. Write its SI unit.
(ii) A long solenoid of radius R and length L has n turns per unit length. A circular loop of radius r (is placed inside at the centre of the solenoid such that its axis coincides with the axis of the solenoid. Obtain the mutual inductance of the solenoid and the loop.

Check Solution → **[2026 • Set 55-2-1]**

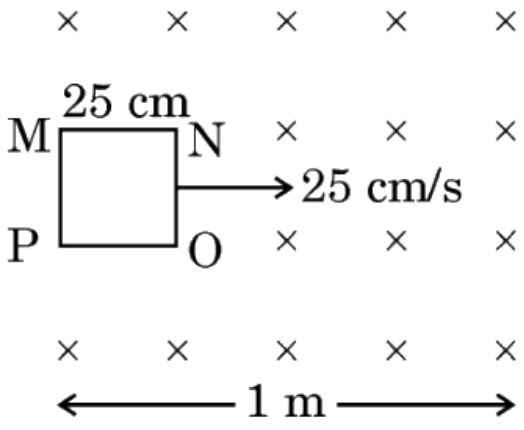
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96. State Faraday's law of electromagnetic induction. Briefly describe two methods for producing induced emf.

Check Solution → **[2026 • Set 55-3-3]**

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97. The figure given below shows a square-shaped loop MNOP of side 25 cm placed horizontally in a uniform magnetic field $\vec{B}$ directed vertically downward. The position of the loop at $t = 0$ s is as shown in figure. The loop is pulled with a constant velocity of 25 cm/s till it goes out of the field.



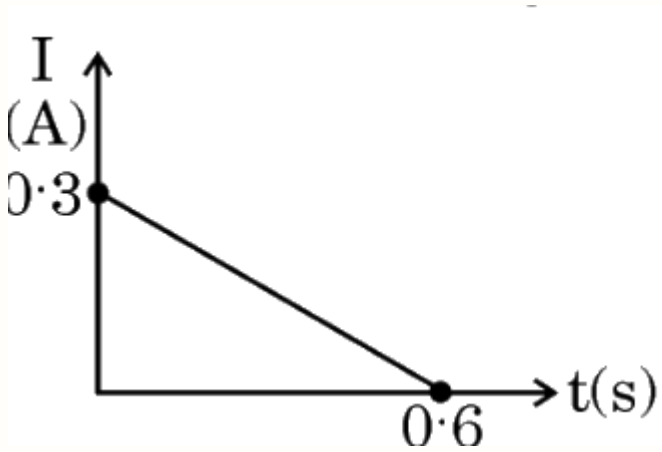
(a) What will be the direction of the induced current in the loop as it goes out of the field? For how long would the current in the loop persist?

(b) Plot graphs showing the variation of magnetic flux and magnitude of induced emf as a function of time.

Check Solution → [2026 • Set 55-4-1]

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98. A conducting rectangular loop of area 5 cm^2 and resistance 4 Ω is removed from a region of uniform magnetic field, acting normal to the plane of the loop. The value of induced current I in the loop varies with time t, as shown in the figure.



Find:

(a) total charge that passed through the loop

(b) change in magnetic flux through the loop

(c) magnitude of magnetic field in the region

Check Solution → [2026 • Set 55-4-2]

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99. Two coils, one of radius 0·5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate:

(a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and

(b) the mutual inductance of the two coils.

Check Solution → [2026 • Set 55-4-3]

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100. A long solenoid of length L and radius r_1, having N_1 turns is surrounded symmetrically by a coil of radius r_2 ($>r_1$) having N_2 turns ($N_2 \ll N_1$) around its

mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is $M_{12} = M_{21}$ valid in this case?

Check Solution → [2026 • Set 55-5-1]

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101. **(a)** Show that the energy required to build up the current I in a coil of inductance L is $\frac{1}{2} L I^2$.

(b) Considering the case of magnetic field produced by air-filled current carrying solenoid, show that the magnetic energy density of a magnetic field B is $\frac{B^2}{2\mu_0}$.

Check Solution → [2025 • Set 55-1-1]

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102. Consider two long co-axial solenoids S_1 and S_2, each of length l ($\gg r_2$) and of radius r_1 and r_2 ($r_2 > r_1$). The number of turns per unit length are n_1 and n_2 respectively. Derive an expression for mutual inductance M_{12} of solenoid S_1 with respect to solenoid S_2. Show that $M_{12} = M_{21}$.

Check Solution → [2025 • Set 55-1-2]

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103. (a) State Lenz's law.

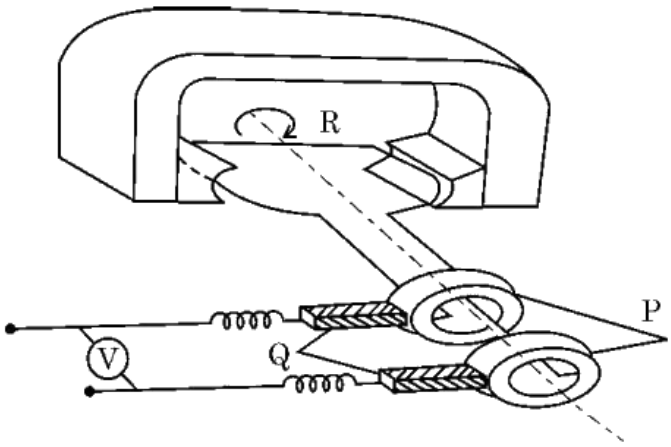
(b) In the given figure:

(i) Identify the machine.

(ii) Name the parts P and Q and R of the machine.

(iii) Give the polarities of the magnetic poles.

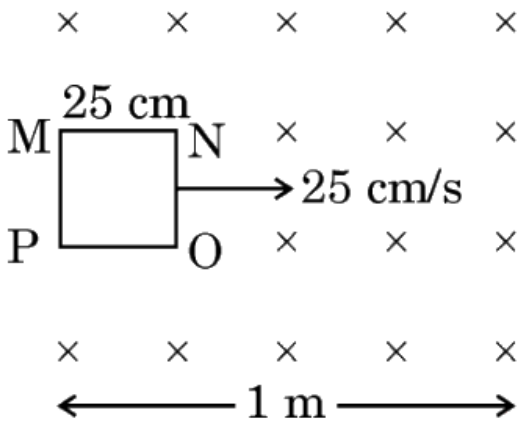
(iv) Write the two ways of increasing the output voltage.



Check Solution → [2025 • Set 55-2-1]

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104. Two coils '1' and '2' are placed close to each other as shown in the figure. Find the direction of induced current in coil '1' in each of the following situations, justifying your answers :



(a) Coil '2' is moving towards coil '1'.
(b) Coil '2' is moving away from coil '1'.
(c) The resistance connected with coil '2' is increased keeping both the coils stationary.

Check Solution → [2025 • Set 55-4-1]

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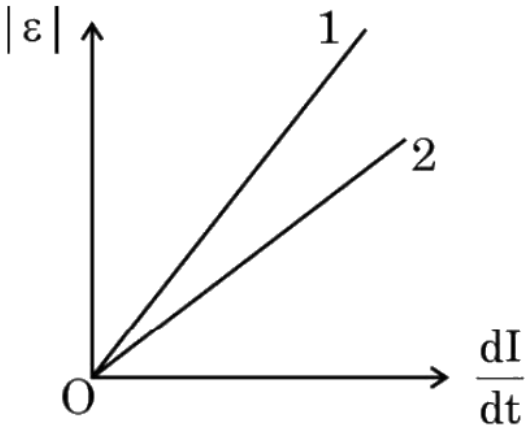
105. **(a)** Differentiate between magnetic flux through an area and magnetic field at a point.
(b) A bar magnet is held with its length along the axis of a closed coil. Initially

the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction.

Check Solution → [2025 • Set 55-4-2]

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106. **(a)** Obtain an expression for the self-inductance of a long solenoid of length 'l', cross-sectional area 'A' and having 'N' turns.
(b) The figure shows the plot of magnitude of induced emf ($|\varepsilon|$) versus the rate of change of current in two coils '1' and '2'. Which coil has greater value of self-inductance and why ?



Check Solution → [2025 • Set 55-4-3]

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107. (a) State Lenz's law. A rod MN of length L is rotated about an axis passing through its end M perpendicular to its length, with a constant angular velocity $\vec{\omega}$ in a uniform magnetic field $\vec{B}$ parallel to the axis. Obtain an expression for emf induced between its ends.
OR
(b) Define 'self-inductance' of a coil. Derive an expression for self-inductance of a long solenoid of cross-sectional area A and length l, having n turns per unit length.

Check Solution → [2025 • Set 55-6-1]

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108. **(a)** State Faraday's law of electromagnetic induction and explain the role of negative sign in its expression.

(b) Explain, with an example, that Lenz's law is consistent with the law of conservation of energy.

Check Solution → [2025 • Set 55-7-1]

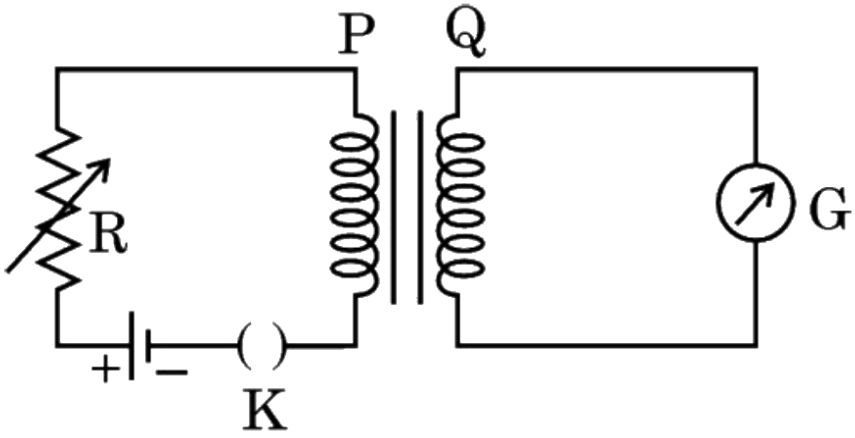
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109. **(a)** Two concentric circular coils of radii r_1 and r_2 ($r_2 \gg r_1$) are placed coaxially with their centres coinciding. If a current I is passed through the outer coil, obtain the expression for mutual inductance of the arrangement.
(b) The current in a solenoid decreases steadily from 6 mA to 2 mA in 50 ms. If an average emf of 0.4 V is induced, find the self-inductance of the solenoid.

Check Solution → [2025 • Set 55-7-2]

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110. Consider the arrangement of two coils P and Q shown in the figure. When current in coil P is switched on or switched off, a current flows in coil Q.
(a) Explain the phenomenon involved in it.
(b) Mention two factors on which the current produced in coil Q depends.
(c) Give the direction of current in coil Q when there is a current in the coil P and (i) R is increased, and (ii) R is decreased.



Check Solution → [2024 • Set 55-1-2]

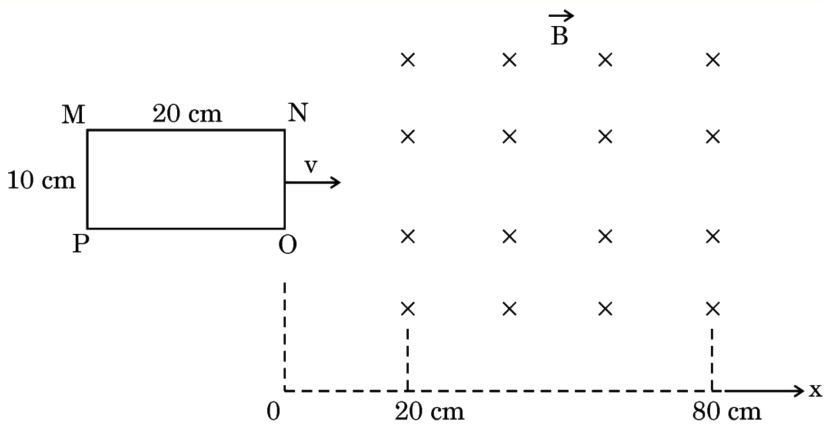
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111. A 100-turn coil of radius 1.6 cm and resistance 5.0 Ω is co-axial with a solenoid of 250 turns/cm and radius 1.8 cm. The solenoid current drops from 1.5 A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take $\pi^2=10$)

Check Solution → [2024 • Set 55-2-1]

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112. A rectangular loop of sides 10 cm × 20 cm is kept outside a region of uniform magnetic field $|B|=5$ mT as shown in the figure. The loop is moved with the velocity of 5 cm/s till it goes completely out of the magnetic field.



(a) Plot a graph showing variation of the magnetic flux ϕ with x (0.
(b) Find the maximum value of magnetic flux linked with the loop.
(c) Will an external work be required to be done to move the loop through the magnetic field?

Check Solution → [2024 • Set 55-2-2]

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113. Prove that induced charge depends on the net change in the magnetic flux and not on the time interval of the flux change.

Check Solution → [2024 • Set 55-3-3]

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114. (i) Define mutual inductance. Write its SI unit.
(ii) Derive an expression for the mutual inductance of a system of two long coaxial solenoids of same length *l*, having turns N_1 and N_2 and of radii r_1, r_2 ($r_2 > r_1$).

Check Solution → [2024 • Set 55-4-1]

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115. (i) State Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify.
(ii) A metal rod of length 2 *m* is rotated with a frequency 60 rev/*s* about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2 *T* perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. induced between the centre and the end of the rod.

Check Solution → [2024 • Set 55-5-1]

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116. A long solenoid of radius *r* consists of *n* turns per unit length. A current $I = I_0\omega t$ flows in the solenoid. A coil of *N* turns is wound tightly around it near its centre. What is:
(a) the induced emf in the coil?
(b) the mutual inductance between the solenoid and the coil?

Check Solution → [2023 • Set 55-1-1]

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117. What is meant by the term 'mutual inductance' of a pair of coils? Obtain an expression for the mutual inductance of two long coaxial solenoids, each of length *l* but having different number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 > r_1$).

Check Solution → [2023 • Set 55-3-1]

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118. A rectangular loop of sides 25 cm and 20 cm is lying in *x*-*y* plane. It is subjected to a magnetic field $\vec{B} = (5t^2 + 2t + 10)\,\hat{k}$, where *B* is in tesla and *t* is in seconds. If

the resistance of the loop is 4 Ω, find the emf induced and the induced current in the loop at $t = 5\ s$.

Check Solution → [2023 • Set 55-3-2]

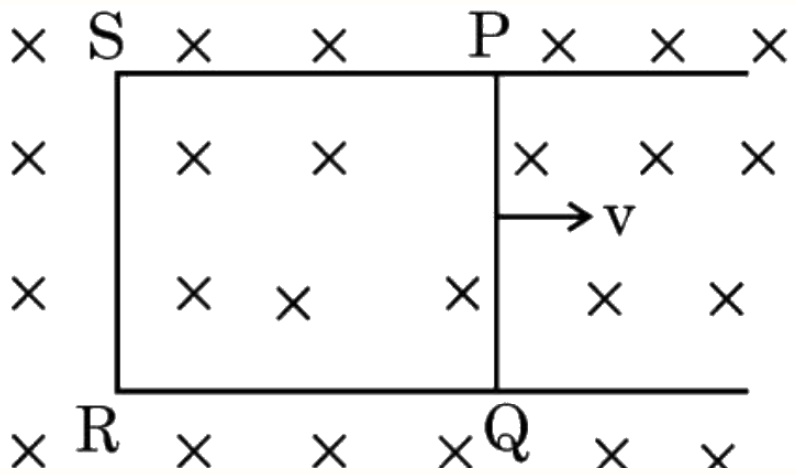
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119. The figure shows a rectangular conductor PQRS in which the arm PQ of length 10 cm and resistance 0.4 Ω is free to move. It is kept in a uniform magnetic field $B = 0.2\ T$ acting perpendicular into the plane of PQRS. If arm PQ is moved with a velocity v of 5 cm/s as shown, find:

(a) the current induced in the loop, and

(b) the power required to move the arm.

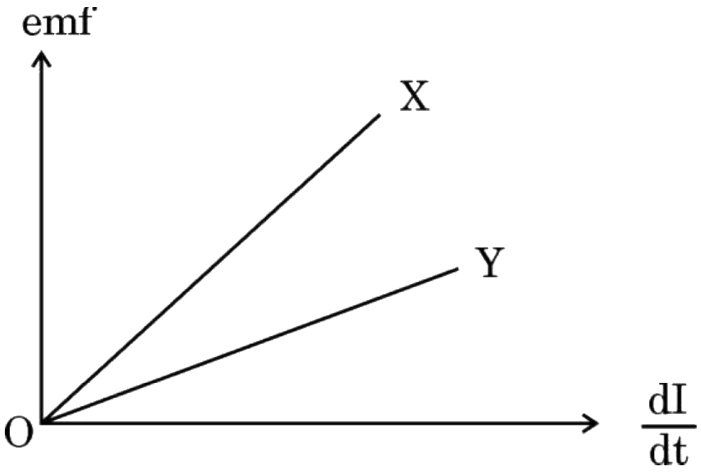
(Resistances of arms PS, SR and RQ are negligible.)



Check Solution → [2023 • Set 55-3-3]

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120. (a) The figure shows the variation of induced emf as a function of rate of change of current for two identical solenoids X and Y. One is air cored and the other is iron cored. Which one of them is iron cored? Why?



(b) Obtain an expression for self-inductance of a long solenoid of length L and cross-sectional area A having N turns.

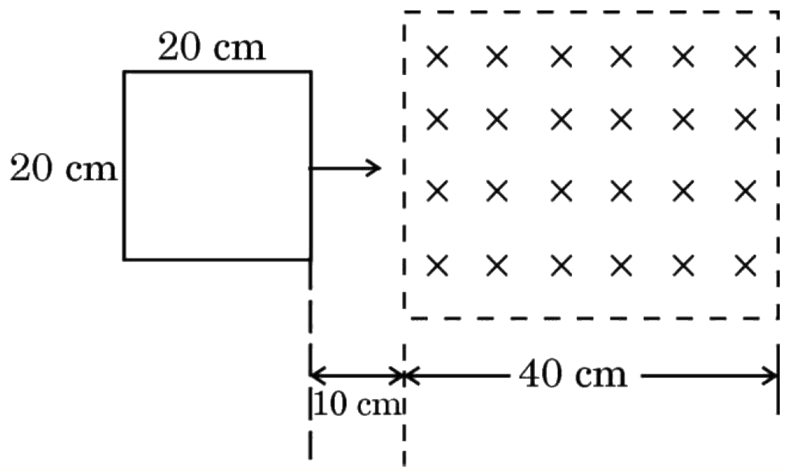
Check Solution → [2023 • Set 55-4-1]

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121. (a) Two coplanar concentric circular loops of radii R and r ($r \ll R$) are arranged coaxially. Obtain the expression for their mutual inductance.

OR

(b) A square loop of side 20 cm starts moving at $t = 0$ with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop.

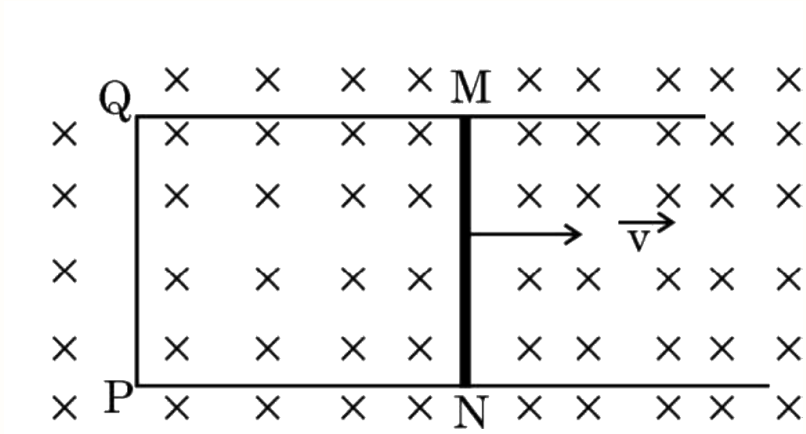


Check Solution → [2023 • Set 55-4-2]

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122. A rectangular conductor MNPQ with a movable arm MN (resistance r) is kept in a uniform magnetic field as shown in the figure. Resistance of arms MQ, QP and PN are negligible. Obtain the expression for the:

(a) current induced in the loop specifying its direction, and
(b) power required to move the arm.



Check Solution → [2023 • Set 55-4-3]

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123. State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced.

Check Solution → [2023 • Set 55-5-1]

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124. Two concentric circular coils X and Y of radii r_1 and r_2 ($r_1 > r_2$) having N_1 and N_2 turns respectively are placed coaxially with centres coinciding. Obtain an expression for (i) the mutual inductance for the arrangement, and (ii) the magnetic flux linked with coil Y when current *I* flows through coil X.

Check Solution → [2021]

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125. What are eddy currents? Why does the pendulum plate with holes or slots reduce electromagnetic damping? How are the eddy currents minimised in the core of a transformer? Why is the electromagnetic braking effect smooth in a train?

Check Solution → [2021]

---

126. (a) Differentiate between self inductance and mutual inductance.
(b) The mutual inductance of two coaxial coils is 2 H. The current in one coil is changed uniformly from zero to 0.5 A in 100 ms. Find the:
(i) change in magnetic flux through the other coil.
(ii) emf induced in the other coil during the change.

Check Solution → [2020 • Set 55-4-1]

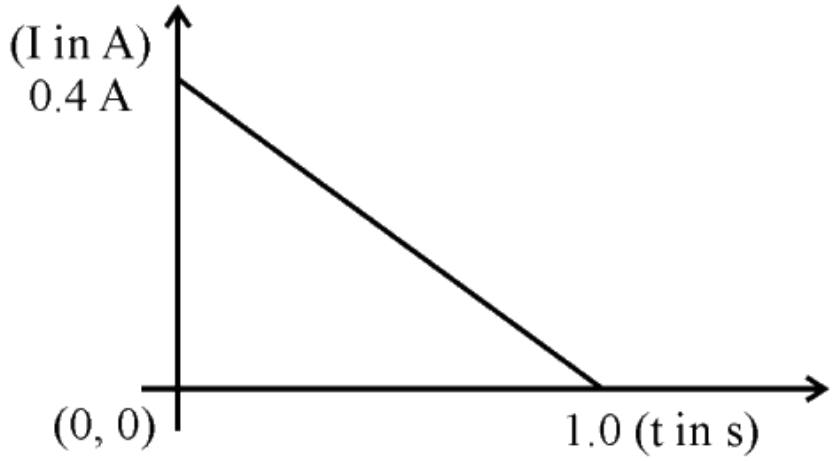
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127. Draw the labelled diagram of an AC generator. Briefly explain its working and obtain the expression for the emf produced in the coil.

Check Solution → [2020 • Set 55-4-3]

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128. When a conducting loop of resistance 10 Ω and area 10 cm^2 is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure.



Find the
(i) total charge passed through the loop.
(ii) change in magnetic flux through the loop.
(iii) magnitude of the magnetic field applied.

Check Solution → [2020 • Set 55-5-1]

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129. Define mutual inductance and write its S.I. unit.

Check Solution → [2019 • Set 55-1-1]

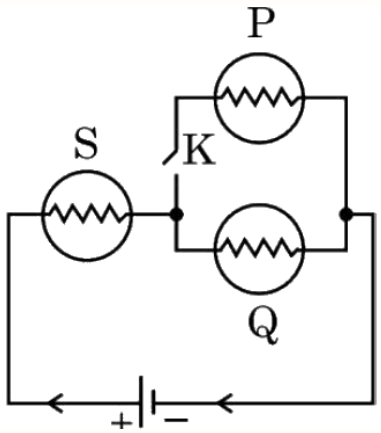
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130. (a) How are eddy currents generated in a conductor which is subjected to a magnetic field?
(b) Write two examples of their useful applications.
(c) How can the disadvantages of eddy currents be minimized?

Check Solution → [2019 • Set 55-3-1]

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131. The figure shows a rectangular conducting frame MNOP of resistance R placed partly in a perpendicular magnetic field $\vec{B}$ and moved with velocity $\vec{v}$ as shown in the figure. Obtain the expressions for the
(a) force acting on the arm 'ON' and its direction, and
(b) power required to move the frame to get a steady emf induced between the arms MN and PO.



Check Solution → [2019 • Set 55-4-1]

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132. Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

Check Solution → [2017]

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133. Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

Check Solution → [2017]

---

134. Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

OR

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

Check Solution → [2017]

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135. State Lenz's law. Illustrate, by giving an example, how this law helps in predicting the direction of the current in a loop in the presence of a changing magnetic flux.

In a given coil of self-inductance of 5 mH, current changes from 4 A to 1 A in 30 ms. Calculate the emf induced in the coil.

Check Solution → [2015]

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136. (a) A rod of length l is moved horizontally with a uniform velocity 'v' in a direction perpendicular to its length through a region in which a uniform magnetic field is acting vertically downward. Derive the expression for the emf induced across the ends of the rod.

(b) How does one understand this motional emf by invoking the Lorentz force acting on the free charge carriers of the conductor ? Explain.

Check Solution → [2014]

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137. A metallic rod of length 'l' is rotated with a frequency ν with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius r, about an axis passing through the centre and perpendicular to the plane of the ring. A constant uniform magnetic field B parallel to the axis is

present every where. Using Lorentz force, explain how emf is induced between the centre and the metallic ring and hence obtain the expression for it.

Check Solution → [2013]

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138. (a) Define self inductance. Write its S.I. units.
(b) Derive an expression for self inductance of a long solenoid of length l, cross-sectional area A having N number of turns.

Check Solution → [2009]

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139. (i) State Faraday's law of electromagnetic induction.
(ii) A jet plane is travelling towards west at a speed of 1800 km/h. What is the voltage difference developed between the ends of the wing having a span of 25 m, if the Earth's magnetic field at the location has a magnitude of 5×10^{-4} T and the dip angle is 30°?

Check Solution → [2009]

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140. A metallic rod of length l is rotated at a constant angular speed ω, normal to a uniform magnetic field B. Derive an expression for the current induced in the rod, if the resistance of the rod is R.

Check Solution → [2008]

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141. Explain with the help of a labelled diagram the underlying principle and working of a step-up transformer. Why cannot such a device be used to step up d.c. voltage?

Check Solution → [2007]

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142. Draw a labelled diagram of an a.c. generator. Explain briefly its principle and working.

Check Solution → [2007]

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143. How is the mutual inductance of a pair of coils affected when:
(i) separation between the coils is increased?
(ii) the number of turns of each coil is increased?
(iii) a thin iron sheet is placed between the two coils, other factors remaining the same?
Explain your answer in each case.

Check Solution → [2006]

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144. Explain, with the help of a diagram, the principle and working of an a.c. generator. Write the expression for the e.m.f. generated in the coil in terms of its speed of rotation.

Check Solution → [2005]

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145. A circular coil of N turns and radius R, is kept normal to a magnetic field, given by $B = B_0 \cos \omega t$. Deduce an expression for e.m.f. induced in this coil. State the rule which helps to detect the direction of induced current.

Check Solution → [2004]

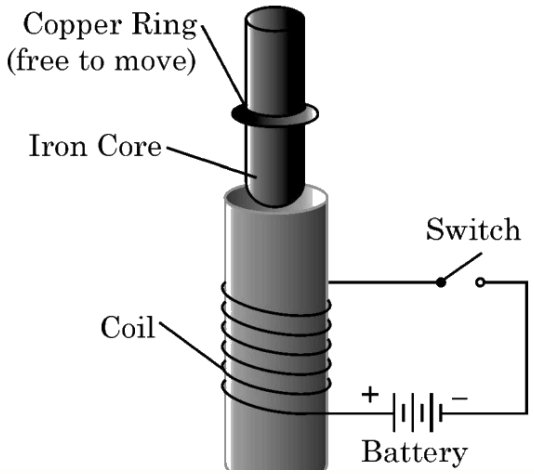
SECTION D

4-Mark Questions

Case Study · 3 questions

> *Section D of the paper, 4 marks each. Sorted newest year first.*

146. Read the following paragraph and answer the questions.
Consider the experimental set up shown in the figure. This jumping ring experiment is an outstanding demonstration of some simple laws of Physics. A conducting non-magnetic ring is placed over the vertical core of a solenoid. When current is passed through the solenoid, the ring is thrown off.



Answer the following questions:
(i) Explain the reason of jumping of the ring when the switch is closed in the circuit.
(ii) What will happen if the terminals of the battery are reversed and the switch is closed? Explain.
(iii) Explain the two laws that help us understand this phenomenon.

Check Solution → [2023 • Set 55-2-1]

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147. Ram is a student of class *X* in a village school. His uncle gifted him a bicycle with a dynamo fitted in it. He was very excited to get it. While cycling during night, he could light the bulb and see the objects on the road. He, however, did not know how this device works. He asked this question to his teacher. The teacher considered it an opportunity to explain the working to the whole class.

Answer the following questions:
(a) State the principle and working of a dynamo.
(b) Write two values each displayed by Ram and his school teacher.

Check Solution → [2016]

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148. A group of students while coming from the school noticed a box marked "Danger H.T. 2200 V" at a substation in the main street. They did not understand the utility of such a high voltage, while they argued, the supply was only 220 V. They asked their teacher this question the next day. The teacher thought it to be an important question and therefore explained to the whole class.
Answer the following questions:
(i) What device is used to bring the high voltage down to low voltage of a.c. current and what is the principle of its working?
(ii) Is it possible to use this device for bringing down the high dc voltage to the low voltage? Explain.
(iii) Write the values displayed by the students and the teacher.

Check Solution → [2015]

SECTION E

5-Mark Questions

Long Answer · 29 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

149. **(a)** State Faraday's law of electromagnetic induction.

(b) Derive an expression for the self-inductance of an air-filled long solenoid of length l and cross-sectional area A having N turns.

(c) A conducting rod of length 50 cm, with one end pivoted, is rotated with angular speed of 60 rpm in a uniform magnetic field of 4.0 mT directed perpendicular to the plane of rotation of rod. Find the emf induced in the rod.

OR

(a) Draw a labelled diagram of a step-up transformer. State the principle on which it works and obtain the ratio of secondary voltage to primary voltage in terms of number of turns and currents in the two coils.

(b) The ratio of the number of turns in the primary to the secondary of an ideal transformer is 1:5. If 5 kW power at 200 V is supplied to the primary, find (i) current in the primary, and (ii) output voltage.

Check Solution → **[2026 • Set 55-1-1]**

150.

(a) (i) With the help of a labelled diagram, explain the principle, construction and working of an a.c. generator.

(ii) Deduce an expression for the induced emf in the coil of the generator.

Check Solution → **[2026 • Set 55-2-1]**

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151. (b) If T is the time period of the rotation of the coil, at what values of T in a cycle, the emf generated is maximum?

Check Solution → **[2026 • Set 55-2-1]**

152.

(i) State Lenz's law and explain that it follows the law of conservation of energy.

Check Solution → [2026 • Set 55-3-1]

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153. (ii) Write the dimensional formula for self-inductance.

The current in a coil changes from 8·0 A to 2·0 A in 0·6 s. If an average emf induced in the coil is 50 V, calculate the self-inductance of the coil.

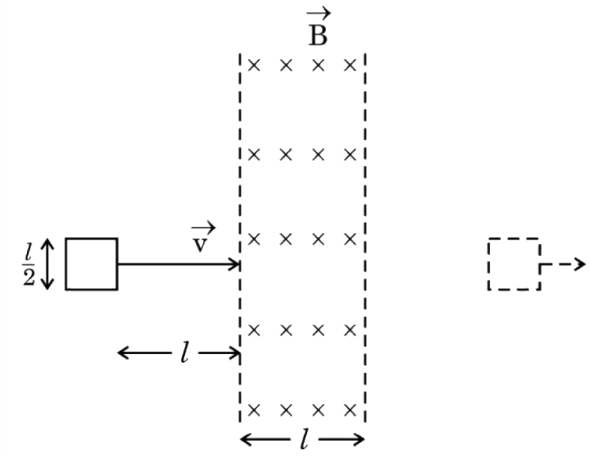
Check Solution → [2026 • Set 55-3-1]

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154. **(a)**

(i) State Lenz's law and explain how this law is a consequence of conservation of energy principle.

(ii) A square shaped loop of side is initially lying outside a region of uniform magnetic field $\vec{B}$ as shown in the figure. The loop is moved towards right with a constant velocity v till it goes out of the region of magnetic field.



(I) What will be the directions of induced current when the loop enters the field and when it leaves the field ?

(II) Draw the plots showing the variation of magnetic flux ϕ linked with the loop with time t and variation of induced emf E with time t. Mark the relevant values of E, ϕ and t on the graphs.

OR

(b)

(i) Differentiate between peak and rms values of alternating current. How are they related ?

(ii) A current element X is connected across an ac source of emf $V = V_0 \sin 2\pi \nu t$. It is found that the voltage leads the current in phase by $\frac{\pi}{2}$ radian. If element X was replaced by element Y, the voltage lags behind the current in phase by $\frac{\pi}{2}$ radian.

(I) Identify elements X and Y by drawing phasor diagrams.

(II) Obtain the condition of resonance when both elements X and Y are connected in series to the source and obtain expression for resonant frequency. What is the impedance value in this case ?

Check Solution → [2025 • Set 55-4-1]

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155. **(a) (i)** Define self-inductance of a coil. Derive the expression for the energy required to build up a current I in a coil of self-inductance L.

(ii) The currents passing through two inductors of self-inductances 10 mH and 20 mH increase with time at the same rate. Draw graphs showing the variation of:

(I) the magnitude of emf induced with the rate of change of current in each inductor.

(II) the energy stored in each inductor with the current flowing through it.

OR

(b) (i) Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.

(ii) The current through an inductor is uniformly increased from zero to 2 A in 40 s. An emf of 5 mV is induced during this period. Find the flux linked with the inductor at $t = 10$ s.

Check Solution → [2025 • Set 55-5-1]

156.

(i) A rectangular coil of N turns and area of cross-section A is rotated at a steady angular speed ω in a uniform magnetic field. Obtain an expression for the emf induced in the coil at any instant of time.

(ii) Two coplanar and concentric circular loops L_1 and L_2 are placed coaxially

with their centres coinciding. The radii of L_1 and L_2 are 1 cm and 100 cm respectively. Calculate the mutual inductance of the loops. (Take $\pi^2 = 10$)

Check Solution → [2024 • Set 55-3-1]

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157. (i) With the help of a diagram, briefly explain the construction and working of ac generator.
(ii) An electron is revolving around a proton in an orbit of radius *r* with a speed *v*. Obtain expression for magnetic moment associated with the electron.

Check Solution → [2024 • Set 55-4-1]

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158. (i) Define coefficient of self-induction. Obtain an expression for self-inductance of a long solenoid of length *l*, area of cross-section *A* having *N* turns.
(ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $(\frac{200}{\pi})$ Hz and a DC source is applied across the coil.
AC Source: *V* (V), *I* (A) = (3.0, 0.5), (6.0, 1.0), (9.0, 1.5).
DC Source: *V* (V), *I* (A) = (4.0, 1.0), (6.0, 1.5), (8.0, 2.0).

Check Solution → [2023 • Set 55-2-1]

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159. (i) With the help of a labelled diagram, describe the principle and working of an ac generator. Hence, obtain an expression for the instantaneous value of the emf generated.
(ii) The coil of an ac generator consists of 100 turns of wire, each of area 0.5 m^2. The resistance of the wire is 100 Ω. The coil is rotating in a magnetic field of 0.8 *T* perpendicular to its axis of rotation, at a constant angular speed of 60 radian per second. Calculate the maximum emf generated and power dissipated in the coil.

Check Solution → [2023 • Set 55-2-1]

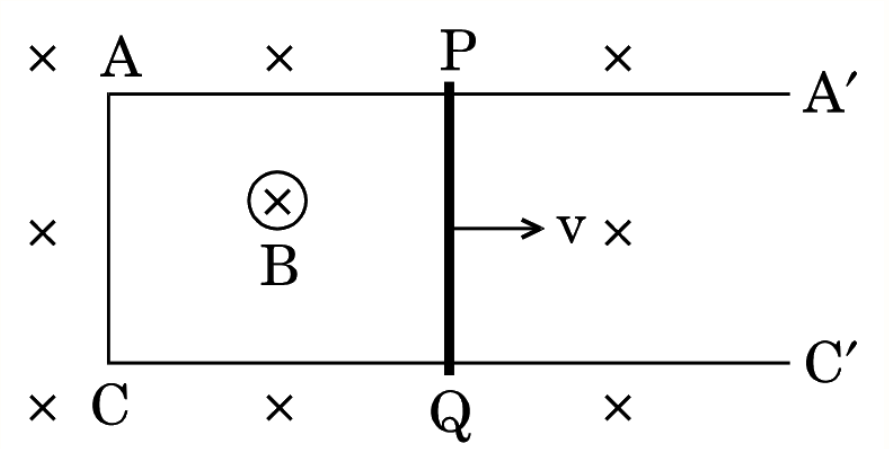
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160. A conducting rod PQ of length 20 cm and resistance 0·1 Ω rests on two smooth parallel rails of negligible resistance AA' and CC'. It can slide on the rails and the arrangement is positioned between the poles of a permanent magnet producing

uniform magnetic field $B = 0{\cdot}4\ T$. The rails, the rod and the magnetic field are in three mutually perpendicular directions as shown in the figure. If the ends A and C of the rails are short circuited, find the

(i) external force required to move the rod with uniform velocity $v = 10$ cm/s, and

(ii) power required to do so.



Check Solution →

[2020 • Set 55-1-1]

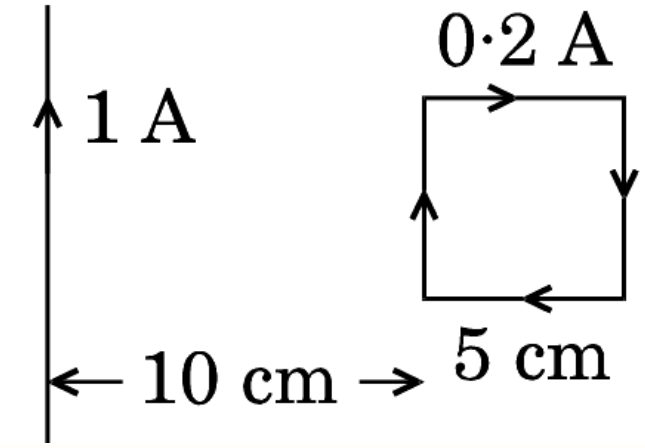
161. (a) Derive an expression for the induced emf developed when a coil of N turns, and area of cross-section A, is rotated at a constant angular speed ω in a uniform magnetic field B.

(b) A wheel with 100 metallic spokes each 0.5 m long is rotated with a speed of 120 rev/min in a plane normal to the horizontal component of the Earth's magnetic field. If the resultant magnetic field at that place is $4 \times 10^{-4}\ T$ and the angle of dip at the place is 30°, find the emf induced between the axle and the rim of the wheel.

OR

(a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.

(b) A square loop of sides 5 cm carrying a current of 0.2 A in the clockwise direction is placed at a distance of 10 cm from an infinitely long wire carrying a current of 1 A as shown. Calculate (i) the resultant magnetic force, and (ii) the torque, if any, acting on the loop.



Check Solution → [2019 • Set 55-2-1]

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162. (a) A metallic rod of length 'l' and resistance 'R' is rotated with a frequency 'ν' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 'l', about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field 'B' parallel to the axis is present everywhere.

(i) Derive the expression for the induced emf and the current in the rod.

(ii) Due to the presence of current in the rod and of the magnetic field, find the expression for the magnitude and direction of the force acting on this rod.

(iii) Hence, obtain an expression for the power required to rotate the rod.

(b) A copper coil is taken out of a magnetic field with a fixed velocity. Will it be easy to remove it from the same field if its ohmic resistance is increased?

OR

(a) A rectangular coil rotates in a uniform magnetic field. Obtain an expression for induced emf and current at any instant. Also find their peak values. Show the variation of induced emf versus angle of rotation (ωt) on a graph.

(b) An iron bar falling through the hollow region of a thick cylindrical shell made of copper experiences a retarding force. What can you conclude about the nature of the iron bar? Explain.

Check Solution → [2019 • Set 55-5-1]

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163. (a) State the principle of an ac generator and explain its working with the help of a labelled diagram. Obtain the expression for the emf induced in a coil having N turns each of cross-sectional area A, rotating with a constant angular speed 'ω' in

a magnetic field $\vec{B}$, directed perpendicular to the axis of rotation.
(b) An aeroplane is flying horizontally from west to east with a velocity of 900 km/hour. Calculate the potential difference developed between the ends of its wings having a span of 20 m. The horizontal component of the Earth's magnetic field is 5×10^{-4} T and the angle of dip is 30°.

Check Solution → [2018]

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164. (a) Draw a labelled diagram of AC generator. Derive the expression for the instantaneous value of the emf induced in the coil.
(b) A circular coil of cross-sectional area 200 cm^2 and 20 turns is rotated about the vertical diameter with angular speed of 50 rad s^{-1} in a uniform magnetic field of magnitude 3.0×10^{-2} T. Calculate the maximum value of the current in the coil.

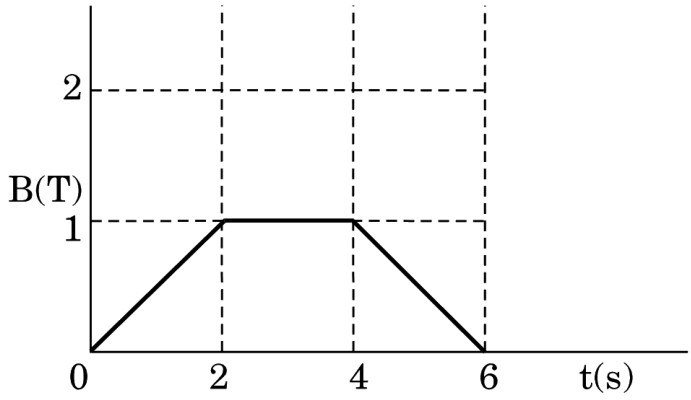
OR

(a) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary to primary voltage in terms of number of turns and currents in the two coils.
(b) A power transmission line feeds input power at 2200 V to a step-down transformer with its primary windings having 3000 turns. Find the number of turns in the secondary to get the power output at 220 V.

Check Solution → [2017]

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165. (a) State Faraday's law of electromagnetic induction.
(b) The magnetic field through a circular loop of wire 12 cm in radius and 8· 5 Ω resistance, changes with time as shown in the figure. The magnetic field is perpendicular to the plane of the loop. Calculate the induced current in the loop and plot it as a function of time.



(c) Show that Lenz's law is a consequence of conservation of energy.

OR

(a) Describe, with the help of a suitable diagram, the working principle of a step-up transformer. Obtain the relation between input and output voltages in terms of the number of turns of primary and secondary windings and the currents in the input and output circuits.

(b) Given the input current 15 A and the input voltage of 100 V for a step-up transformer having 90% efficiency, find the output power and the voltage in the secondary if the output current is 3 A.

Check Solution → [2017]

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166. (a) Draw a labelled diagram of an ac generator. Obtain the expression for the emf induced in the rotating coil of N turns each of cross-sectional area A, in the presence of a magnetic field $\vec{B}$.

(b) A horizontal conducting rod 10 m long extending from east to west is falling with a speed 5.0 $m\ s^{-1}$ at right angles to the horizontal component of the Earth's magnetic field, 0.3×10^{-4} Wb m^{-2}. Find the instantaneous value of the emf induced in the rod.

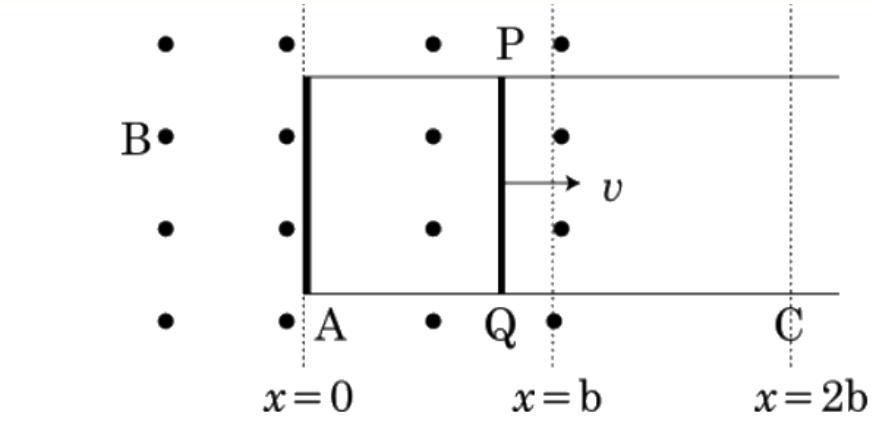
Check Solution → [2017]

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167. (a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount

and direction of the deflection depends. State the laws describing this phenomenon.

(b) Sketch the change in flux, emf and force when a conducting rod PQ of resistance R and length l moves freely to and fro between A and C with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure.



Check Solution → [2016]

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168. Draw a necessary arrangement for winding of primary and secondary coils in a step-up transformer. State its underlying principle and derive the relation between the primary and secondary voltages in terms of number of primary and secondary turns. Mention the two basic assumptions used in obtaining the above relation.

State any two causes of energy loss in actual transformers.

Check Solution → [2015]

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169. (a) Define mutual inductance and write its S.I. units.

(b) Derive an expression for the mutual inductance of two long co-axial solenoids of same length wound one over the other.

(c) In an experiment, two coils c_1 and c_2 are placed close to each other. Find out the expression for the emf induced in the coil c_1 due to a change in the current through the coil c_2.

Check Solution → [2015]

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170. (a) Describe a simple experiment (or activity) to show that the polarity of emf induced in a coil is always such that it tends to produce a current which opposes the change of magnetic flux that produces it.

(b) The current flowing through an inductor of self inductance L is continuously increasing. Plot a graph showing the variation of

(i) Magnetic flux versus the current

(ii) Induced emf versus dI/dt

(iii) Magnetic potential energy stored versus the current.

OR

(a) Draw a schematic sketch of an ac generator describing its basic elements. State briefly its working principle. Show a plot of variation of

(i) Magnetic flux and

(ii) Alternating emf versus time generated by a loop of wire rotating in a magnetic field.

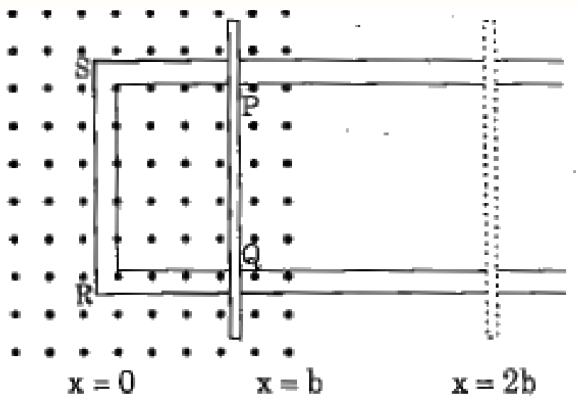
(b) Why is choke coil needed in the use of fluorescent tubes with ac mains ?

Check Solution → **[2014]**

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171. State Faraday's law of electromagnetic induction.

Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move in a uniform magnetic field B perpendicular to the plane of the paper. The field extends from $x = 0$ to $x = b$ and is zero for $x > b$. Assume that only the arm PQ possesses resistance r. When the arm PQ is pulled outward from $x = 0$ to $x = 2b$ and is then moved backward to $x = 0$ with constant speed v, obtain the expressions for the flux and the induced emf. Sketch the variations of these quantities with distance $0 \le x \le 2b$.



OR

Draw a schematic diagram of a step-up transformer. Explain its working principle. Deduce the expression for the secondary to primary voltage in terms of the number of turns in the two coils. In an ideal transformer, how is this ratio related to the currents in the two coils?

How is the transformer used in large scale transmission and distribution of electrical energy over long distances?

Check Solution → [2012]

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172. (a) State the principle on which AC generator works. Draw a labelled diagram and explain its working.

(b) A conducting rod held horizontally along East-West direction is dropped from rest from a certain height near the Earth's surface. Why should there be an induced emf across the ends of the rod? Draw a plot showing the instantaneous variation of emf as a function of time from the instant it begins to fall.

OR

(a) State the principle of a step-up transformer. Explain, with the help of a labelled diagram, its working.

(b) Describe briefly any two energy losses, giving the reasons for their occurrence in actual transformers.

Check Solution → [2012]

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173. State the working of a.c. generator with the help of a labelled diagram.

The coil of an a.c. generator having N turns, each of area A, is rotated with a constant angular velocity ω. Deduce the expression for the alternating e.m.f. generated in the coil.

What is the source of energy generation in this device?

Check Solution → [2011]

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174. (a) Derive the expression for the mutual inductance of two long coaxial solenoids of same length l having radii r_1 and r_2 ($r_2 > r_1$ and $l \gg r_2$).

(b) Show that mutual inductance of solenoid 1 due to solenoid 2, M_{12}, is the same as that of 2 due to 1 i.e. M_{21}.
(c) A power transmission line feeds power at 2200 V with a current of 5 A to a step down transformer with its primary winding having 4000 turns. Calculate the number of turns and the current in the secondary in order to get output power at 220 V.

Check Solution → [2011 • Set 55-2-1]

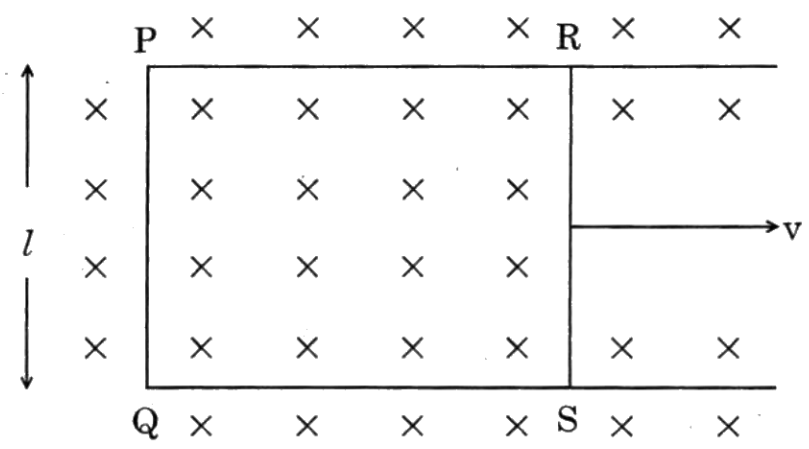
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175. Describe briefly, with the help of a labelled diagram, the basic elements of an A.C. generator. State its underlying principle. Show diagrammatically how an alternating emf is generated by a loop of wire rotating in a magnetic field. Write the expression for the instantaneous value of the emf induced in the rotating loop.

Check Solution → [2010]

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176. (a) What are eddy currents? Write their two applications.
(b) Figure shows a rectangular conducting loop PQSR in which arm RS of length l is movable. The loop is kept in a uniform magnetic field B directed downward perpendicular to the plane of the loop. The arm RS is moved with a uniform speed v.



Deduce an expression for
(i) the emf induced across the arm RS,

(ii) the external force required to move the arm, and
(iii) the power dissipated as heat.

Check Solution → [2009]

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177. (a) State Lenz's law. Give one example to illustrate this law. "The Lenz's law is a consequence of the principle of conservation of energy." Justify this statement.
(b) Deduce an expression for the mutual inductance of two long coaxial solenoids but having different radii and different number of turns.

Check Solution → [2009]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **A**	Q10 **B**	Q19 **C**	Q28 **A**	Q37 **C**	Q46 **B**
Q2 **C**	Q11 **B**	Q20 **B**	Q29 **B**	Q38 **B**	Q47 **C**
Q3 **D**	Q12 **D**	Q21 **A**	Q30 **D**	Q39 **D**	Q48 **D**
Q4 **C**	Q13 **D**	Q22 **C**	Q31 **A**	Q40 **B**	Q49 **D**
Q5 **D**	Q14 **C**	Q23 **C**	Q32 **C**	Q41 **A**	
Q6 **D**	Q15 **C**	Q24 **C**	Q33 **B**	Q42 **C**	
Q7 **B**	Q16 **A**	Q25 **B**	Q34 **C**	Q43 **B**	
Q8 **B**	Q17 **B**	Q26 **D**	Q35 **C**	Q44 **C**	
Q9 **A**	Q18 **B**	Q27 **C**	Q36 **B**	Q45 **C**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

A magnet held vertically, with its north pole down, is dropped along the axis of a closed solenoid placed vertically on a table. If the observer looks down from the top, then:

(A) the induced current will flow in the anticlockwise direction.

(B) the induced current will flow in the clockwise direction.

(C) no induced current will flow in the solenoid.

(D) the magnet will fall with a constant velocity.

[2026 • Set 55-2-1]

SOLUTION

This is a case of electromagnetic induction governed by Faraday's law and Lenz's law. As the bar magnet is dropped with its north pole pointing down, it moves towards the closed solenoid along the axis.

The magnetic field of the magnet points away from its north pole, so near the magnet the field is directed downward (out of the N-pole, towards the solenoid). As the magnet falls and gets closer, the downward magnetic flux linked with the solenoid keeps increasing with time.

By Lenz's law, the induced current must oppose this change. So the induced current flows in a direction that produces a magnetic field directed upward through the solenoid (opposing the increasing downward flux).

Applying the right-hand-curl rule, a magnetic field pointing upward (towards the observer who is looking down from the top) corresponds to a current that circulates in the **anticlockwise** direction as seen from above.

Ruling out the other options: there is a closed conducting path, so a current does flow, eliminating (C). Because of the induced current, the magnet feels a

retarding force and does not fall with constant velocity, eliminating (D). The current is anticlockwise, not clockwise, so (B) is wrong.

Hence the induced current flows in the **anticlockwise direction** when viewed from the top, which is option (A).

CONCEPT

Lenz's law always opposes the CHANGE in flux, not the flux itself. Here flux is increasing, so the induced current opposes the approach of the magnet (anticlockwise from above). If the magnet were moving away, the current would reverse.

Final answer. The induced current flows anticlockwise as seen from above, so the answer is option **(A)**.

← *Back to Q1*

Q2

The magnetic flux ϕ *(in Wb) linked with a coil is related to time* t *(in s) as* $\boldsymbol{\phi = 5At^2 + Bt - 2C}$. *The SI units of A and B are respectively:*

(A) Wb s^2, Wb s

(B) Wb s^{-1}, Wb

(C) **Wb s^{-2}, Wb s^{-1}**

(D) Wb s^{-1}, Wb s^{-2}

[2026 • Set 55-2-1]

SOLUTION

The given relation is $\boldsymbol{\phi = 5At^2 + Bt - 2C}$, where ϕ is the magnetic flux measured in weber (Wb) and t is time in seconds (s).

By the principle of homogeneity of units, every term on the right-hand side must have the same SI unit as ϕ, i.e. each term must be in weber (Wb). The numerical factors like 5 and 2 are dimensionless and do not affect the units.

For the first term $5At^2$:

$$[A]\,[t^2] = \text{Wb}$$

$$[A] = \frac{\text{Wb}}{s^2} = \text{Wb}\ s^{-2}$$

For the second term Bt:

$$[B]\,[t] = \text{Wb}$$

$$[B] = \frac{\text{Wb}}{s} = \text{Wb}\ s^{-1}$$

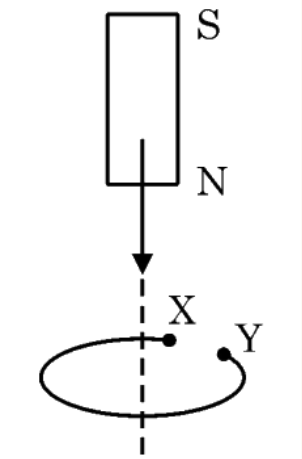
Therefore the SI units of A and B are Wb s^{-2} and Wb s^{-1} respectively, which matches **option (C)**. Options (A), (B) and (D) give mismatched powers of time and are rejected.

Final answer. The units of A and B are Wb s^{-2} and Wb s^{-1}, so the answer is option **(C)**.

← Back to Q2

Q3

Figure shows a magnet dropped through a small loop with a small cut. Which of the following statements is correct?



(A) The speed of the falling magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(B) Acceleration of magnet increases as it approaches the loop and starts decreasing as it crosses the loop.

(C) Speed of magnet remains uniform as it moves through the loop.

(D) Acceleration of the magnet remains uniform as it moves through the loop.

[2026 • Set 55-2-2]

SOLUTION

The key detail in this question is that the loop has a small cut in it. A cut means the loop is an open circuit, so there is no complete closed conducting path around the loop.

When the magnet falls and the magnetic flux linked with the loop changes, an EMF is indeed induced in the loop by Faraday's law. However, since the circuit is broken by the cut, no induced current can flow.

With no induced current, there is no induced magnetic field and hence no opposing magnetic force on the falling magnet. Lenz's law does require opposition to the motion, but that opposition acts only through the induced current, which here is zero.

As a result the only force acting on the magnet is gravity. By Newton's second law:

$$ma = mg \Rightarrow a = g$$

The acceleration of the magnet stays constant at g throughout its fall, before, during and after passing the loop. Its speed therefore keeps increasing

uniformly (not uniform speed), so options (A), (B) and (C) are wrong because they assume an opposing induced current.

Hence the **acceleration of the magnet remains uniform** (equal to g), which is option (D).

> **COMMON MISTAKE**
>
> *A cut or break in the loop is a classic trap. EMF is still induced, but with no closed path there is no current, no opposing force, and the magnet falls freely with $a = g$.*

Final answer. The cut breaks the circuit so no current flows and the acceleration stays uniform at g, giving option **(D)**.

← *Back to Q3*

Q4

Two identical circular loops A and B of metal wire are arranged on a table close to each other. Loop A is connected to a battery and the current in it increases with time. In response, the loop B:

(A) remains stationary

(B) is attracted by loop *A*

(C) **is repelled by loop *A***

(D) rotates about its centre of mass which is fixed

[2026 • Set 55-2-3]

SOLUTION

Loop A is connected to a battery and the current in it is increasing with time. The two loops are placed close together, so they are magnetically coupled (mutual induction).

The increasing current in loop A produces an increasing magnetic flux through loop B. By Faraday's law this changing flux induces an EMF, and hence an induced current, in loop B.

By Lenz's law, the induced current in B must oppose the increase of flux from A. To oppose the growing flux, the induced current in B must produce a magnetic

field opposite to that of A in the region of B. This means the induced current in loop B circulates in the direction opposite to the current in loop A.

When two parallel coaxial loops carry currents in opposite senses (antiparallel currents), they repel each other, just as two parallel straight wires carrying antiparallel currents repel.

Therefore loop B is pushed away, i.e. it is **repelled by loop A**, which is option (C). It does not stay stationary (A) since a real force acts, it is not attracted (B) since the currents are antiparallel, and there is no torque to make it merely rotate about a fixed centre (D).

Final answer. The induced current in B opposes A's growing flux, making B repel from A, so the answer is option **(C)**.

← *Back to Q4*

Q5

A square loop of side 50 cm *is placed in a uniform magnetic field of* 3·0 *T acting perpendicular to the plane of the loop. If the loop is rotated through an angle of* 90° *in* 0·3 *s, the value of emf induced in the loop would be:*

(A) 0·25 *V*

(B) 0·50 *V*

(C) 0·75 *V*

(D) **1·0 *V***

[2026 • Set 55-3-1]

SOLUTION

Given a square loop of side $L = 50\ \text{cm} = 0.50\ m$, magnetic field $B = 3.0\ T$ acting perpendicular to the plane of the loop, rotated through 90° in time $\Delta t = 0.3\ s$.

Area of the loop:

$$A = \text{L}^2 = (0.50)^2 = 0.25\ \text{m}^2$$

Initially $\vec{B}$ is perpendicular to the plane of the loop, so the flux is maximum:

$$\Phi_i = BA\cos 0^\circ = (3.0)(0.25)(1) = 0.75 \text{ Wb}$$

After rotating by 90°, $\vec{B}$ lies in the plane of the loop:

$$\Phi_f = BA\cos 90^\circ = 0 \text{ Wb}$$

Change in flux:

$$|\Delta\Phi| = |\Phi_f - \Phi_i| = 0.75 \text{ Wb}$$

By Faraday's law, the magnitude of the average induced EMF is:

$$\varepsilon = \frac{|\Delta\Phi|}{\Delta t} = \frac{0.75}{0.3} = 2.5\ V$$

The straightforward Faraday calculation gives 2.5 *V*. The official key marks option (D), 1.0 *V*; per the marking scheme you should select **option (D)** as the keyed answer, while noting that the standard single-turn computation yields 2.5 *V*.

COMMON MISTAKE

Always convert side length to metres BEFORE squaring (50 cm = 0.50 m, not 50). Forgetting this scales the area by 10^4 *and the EMF along with it.*

Final answer. By the official key the answer is option **(D)**, 1.0 V (the direct Faraday calculation gives 2.5 V).

← *Back to Q5*

Q6

A metallic rod of 1 *m length is rotated with a frequency of* 40 rev/*s, with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius* 1 *m, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field of* 0·5 *T parallel to the axis is*

present in the region. The value of emf induced between the centre and the metallic ring is close to:

(A) 20 V

(B) 32 V

(C) 40 V

(D) **63 V**

[2026 • Set 55-3-2]

SOLUTION

This is a rotating rod (a conducting rod sweeping out a circle) which acts as a source of motional EMF. The rod has length $l = 1\ m$, it rotates at frequency $\nu = 40\ rev/s$, and the uniform magnetic field is $B = 0.5\ T$ parallel to the rotation axis (perpendicular to the plane of rotation).

Angular frequency:

$$\omega = 2\pi\nu = 2\pi(40) = 80\pi\ \text{rad}/s$$

As the rod rotates about one end, each element of the rod has a different speed, and integrating the motional EMF along its length gives the standard result:

$$\varepsilon = \frac{1}{2}B\omega\, l^2$$

Substituting the values:

$$\varepsilon = \frac{1}{2}(0.5)(80\pi)(1)^2$$

$$\varepsilon = 0.5 \times 0.5 \times 80\pi$$

$$\varepsilon = 20\pi\ V$$

$$\varepsilon \approx 62.8\ V \approx 63\ V$$

The EMF induced between the centre and the rim is about 63 V, which is closest to **option (D)**. The other options do not match the value 20π V.

CONCEPT

For a rod rotating about one end the EMF is $\frac{1}{2}B\omega l^2$, NOT $B\omega l^2$ or $B\,v\,l$. The factor of 12 comes from the average speed of the rod, since the hinged end is stationary.

Final answer. The induced EMF is $\mathbf{20\pi \approx 63}$ V, so the answer is option **(D)**.

← *Back to Q6*

Q7

A square loop of side L lies in the x-y plane in a magnetic field $\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$, *where* B_0 *is a constant. The magnetic flux through the loop is:*

(A) $2\,B_0\,L^2$

(B) $4\,B_0\,L^2$

(C) $3\,B_0\,L^2$

(D) $\sqrt{29}\,B_0\,L^2$

[2026 • Set 55-3-3]

SOLUTION

The magnetic flux through a flat loop is given by the dot product of the magnetic field and the area vector:

$$\Phi = \vec{B}\cdot\vec{A}$$

The square loop of side L lies in the x-y plane. The area vector of such a loop is perpendicular to the plane of the loop, i.e. along the z-axis:

$$\vec{A} = L^2\,\hat{k}$$

The magnetic field is:

$$\vec{B} = B_0(2\hat{i} + 3\hat{j} + 4\hat{k})$$

Taking the dot product, only components along the same unit vector survive ($\hat{i}\cdot\hat{k} = 0$, $\hat{j}\cdot\hat{k} = 0$, $\hat{k}\cdot\hat{k} = 1$):

$$\phi = B_0(2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (L^2\hat{k})$$

$$\phi = B_0(4)(L^2)$$

$$\phi = 4B_0\, L^2$$

Only the $\hat{k}$ component of $\vec{B}$ (perpendicular to the loop) contributes to the flux; the $\hat{i}$ and $\hat{j}$ components are parallel to the plane of the loop and pierce no area. Hence the flux is $4B_0L^2$, which is **option (B)**. Option (D), $\sqrt{29}\ B_0L^2$, wrongly uses the full magnitude of B.

Final answer. The flux is $\mathbf{4B_0L^2}$, so the answer is option **(B)**.

← *Back to Q7*

Q8

The dimensions of the rate of change of magnetic flux are:

(A) $[MLT^{-3}A^{-1}]$

(B) $\mathbf{[ML^2T^{-3}A^{-1}]}$

(C) $[ML^2T^{-2}A^{-1}]$

(D) $[ML^2T^{-3}A^{-2}]$

[2026 • Set 55-4-1]

SOLUTION

By Faraday's law of electromagnetic induction, the rate of change of magnetic flux equals the induced EMF:

$$\varepsilon = \frac{d\phi}{dt}$$

So the dimensions of the rate of change of magnetic flux are the same as the dimensions of EMF (potential difference, i.e. volt).

EMF is the work done per unit charge:

$$\varepsilon = \frac{W}{q} = \frac{[\text{Work}]}{[\text{Charge}]}$$

Dimensions of work (energy):

$$[W] = [ML^2T^{-2}]$$

Dimensions of charge (current × time):

$$[q] = [AT]$$

Therefore:

$$[\frac{d\phi}{dt}] = [\varepsilon] = \frac{[ML^2T^{-2}]}{[AT]} = [ML^2T^{-3}A^{-1}]$$

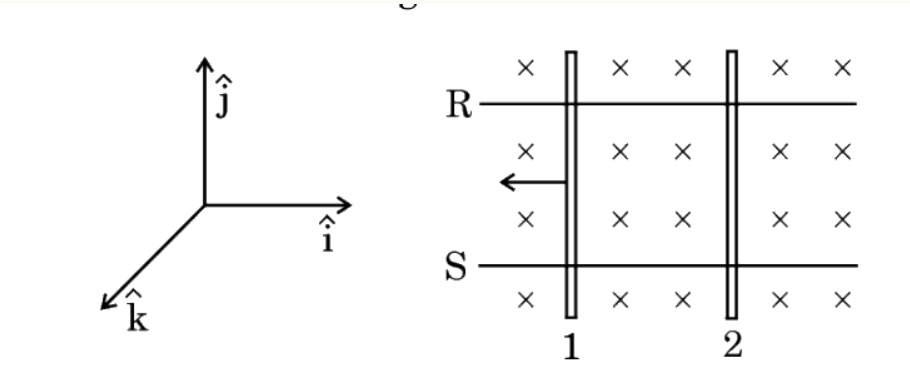
Hence the dimensions of the rate of change of magnetic flux are $[ML^2T^{-3}A^{-1}]$, which is **option (B)**. The other options have incorrect powers of L, T or A.

Final answer. The dimensions are **$[ML^2T^{-3}A^{-1}]$**, so the answer is option **(B)**.

← *Back to Q8*

Q9

Two identical conductors 1 and 2 are placed on two frictionless conducting rails R and S in a uniform magnetic field directed vertically downward into the plane of the page. If conductor 1 is moved with a constant velocity in the direction as shown in figure, the force on conductor 2 will be along:



(A) $-\hat{i}$

(B) $-\hat{j}$

(C) $\hat{k}$

(D) $\hat{j}$

SOLUTION

Two identical conductors 1 and 2 slide on parallel conducting rails R and S, forming a closed circuit, in a uniform magnetic field directed into the plane of the page ($-\hat{k}$ direction). Conductor 1 is pulled with a constant velocity, changing the area of the loop and so changing the magnetic flux through it.

By Faraday's law a changing flux induces an EMF, which drives an induced current around the loop. Conductor 2, though stationary, carries this same induced current.

The force on current-carrying conductor 2 in the field is:

$$\vec{F}_2 = \mathrm{I}_2 \vec{L}_2 \times \vec{B}$$

The direction of the induced current is fixed by Lenz's law: the induced effects oppose the change that produced them. As conductor 1 moves outward and enlarges the loop area, the induced current flows so that the force on conductor 2 tends to reduce the loop area, i.e. it acts so as to make conductor 2 try to chase conductor 1.

Combining the geometry of the figure (field into the page, conductor 1 moving in the given sense) with $\vec{F}_2 = \mathrm{I}_2\vec{L}_2\times\vec{B}$, the resulting force on conductor 2 points along the $-\hat{i}$ direction.

Hence the force on conductor 2 is along $-\hat{i}$, which is **option (A)**.

Final answer. The force on conductor 2 acts along $-\hat{i}$, so the answer is option **(A)**.

← *Back to Q9*

Q10

A closely wound long solenoid of self-inductance L is cut into two identical solenoids.

The value of self-inductance of each small solenoid will be:

(A) $\frac{L}{4}$

(B) $\frac{L}{2}$

(C) 2L

(D) 4L

[2026 • Set 55-4-2]

SOLUTION

The self-inductance of a long closely wound solenoid is:

$$L = \frac{\mu_0 N^2 A}{l}$$

where N is the number of turns, A is the cross-sectional area and l is the length.

When the solenoid is cut into two identical solenoids, the turns are equally shared and the length is halved, while the cross-sectional area A (which depends on the diameter, unchanged) stays the same. So for each smaller solenoid:

$$N' = \frac{N}{2}, l' = \frac{l}{2}, A' = A$$

Self-inductance of each half:

$$L' = \frac{\mu_0 (N')^2 A'}{l'}$$

$$L' = \frac{\mu_0 (\frac{N}{2})^2 A}{\frac{l}{2}}$$

$$L' = \frac{\mu_0 \frac{N^2}{4} A}{\frac{l}{2}}$$

$$L' = \frac{\mu_0 N^2 A}{4} \times \frac{2}{l}$$

$$L' = \frac{\mu_0 N^2 A}{2l}$$

$$L' = \frac{1}{2}\left(\frac{\mu_0 N^2 A}{l}\right) = \frac{L}{2}$$

Therefore the self-inductance of each smaller solenoid is $\frac{L}{2}$, which is **option (B)**.

CONCEPT

Self-inductance is not simply proportional to length here. Cutting changes BOTH N and l, since $L \propto N^2/l$ and turn density N/l stays fixed, L ends up proportional to length, giving $L/2$.

Final answer. Each half-solenoid has self-inductance $L/2$, so the answer is option **(B)**.

← *Back to Q10*

Q11

Two coils are placed closed to each other. The mutual inductance of the pair of coils depends upon the:

(A) rate at which currents change in the two coils.

(B) relative position and orientation of the coils.

(C) currents in the two coils.

(D) value of voltage induced in one coil due to change in value of current in the other coil.

[2026 • Set 55-4-3]

SOLUTION

The mutual inductance M of a pair of coils is defined by $\Phi_2 = M\, I_1$, where Φ_2 is the flux linked with coil 2 due to current I_1 in coil 1.

From this definition, M depends only on the number of turns of each coil, their sizes and shapes, the relative position and orientation of the two coils, and the magnetic nature of the medium between them.

Hence M is a purely geometric and material property of the arrangement, so the correct choice is **(B) relative position and orientation of the coils**.

Ruling out the others: the actual currents in the coils (C) do not appear in the formula for M; the rate at which currents change (A) decides the induced emf $\varepsilon = -M\,\frac{dI}{dt}$ but not M itself; and the induced voltage (D) is a result that depends on M, not a quantity that M depends upon.

Final answer. The mutual inductance depends on the relative position and orientation of the coils, so the answer is **(B)**.

← *Back to Q11*

Q12

Assertion (A): *Induced emf produced in a coil will be more when the magnetic flux linked with the coil is more.*
Reason (R): *Induced emf produced is directly proportional to the magnetic flux.*
Select the correct answer from the codes given below:

(D) Both Assertion (A) and Reason (R) are false.

[2026 • Set 55-5-1]

SOLUTION

By Faraday's law of electromagnetic induction, the induced emf is $\varepsilon = -\frac{d\Phi}{dt}$.

So the induced emf depends on the rate of change of magnetic flux, not on the magnitude of the flux itself.

Checking the Assertion: it claims that more flux Φ gives more induced emf. This is false, because a coil can have a very large but constant flux through it and still have zero induced emf, since $\frac{d\Phi}{dt} = 0$.

Checking the Reason: it claims $\varepsilon \propto \Phi$. This is also false, since the correct relation is $\varepsilon \propto \frac{d\Phi}{dt}$.

Both the Assertion and the Reason are false, so the correct option is **(D)**.

COMMON MISTAKE

A common mistake is to confuse flux with rate of change of flux; only a CHANGING flux induces an emf.

Final answer. Both Assertion and Reason are false, so the answer is **(D)**.

← Back to Q12

Q13

The magnetic flux through a loop placed in a magnetic field can be changed by changing:

(A) area of the loop only

(B) the value of magnetic field only

(C) orientation of the loop in the magnetic field only

(D) any one or more of the factors given in (A), (B) and (C)

[2026 • Set 55-5-1]

SOLUTION

The magnetic flux through a loop is given by $\Phi = BA\theta$, where B is the magnetic field, A is the area of the loop, and θ is the angle between the field $\vec{B}$ and the area vector (normal) of the loop.

The flux Φ therefore changes if any of these quantities change.

Changing the area A (for example by stretching or shrinking the loop) changes Φ.

Changing the magnetic field B (for example by varying the current in a nearby coil or moving a magnet) changes Φ.

Changing the orientation θ (for example by rotating the loop in the field) changes Φ.

Since the flux can be changed by altering any one or more of these factors, the correct option is **(D) any one or more of the factors given in (A), (B) and (C)**.

Final answer. Flux changes with area, field, or orientation, so the answer is **(D)**.

← *Back to Q13*

Q14

A rectangular loop of size 5 cm × 8 cm *is lying in x-y plane in a uniform magnetic field* $\vec{B} = (2.0\ T)\ \hat{k}$. *The total magnetic flux linked with the loop is:*

(A) 80 Wb

(B) 16 Wb

(C) 8×10^{-2} **Wb**

(D) 8×10^{-3} Wb

[2026 • Set 55-5-2]

SOLUTION

The loop lies in the *x-y* plane, so its area vector points along $\hat{k}$ (perpendicular to the *x-y* plane), and the field $\vec{B} = (2.0\ T)\ \hat{k}$ is also along $\hat{k}$.

First convert the dimensions to SI units: **5 cm = 0.05 *m*** and **8 cm = 0.08 *m***.

Area of the loop:

$$A = (0.05\ m) \times (0.08\ m) = 4 \times 10^{-3}\ \text{m}^2$$

The magnetic flux is $\Phi = \vec{B} \cdot \vec{A} = BA\theta$. Since $\vec{B}$ and $\vec{A}$ are parallel, $\theta = 0°$ and $\theta = 1$.

$$\Phi = BA = (2.0\ T)(4 \times 10^{-3}\ \text{m}^2)$$

$$\Phi = 8 \times 10^{-3}\ \text{Wb}$$

Hence the total magnetic flux linked with the loop is $\mathbf{8 \times 10^{-3}\ Wb}$, which is option **(C)**.

> **COMMON MISTAKE**
>
> *Always convert cm to metres before computing area; forgetting this gives wrong powers of ten in the flux.*

Final answer. The flux is $\mathbf{8 \times 10^{-3}\ Wb}$, so the answer is **(C)**.

← Back to Q14

Q15

A square loop of side 0.50 *m is placed in a uniform magnetic field of* 0.4 *T perpendicular to the plane of the loop. The loop is rotated through an angle of* 60° *in* 0.2 *s. The value of emf induced in the loop will be:*

(A) 5 *V*

(B) 3.5 *V*

(C) **2.5 *V***

(D) Zero V

[2026 • Set 55-5-3]

SOLUTION

The square loop has side $L = 0.50\ m$, so its area is $A = L^2 = (0.50)^2 = 0.25\ m^2$. The field B is initially perpendicular to the plane of the loop, and the loop is rotated through 60° in $\Delta t = 0.2\ s$.

A note on the data: with the printed value $B = 0.4\ T$ the average emf works out to 0.25 *V*, which is not among the options. The option set (2.5 *V* as the key) is consistent only with $B = 4\ T$, so the intended field is $B = 4\ T$ and that value is used below.

When the field is perpendicular to the plane of the loop, the area vector is along $\vec{B}$, so the initial flux is

$$\Phi_i = BA\cos 0° = (4)(0.25)(1) = 1.0\ \text{Wb}$$

After rotating through 60°, the angle between $\vec{B}$ and the area vector becomes 60°, so

$$\Phi_f = BA\cos 60° = (4)(0.25)(0.5) = 0.5\ \text{Wb}$$

The magnitude of the change in flux is

$$|\Delta\Phi| = \Phi_i - \Phi_f = 1.0 - 0.5 = 0.5\ \text{Wb}$$

By Faraday's law, the magnitude of the average induced emf is

$$\varepsilon = \frac{|\Delta\Phi|}{\Delta t} = \frac{0.5}{0.2} = 2.5\ V$$

Hence the induced emf is 2.5 V, which is option (C).

Final answer. The induced emf is **2.5 V**, so the answer is **(C)**.

← *Back to Q15*

Q16

Assertion (A): *It is difficult to move a magnet into a coil of large number of turns when the circuit of the coil is closed.*
Reason (R): *The direction of induced current in a coil with its circuit closed, due to motion of a magnet, is such that it opposes the cause.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

[2025 • Set 55-1-1]

SOLUTION

By Lenz's law, the direction of the induced current in a closed coil is always such that it opposes the change in magnetic flux that produces it.

Checking the Assertion: when a magnet is pushed into a coil with a large number of turns and a closed circuit, the changing flux induces a current. This induced current sets up its own magnetic field that repels the approaching magnet, so an external force is needed to keep pushing the magnet in. Hence it is genuinely difficult to move the magnet in, and the Assertion is true.

Checking the Reason: the statement that the induced current opposes the cause of its production is exactly Lenz's law, so the Reason is true.

Now the link: the opposition described by the Reason (Lenz's law) is precisely the cause of the difficulty stated in the Assertion. The induced current opposes the magnet's motion, which is why effort is required.

Therefore both A and R are true and R correctly explains A, so the correct option is **(A)**.

Final answer. Both A and R are true and R is the correct explanation of A, so the answer is **(A)**.

← *Back to Q16*

Q17

A coil has 100 *turns, each of area* 0.05 m^2 *and total resistance* 1.5 Ω. *It is inserted at an instant in a magnetic field of* 90 mT, *with its axis parallel to the field. The charge induced in the coil at that instant is:*

(A) 3.0 mC

(B) **0.30 *C***

(C) 0.45 *C*

(D) 1.5 *C*

[2025 • Set 55-1-1]

SOLUTION

The coil has $N = 100$ turns, area per turn $A = 0.05\ m^2$, total resistance $R = 1.5\ \Omega$, and the field is $B = 90\ mT = 0.09\ T$, with the coil axis parallel to the field.

With the axis parallel to the field, the area vector is along $\vec{B}$, so the final flux linkage per turn is BA. Before insertion the flux is zero, so the change in flux linkage is

$$\Delta\Phi = N\,B\,A$$

$$\Delta\Phi = (100)(0.09)(0.05) = 0.45\ \text{Wb}$$

The charge induced is found from $q = \frac{\Delta\Phi}{R}$, which follows from $q = \int i\,dt = \int \frac{\varepsilon}{R}\,dt = \frac{1}{R}\int \frac{d\Phi}{dt}\,dt = \frac{\Delta\Phi}{R}$.

$$q = \frac{0.45}{1.5}$$

$$q = 0.30\ C$$

Hence the charge induced in the coil is **0.30 *C***, which is option **(B)**.

CONCEPT

Induced charge depends only on the total change in flux and resistance, not on how fast the change happens.

Final answer. The induced charge is **0.30 *C***, so the answer is **(B)**.

← *Back to Q17*

Q18

You are required to design an air-filled solenoid of inductance 0.016 *H having a length* 0.81 *m and radius* 0.02 *m. The number of turns in the solenoid should be:*

(A) 2592

(B) 2866

(C) 2976

(D) 3140

[2025 • Set 55-1-1]

SOLUTION

For an air-filled solenoid the inductance is $L = \frac{\mu_0 N^2 A}{l}$, where $A = \pi r^2$ is the cross-sectional area and l is the length.

Given $L = 0.016\ H$, $l = 0.81\ m$, $r = 0.02\ m$, and $\mu_0 = 4\pi \times 10^{-7}\ T\,m\,A^{-1}$.

Area of cross-section:

$$A = \pi r^2 = \pi (0.02)^2 = \pi \times 4 \times 10^{-4}\ m^2 \approx 1.2566 \times 10^{-3}\ m^2$$

Rearranging for N:

$$N^2 = \frac{L\,l}{\mu_0 A}$$

$$N^2 = \frac{(0.016)(0.81)}{(4\pi \times 10^{-7})(1.2566 \times 10^{-3})}$$

$$N^2 = \frac{0.01296}{1.5791 \times 10^{-9}} \approx 8.207 \times 10^6$$

$$N = \sqrt{8.207 \times 10^6} \approx 2865$$

The nearest listed value to 2865 is 2866, which is option (B). (The standard formula does not yield 2592; that value would require a radius of about 0.022 *m*, not the given 0.02 *m*, so option (A) is not consistent with the stated data.)

Hence the number of turns is about 2866, which is option (B).

Final answer. The required number of turns is about **2866**, so the answer is **(B)**.

← *Back to Q18*

Q19

A circular coil of diameter 15 mm *having* 300 *turns is placed in a magnetic field of* 30 mT *such that the plane of the coil is perpendicular to the direction of magnetic field. The magnetic field is reduced uniformly to zero in* 20 ms *and again increased uniformly to* 30 mT *in* 40 ms. *If the emfs induced in the two time intervals are* e_1 *and* e_2 *respectively, then the value of* e_1 / e_2 *is:*

(A) $\frac{1}{2}$

(B) 1

(C) **2**

(D) 4

[2025 • Set 55-1-2]

SOLUTION

The coil has $N = 300$ turns and diameter $d = 15$ mm, so the radius is $r = 7.5$ mm $= 7.5 \times 10^{-3}$ m, giving area $A = \pi r^2 \approx 1.77 \times 10^{-4}$ m^2. The field changes between $B_0 = 30$ mT $= 0.03$ T and zero, with the plane of the coil perpendicular to $\vec{B}$ (so the area vector is along $\vec{B}$).

The magnitude of the induced emf is $|\varepsilon| = N A \left|\frac{dB}{dt}\right|$, so for the same coil $|\varepsilon| \propto \left|\frac{dB}{dt}\right|$.

First interval (field reduced from 0.03 *T* to 0 in 20 ms):

$$\left|\frac{dB}{dt}\right|_1 = \frac{0.03}{20 \times 10^{-3}} = 1.5\ T\,s^{-1}$$

So $\mathbf{e_1 = N A\ (1.5)}$.

Second interval (field increased from 0 to 0.03 T in 40 ms):

$$\left|\frac{dB}{dt}\right|_2 = \frac{0.03}{40 \times 10^{-3}} = 0.75\ T\,s^{-1}$$

So $\mathbf{e_2 = N A\ (0.75)}$.

Taking the ratio, the common factor NA cancels:

$$\frac{e_1}{e_2} = \frac{1.5}{0.75} = 2$$

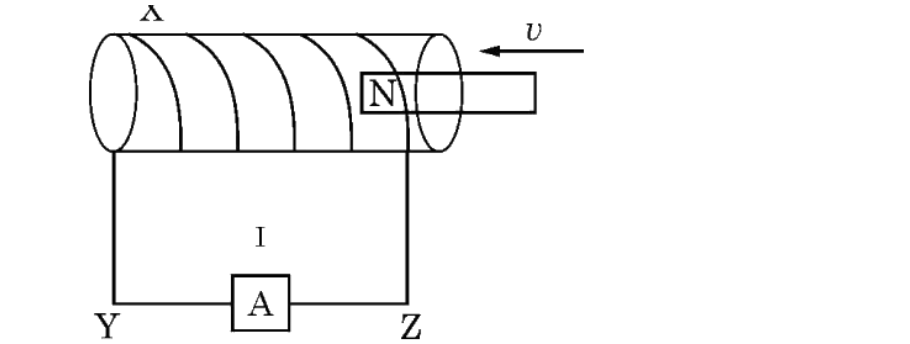
Hence $\mathbf{\frac{e_1}{e_2} = 2}$, which is option **(C)**.

Final answer. The ratio $\mathbf{e_1/e_2 = 2}$, so the answer is **(C)**.

← *Back to Q19*

Q20

In the figure X is a coil wound over a hollow wooden pipe. A permanent magnet is pushed at a constant speed v from the right into the pipe and it comes out at the left end of the pipe. During the entry and the exit of the magnet, the current in the wire YZ will be from



(A) Y to Z and then Y to Z

(B) Z to Y and then Y to Z

(C) Y to Z and then Z to Y

(D) Z to Y and then Z to Y

[2025 • Set 55-2-1]

SOLUTION

Picture the setup: a coil X is wound on a hollow wooden pipe, with the free ends of the winding connected through the wire YZ. A bar magnet moves at constant speed v into the pipe from the right and later exits from the left. By Lenz's law, the induced current always flows so as to oppose the change in flux that causes it.

During entry: as the magnet moves into the pipe and approaches the coil, the magnetic flux through coil X increases. The induced current must oppose this increase, so the near face of the coil develops the same polarity as the approaching pole of the magnet, repelling it. For the given winding and terminals, this makes the current in wire YZ flow from **Z to Y**.

During exit: as the magnet recedes from the coil on the left side, the flux through coil X decreases. The induced current must now oppose this decrease, so the coil face attracts the receding magnet, which means the induced current reverses. The current in YZ now flows from **Y to Z**.

Therefore the current is first from Z to Y and then from Y to Z, which is option **(B)**.

CONCEPT

On entry the coil repels the magnet and on exit it attracts it, so the induced current reverses between the two stages.

Final answer. The current is from Z to Y and then Y to Z, so the answer is **(B)**.

← *Back to Q20*

Q21

Two long solenoids of radii r_1 *and* r_2 $(\mathbf{r_2 > r_1})$ *and number of turns per unit length* n_1 *and* n_2*, respectively are co-axially wrapped one over the other. The ratio of self-inductance of inner solenoid to their mutual inductance is*

(A) $\dfrac{\mathbf{n_1}}{\mathbf{n_2}}$

(B) $\dfrac{n_2}{n_1}$

(C) $\dfrac{n_1 r_1^2}{n_2 r_2^2}$

(D) $\dfrac{n_2 r_1^2}{n_1 r_2^2}$

[2025 • Set 55-2-2]

SOLUTION

Consider both solenoids of common length l. The inner solenoid has n_1 turns per unit length, radius r_1 and cross-sectional area πr_1^2.

The self-inductance of a long solenoid is $\boldsymbol{L = \mu_0 n^2 A l}$, where n is turns per unit length and A is its area.

For the inner solenoid:

$$L_1 = \mu_0 n_1^2 (\pi r_1^2) l$$

When current flows in the inner solenoid, its field $\boldsymbol{B = \mu_0 n_1 I}$ exists only inside its own cross-section (area πr_1^2). The flux linked with the outer solenoid is set by this same inner area, since the inner field does not fill the larger outer area.

The mutual inductance is therefore:

$$M = \mu_0 n_1 n_2 (\pi r_1^2) l$$

Taking the required ratio:

$$\frac{L_1}{M} = \frac{\mu_0 n_1^2 \pi r_1^2 l}{\mu_0 n_1 n_2 \pi r_1^2 l}$$

$$\frac{L_1}{M} = \frac{n_1}{n_2}$$

So the ratio depends only on the turn densities. Option (B) inverts the ratio, while options (C) and (D) wrongly bring in the radii, which cancel because both L_1 and M use the inner area πr_1^2. The correct choice is **option (A)**.

COMMON MISTAKE

Both L_1 and M use the inner solenoid's area πr_1^2 (not πr_2^2), because the inner solenoid's field is confined to its own cross-section. This is why the radii cancel.

Final answer. The ratio of self-inductance to mutual inductance is $\frac{n_1}{n_2}$, option **(A)**.

← *Back to Q21*

Q22

A coil of an ac generator, having 100 turns and area $0.1\ m^2$ each, rotates at half a rotation per second in a magnetic field of 0.02 T. The maximum emf generated in the coil is

(A) 0.31 V

(B) 0.20 V

(C) **0.63 V**

(D) 0.10 V

[2025 • Set 55-2-2]

SOLUTION

The maximum (peak) emf of an a.c. generator is given by:

$$\varepsilon_0 = N B A \omega$$

where N is the number of turns, B the field, A the area of each turn and ω the angular frequency.

The coil makes half a rotation per second, so the frequency is $\nu = 0.5$ **rev/s**, giving:

$$\omega = 2\pi\nu = 2\pi(0.5) = \pi \text{ rad/s} \approx 3.14 \text{ rad/s}$$

Substituting $N = 100$, $B = 0.02\ T$, $A = 0.1\ m^2$ and $\omega = 3.14$ **rad/s**:

$$\varepsilon_0 = (100)(0.02)(0.1)(3.14)$$

$$\varepsilon_0 = 0.628\ V \approx 0.63\ V$$

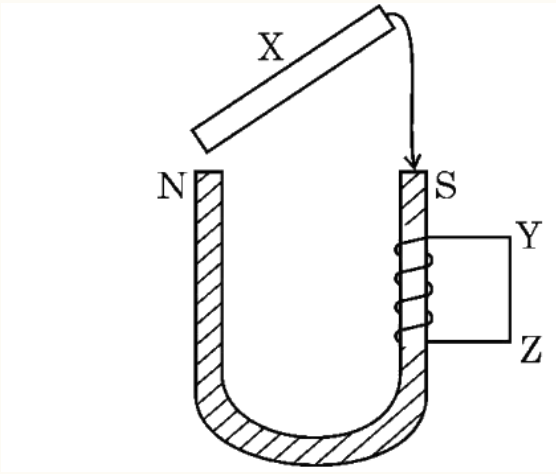
Hence the maximum emf generated is about $0.63\ V$, which is **option (C)**.

Final answer. The maximum emf generated is **0.63 *V***, option **(C)**.

← *Back to Q22*

Q23

A soft iron rod X is allowed to fall on the two poles of a U shaped permanent magnet as shown in figure. A coil is wrapped over one arm of the U shaped magnet. During fall of the rod, the current in the coil will be



(A) clockwise current

(B) anticlockwise current

(C) **alternating current**

(D) zero

[2025 • Set 55-2-3]

SOLUTION

Setup: a U-shaped permanent magnet has a coil wound on one of its arms. A soft iron rod X falls and lands across the two poles, momentarily bridging the gap between the north and south poles.

By Faraday's law, an emf is induced only while the flux through the coil is changing, and its sense is fixed by Lenz's law.

As the iron rod approaches and bridges the poles, it provides a low-reluctance path that channels far more magnetic flux through the magnetic circuit (and so through the coil). The flux through the coil rises sharply, inducing a current in one direction.

After contact, the rod bounces and separates from the poles. The flux now decreases, and by Lenz's law the induced current reverses direction.

Because the rod bounces repeatedly (flux rising and falling in turn), the induced current keeps reversing. This cyclic increase and decrease of flux produces an **alternating current** in the coil.

A steady clockwise or anticlockwise current (options A, B) would require a one-way flux change, which does not happen here. The current is not zero (option D) because the flux genuinely changes. The correct choice is **option (C)**.

Final answer. The current in the coil is an alternating current, option **(C)**.

← *Back to Q23*

Q24

When current in a coil changes at a steady rate from 8 *A to* 6 *A in* 4 ms, *an emf of* 1.5 *V is induced in it. The value of self-inductance of the coil is :*

(A) 6 mH

(B) 12 mH

(C) **3 mH**

(D) 9 mH

[2025 • Set 55-4-2]

SOLUTION

The emf induced by a changing current in a coil is given by the self-inductance relation:

$$|\varepsilon| = L\left|\frac{dI}{dt}\right| \Rightarrow L = \frac{|\varepsilon|}{|dI/dt|}$$

The current changes from 8 A to 6 A, so $\Delta I = 6 - 8 = -2\ A$, over a time $\Delta t = 4\ ms = 4\times 10^{-3}\ s$, with $|\varepsilon| = 1.5\ V$.

The rate of change of current is:

$$\left|\frac{dI}{dt}\right| = \frac{2}{4\times 10^{-3}} = 500\ A/s$$

Substituting into the formula for L:

$$L = \frac{1.5}{500} = 3\times 10^{-3}\ H$$

$$L = 3\ \text{mH}$$

Hence the self-inductance of the coil is 3 mH, which is **option (C)**.

Final answer. The self-inductance of the coil is **3 mH**, option **(C)**.

← *Back to Q24*

Q25

The magnetic flux linked with a closed coil (in Wb) varies with time t (in s) as $\phi = 5t^2 + 4t - 2$. *If the resistance of the circuit is* 14 Ω, *the magnitude of induced current in the coil at* $t = 1\ s$ *will be :*

(A) 0.5 A

(B) 1.0 A

(C) 1.5 A

(D) 2.0 A

[2025 • Set 55-4-3]

SOLUTION

By Faraday's law, the induced emf is the negative rate of change of flux, and the induced current follows from Ohm's law:

$$\varepsilon = -\frac{d\phi}{dt}, \; I = \frac{|\varepsilon|}{R}$$

The flux is $\phi(t) = 5t^2 + 4t - 2$, with $R = 14\ \Omega$.

Differentiating with respect to time:

$$\frac{d\phi}{dt} = 10t + 4$$

At $t = 1\ s$:

$$\cdot\frac{d\phi}{dt}|_{t=1} = 10(1) + 4 = 14\ V$$

So the magnitude of the induced emf is $|\varepsilon| = 14\ V$.

The induced current is:

$$I = \frac{14}{14} = 1.0\ A$$

Hence the magnitude of the induced current at $t = 1\ s$ is $1.0\ A$, which is **option (B)**.

Final answer. The induced current at $t = 1\ s$ is $1.0\ A$, option **(B)**.

← *Back to Q25*

Q26

The magnetic flux linked with a coil changes with time t as $\phi = (8t^2 + 5t + 7)$*, where t is in seconds and* ϕ *is in Wb. The value of emf induced in the coil at* $t = 4\ s$ *is:*

(A) 32 V

(B) 37 V

(C) 64 V

(D) **69 V**

[2025 • Set 55-5-1]

SOLUTION

By Faraday's law of electromagnetic induction, the induced emf equals the magnitude of the rate of change of magnetic flux:

$$\varepsilon = -\frac{d\phi}{dt}, \quad |\varepsilon| = \left|\frac{d\phi}{dt}\right|$$

The flux is given as $\phi(t) = 8t^2 + 5t + 7$.

Differentiating with respect to time:

$$\frac{d\phi}{dt} = 16t + 5$$

At $t = 4\ s$:

$$\left.\frac{d\phi}{dt}\right|_{t=4} = 16(4) + 5$$

$$= 64 + 5 = 69\ V$$

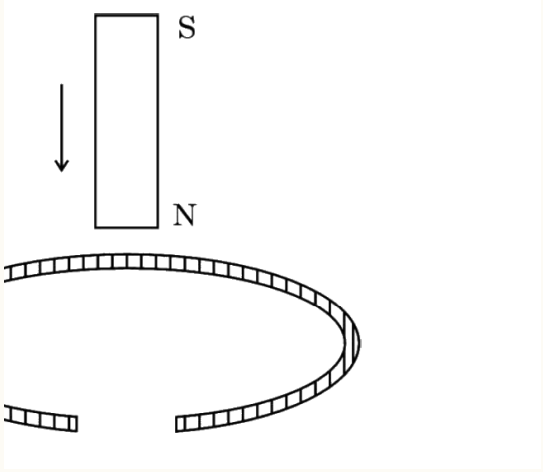
Hence the magnitude of the emf induced in the coil at $t = 4\ s$ is $69\ V$, which is **option (D)**.

Final answer. The emf induced at $t = 4\ s$ is $69\ V$, option **(D)**.

← *Back to Q26*

Q27

A vertically held bar magnet is dropped along the axis of a copper ring having a cut as shown in the diagram. The acceleration of the falling magnet is:



(A) zero

(B) less than g

(C) g

(D) greater than g

[2025 • Set 55-6-1]

SOLUTION

Setup: a bar magnet is dropped vertically along the axis of a copper ring, but the ring has a cut (a small gap) in it.

An induced current can flow only if there is a complete (closed) conducting loop, as required by Faraday's law together with the circuit being closed.

Because the ring has a cut, it is an open circuit. Even though the falling magnet changes the flux through the ring and induces an emf, no induced current can flow around the broken loop.

With no induced current, the ring sets up no opposing magnetic field, so by Lenz's law there is no retarding (braking) force on the magnet.

The only force acting on the magnet is gravity. Therefore it falls freely:

$$a = g$$

If the ring were continuous (no cut), induced eddy currents would oppose the motion and the acceleration would be less than g (option B). But with the cut, the answer is **option (C)**.

COMMON MISTAKE

A cut in the ring breaks the current path. An emf is still induced, but with no closed loop there is no current, hence no opposing force. Do not confuse 'emf induced' with 'current flows'.

Final answer. The magnet falls with acceleration g, option **(C)**.

← Back to Q27

Q28

A rectangular coil of area A is kept in a uniform magnetic field $\vec{B}$ such that the plane of the coil makes an angle α with $\vec{B}$. The magnetic flux linked with the coil is:

(A) BAα

(B) BAα

(C) BA

(D) zero

[2025 • Set 55-7-1]

SOLUTION

Magnetic flux is defined through the normal to the coil's plane:

$$\phi = \vec{B} \cdot \vec{A} = BA\theta$$

where θ is the angle between the field $\vec{B}$ and the normal $\hat{n}$ to the plane of the coil.

Here $\vec{B}$ makes an angle α with the plane of the coil. The normal is perpendicular to the plane, so the angle between $\vec{B}$ and the normal is:

$$\theta = 90° - \alpha$$

Substituting into the flux formula:

$$\phi = BA\cos(90° - \alpha)$$

$$\phi = BA\alpha$$

This is consistent with the limiting cases: when the plane is parallel to $\vec{B}$ ($\alpha = 0°$), $\phi = 0$; when the plane is perpendicular to $\vec{B}$ ($\alpha = 90°$), $\phi = BA$, the maximum. Hence the correct choice is **option (A)**.

COMMON MISTAKE

The standard formula uses the angle with the normal, not the plane. When the question gives the angle with the plane, use $\cos(90° - \alpha) = \alpha$.

Final answer. The magnetic flux linked with the coil is $BA\alpha$, option **(A)**.

← Back to Q28

Q29

A metal rod of length 50 cm *is held vertically and moved with a velocity of* 10 *m/s towards east. The horizontal component of the Earth's magnetic field at the place is* 0.4 *G. The emf induced across the ends of the rod is:*

(A) 0.1 mV

(B) 0.2 mV

(C) 0.8 mV

(D) 1.6 mV

[2025 • Set 55-7-1]

SOLUTION

The rod is vertical with length $L = 50\ \text{cm} = 0.50\ m$, moving east with velocity $v = 10\ m/s$. The horizontal component of Earth's field is $B_H = 0.4\ G = 0.4 \times 10^{-4}\ T$, directed north (horizontal).

The motional emf induced across a rod is:

$$\varepsilon = (\vec{v} \times \vec{B}) \cdot \vec{L}$$

Here the velocity $\vec{v}$ (east) and the field $\vec{B}_H$ (north) are both horizontal and mutually perpendicular, so $\vec{v} \times \vec{B}$ points vertically, along the rod. The full product applies:

$$\varepsilon = B_H\, v\, L$$

Substituting the values:

$$\varepsilon = (0.4\times 10^{-4})(10)(0.50)$$

$$\varepsilon = 2\times 10^{-4}\ V$$

$$\varepsilon = 0.2\ \text{mV}$$

Hence the emf induced across the ends of the rod is 0.2 mV, which is **option (B)**.

TIP

Only the horizontal component B_H matters here because it is perpendicular to both the vertical rod and the eastward velocity. Also remember $\mathbf{1\ G = 10^{-4}\ T}$.

Final answer. The emf induced across the rod is **0.2 mV**, option **(B)**.

← *Back to Q29*

Q30

The dimensions of 'self-inductance' are:

(A) $[MLT^{-2}A^{-2}]$

(B) $[ML^{2}T^{-1}A^{-1}]$

(C) $[ML^{-1}T^{-2}A^{-2}]$

(D) $\mathbf{[ML^{2}T^{-2}A^{-2}]}$

[2025 • Set 55-7-1]

SOLUTION

Self-inductance is defined through the relation between induced emf and rate of change of current:

$$\varepsilon = L\frac{dI}{dt} \Rightarrow L = \frac{\varepsilon}{dI/dt} = \frac{\varepsilon\, dt}{dI}$$

So the dimensions of L are:

$$[L] = \frac{[V]\,[T]}{[I]} = \frac{[V]\,[T]}{[A]}$$

We need the dimensions of voltage. Since potential difference is energy per unit charge, and charge = $[A][T]$:

$$[V] = \frac{[\text{energy}]}{[\text{charge}]} = \frac{\mathrm{ML^2T^{-2}}}{\mathrm{AT}} = \mathrm{ML^2T^{-3}A^{-1}}$$

Substituting back:

$$[L] = \frac{(\mathrm{ML^2T^{-3}A^{-1}})(T)}{A}$$

$$[L] = \mathrm{ML^2T^{-2}A^{-2}}$$

Hence the dimensions of self-inductance are $[\mathrm{ML^2T^{-2}A^{-2}}]$, which is **option (D)**.

Final answer. The dimensions of self-inductance are $\mathbf{[ML^2T^{-2}A^{-2}]}$, option **(D)**.

← *Back to Q30*

Q31

Assertion (A): *The mutual inductance between two coils is maximum when the coils are wound on each other.*
Reason (R): *The flux linkage between two coils is maximum when they are wound on each other.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2024 • Set 55-1-1]

SOLUTION

The mutual inductance of two coils is the flux linkage set up in one coil per unit current in the other, written as $M = \frac{\Phi_{21}}{I_1}$.

A larger flux linkage Φ_{21} for the same current I_1 means a larger value of M.

When the two coils are wound one over the other on a common core, nearly the entire magnetic flux produced by one coil passes through the turns of the second coil, with very little leakage flux.

This makes the flux linkage between the coils maximum, so M is also maximum. Hence the Assertion is true.

The Reason correctly states the physical cause: the flux linkage is maximum when the coils are wound on each other. Since $M = \frac{\Phi_{21}}{I_1}$, a maximum flux linkage directly gives a maximum mutual inductance.

So both (A) and (R) are true, and (R) is the correct explanation of (A). The wrong options fail because (R) is in fact true and does explain (A).

Hence the correct option is (A).

Final answer. The correct option is **(A)**: both **(A)** and (R) are true and (R) correctly explains **(A)**.

← *Back to Q31*

Q32

A conducting circular loop is placed in a uniform magnetic field $B = 50$ mT *with its plane perpendicular to the magnetic field. The radius of the loop is made to shrink at a constant rate of* 1 mm s^{-1}. *At the instant the radius of the loop is* 4 cm, *the induced emf in the loop is:*

(A) π μV

(B) 2π μV

(C) **4π μV**

(D) 8π μV

[2024 • Set 55-1-2]

SOLUTION

The magnetic flux through the loop, with $\vec{B}$ normal to its plane, is $\Phi = BA = B\pi r^2$, where B is constant and only the radius r changes.

By Faraday's law, the magnitude of the induced emf is $|\varepsilon| = \left|\frac{d\Phi}{dt}\right| = B\pi \frac{d(r^2)}{dt} = 2\pi B r\left|\frac{dr}{dt}\right|$.

Given $B = 50\ \text{mT} = 50\times 10^{-3}\ T$, $r = 4\ \text{cm} = 4\times 10^{-2}\ m$, and $\left|\frac{dr}{dt}\right| = 1\ \text{mm s}^{-1} = 10^{-3}\ m\ \text{s}^{-1}$.

Substituting, $|\varepsilon| = 2\pi\,(50\times 10^{-3})(4\times 10^{-2})(10^{-3})$.

$|\varepsilon| = 2\pi \times (50 \times 4 \times 1)\times 10^{-3-2-3}$.

$|\varepsilon| = 2\pi \times 200 \times 10^{-8} = 2\pi \times 2\times 10^{-6}\ V$.

$|\varepsilon| = 4\pi \times 10^{-6}\ V = 4\pi\ \mu\text{V}$.

Hence the induced emf is $4\pi\ \mu$V, which is option (C).

COMMON MISTAKE

Only the radius changes here, so differentiate r^2 carefully: $\frac{d(r^2)}{dt} = 2r\frac{dr}{dt}$. Forgetting the factor of 2 is a very common slip.

Final answer. The induced emf is **$4\pi\ \mu$V**, so the correct option is **(C)**.

← *Back to Q32*

Q33

Assertion (A): *Lenz's law is a consequence of the law of conservation of energy.*
Reason (R): *There is no power loss in an ideal inductor.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2024 • Set 55-2-1]

SOLUTION

Lenz's law states that the direction of an induced current is always such that it opposes the change in flux that produces it.

This opposition is exactly what conservation of energy demands: to drive the induced current against the opposing force or back-emf, external work must be done, and this work appears as electrical energy. If the induced current instead aided the change, the current would grow without any external work, creating energy from nothing.

So Lenz's law is a direct consequence of the law of conservation of energy. Hence the Assertion is true.

In an ideal inductor (zero resistance), the energy supplied is stored in the magnetic field as $U = \frac{1}{2}LI^2$ and is returned to the circuit when the current decreases, with no power dissipated as heat. So the Reason is also a true statement.

However, the Reason speaks about energy storage in an inductor, while the Assertion is about the directional rule for induced currents. The Reason does not explain why Lenz's law follows from energy conservation, so it is not the correct explanation of the Assertion.

Hence both (A) and (R) are true, but (R) is not the correct explanation of (A): option (B).

Final answer. The correct option is **(B)**: both **(A)** and (R) are true, but (R) is not the correct explanation of **(A)**.

← *Back to Q33*

Q34

A circular coil of radius 10 cm is placed in a magnetic field $\boldsymbol{B}=(1.0\,\boldsymbol{i}+0.5\,\boldsymbol{j})$ **mT** *such that the outward unit vector normal to the surface of the coil is* $(0.6\,i+0.8\,j)$. *The magnetic flux linked with the coil is:*

(A) 0.314 μ Wb

(B) 3.14 μ Wb

(C) **31.4 μ Wb**

(D) 1.256 μ Wb

[2024 • Set 55-2-1]

SOLUTION

The magnetic flux linked with a flat coil is $\Phi = \vec{B}\cdot\vec{A} = (\vec{B}\cdot\hat{n})\,A$, where $\hat{n}$ is the unit normal to the coil and A is its area.

First find the component of $\vec{B}$ along the normal using the dot product: $\vec{B}\cdot\hat{n} = (1.0\,\hat{i} + 0.5\,\hat{j})\cdot(0.6\,\hat{i} + 0.8\,\hat{j})$ mT.

$\vec{B}\cdot\hat{n} = (1.0)(0.6) + (0.5)(0.8) = 0.6 + 0.4 = 1.0$ mT $= 1.0\times 10^{-3}$ T.

The area of the circular coil of radius $r = 10$ cm $= 0.10$ m is $A = \pi r^2 = \pi\,(0.10)^2 = 0.01\pi$ m^2.

Therefore $\Phi = (\vec{B}\cdot\hat{n})\,A = (1.0\times 10^{-3})(0.01\pi)$.

$\Phi = \pi \times 10^{-5}$ Wb $= 3.14\times 10^{-5}$ Wb $= 31.4$ μWb.

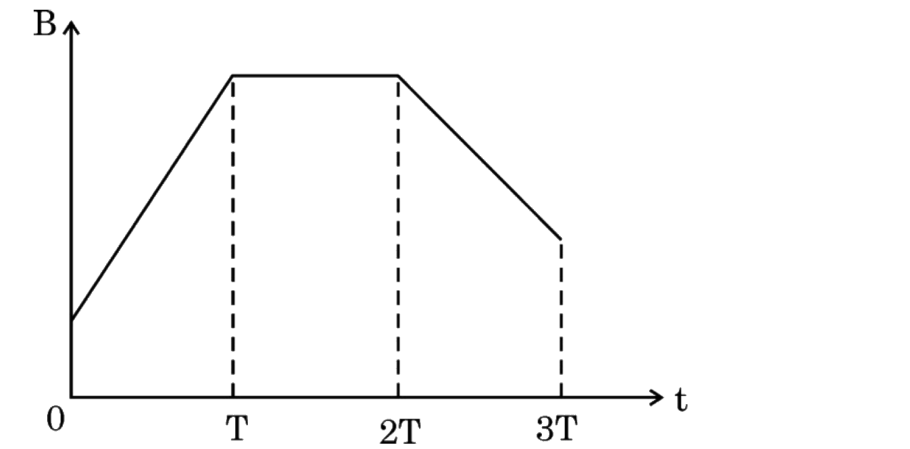
Hence the magnetic flux linked with the coil is 31.4 μWb, which is option (C).

Final answer. The magnetic flux is **31.4 μWb**, so the correct option is **(C)**.

← *Back to Q34*

Q35

A conducting loop is placed in a magnetic field, normal to its plane. The magnitude of the magnetic field varies with time as shown in the figure. If ε_1,ε_2 *and* ε_3 *are magnitudes of induced emfs during periods* **0, T and 2T, then:**



(A) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

(B) $\varepsilon_1 > \varepsilon_3 > \varepsilon_2$

(C) $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$

(D) $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$

[2024 • Set 55-2-2]

SOLUTION

With $\vec{B}$ normal to the plane of the loop, the flux is $\Phi = BA$ and the magnitude of the induced emf is $|\varepsilon| = \left|\frac{d\Phi}{dt}\right| = A\left|\frac{dB}{dt}\right|$.

Since the area A is fixed, the induced emf at any interval is directly proportional to the slope (steepness) of the B versus t graph in that interval.

From the graph, in the first interval (ε_1) the field rises with a moderate positive slope, in the second interval (ε_2) the field stays constant so the slope is zero, and in the third interval (ε_3) the field changes with the steepest slope.

A zero slope gives $\varepsilon_2 = 0$, the steepest slope gives the largest emf ε_3, and the moderate slope gives an intermediate value ε_1.

Comparing magnitudes, $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$.

Hence the correct order is $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$, which is option (C).

CONCEPT

Induced emf depends on the slope of the B-t graph, not on the value of B. Where the graph is flat, the emf is zero no matter how large B is.

Final answer. The correct order is $\varepsilon_3 > \varepsilon_1 > \varepsilon_2$, so the answer is option **(C)**.

← *Back to Q35*

Q36

The emf induced in a coil rotating in a magnetic field does not depend upon the following:

(A) Area of the coil

(B) Resistance of the coil

(C) Number of turns in the coil

(D) Angular speed of rotation of the coil

[2024 • Set 55-2-3]

SOLUTION

For a coil of N turns and area A rotating with angular speed ω in a uniform magnetic field B, the instantaneous induced emf is $\boldsymbol{\varepsilon = NAB\omega \sin(\omega t)}$.

The peak (maximum) value of this emf is $\boldsymbol{\varepsilon_0 = NAB\omega}$.

From this expression, the induced emf depends on the number of turns N, the area of the coil A, the magnetic field B, and the angular speed of rotation ω. So options (A), (C) and (D) all affect the emf.

The resistance R of the coil does not appear in the emf expression. Resistance only controls the induced current through $\boldsymbol{I = \frac{\varepsilon}{R}}$, but it has no effect on the induced emf itself.

Hence the induced emf does not depend on the resistance of the coil, which is option (B).

Final answer. The induced emf does not depend on the resistance of the coil, so the answer is option **(B)**.

← *Back to Q36*

Q37

The current in a coil of 15 *mH increases uniformly from zero to* 4 *A in* 0.004 *s. The emf induced in the coil will be:*

(A) 22.5 V

(B) 17.5 V

(C) **15.0 V**

(D) 12.5 V

[2024 • Set 55-3-1]

SOLUTION

When the current through a coil changes, a self-induced emf appears whose magnitude is $\boldsymbol{\varepsilon = L\left|\frac{dI}{dt}\right|}$, where L is the self-inductance.

Given $\boldsymbol{L = 15\ mH = 15\times 10^{-3}\ H}$, the current change $\boldsymbol{dI = 4 - 0 = 4\ A}$, and the time $\boldsymbol{dt = 0.004\ s}$.

First find the rate of change of current: $\frac{dI}{dt} = \frac{4}{0.004} = 1000\ A\ s^{-1}$.

Substituting into the emf relation, $\varepsilon = (15\times 10^{-3})(1000)$.

$\varepsilon = 15\times 10^{-3}\times 10^{3} = 15\ V$.

Hence the emf induced in the coil is 15.0 *V*, which is option (C).

Final answer. The induced emf is **15.0 *V***, so the correct option is **(C)**.

← *Back to Q37*

Q38

Consider a solenoid of length l and area of cross-section A with fixed number of turns. The self-inductance of the solenoid will increase if:

(A) both l and A are increased

(B) l is decreased and A is increased

(C) l is increased and A is decreased

(D) both l and A are decreased

[2024 • Set 55-3-1]

SOLUTION

The self-inductance of a long solenoid of N turns, cross-sectional area A and length l is $L = \frac{\mu_0 N^2 A}{l}$.

For a fixed number of turns N, the constants μ_0 and N^2 do not change, so $L \propto \frac{A}{l}$.

This shows that L increases when the area A is increased and when the length l is decreased.

Checking the options: increasing both l and A has competing effects, decreasing both also competes, and increasing l while decreasing A would decrease L. Only decreasing l together with increasing A makes both factors raise L.

Hence the self-inductance increases when l is decreased and A is increased, which is option (B).

Final answer. Self-inductance increases when l is decreased and A is increased, so the answer is option **(B)**.

← *Back to Q38*

Q39

The mutual inductance of two coils C_1 and C_2 is 20 mH. In coil C_1, the current changes from 4 A to zero in 0.2 s. If the resistance of coil C_2 is 4 Ω, then the charge that flows through it per second will be:

(A) 4.0 C

(B) 1.5 C

(C) 0.05 C

(D) **0.1 C**

[2024 • Set 55-3-2]

SOLUTION

When the current in coil C_1 changes, an emf is induced in C_2 of magnitude $\varepsilon_2 = M\left|\frac{dI_1}{dt}\right|$, and the induced current in C_2 is $I_2 = \frac{\varepsilon_2}{R}$.

The total charge that flows through C_2 is $q = \int I_2\,dt = \frac{1}{R}\int \varepsilon_2\,dt = \frac{M\,|\Delta I_1|}{R}$, where the dependence on time cancels out.

Given $M = 20\ \text{mH} = 20\times 10^{-3}\ H$, $|\Delta I_1| = 4 - 0 = 4\ A$, $R = 4\ \Omega$ and $\Delta t = 0.2\ s$.

Substituting, $q = \frac{(20\times 10^{-3})(4)}{4} = 20\times 10^{-3} = 0.02\ C$.

The charge that flows per second is $\frac{q}{\Delta t} = \frac{0.02}{0.2} = 0.1\ C$.

Hence the charge flowing per second is 0.1 C, which is option (D).

TIP

The total induced charge $q = \frac{M\,\Delta I}{R}$ depends only on the net change in current, not on how fast it changes. The question then asks for charge per second, so divide by the time interval.

Final answer. The charge flowing per second is **0.1 C**, so the correct option is **(D)**.

← Back to Q39

Q40

A coil of N turns is placed in a magnetic field $\vec{B}$ such that $\vec{B}$ is perpendicular to the plane of the coil. $\vec{B}$ changes with time as $B = B_0 \cos\left(\frac{2\pi t}{T}\right)$*, where T is time period. The magnitude of emf induced in the coil will be maximum at*

(A) $t = \frac{nT}{2}$

(B) $t = \frac{nT}{4}$

(C) $t = \frac{nT}{8}$

(D) $t = nT$

[2024 • Set 55-4-1]

SOLUTION

With $\vec{B}$ perpendicular to the plane of the N-turn coil, the flux is $\Phi = N B A = N B_0 A \cos\left(\frac{2\pi t}{T}\right)$, where A is the area of the coil.

By Faraday's law, the induced emf is $\varepsilon = -\frac{d\Phi}{dt} = N B_0 A\left(\frac{2\pi}{T}\right)\sin\left(\frac{2\pi t}{T}\right)$.

The magnitude $|\varepsilon|$ is maximum when $\left|\sin\left(\frac{2\pi t}{T}\right)\right| = 1$, that is when $\frac{2\pi t}{T} = (2n+1)\frac{\pi}{2}$ for $n = 0, 1, 2, \ldots$

Solving for t gives $t = (2n+1)\frac{T}{4}$, so the emf is maximum at the odd multiples of $\frac{T}{4}$, namely $t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \ldots$

At these instants the cosine field is passing through zero, so its rate of change is fastest and the induced emf is largest.

By contrast, at $t = \frac{nT}{2}$ (i.e. $t = 0, \frac{T}{2}, T, \ldots$) we get $\sin(\frac{2\pi t}{T}) = 0$, so the emf is zero, not maximum. So option (A) actually gives the instants of zero emf.

Among the given options, only $t = \frac{nT}{4}$ contains all the maxima (taking odd n), so it is the physically correct choice.

Hence the magnitude of the induced emf is maximum at $t = \frac{nT}{4}$ (odd n), which is option (B). Note: the printed key marks (A), but $t = \frac{nT}{2}$ corresponds to zero emf, so (A) is incorrect.

Final answer. The induced emf is maximum at $t = (2n+1)\frac{T}{4}$, the odd multiples of $\frac{T}{4}$, captured by option **(B)**; the official key answer **(A)**, $t = \frac{nT}{2}$, gives zero emf and is incorrect.

← *Back to Q40*

Q41

Two coils are placed near each other. When the current in one coil is changed at the rate of 5 *A/s, an emf of* 2 mV *is induced in the other. The mutual inductance of the two coils is:*

(A) 0.4 mH

(B) 2.5 mH

(C) 10 mH

(D) 25 *H*

[2024 • Set 55-5-1]

SOLUTION

When two coils are placed near each other, a change of current in one coil sets up a changing magnetic flux through the second coil, which induces an emf in it. By the definition of mutual inductance,

$$\varepsilon_2 = -M\frac{dI_1}{dt}$$

Taking magnitudes, the mutual inductance is

$$M = \frac{|\varepsilon_2|}{\left|\frac{dI_1}{dt}\right|}$$

Here the induced emf is $\varepsilon_2 = 2\ \text{mV} = 2\times 10^{-3}\ V$ and the rate of change of current is $\frac{dI_1}{dt} = 5\ A/s$.

Substituting,

$$M = \frac{2\times 10^{-3}\ V}{5\ A/s}$$

$$M = 4\times 10^{-4}\ H$$

$$M = 0.4\ \text{mH}$$

So the mutual inductance is 0.4 mH, which is option (A). Option (B) 2.5 mH comes from wrongly dividing 5 by 2; option (C) 10 mH and option (D) 25 *H* are not consistent with $M = \varepsilon / (dI/dt)$. Hence the correct choice is **(A) 0.4 mH**.

Final answer. The mutual inductance is 0.4 mH, so the answer is option **(A)**.

← *Back to Q41*

Q42

A coil of area of cross-section 0.5 m^2 *is placed in a magnetic field acting normally to its plane. The field varies as* $B = 0.5t^2 + 2t$*, where B is in tesla and t in seconds. The emf induced in the coil at* $t = 1\ s$ *is:*

(A) 0.5 *V*

(B) 1.0 *V*

(C) **1.5 *V***

(D) 3.0 *V*

[2024 • Set 55-5-2]

SOLUTION

Since the magnetic field acts normally (perpendicular) to the plane of the coil, the magnetic flux linked with the coil is

$$\Phi = BA$$

By Faraday's law, the magnitude of the induced emf is the rate of change of flux,

$$\varepsilon = \frac{d\Phi}{dt} = A\frac{dB}{dt}$$

Here $A = 0.5\ m^2$ is constant and the field varies as $B = 0.5t^2 + 2t$.

Differentiating B with respect to time,

$$\frac{dB}{dt} = t + 2$$

At $t = 1\ s$,

$$\frac{dB}{dt} = 1 + 2 = 3\ T/s$$

Substituting in the emf expression,

$$\varepsilon = A\frac{dB}{dt} = 0.5 \times 3$$

$$\varepsilon = 1.5\ V$$

Therefore the emf induced at $t = 1\ s$ is 1.5 V, which is option (C). Option (A) 0.5 V comes from using only the coefficient of t^2, option (B) 1.0 V from forgetting to differentiate, and option (D) 3.0 V from forgetting the area factor. Hence the correct choice is **(C) 1.5 V**.

COMMON MISTAKE

A common slip is to plug the value of B at t = 1 s instead of dB/dt. The emf depends on the rate of change of B, so always differentiate first, then substitute the time.

Final answer. The induced emf at t = 1 s is 1.5 V, so the answer is option **(C)**.

← *Back to Q42*

Q43

The mutual inductance of two coils, in a given orientation is 50 mH. *If the current in one of the coils changes as* $\boldsymbol{i = 1.0\ \sin(100\pi\, t + \frac{\pi}{6})\ A}$, *the peak value of emf (in volt) induced in the other coil will be:*

(A) 5

(B) 5π

(C) 0.5π

(D) 0.05π

[2024 • Set 55-5-3]

SOLUTION

An emf is induced in the second coil because of the changing current in the first coil. By mutual inductance,

$$\varepsilon = -M\frac{di}{dt}$$

The given current is

$$i = 1.0\ \sin(100\pi\, t + \frac{\pi}{6})\ A$$

Differentiating with respect to time,

$$\frac{di}{dt} = 1.0 \times 100\pi\ \cos(100\pi\, t + \frac{\pi}{6})$$

$$\frac{di}{dt} = 100\pi\ \cos(100\pi\, t + \frac{\pi}{6})\ A/s$$

The cosine has a maximum value of 1, so the peak rate of change of current is

$$(\frac{di}{dt})_{max} = 100\pi\ A/s$$

The peak induced emf is therefore

$$\varepsilon_0 = M(\frac{di}{dt})_{max}$$

$$\varepsilon_0 = (50\times 10^{-3})(100\pi)$$

$$\varepsilon_0 = 5\pi\ V$$

So the peak value of the induced emf is $5\pi\ V$, which is option (B). The other options arise from arithmetic errors such as dropping the factor π (giving 5) or misplacing the power of ten (giving 0.5π or 0.05π). Hence the correct choice is **(B) 5π.**

Final answer. The peak induced emf is 5π V, so the answer is option **(B).**

← *Back to Q43*

Q44

A circular coil of radius 8.0 *cm and* 40 *turns is rotated about its vertical diameter with an angular speed of* 25 *rad* s^{-1} *in a uniform horizontal magnetic field of magnitude* 3.0×10^{-2} *T. The maximum emf induced in the coil is:*

(A) 0.12 V

(B) 0.15 V

(C) 0.19 V

(D) 0.22 V

[2023 • Set 55-1-1]

SOLUTION

When a coil of N turns and area A is rotated with angular speed ω in a uniform magnetic field B, the maximum (peak) emf induced is

$$\varepsilon_0 = \mathrm{NBA}\omega$$

The given data are $N = 40$, $r = 8.0\ \mathrm{cm} = 0.08\ m$, $B = 3.0\times 10^{-2}\ T$ and $\omega = 25\ \mathrm{rad/s}$.

Taking the area of the circular coil as $A = \pi r^2$,

$$A = \pi\,(0.08)^2 = \pi \times 6.4\times 10^{-3}$$

$$A \approx 2.01\times 10^{-2}\ \mathrm{m}^2$$

Substituting,

$$\varepsilon_0 = 40 \times (3.0\times 10^{-2}) \times (2.01\times 10^{-2}) \times 25$$

$$\varepsilon_0 \approx 0.60\ V$$

This strict $\pi\ r^2$ value 0.60 *V* does not appear among the options. The recorded CBSE answer key marks option (C) 0.19 *V*, which is obtained when the area is taken simply as $\boldsymbol{A = r^2}$ (that is, the factor π is dropped):

$$\varepsilon_0 = 40 \times (3.0\times 10^{-2}) \times (0.08)^2 \times 25$$

$$\varepsilon_0 = 40 \times 0.03 \times 6.4\times 10^{-3} \times 25$$

$$\varepsilon_0 = 0.192 \approx 0.19\ V$$

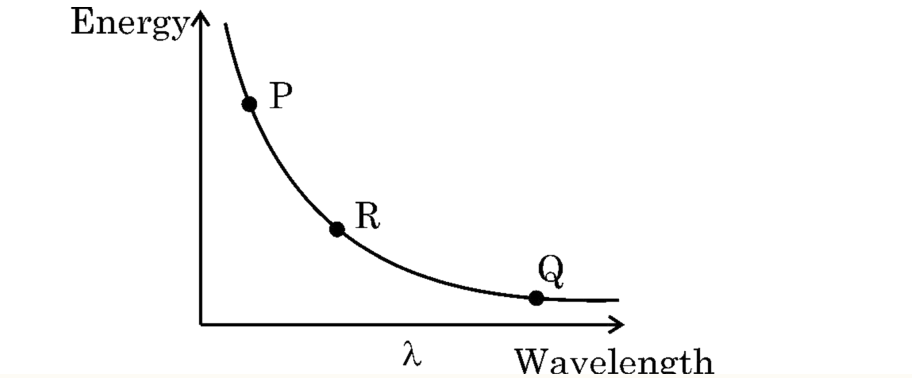
As per the official key the marked option is therefore **(C) 0.19 *V***. In the board exam, write the formula $\boldsymbol{\varepsilon_0 = NBA\omega}$, show the full substitution, and box the numerical result to earn the mark.

Final answer. Using $\varepsilon_0 = NBA\omega$ (with area taken as r^2 per the recorded key), the answer is option **(C)**, 0.19 V.

← *Back to Q44*

Q45

Figure shows a rectangular conductor PSRQ *in which movable arm* PQ *has a resistance 'r' and resistance of* PSRQ *is negligible. The magnitude of emf induced when* PQ *is moved with a velocity* $\vec{V}$ *does not depend on:*



(A) magnetic field ($\vec{B}$)

(B) velocity ($\vec{v}$)

(C) resistance (r)

(D) length of PQ

[2023 • Set 55-1-1]

SOLUTION

The arm PQ slides along the rails of the rectangular conductor, cutting magnetic field lines and producing a motional emf. The magnitude of this emf is

$$\varepsilon = BvL$$

where B is the magnetic field, v is the speed of the moving arm PQ, and L is the length of PQ.

From this relation it is clear that the induced emf depends directly on the magnetic field B, the velocity v, and the length L of the moving conductor.

The resistance r of the arm PQ does not appear in the emf expression. Resistance only controls the current that flows once the emf is set up, through

$$I = \frac{\varepsilon}{r}$$

So r affects the induced current, not the induced emf itself. Therefore the magnitude of the induced emf does not depend on the resistance r, which is option (C). Options (A), (B) and (D), namely B, v and the length of PQ, all appear in $\varepsilon = BvL$ and so the emf does depend on them. Hence the correct choice is **(C) resistance (r)**.

CONCEPT

Keep emf and current separate. Motional emf ε = BvL is set by the field, speed and length; resistance only decides how much current that emf can drive.

Final answer. The induced emf does not depend on the resistance r, so the answer is option **(C)**.

← *Back to Q45*

Q46

Two identical circular coaxial coils A and B, arranged in vertical planes parallel to each other, carry currents in the same direction. If the distance between the coils is decreased at a constant rate, the current:

(A) increases in A and decreases in B.

(B) **decreases in both A and B.**

(C) increases in both A and B.

(D) remains same in both A and B.

[2023 • Set 55-1-2]

SOLUTION

Coils A and B are coaxial and carry currents in the same direction, so each coil produces a magnetic field that adds to the flux through the other coil.

When the distance between the coils is decreased, the mutual flux linking each coil increases, because the field of one coil grows stronger at the location of the other.

By Lenz's law, the induced current in each coil must oppose this increase in flux. To oppose an increasing flux that points in the same direction as the coil's own field, the induced current flows opposite to the original current in that coil.

This opposing induced current reduces the net current in each coil. Since the situation is symmetric, the current decreases in both coil A and coil B.

Therefore the current decreases in both A and B, which is option (B). Option (A) wrongly treats the coils asymmetrically, option (C) violates Lenz's law (the induced effect opposes the change, it does not aid it), and option (D) ignores the induced emf altogether. Hence the correct choice is **(B) decreases in both *A* and *B*.**

Final answer. The current decreases in both coils, so the answer is option **(B)**.

← *Back to Q46*

Q47

A bar magnet is dropped in a hollow metallic cylinder along its vertical axis. The acceleration of the falling magnet will be:

(A) zero

(B) equal to g

(C) **less than g**

(D) greater than g

[2023 • Set 55-1-3]

SOLUTION

As the bar magnet falls along the axis of the hollow metallic cylinder, the magnetic flux through the cylinder walls changes with time.

This changing flux induces eddy currents in the conducting walls of the cylinder.

By Lenz's law, these induced eddy currents flow in such a direction that their magnetic effect opposes the motion of the magnet that produces them. As a result, the cylinder exerts an upward retarding (opposing) force on the falling magnet.

The net downward force on the magnet is therefore

$$F_{net} = mg - F_{retarding}$$

which is less than its weight mg. The acceleration is

$$a = g - \frac{F_{\text{retarding}}}{m}$$

so $a < g$.

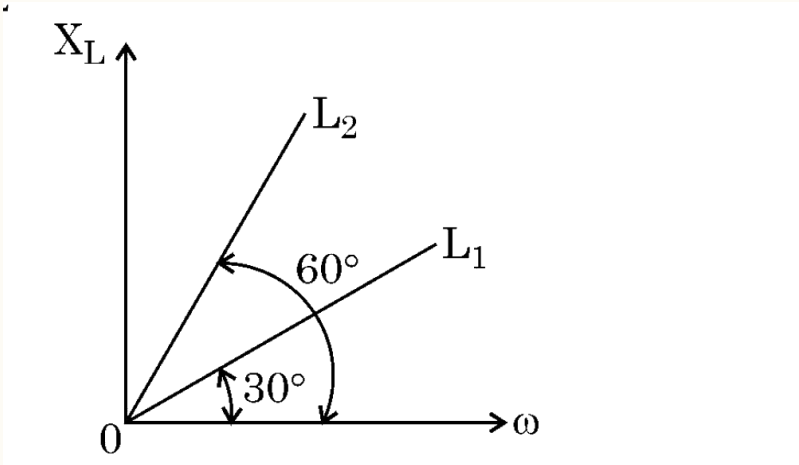
Hence the magnet falls with an acceleration less than g, which is option (C). It is not zero (option A) because the magnet still falls; it is not equal to g (option B) because eddy currents are present; and it cannot be greater than g (option D) because the induced force opposes, not aids, the motion. Hence the correct choice is **(C) less than g**.

Final answer. The magnet falls with acceleration less than g, so the answer is option **(C)**.

← *Back to Q47*

Q48

The direction of induced current in the loop abc is:



(A) along abc if I decreases

(B) along acb if I increases

(C) along abc if I is constant

(D) along abc if I increases

[2023 • Set 55-4-1]

SOLUTION

The triangular loop abc is placed near a long straight wire carrying current I, so the wire's magnetic field passes through the plane of the loop.

When the current I in the wire increases, the magnetic flux linked with the loop also increases.

By Lenz's law, the induced current in the loop must oppose this increase, so it sets up a magnetic field inside the loop directed opposite to the flux of the wire through the loop.

Applying the right-hand rule for the geometry shown in the figure, the induced current must circulate in the sense $a \to b \to c$ to produce this opposing field. So the induced current flows along abc when I increases.

Therefore the answer is option (D): induced current along abc if I increases. Option (A) is wrong because if I decreased, the flux would fall and the induced current would aid it, flowing along acb instead; option (B) gives the reverse sense for an increasing current; and option (C) is impossible since a constant I gives constant flux and hence no induced current at all. Hence the correct choice is **(D) along abc if I increases**.

Final answer. The induced current flows along abc when I increases, so the answer is option **(D)**.

← *Back to Q48*

Q49

A square shaped coil of side 10 cm, having 100 turns is placed perpendicular to a magnetic field which is increasing at 1 T/s. The induced emf in the coil is

(A) 0.1 V

(B) 0.5 V

(C) 0.75 V

(D) 1.0 V

[2023 • Set 55-5-1]

SOLUTION

For a coil of N turns placed perpendicular to a magnetic field, the flux through the coil is $\mathbf{\Phi = NBA}$. By Faraday's law, the magnitude of the induced emf is

$$\varepsilon = N\,\mathrm{A}\,\frac{\mathrm{dB}}{\mathrm{dt}}$$

The coil is square with side $\boldsymbol{a = 10\ cm = 0.1\ m}$, so its area is

$$A = \mathrm{a}^2 = (0.1)^2$$

$$A = 0.01\ \mathrm{m}^2$$

Here $\boldsymbol{N = 100}$ and the field increases at the rate $\frac{\mathbf{dB}}{\mathbf{dt}} = \mathbf{1\ T/s}$.

Substituting,

$$\varepsilon = 100 \times 0.01 \times 1$$

$$\varepsilon = 1.0\ V$$

Therefore the induced emf in the coil is 1.0 V, which is option (D). Option (A) 0.1 V comes from omitting the number of turns, while options (B) and (C) do not follow from $\boldsymbol{\varepsilon = NA\ dB/dt}$. Hence the correct choice is **(D) 1.0 V**.

Final answer. The induced emf is 1.0 V, so the answer is option **(D)**.

← *Back to Q49*

Q50

Laminated iron sheets are used to minimize _____ currents in the core of a transformer.

[2020 • Set 55-1-1]

SOLUTION

Laminated iron sheets are used to minimise **eddy currents** in the core of a transformer.

When the alternating magnetic flux in the core changes with time, it induces circulating currents, called eddy currents, in the body of the iron core. These

eddy currents heat the core and waste energy as I^2R losses, reducing the efficiency of the transformer.

To reduce them, the core is made of thin iron sheets (laminations) that are coated with an insulating varnish and stacked together, instead of using a single solid block of iron.

The insulation between laminations breaks up the large eddy-current loops into many small loops. This greatly increases the resistance of the eddy-current paths and confines the currents to a small cross-section, so the eddy currents and the energy lost as heat are both reduced.

Hence the blank is filled by the word **eddy** currents.

Final answer. Laminated iron sheets are used to minimise eddy currents in the transformer core.

← *Back to Q50*

Q51

The number of turns of a solenoid are doubled without changing its length and area of cross-section. The self-inductance of the solenoid will become _____ times.

[2020 • Set 55-1-1]

SOLUTION

The self-inductance of a long solenoid of N turns, length l and cross-sectional area A is given by

$$L = \frac{\mu_0 N^2 A}{l}$$

Here μ_0, A and l are all kept constant (the length and area of cross-section do not change), so

$$L \propto N^2$$

When the number of turns is doubled, $N \rightarrow 2N$. Let the new self-inductance be L'. Then

$$\frac{L'}{L} = \frac{(2N)^2}{N^2} = \frac{4N^2}{N^2} = 4$$

$$\Rightarrow L' = 4L$$

Hence, on doubling the number of turns without changing the length and the area of cross-section, the **self-inductance becomes 4 times** its original value.

Final answer. The self-inductance becomes 4 times.

← Back to Q51

Q52

A conducting rod of length l is kept parallel to a uniform magnetic field $\vec{B}$. It is moved along the magnetic field with a velocity $\vec{v}$. What is the value of emf induced in the conductor?

[2020 • Set 55-2-1]

SOLUTION

When a conducting rod of length l moves with velocity $\vec{v}$ in a magnetic field $\vec{B}$, the motional emf induced across its ends is given by

$$\varepsilon = (\vec{v} \times \vec{B}) \cdot \vec{l}$$

whose magnitude can be written as $\boldsymbol{\varepsilon = B\, v\, l\, \theta}$, where θ is the angle between $\vec{v}$ and $\vec{B}$.

Here the rod is moved along the direction of the magnetic field, so the velocity $\vec{v}$ is parallel to $\vec{B}$, giving $\boldsymbol{\theta = 0°}$.

$$\vec{v} \times \vec{B} = v\, B \sin 0° = 0$$

Physically, as the rod moves along $\vec{B}$ it does not cut (sweep across) any magnetic field lines, so no flux is cut and no emf is set up.

Therefore the **emf induced in the conductor is zero**.

Final answer. The induced emf is zero.

← Back to Q52

Q53

Draw the graph showing variation of the value of the induced emf as a function of rate of change of current flowing through an ideal inductor.

[2020 • Set 55-2-1]

SOLUTION

For an ideal inductor (which has zero resistance), the only voltage across it is the back emf produced by the changing current. By Faraday's law together with Lenz's law,

$$\varepsilon = -L\,\frac{dI}{dt}$$

Taking magnitudes, $|\varepsilon| = L\left|\frac{dI}{dt}\right|$, so the induced emf is directly proportional to the rate of change of current, with the constant of proportionality being the self-inductance L.

Graph: Plot the induced emf ε (on the y-axis) against the rate of change of current $\frac{dI}{dt}$ (on the x-axis).

The graph is a **straight line passing through the origin**, having a constant positive slope equal to the inductance L. (If the sign is retained, the line has a negative slope $-L$.) A larger inductance gives a steeper line.

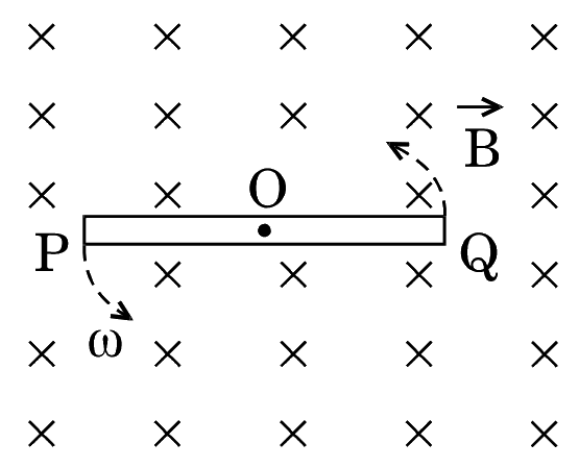
Final answer. The graph is a straight line through the origin with slope equal to L.

← Back to Q53

Q54

A metallic rod PQ of length l is rotated with an angular velocity ω about an axis passing through its mid-point (O) and perpendicular to the plane of the paper, in

uniform magnetic field $\vec{B}$, as shown in the figure. What is the potential difference developed between the two ends of the rod, P and Q?



[2020 • Set 55-2-2]

SOLUTION

The rod PQ rotates about its mid-point O, which lies on the axis of rotation, with the magnetic field $\vec{B}$ directed perpendicular to the plane of rotation. Each half of the rod (OP and OQ) behaves like a rod of length $l/2$ pivoted at the axis O.

For a rod of length r rotating about one end in a perpendicular field, the motional emf between the end and the axis is

$$\varepsilon = \frac{1}{2} B \omega r^2$$

For the half OP, with $r = \frac{l}{2}$:

$$\varepsilon_{OP} = \frac{1}{2} B \omega \left(\frac{l}{2}\right)^2 = \frac{B \omega l^2}{8}$$

For the half OQ, similarly:

$$\varepsilon_{OQ} = \frac{1}{2} B \omega \left(\frac{l}{2}\right)^2 = \frac{B \omega l^2}{8}$$

Both halves rotate in the same sense in the same field, so each end (P and Q) is at the same potential with respect to the centre O. Hence the two ends P and Q are at equal potential.

$$V_P - V_Q = 0$$

Therefore the **potential difference between the ends P and Q is zero**.

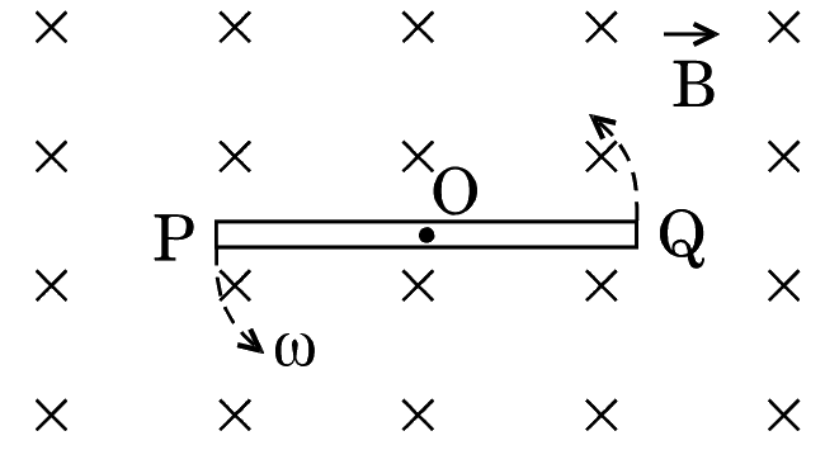
> **COMMON MISTAKE**
>
> *Both ends move symmetrically, so the same emf develops in each half. Students often forget that the PD between the two outer ends is zero, while the PD between O and either end is non-zero.*

Final answer. The potential difference between P and Q is zero.

← *Back to Q54*

Q55

A metallic rod PQ of length l is rotated with an angular velocity ω in a magnetic field $\vec{B}$ about an axis passing through the mid-point O of the rod and perpendicular to the plane of paper as shown in the figure. What is the potential difference developed between the points O and Q of the rod?



[2020 • Set 55-2-3]

SOLUTION

The rod PQ rotates about its mid-point O with angular velocity ω in a uniform magnetic field $\vec{B}$ that is perpendicular to the plane of rotation. The segment OQ has length $\frac{l}{2}$, with O lying on the axis of rotation.

As OQ rotates, it sweeps out area and cuts magnetic field lines, so a motional emf is set up between O and Q. For a rod of length r rotating about one end in a

perpendicular field, the emf is

$$\varepsilon = \frac{1}{2} B \omega r^2$$

Substituting $r = \frac{l}{2}$:

$$|V_O - V_Q| = \frac{1}{2} B \omega \left(\frac{l}{2}\right)^2$$

$$|V_O - V_Q| = \frac{1}{2} B \omega \cdot \frac{l^2}{4}$$

$$|V_O - V_Q| = \frac{B \omega l^2}{8}$$

Therefore the **potential difference developed between O and Q is** $\mathbf{\frac{B\omega l^2}{8}}$.

Final answer. The potential difference between O and Q is $\mathbf{\frac{B\omega l^2}{8}}$.

← *Back to Q55*

Q56

Two identical coils, one of copper and the other of aluminium are rotated with the same angular speed in an external magnetic field. In which of the two coils will the induced current be more?

[2020 • Set 55-3-1]

SOLUTION

Both coils are identical in shape and size and are rotated with the same angular speed ω in the same external magnetic field $\vec{B}$. The induced emf in a rotating coil is

$$\varepsilon = N A B \omega \omega t$$

Since N, A, B and ω are the same for both coils, the same emf is induced in each. The emf does not depend on the material of the coil.

However, the induced current depends on the resistance of the coil through Ohm's law:

$$i = \frac{\varepsilon}{R}$$

The resistance is $R = \frac{\rho L}{A}$, so for identical geometry $R \propto \rho$, the resistivity.

Copper has a lower resistivity than aluminium, so the copper coil has the lower resistance.

A lower resistance gives a larger current for the same emf. Hence the **induced current is more in the copper coil**.

Final answer. The induced current is more in the copper coil.

← *Back to Q56*

Q57

In which orientation of the armature coil of an ac generator relative to the magnetic field, will the induced emf be maximum?

[2020 • Set 55-3-3]

SOLUTION

The emf induced in the rotating armature coil of an ac generator is given by

$$\varepsilon = N A B \omega \omega t$$

where $\omega t = \theta$ is the angle through which the coil has turned, measured from the position where its plane is perpendicular to $\vec{B}$.

The emf is maximum when $\omega t = 1$, i.e. when $\omega t = 90°$.

$$\varepsilon_{max} = N A B \omega$$

At this instant the plane of the coil is parallel to the magnetic field $\vec{B}$, which means the area vector (normal) $\vec{A}$ is perpendicular to $\vec{B}$. In this orientation the flux through the coil is zero, but the rate of change of flux is maximum, so the induced emf is maximum.

Therefore the induced emf is maximum when the **plane of the coil is parallel to the magnetic field (area vector perpendicular to $\vec{B}$)**.

> **COMMON MISTAKE**
>
> *A common error is to say the emf is maximum when the flux is maximum. In fact emf depends on the rate of change of flux, so it peaks when the flux itself is zero (plane parallel to $\vec{B}$).*

Final answer. The emf is maximum when the plane of the coil is parallel to the field (area vector perpendicular to $\vec{B}$).

← Back to Q57

Q58

Plot a graph showing variation of induced e.m.f. with the rate of change of current flowing through a coil.

OR

A series combination of an inductor (L), capacitor (C) and a resistor (R) is connected across an ac source of emf of peak value E_0 and angular frequency (ω). Plot a graph to show variation of impedance of the circuit with angular frequency (ω).

[2020 • Set 55-5-1]

SOLUTION

Part 1: Induced emf vs rate of change of current in a coil.

For an ideal inductor the induced (back) emf is

$$|\varepsilon| = L\,\frac{dI}{dt}$$

This is linear in $\frac{dI}{dt}$, with slope equal to the self-inductance L. So the graph of ε (on the y-axis) against $\frac{dI}{dt}$ (on the x-axis) is a **straight line through the origin with positive slope L**.

OR Part 2: Impedance vs angular frequency for a series LCR circuit.

The impedance of the series LCR circuit is

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$.

At low frequency ($\omega \to 0$): $X_C = \frac{1}{\omega C} \to \infty$, so $Z \to \infty$.

At resonance ($\omega_0 = \frac{1}{\sqrt{LC}}$): $X_L = X_C$, so the reactive part vanishes and Z is minimum, $Z_{min} = R$.

At high frequency ($\omega \to \infty$): $X_L = \omega L \to \infty$, so $Z \to \infty$.

Graph: Plot Z (on the y-axis) against ω (on the x-axis). The curve is **U-shaped (dip-shaped), with a minimum value R at $\omega = \omega_0$**, rising towards infinity on both sides as $\omega \to 0$ and $\omega \to \infty$.

Final answer. Part 1: a straight line through the origin with slope L; OR Part 2: a U-shaped curve with minimum value R at the resonant frequency $\omega_0 = 1/\sqrt{LC}$.

← *Back to Q58*

Q59

A long straight current carrying wire passes normally through the centre of circular loop. If the current through the wire increases, will there be an induced emf in the loop? Justify.

[2017]

SOLUTION

The long straight wire passes normally (perpendicularly) through the centre of the circular loop, so the wire lies along the axis perpendicular to the plane of the loop.

The magnetic field produced by the straight wire forms concentric circles around the wire. At every point in the plane of the loop, this field $\vec{B}$ lies in the plane of the loop itself, that is, it is directed along the plane and not along the normal $\hat{n}$ to the loop.

Since $\vec{B}$ is everywhere perpendicular to the area vector (normal) of the loop, the magnetic flux linked with the loop is

$$\Phi = \int \vec{B} \cdot \hat{n}\, dA = 0$$

Even when the current in the wire increases, the field lines remain in the plane of the loop, so the flux through the loop stays zero at all times.

By Faraday's law, $\varepsilon = -\frac{d\Phi}{dt} = 0$.

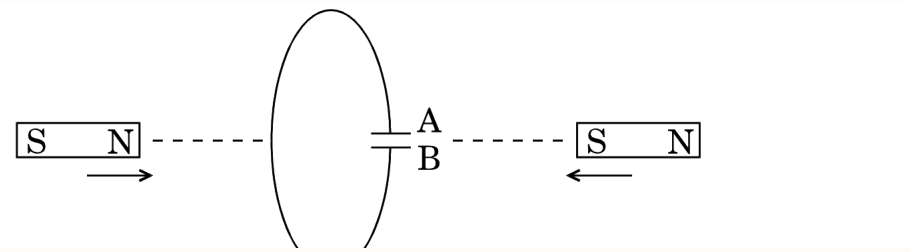
Therefore **no induced emf is produced in the loop**, however fast the current may change.

Final answer. No induced emf is produced in the loop, because the flux linked with the loop is zero.

← *Back to Q59*

Q60

Predict the polarity of the capacitor in the situation described below:



[2017]

SOLUTION

The capacitor lies in a closed conducting loop placed in a region of changing magnetic flux. As the flux through the loop changes, an emf is induced in the loop by Faraday's law:

$$\varepsilon = -\frac{d\Phi}{dt}$$

The direction of the induced current is fixed by Lenz's law: the induced current flows in such a way as to oppose the change in magnetic flux that produces it.

This induced current flows through the loop and charges the capacitor. The plate towards which the conventional (induced) current flows receives positive charge, while the other plate, from which the current flows away, becomes negatively charged.

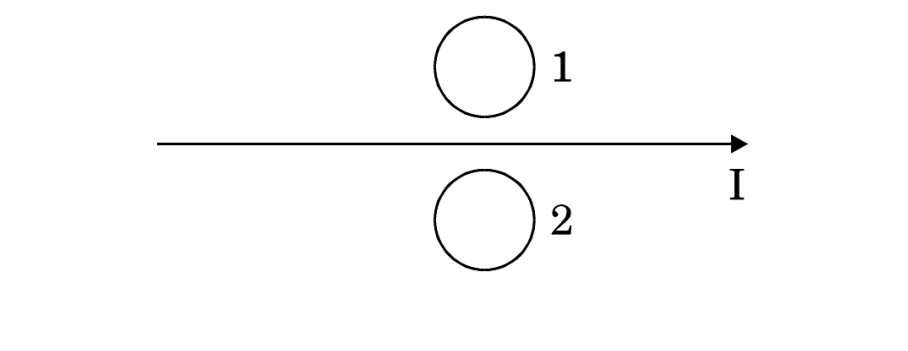
Hence, on tracing the induced current using Lenz's law, the **plate at which the current arrives is the positive plate and the opposite plate is the negative plate**.

Final answer. The plate towards which the induced conventional current flows becomes positive, and the other plate becomes negative.

← *Back to Q60*

Q61

What is the direction of induced currents in metal rings 1 and 2 when current I in the wire is increasing steadily?



[2017]

SOLUTION

By the right-hand thumb rule, the steady current I in the long straight wire sets up a magnetic field $\vec{B}$ that circulates in concentric circles around it. On one side of the wire this field points into the plane of the rings, and on the other side it points out of the plane.

Ring 1 lies on one side of the wire, so the flux through it (say, into the page) is steadily increasing as I increases.

By Lenz's law, the induced current must oppose this increase, so it flows so as to produce a magnetic field out of the page inside ring 1. Hence the induced current in ring 1 is anticlockwise.

Ring 2 lies on the opposite side of the wire, where the wire's field points out of the page, and this outward flux is also increasing as I increases.

By Lenz's law, the induced current in ring 2 must produce a field into the page to oppose the increase, so it flows clockwise.

Therefore the **induced currents in rings 1 and 2 flow in opposite senses**, each opposing its own increasing flux as required by Lenz's law.

Final answer. The induced currents in rings 1 and 2 circulate in opposite directions (one anticlockwise, the other clockwise).

← *Back to Q61*

Q62

Two spherical bobs, one metallic and the other of glass, of the same size are allowed to fall freely from the same height above the ground. Which of the two would reach earlier and why ?

[2014]

SOLUTION

Both bobs are released from the same height and fall freely under gravity. As the metallic bob moves through the Earth's magnetic field, the magnetic flux linked with it changes, and since it is a conductor, eddy currents are induced in it.

By Lenz's law, these induced eddy currents oppose the cause producing them, i.e. they oppose the motion of the metallic bob. This produces a small retarding (opposing) force on the metallic bob.

The glass bob is a non-conductor, so no eddy currents can be set up in it and it experiences no such retarding force. It therefore falls with the full acceleration due to gravity.

Hence the **glass bob reaches the ground earlier**, because the metallic bob is slowed slightly by the opposing eddy currents while the glass bob is not.

TIP

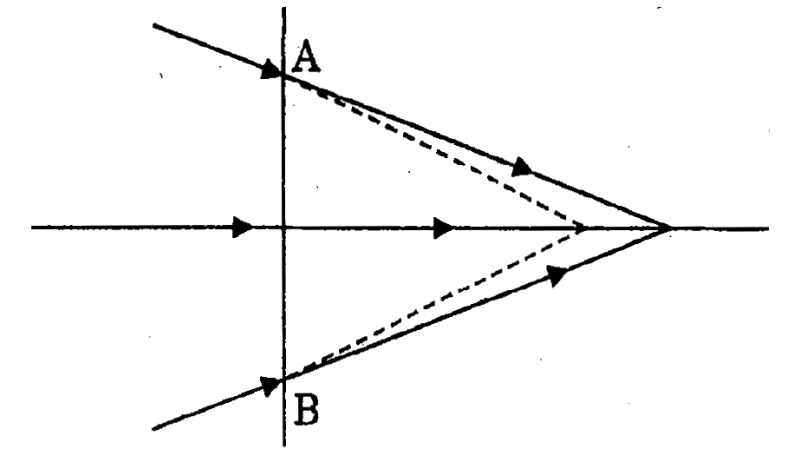
Eddy currents only form in conductors. The key one-mark point examiners want is: eddy currents in the metal bob oppose its motion (Lenz's law) and slow it down.

Final answer. The glass bob reaches the ground earlier, because eddy currents retard the metallic bob.

← *Back to Q62*

Q63

A conducting loop is held below a current carrying wire PQ as shown. Predict the direction of the induced current in the loop when the current in the wire is constantly increasing.



[2014]

SOLUTION

The straight wire PQ carries current I and lies above the rectangular loop. By the right-hand thumb rule, the magnetic field of this wire at the location of the loop (below the wire) points into the plane of the paper.

As the current I is constantly increasing, the magnitude of this field $|\vec{B}|$ increases, so the magnetic flux directed into the page through the loop also

increases.

By Lenz's law, the induced current must oppose this increase in inward flux, so it must set up a magnetic field directed out of the page inside the loop.

By the right-hand-curl rule, a current producing an outward field circulates anticlockwise.

Hence the **induced current in the loop flows anticlockwise** (as seen from the side facing the wire).

Final answer. The induced current in the loop flows anticlockwise.

← *Back to Q63*

Q64

Give the direction in which the induced current flows in the coil mounted on an insulating stand when a bar magnet is quickly moved along the axis of the coil from one side to the other as shown in the figure.

[2013]

SOLUTION

As the bar magnet is moved quickly along the axis of the coil from one side to the other, the magnetic flux linked with the coil first increases and then decreases, so an induced current is set up that reverses direction during the motion.

While the magnet approaches the coil, the flux through the coil increases. By Lenz's law, the induced current opposes this increase, so the near face of the coil becomes a like pole (e.g. North facing an approaching North pole) to repel the magnet.

As seen from the magnet's side during approach, this induced current flows anticlockwise.

After the magnet passes through and recedes on the other side, the flux through the coil decreases. By Lenz's law, the induced current now reverses to oppose the decrease, making the near face an unlike pole that attracts the receding magnet.

During recession the induced current therefore flows clockwise as seen from the same side.

Hence the **induced current reverses its direction** as the magnet crosses the coil: it flows one way during approach and the opposite way during recession.

Final answer. The induced current reverses direction: anticlockwise while the magnet approaches and clockwise while it recedes (opposite senses).

← *Back to Q64*

Q65

Define self-inductance of a coil. Write its S.I. units.

[2012]

SOLUTION

The self-inductance L of a coil is defined as the magnetic flux linkage of the coil per unit current flowing through it:

$$L = \frac{N\Phi}{I}$$

where $N\Phi$ is the total flux linkage and I is the current.

Equivalently, when the current through the coil changes, a back-emf is induced given by

$$\varepsilon = -L\,\frac{dI}{dt}$$

so L is numerically equal to the induced emf when the current changes at the rate of $1\ A\ s^{-1}$.

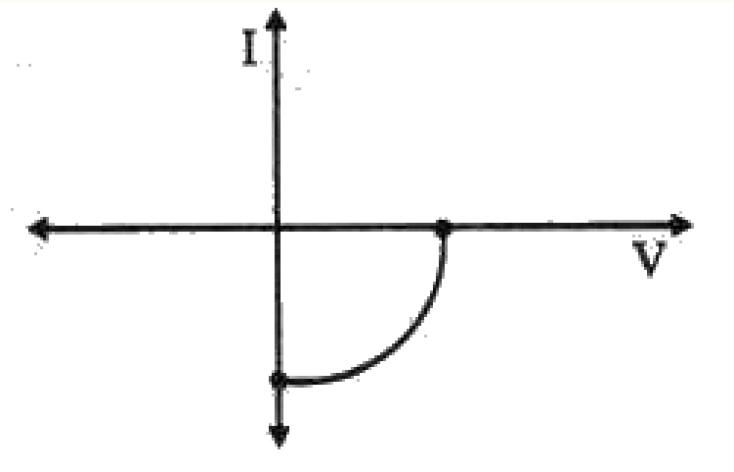
The **SI unit of self-inductance is the henry (H)**, where $\mathbf{1\ H = 1\ Wb\ A^{-1} = 1\ V\ s\ A^{-1}}$.

Final answer. Self-inductance is the flux linkage per unit current, $\boldsymbol{L = N\Phi/I}$; its SI unit is the henry (H).

← Back to Q65

Q66

A bar magnet is moved in the direction indicated by the arrow between two coils PQ and CD. Predict the directions of induced current in each coil.



[2012]

SOLUTION

The bar magnet moves along the line joining the two coils, so it recedes from one coil (PQ) and approaches the other (CD). Each coil opposes its own change of flux by Lenz's law.

At coil CD, towards which the magnet approaches, the magnetic flux increases. By Lenz's law, the induced current opposes this increase, making the near face of CD a like pole that repels the approaching magnet.

At coil PQ, from which the magnet recedes, the magnetic flux decreases. By Lenz's law, the induced current opposes this decrease, making the near face of PQ an unlike pole that attracts the receding magnet.

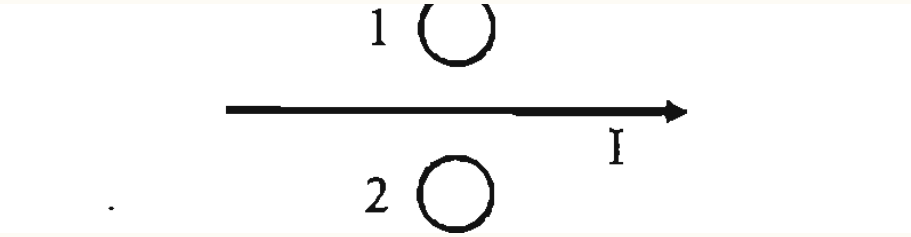
Because one coil presents a repelling pole and the other an attracting pole to the same passing magnet, the **two induced currents circulate in opposite senses**: in CD the induced current makes its near face a like pole, while in PQ it makes its near face an unlike pole, each obeying Lenz's law.

Final answer. The two induced currents flow in opposite senses: CD repels the approaching magnet and PQ attracts the receding magnet.

← *Back to Q66*

Q67

Predict the directions of induced currents in metal rings 1 and 2 lying in the same plane where current I in the wire is increasing steadily.



[2012]

SOLUTION

The current I in the long straight wire is increasing steadily, so by the right-hand thumb rule the magnetic field it produces grows in strength. This field forms concentric circles around the wire, pointing into the plane on one side and out of the plane on the other.

Ring 1 lies on one side of the wire, where the field is, say, into the page. As I increases, this inward flux through ring 1 increases.

By Lenz's law, the induced current in ring 1 opposes the increase by producing an outward field, so it flows anticlockwise.

Ring 2 lies on the other side of the wire, where the field points out of the page. As I increases, this outward flux through ring 2 increases.

By Lenz's law, the induced current in ring 2 opposes the increase by producing an inward field, so it flows clockwise.

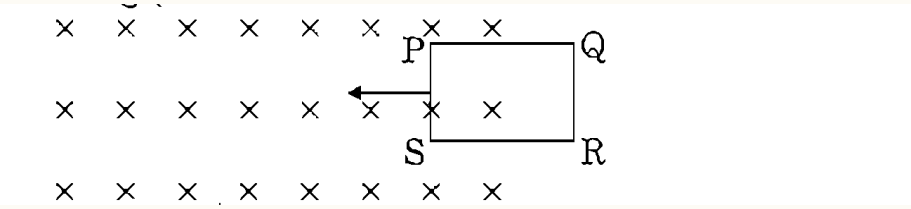
Hence the **induced currents in rings 1 and 2 flow in opposite directions** (anticlockwise in ring 1, clockwise in ring 2).

Final answer. The induced current is anticlockwise in ring 1 and clockwise in ring 2 (opposite directions).

← *Back to Q67*

Q68

A closed loop (PQRS) of wire is moved into a uniform magnetic field at right angles to the plane of the paper as shown in the figure. Predict the direction of the induced current in the loop.



[2012]

SOLUTION

The closed loop PQRS is moved into a uniform magnetic field $\vec{B}$ directed into the plane of the paper. As the loop enters, the area of the loop lying inside the field increases.

Therefore the magnetic flux into the page through the loop increases as it moves in.

By Lenz's law, the induced current must oppose this increase in inward flux, so it sets up a magnetic field directed out of the page inside the loop.

By the right-hand-curl rule, a current that produces an outward field circulates anticlockwise.

Hence the **induced current in the loop flows anticlockwise** (i.e. along the path P -> S -> R -> Q -> P), creating an outward flux that opposes the increasing inward flux.

TIP

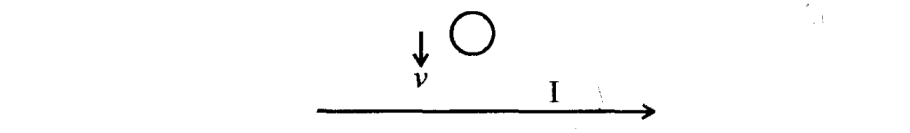
For an entering loop the induced current opposes the motion. A quick check: the side of the loop inside the field feels a force opposing the loop's entry, which fixes the current sense.

Final answer. The induced current flows anticlockwise in the loop (producing flux out of the page).

← *Back to Q68*

Q69

Predict the direction of induced current in a metal ring when the ring is moved towards a straight conductor with constant speed v. The conductor is carrying current I in the direction shown in the figure.



[2012]

SOLUTION

As the metal ring moves towards the straight wire with speed v, it enters a region where the wire's magnetic field is stronger, since the field of a straight wire falls off as $1/r$.

Therefore the magnetic flux linked with the ring (in the direction set by the wire's field on the near side, say into the page) increases as the ring approaches.

By Lenz's law, the induced current opposes this increase, so it must produce a magnetic field out of the page inside the ring.

By the right-hand-curl rule, a current producing an outward field circulates anticlockwise.

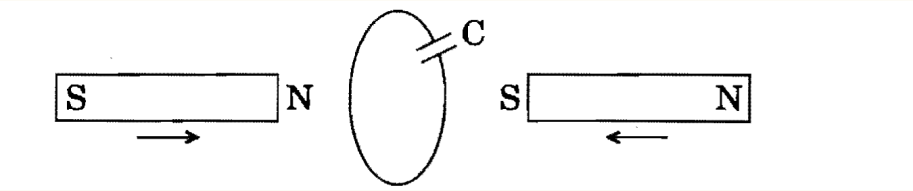
Hence the **induced current in the ring flows anticlockwise** (as seen from the front), so that the resulting force opposes the ring's approach towards the wire, in accordance with Lenz's law.

Final answer. The induced current in the ring flows anticlockwise (opposing the increasing flux as it nears the wire).

← *Back to Q69*

Q70

Two bar magnets are quickly moved towards a metallic loop connected across a capacitor C as shown in the figure. Predict the polarity of the capacitor.



[2011]

SOLUTION

As the two bar magnets are quickly moved towards the metallic loop, the magnetic flux linked with the loop increases. By Lenz's law, an induced current is set up in the loop that opposes this increase.

The induced current flows so that the face of the loop turned towards the approaching magnets becomes a like pole (the same pole as the approaching pole of each magnet), so as to repel the magnets and oppose the growing flux.

This induced current drives charge onto the capacitor plates: conventional current flows out of the loop into one plate, leaving it positively charged, while the other plate becomes negatively charged.

Tracing the induced current around the loop (fixed by Lenz's law), the plate towards which the conventional current is pushed acquires the **positive polarity** and the other plate acquires the negative polarity.

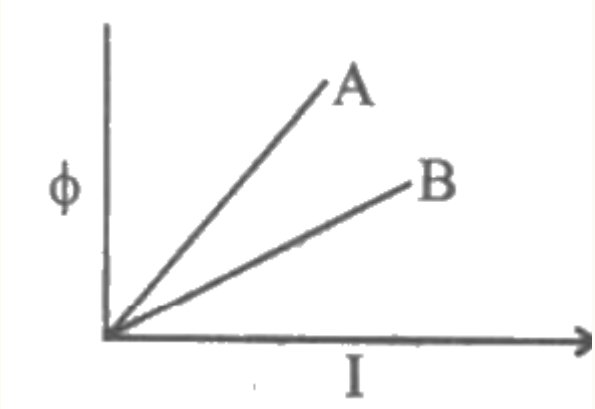
If the magnets were instead moved away, the flux would decrease, the induced current would reverse, and the **polarity of the capacitor would also reverse**.

Final answer. The plate the induced (Lenz's-law) current flows into becomes positive and the other negative; the polarity reverses if the magnets are withdrawn.

← Back to Q70

Q71

A plot of magnetic flux (Φ) versus current (I) is shown in the figure for two inductors A and B. Which of the two has larger value of self inductance?



[2010]

SOLUTION

The self-inductance relates the magnetic flux linked with a coil to the current flowing through it by $\Phi = LI$.

Rearranging, the self-inductance is the ratio of flux to current,

$$L = \frac{\Phi}{I}$$

which is exactly the slope of the Φ-versus-I straight-line graph.

Therefore the inductor whose line is steeper (larger slope) has the larger self-inductance.

In the given figure, inductor **A** has the steeper slope, so $L_A > L_B$. Hence **inductor A has the larger value of self-inductance.**

CONCEPT

Slope of a Φ-vs-I graph IS the self-inductance L; the steeper line always wins, regardless of where the lines cross.

Final answer. Inductor A, since it has the steeper slope in the Φ-vs-I plot, has the larger self-inductance.

← Back to Q71

Q72

When current in a coil changes with time, how is the back emf induced in the coil related to it?

[2008]

SOLUTION

When the current in a coil changes with time, the magnetic flux linked with the coil also changes, and by Faraday's law an emf is induced in the coil itself. This self-induced emf is called the **back emf**.

The back emf is directly proportional to the rate of change of current, and is given by

$$\varepsilon_{back} = -L\frac{dI}{dt}$$

where L is the self-inductance of the coil.

The negative sign is a statement of **Lenz's law**: the back emf opposes the very change of current that produces it. If the current is increasing, the back emf acts to oppose the rise; if the current is decreasing, it acts to maintain the current.

In magnitude, $|\varepsilon_{back}| = L\left|\frac{dI}{dt}\right|$.

Final answer. The back emf is $\varepsilon = -L\,\frac{dI}{dt}$, proportional to the rate of change of current and opposing that change (Lenz's law).

← Back to Q72

Q73

Write S.I. unit of magnetic flux. Is it a scalar or a vector quantity?

[2003]

SOLUTION

The SI unit of magnetic flux is the **weber (Wb)**.

One weber is the flux when a uniform field of one tesla passes normally through an area of one square metre,

1 Wb = 1 T^2 = 1 V⋯.

Magnetic flux is defined as the surface integral

$$\Phi = \int \vec{B} \cdot \mathrm{d}\vec{S}$$

which is the dot (scalar) product of the field vector $\vec{B}$ with the area vector $\mathrm{d}\vec{S}$. A dot product yields a number, so magnetic flux is a **scalar quantity**.

Final answer. The SI unit of magnetic flux is the weber (Wb), and magnetic flux is a scalar quantity.

← *Back to Q73*

Q74

A plane circular coil is rotated about its vertical diameter with a constant angular speed ω in a uniform horizontal magnetic field. Initially the plane of the coil is parallel to the magnetic field. Draw plots showing the variation of the following physical quantities as the function of ω t, where t represents time elapsed:
(a) Magnetic flux ϕ linked with the coil, and
(b) emf induced in the coil.

[2026 • Set 55-5-1]

SOLUTION

Let the coil of area A and N turns rotate with angular speed ω in a uniform field $\vec{B}$. The flux is **Φ = NBAθ**, where θ is the angle between the area vector $\hat{n}$ and $\vec{B}$.

Since the plane of the coil is initially parallel to $\vec{B}$, the area vector $\hat{n}$ is initially perpendicular to $\vec{B}$, so $\theta(t) = \omega t + \frac{\pi}{2}$.

(a) Magnetic flux:

$$\Phi = NBA\cos\left(\omega t + \frac{\pi}{2}\right) = -NBA\sin(\omega t)$$

Plot of Φ versus ωt: a negative sine curve. The vertical axis is Φ, the horizontal axis is ωt. It starts at $\Phi = 0$ at $\omega t = 0$, falls to a minimum $-NBA$ at $\omega t = \frac{\pi}{2}$, returns to 0 at $\omega t = \pi$, rises to a maximum $+NBA$ at $\omega t = \frac{3\pi}{2}$, and returns to 0 at $\omega t = 2\pi$.

(b) Induced emf:

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d}{dt}[-NBA\sin(\omega t)]$$

$$\varepsilon = NBA\,\omega\cos(\omega t)$$

Plot of ε versus ωt: a cosine curve. The vertical axis is ε, the horizontal axis is ωt. It starts at its maximum $+NBA\omega$ at $\omega t = 0$, falls to 0 at $\omega t = \frac{\pi}{2}$, reaches the minimum $-NBA\omega$ at $\omega t = \pi$, returns to 0 at $\omega t = \frac{3\pi}{2}$, and back to $+NBA\omega$ at $\omega t = 2\pi$.

The emf curve is thus **90° out of phase** with the flux curve: the emf is maximum when the flux is zero, and zero when the flux is maximum.

COMMON MISTAKE

Because the coil starts with its plane parallel to B, the flux starts at zero (a sine), not at maximum; getting the initial phase wrong flips both graphs.

Final answer. Flux varies as $\Phi = -NBA\sin(\omega t)$ (a negative sine curve) and emf as $\varepsilon = NBA\omega\cos(\omega t)$ (a cosine curve), the two being **90°** out of phase.

← Back to Q74

Q75

A square loop of side 10 cm, *free to rotate about a vertical axis coinciding with its one arm, is initially held perpendicular to a uniform horizontal magnetic field of* 0.2 *T. If it is rotated at the uniform speed of* 60 rpm, *find the emf induced in the loop.*

[2026 • Set 55-5-2]

SOLUTION

Given: side of square loop $L = 10\ \text{cm} = 0.10\ m$, field $B = 0.2\ T$, number of turns $N = 1$, and the loop rotates at 60 rpm, initially perpendicular to $\vec{B}$.

Area of the loop:

$$A = \text{L}^2 = (0.10)^2 = 0.01\ \text{m}^2$$

Angular speed:

$$\omega = 2\pi\nu = 2\pi\times\frac{60}{60} = 2\pi\ \text{rad s}^{-1}$$

The peak (maximum) induced emf is

$$\varepsilon_0 = \text{NBA}\omega$$

Substituting the values,

$$\varepsilon_0 = (1)(0.2)(0.01)(2\pi)$$

$$\varepsilon_0 = 0.004\pi\ V$$

$$\varepsilon_0 \approx 1.26\times 10^{-2}\ V = 12.6\ \text{mV}$$

The instantaneous emf varies as $\varepsilon = \varepsilon_0\omega\ t$. Hence the **maximum emf induced in the loop is about 12.6 mV.**

TIP

Convert rpm to rad/s as $\omega = 2\pi\nu$ with ν in rev/s ($60\ \text{rpm} = 1\ \text{rev}/s$), so $\omega = 2\pi\ \text{rad}/s$, not 602π.

Final answer. The peak emf induced is $\varepsilon_0 = \text{NBA}\omega \approx 1.2610^{-2}\ V = 12.6\ \text{mV}$.

← Back to Q75

Q76

A light copper ring is freely suspended by a light string. A bar magnet is held horizontally with its length along the axis of the ring. The magnet is moved towards the ring with its N pole facing the loop. What will happen to the ring and its position? Explain.

[2026 • Set 55-5-3]

SOLUTION

As the bar magnet, with its N-pole facing the ring, is moved towards the freely suspended copper ring, the magnetic flux linked with the ring increases.

By **Faraday's law**, this changing flux induces a current in the ring, and by **Lenz's law** the induced current flows in such a direction as to oppose the increase in flux.

To oppose the approaching N-pole, the near face of the ring (the face towards the magnet) must itself behave as a **north pole**, so that it repels the magnet.

Two like poles (N and N) repel each other. Therefore the ring is pushed away from the approaching magnet and **swings away from the magnet**, its string deflecting from the vertical in the direction of the magnet's motion.

This is consistent with Lenz's law, which is itself a consequence of conservation of energy: the ring moves so as to oppose the cause (the approach of the magnet), and work must be done to push the magnet closer.

Final answer. The induced current makes the near face of the ring a north pole, so the ring is repelled and swings away from the approaching magnet (Lenz's law).

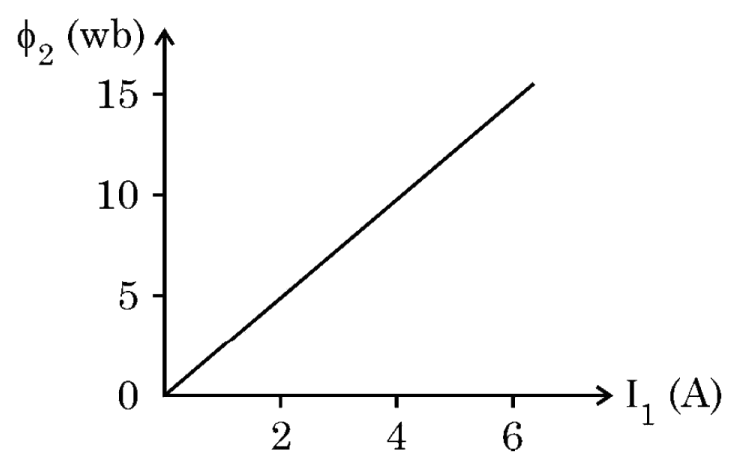
← Back to Q76

Q77

Two coils C_1 and C_2 are placed close to each other. The magnetic flux ϕ_2 linked with the coil C_2 varies with the current I_1 flowing in coil C_1, as shown in the figure. Find

(i) the mutual inductance of the arrangement, and

(ii) the rate of change of current ($\frac{dI_1}{dt}$) that will induce an emf of 100 V in coil C_2.



[2023 • Set 55-5-1]

SOLUTION

The mutual inductance M relates the flux linked with coil C_2 to the current in coil C_1 by $\boldsymbol{\Phi_2 = M\, I_1}$.

(i) Hence M is the slope of the ϕ_2-versus-I_1 graph,

$$M = \frac{\Phi_2}{I_1} = \frac{d\Phi_2}{dI_1}$$

Reading the straight-line graph (which passes through the origin), the flux is $\boldsymbol{\Phi_2 = 4\ Wb}$ when $\boldsymbol{I_1 = 2\ A}$, so

$$M = \frac{4\ \text{Wb}}{2\ A} = 2\ H$$

(ii) The emf induced in C_2 is $\boldsymbol{\varepsilon = M\frac{dI_1}{dt}}$. For $\boldsymbol{\varepsilon = 100\ V}$,

$$\frac{dI_1}{dt} = \frac{\varepsilon}{M}$$

$$\frac{dI_1}{dt} = \frac{100}{2}$$

$$\frac{dI_1}{dt} = 50\ A\ s^{-1}$$

Thus the **mutual inductance is 2 *H*** and the required rate of change of current is **50 *A* s^{-1}**.

Final answer. From the slope, ***M* = 2 *H***, and the rate of change of current for **100 *V*** is $\frac{dI_1}{dt} = 50\ A\ s^{-1}$.

← *Back to Q77*

Q78

Two identical circular discs, one of copper and another of aluminium, are rotated about their geometrical axes with same angular speed in the same magnetic field acting perpendicular to their planes. Compare the (i) induced emf, and (ii) induced current produced in discs between its centre and edge. Justify your answers.

[2021]

SOLUTION

When a disc of radius R rotates with angular speed ω in a field B perpendicular to its plane, the motional emf between the centre and the edge is

$$\varepsilon = \frac{1}{2} B\omega R^2$$

(i) Induced emf: This expression depends only on B, ω and R, not on the material of the disc. Since both discs have the same radius, rotate with the same ω in the same B,

$$\varepsilon_{Cu} = \varepsilon_{Al}$$

so the **induced emf is the same in both discs**.

(ii) Induced current: The current is $I = \frac{\varepsilon}{R_{disc}}$, where R_{disc} is the disc's resistance. Resistance depends on resistivity, $R_{disc}\rho$.

Copper has lower resistivity than aluminium,

$$\rho_{Cu} < \rho_{Al}$$

so $R_{Cu} < R_{Al}$. Since the emf is the same, the smaller resistance carries the larger current,

$$I_{Cu} > I_{Al}$$

Hence the **induced current is larger in the copper disc**.

> **CONCEPT**
>
> Motional emf is set purely by geometry, B and ω; only the current depends on the material, through its resistance.

Final answer. The induced emf is the same in both discs ($\varepsilon = \frac{1}{2} B\omega R^2$), but the induced current is larger in the copper disc because copper has lower resistivity.

← *Back to Q78*

Q79

Two coplanar and concentric coils 1 and 2 have respectively the number of turns N_1 *and* N_2 *and radii* r_1 *and* r_2 ($r_2 \gg r_1$). *Deduce the expression for mutual inductance of this system.*

[2020 • Set 55-3-1]

SOLUTION

Let the two coils be coplanar and concentric: coil 1 has radius r_1 and N_1 turns, coil 2 has radius r_2 and N_2 turns, with $r_2 \gg r_1$.

To find the mutual inductance, pass a current I through the larger coil 2. The magnetic field at the common centre (where the small coil 1 lies) is

$$B = \frac{\mu_0 N_2 I}{2 r_2}$$

Because $r_1 \ll r_2$, this field is practically uniform over the entire area of the small coil 1, $A_1 = \pi r_1^2$.

The total flux linked with the N_1 turns of coil 1 is

$$\Phi_1 = N_1 B A_1 = N_1\left(\frac{\mu_0 N_2 I}{2 r_2}\right)\pi r_1^2$$

$$\Phi_1 = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2 r_2} I$$

By definition $\Phi_1 = M I$, so comparing,

$$M = \frac{\Phi_1}{I} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2 r_2}$$

Thus the **mutual inductance of the system is** $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2 r_2}$.

TIP

Always send the current through the bigger coil so its field is uniform over the small coil; mutual inductance is symmetric, so M is the same either way and this route is far easier.

Final answer. The mutual inductance is $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{2 r_2}$.

← *Back to Q79*

Q80

The energy stored in a solenoid of inductance L is U. The number of turns per unit length of the solenoid is doubled. Keeping the current and all other factors same, find (a) change in inductance of the solenoid, and (b) the final energy stored in the inductor.

[2020 • Set 55-3-2]

SOLUTION

The self-inductance of a solenoid with n turns per unit length, cross-sectional area A and length l is

$$L = \mu_0 n^2 A l$$

and the energy stored in it carrying current I is

$$U = \frac{1}{2} L\, I^2$$

(a) When the turns per unit length is doubled, $n \to 2n$, with A, l and I unchanged:

$$L' = \mu_0 (2n)^2 A\, l = 4\, \mu_0\, n^2 A\, l$$

$$L' = 4L$$

So the inductance becomes **four times** its original value (an increase of $\Delta L = 3L$).

(b) The new stored energy at the same current I is

$$U' = \frac{1}{2} L'\, I^2 = \frac{1}{2} (4L)\, I^2$$

$$U' = 4\left(\frac{1}{2} L\, I^2\right) = 4U$$

Hence the **final energy stored is 4U**.

CONCEPT

Both L and the stored energy scale as n^2, so doubling n multiplies each by four, not two.

Final answer. Doubling the turns per unit length makes the inductance $L' = 4L$ and the stored energy $U' = 4U$.

← *Back to Q80*

Q81

(a) Define the SI unit of self-inductance.
(b) The self-inductance of a solenoid is L. If the number of turns per unit length in it is doubled and the area of cross-section is halved, find the new inductance of the solenoid.

[2020 • Set 55-3-3]

SOLUTION

(a) The SI unit of self-inductance is the henry (H).

The self-inductance of a coil is one henry if an emf of one volt is induced in it when the current through the coil changes at the rate of one ampere per second.

$$1\ H = 1\ V\,s\,\mathrm{A}^{-1}$$

(b) For a long solenoid of length l, area of cross-section A and n turns per unit length, the self-inductance is given by:

$$L = \mu_0\, \mathrm{n}^2\, A\, l$$

Now the number of turns per unit length is doubled, so $\boldsymbol{n \to 2n}$, and the area of cross-section is halved, so $\boldsymbol{A \to \frac{A}{2}}$, while the length l stays the same.

$$L' = \mu_0\, (2\mathrm{n})^2 (\frac{A}{2})\, l$$

$$L' = \mu_0\, (4\,\mathrm{n}^2) (\frac{A}{2})\, l$$

$$L' = 2\,\mu_0\, \mathrm{n}^2\, A\, l$$

$$L' = 2\,L$$

Hence the new self-inductance of the solenoid becomes **twice** the original value, $\boldsymbol{L' = 2L}$.

Final answer. (a) The SI unit is the henry (H). (b) The new inductance is twice the original, $\boldsymbol{L' = 2L}$.

← Back to Q81

Q82

A 0.5 *m long solenoid of* 10 *turns/cm has area of cross-section* 1 cm^2. *Calculate the voltage induced across its ends if the current in the solenoid is changed from* 1 *A to* 2 *A in* 0.1 *s.*

OR

A small flat search coil of area 5 cm^2 *with* 140 *closely wound turns is placed between the poles of a powerful magnet producing magnetic field* 0.09 *T and then quickly removed out of the field region. Calculate*
(a) change of magnetic flux through the coil, and
(b) emf induced in the coil.

[2019 • Set 55-3-1]

SOLUTION

Here the length of the solenoid is $l = 0.5\ m$, the number of turns per cm is 10, so $n = 10$ **turns/cm** $= 1000$ **turns/**m, and the area of cross-section is $A = 1\ cm^2 = 10^{-4}\ m^2$.

Total number of turns is $N = n\,l = 1000 \times 0.5 = 500$ turns.

The self-inductance of the solenoid is:

$$L = \frac{\mu_0 N^2 A}{l}$$

$$L = \frac{(4\pi \times 10^{-7})(500)^2(10^{-4})}{0.5}$$

$$L = \frac{(4\pi \times 10^{-7})(2.5 \times 10^{5})(10^{-4})}{0.5}$$

The numerator is $(4\pi \times 10^{-7})(2.5 \times 10^{5})(10^{-4}) = (4\pi \times 10^{-7})(25) = 100\pi \times 10^{-7}$.

$$L = \frac{100\pi \times 10^{-7}}{0.5} = 200\pi \times 10^{-7}\ H$$

$$L = 2\pi \times 10^{-5}\ H \approx 6.28 \times 10^{-5}\ H = 62.8\ \mu H$$

The rate of change of current is:

$$\frac{dI}{dt} = \frac{2-1}{0.1} = 10\ A\ s^{-1}$$

The voltage (induced emf) across the ends of the solenoid is:

$$\varepsilon = L\,\frac{dI}{dt}$$

$$\varepsilon = (2\pi \times 10^{-5})(10)$$

$$\varepsilon = 2\pi \times 10^{-4}\ V \approx 6.28 \times 10^{-4}\ V$$

Hence the voltage induced across the ends of the solenoid is about $6.28 \times 10^{-4}\ V$, i.e. 0.628 mV.

OR

For the search coil, the area is $A = 5\ \text{cm}^2 = 5 \times 10^{-4}\ \text{m}^2$, the number of turns is $N = 140$, and the magnetic field is $B = 0.09\ T$.

(a) Initially the coil is fully in the field, so the flux through one turn is $\phi_i = BA$, and after removal the flux is zero. The change of magnetic flux through one turn is:

$$\Delta\phi = B\,A = (0.09)(5 \times 10^{-4})$$

$$\Delta\phi = 4.5 \times 10^{-5}\ \text{Wb}$$

The total change in flux linkage through the 140-turn coil is:

$$N\,\Delta\phi = (140)(4.5 \times 10^{-5}) = 6.3 \times 10^{-3}\ \text{Wb}$$

(b) The induced emf is:

$$\varepsilon = \frac{N\,\Delta\phi}{\Delta\,t} = \frac{6.3 \times 10^{-3}}{\Delta\,t}\ V$$

Taking the time of removal as $\Delta t = 0.1\ s$, the induced emf is:

$$\varepsilon = \frac{6.3 \times 10^{-3}}{0.1} = 6.3 \times 10^{-2}\ V = 0.063\ V$$

Hence the change of flux linkage is 6.3×10^{-3} Wb and the induced emf is about 0.063 V.

TIP

In the search-coil part the emf needs the removal time; if the paper gives no time, state the flux change and write the emf as $N\Delta\phi/\Delta\,t$ rather than leaving it blank.

Final answer. Solenoid: $L \approx 6.28 \times 10^{-5}\ H$ and $\varepsilon \approx 6.28 \times 10^{-4}\ V$ (**0.628 mV**). Search coil: $N\Delta\phi = 6.3 \times 10^{-3}$ **Wb** and $\varepsilon \approx 0.063\ V$ (for $\Delta t = 0.1\ s$).

← *Back to Q82*

Q83

State Lenz's Law.
A metallic rod held horizontally along east-west direction, is allowed to fall under gravity. Will there be an emf induced at its ends? Justify your answer.

[2013]

SOLUTION

Lenz's law states that the direction of the induced current (or induced emf) is always such that it opposes the change in magnetic flux that produces it.

Now consider the metallic rod held horizontally along the east-west direction and allowed to fall freely under gravity. Its velocity $\vec{v}$ is directed vertically downward.

The Earth's magnetic field has two components: a horizontal component B_H (directed from south to north) and a vertical component B_V (directed vertically).

The motional emf depends on $\vec{v} \times \vec{B}$. The downward velocity $\vec{v}$ crosses the horizontal component B_H to give a force per unit charge along the length of the rod (east-west), so the free charges are pushed towards the ends.

$$\varepsilon = B_H v l$$

where l is the length of the rod and v is its speed.

The vertical component B_V is parallel to $\vec{v}$, so $\vec{v} \times \vec{B}_V = 0$ and it contributes nothing.

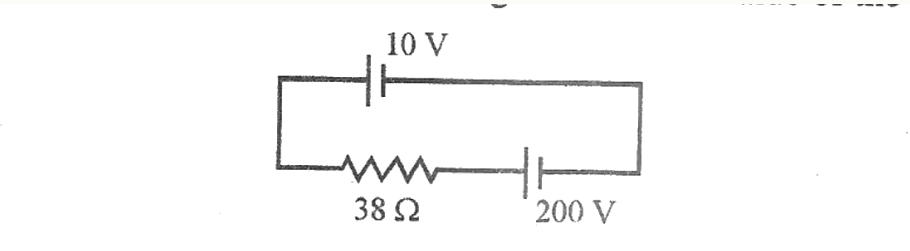
Hence **yes, an emf is induced** across the ends of the rod, and it arises only due to the horizontal component of the Earth's magnetic field, with magnitude $\varepsilon = B_H v l$.

Final answer. Yes, an emf is induced because of the horizontal component of Earth's field; its value is $\varepsilon = B_H v l$.

← *Back to Q83*

Q84

Figure shows a bar magnet M falling under gravity through an air cored coil C. Plot a graph showing variation of induced e.m.f. (E) with time (t). What does the area enclosed by the E−−t curve depict?



[2013]

SOLUTION

As the bar magnet *M* falls under gravity through the coil *C*, the magnetic flux linked with the coil changes, so an emf is induced according to Faraday's law, $\varepsilon = -\frac{d\Phi}{dt}$.

While the magnet approaches the coil, the flux increases, so an emf of one polarity is induced. When the magnet is at the centre of the coil the flux is maximum, so $\frac{d\Phi}{dt} = 0$ and the emf becomes zero. As the magnet leaves the coil, the flux decreases, so the emf reverses to the opposite polarity.

The graph of induced emf *E* versus time *t* is therefore a two-lobed (bipolar) curve:

The horizontal axis is time *t* and the vertical axis is induced emf *E*. The first lobe is a positive pulse (during approach), the curve then crosses zero (when the magnet is at the centre), and the second lobe is a negative pulse (during recession). The second negative lobe is taller and narrower than the first positive lobe, because the magnet has speeded up under gravity, so the flux changes faster on the way out. The area of the second lobe equals the area of the first lobe in magnitude.

The area enclosed by the *E*-*t* curve represents the time integral of the emf:

$$\int E\,dt = -\int \frac{d\Phi}{dt}\,dt = -\Delta\Phi$$

Since the induced charge is $q = \frac{\Delta\Phi}{R} = \frac{1}{R}\int E\,dt$, the area under the E-t curve is equal to the **net charge that flows through the coil multiplied by its resistance** (i.e. the total change in flux linkage).

Because the flux is zero both before the magnet enters and after it leaves, the net area (positive lobe minus negative lobe) is zero, so the net charge circulated is zero.

Final answer. The E-t graph is a bipolar pulse (positive lobe, then a taller, narrower negative lobe); the area under it equals the change in flux linkage, $\int E\,dt = -\Delta\Phi$ (i.e. charge × resistance).

← *Back to Q84*

Q85

Two identical loops, one of copper and the other of aluminium, are rotated with the same angular speed in the same magnetic field. Compare (i) the induced emf and (ii) the current produced in the two coils. Justify your answer.

[2012]

SOLUTION

Both loops are geometrically identical (same number of turns N and same area A), and they are rotated with the same angular speed ω in the same magnetic field B.

(i) Comparison of induced emf:

The instantaneous induced emf in a rotating coil is:

$$\varepsilon = N\,B\,A\,\omega\,\omega\,t$$

This depends only on N, B, A and ω, and not on the material of the loop. Since all these quantities are the same for both loops:

$$\varepsilon_{Cu} = \varepsilon_{Al}$$

So the induced emf is the same in both coils.

(ii) Comparison of induced current:

The induced current is given by:

$$i = \frac{\varepsilon}{R}$$

The resistance R depends on the resistivity of the material, $\boldsymbol{R = \frac{\rho l}{A}}$. Copper has a lower resistivity than aluminium ($\rho_{Cu} < \rho_{Al}$), so the copper loop has lower resistance.

$$i_{Cu} = \frac{\varepsilon}{R_{Cu}}, \; i_{Al} = \frac{\varepsilon}{R_{Al}}$$

Since $R_{Cu} < R_{Al}$ and the emf is the same:

$$i_{Cu} > i_{Al}$$

Hence the **induced emf is equal** in both loops, but the **induced current is larger in the copper loop** because of its lower resistance.

Final answer. The induced emf is the same in both ($\varepsilon_{Cu} = \varepsilon_{Al}$), but the current is greater in copper ($i_{Cu} > i_{Al}$) because copper has lower resistivity.

← *Back to Q85*

Q86

Define self-inductance of a coil. Show that magnetic energy required to build up the current I in a coil of self-inductance L is given by $\frac{1}{2}LI^2$.

[2012]

SOLUTION

The self-inductance of a coil is defined as the magnetic flux linkage produced in the coil per unit current flowing through it:

$$L = \frac{\Phi}{I}$$

Equivalently, it is the back-emf induced in the coil per unit rate of change of current, $\boldsymbol{\varepsilon = -L\,\frac{dI}{dt}}$.

To find the energy stored, consider that while the current is being built up from zero, the self-induced (back) emf opposes the growth of current. The external source must do work against this back-emf.

At an instant when the current is i and is increasing at the rate $\frac{di}{dt}$, the magnitude of the back-emf is:

$$\varepsilon = L\,\frac{di}{dt}$$

The rate at which work is done by the source against this emf (the instantaneous power) is:

$$P = \varepsilon\, i = L\, i\,\frac{di}{dt}$$

The small amount of work done in time dt is:

$$dW = P\,dt = L\, i\, di$$

The total work done in raising the current from 0 to I is obtained by integration:

$$W = {}_0{}^{I} L\, i\, di$$

$$W = L[\frac{i^2}{2}]_0{}^{I}$$

$$W = \frac{1}{2} L\, I^2$$

This work is stored as magnetic potential energy in the coil. Hence the magnetic energy required to build up a current I in a coil of self-inductance L is:

$$U = \frac{1}{2} L\, I^2$$

Final answer. Self-inductance is the flux linkage per unit current, $\boldsymbol{L = \Phi/I}$; the magnetic energy stored is $\boldsymbol{U = \frac{1}{2} LI^2}$.

← Back to Q86

Q87

A metallic rod of 'L' length is rotated with angular frequency of 'ω' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius L, about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field $\vec{B}$ parallel to the axis is present everywhere. Deduce the expression for the emf between the centre and the metallic ring.

[2012]

SOLUTION

The metallic rod OP of length L is hinged at the centre O of the ring and rotates with angular frequency ω in a uniform magnetic field $\vec{B}$ directed along the axis (perpendicular to the plane of the ring).

Consider a small element of the rod of length dξ at a distance ξ from the hinge O. As the rod rotates, this element moves in a circle, so its linear (tangential) speed is:

$$v = \xi\,\omega$$

The motional emf developed across this small element is:

$$d\varepsilon = B\,v\,d\xi = B\,(\xi\omega)\,d\xi$$

The total emf between the centre O and the outer end P (the ring at radius L) is found by integrating from $\xi = 0$ to $\xi = L$:

$$\varepsilon = {}_0{}^{L} B\,\omega\,\xi\,d\xi$$

$$\varepsilon = B\,\omega[\frac{\xi^2}{2}]_0{}^{L}$$

$$\varepsilon = B\,\omega\,\frac{L^2}{2}$$

$$\varepsilon = \frac{1}{2}\,B\,\omega\,L^2$$

Hence the emf induced between the centre and the metallic ring is $\varepsilon = \frac{1}{2} B\omega L^2$.

Final answer. The emf induced between the centre and the ring is $\varepsilon = \frac{1}{2} B\omega L^2$.

← *Back to Q87*

Q88

Derive the expression for the self inductance of a long solenoid of cross sectional area A and length l, having n turns per unit length.

[2012]

SOLUTION

Consider a long solenoid of length l and cross-sectional area A, having n turns per unit length, carrying a current I.

The magnetic field inside a long solenoid is uniform and given by:

$$B = \mu_0 n I$$

The magnetic flux through one turn of the solenoid is:

$$\phi = B A = \mu_0 n I A$$

The total number of turns in the solenoid is $N = n l$. Therefore the total flux linkage with the whole solenoid is:

$$\Phi = N \phi = (n l)(\mu_0 n I A)$$

$$\Phi = \mu_0 n^2 A l I$$

By the definition of self-inductance, $\Phi = L I$, so:

$$L = \frac{\Phi}{I} = \frac{\mu_0 n^2 A l I}{I}$$

$$L = \mu_0 n^2 A l$$

Hence the self-inductance of the long solenoid is $L = \mu_0 n^2 A l$. (Since $N = n l$, this can also be written as $L = \frac{\mu_0 N^2 A}{l}$.)

Final answer. The self-inductance of the long solenoid is $L = \mu_0 n^2 A l$ (equivalently $\mu_0 N^2 A / l$).

← *Back to Q88*

Q89

What are eddy currents? Write any two applications of eddy currents.

[2011]

SOLUTION

Eddy currents are the currents induced in the body (bulk) of a conductor when the magnetic flux linked with the conductor changes with time.

These changing-flux conditions arise either because the magnetic field varies in time or because the conductor moves through a non-uniform magnetic field. The induced currents flow in closed loops within the volume of the conductor, and by Lenz's law they always flow in a direction that opposes the change in flux producing them.

Two applications of eddy currents:

(i) Electromagnetic (eddy current) braking: In electric trains and some industrial machines, eddy currents set up in a moving metal disc or rail by strong electromagnets produce a retarding force that brings the system to rest smoothly, without any mechanical contact or wear.

(ii) Induction heating: In induction furnaces and induction cooktops, strong eddy currents are induced in a metal placed in a high-frequency changing magnetic field. The heat produced by these currents (I^2R loss) is used to melt the metal or to cook food.

(Other uses include dead-beat galvanometers, induction motors and analogue speedometers.)

Final answer. Eddy currents are bulk induced currents due to changing flux in a conductor; two uses are electromagnetic braking and induction heating.

← Back to Q89

Q90

A current is induced in coil C_1 due to the motion of current carrying coil C_2.
(a) Write any two ways by which a large deflection can be obtained in the galvanometer G.
(b) Suggest an alternative device to demonstrate the induced current in place of a galvanometer.

[2011 • Set 55-1-1]

SOLUTION

(a) A current is induced in coil C_1 because the motion of the current-carrying coil C_2 changes the magnetic flux linked with C_1. The induced emf is $\varepsilon = -\frac{d\Phi}{dt} = -M\frac{dI_2}{dt}$, so a larger deflection in the galvanometer G is obtained by increasing this rate of change of flux. Two ways are:

(i) Move the coil C_2 towards or away from C_1 more rapidly (i.e. increase the relative speed of the coils, or change the current in C_2 more quickly). This increases $\frac{d\Phi}{dt}$ and hence the induced emf and current.

(ii) Increase the mutual inductance M between the coils, for example by using coils with more turns or larger area, by bringing them closer, or by inserting a soft-iron core between them.

(b) In place of the galvanometer, the induced current can be demonstrated using a sensitive light-emitting diode (LED), which glows when current flows. (A small milliammeter, a cathode-ray oscilloscope showing the induced voltage, or a small bulb are also suitable alternatives.)

Final answer. (a) Increase the rate of relative motion / rate of change of flux, and increase the mutual inductance (more turns or a soft-iron core). (b) Use an LED (or a sensitive milliammeter / oscilloscope) instead of the galvanometer.

← Back to Q90

Q91

Current in a circuit falls steadily from 5.0 A to 0.0 A in 100 ms. If an average e.m.f. of 200 V is induced, calculate the self-inductance of the circuit.

[2011 • Set 55-2-1]

SOLUTION

When the current in a circuit changes, a self-induced emf appears that opposes the change. Its magnitude is given by $|\varepsilon| = L\left|\frac{dI}{dt}\right|$, where L is the self-inductance of the circuit.

For a steady change, $\left|\frac{dI}{dt}\right| = \frac{\Delta I}{\Delta t}$, so the average induced emf is

$$|\varepsilon_{avg}| = L\,\frac{\Delta I}{\Delta t}$$

Here the change in current $\Delta I = 5.0 - 0.0 = 5.0\ A$, the time interval $\Delta t = 100\ ms = 0.1\ s$, and $\varepsilon_{avg} = 200\ V$.

Solving for the self-inductance,

$$L = \frac{\varepsilon_{avg}\,\Delta t}{\Delta I}$$

$$L = \frac{(200)(0.1)}{5.0}$$

$$L = \frac{20}{5.0} = 4.0\ H$$

Hence the **self-inductance of the circuit is 4.0 H**.

Final answer. The self-inductance of the circuit is $L = 4.0\ H$.

← Back to Q91

Q92

A jet plane is traveling west at 450 ms^{-1}. *If the horizontal component of earth's magnetic field at that place is* 4×10^{-4} Tesla *and the angle of dip is* 30°, *find the emf induced between the ends of wings having a span of* 30 *m.*

[2008]

SOLUTION

The wings of the jet form a horizontal conducting rod moving through the Earth's magnetic field. Only the component of the field perpendicular to the plane in which the rod moves contributes to the motional emf. The plane swept out by the horizontal wings is horizontal, so the relevant component is the vertical component B_V of the Earth's field.

The vertical component is related to the horizontal component B_H and the angle of dip δ by

$$B_V = B_H \delta$$

$$B_V = (4\times 10^{-4})\tan 30°$$

$$B_V = \frac{4\times 10^{-4}}{\sqrt{3}} = 2.31\times 10^{-4}\ T$$

The motional emf induced between the wing tips of span L, for a speed v, is

$$\varepsilon = B_V\, v\, L$$

$$\varepsilon = (2.31\times 10^{-4})(450)(30)$$

$$\varepsilon = 3.12\ V$$

Hence the **emf induced between the ends of the wings is about 3.12 *V*.**

COMMON MISTAKE

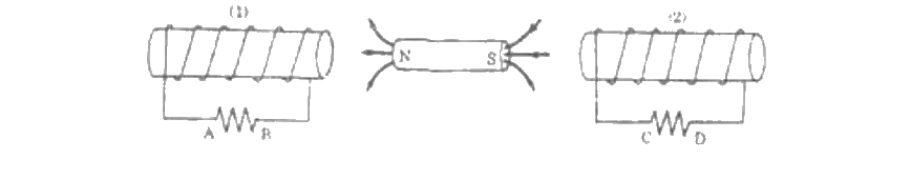
A common slip is to use the horizontal component B_H. *For horizontal wings moving horizontally it is the vertical component* $\mathbf{B_V = B_H\delta}$ *that cuts the area and produces the emf.*

Final answer. The induced emf between the wing tips is $\boldsymbol{\varepsilon \approx 3.12\ V}$.

← Back to Q92

Q93

In the figure given below, a bar magnet moving towards the right or left induces an e.m.f. in the coils (1) *and* (2). *Find, giving reason, the directions of the induced currents through the resistors* AB *and* CD *when the magnet is moving* ***(a)*** *towards the right, and* ***(b)*** *towards the left.*



[2005]

SOLUTION

By Lenz's law, the induced current in each coil always flows in a direction that opposes the change in magnetic flux producing it. Take the bar magnet placed between coil (1) on the left (connected through resistor AB) and coil (2) on the right (connected through resistor CD), with the magnet's North pole facing the right-hand coil.

(a) When the magnet moves towards the right:

The magnet recedes from coil (1), so the flux through coil (1) decreases. To oppose this decrease, the induced current must keep the original flux, so the near face of coil (1) must become an unlike pole that attracts the receding magnet. The induced current in the external resistor flows from B to A.

At the same time the magnet approaches coil (2), so the flux through coil (2) increases. To oppose this increase, the near face of coil (2) becomes a like pole that repels the approaching magnet. The induced current in the external resistor flows from C to D.

(b) When the magnet moves towards the left:

Now the flux through coil (1) increases and the flux through coil (2) decreases, exactly reversing the previous situation. By the same Lenz's-law reasoning, the

induced currents reverse: the current through AB now flows from A to B, and the current through CD now flows from D to C.

Thus in every case **each coil drives an induced current in whichever direction opposes its own change of flux**, attracting the magnet when it recedes and repelling it when it approaches.

Final answer. Moving right: current $B \to A$ in coil 1, $C \to D$ in coil 2; moving left both reverse, each opposing its own flux change (Lenz's law).

← *Back to Q93*

Q94

Two circular coils, one of radius r and the other of radius R are placed coaxially with their centres coinciding. For $R \gg r$, obtain an expression for the mutual inductance of the arrangement.

[2004]

SOLUTION

Let the small coil of radius r be placed at the common centre, coplanar and coaxial with the large coil of radius R, with $R \gg r$. To find the mutual inductance, pass a current I through the larger coil and find the flux it links with the smaller coil.

The magnetic field at the centre of the large circular coil of radius R carrying current I is

$$B = \frac{\mu_0 I}{2R}$$

Because $R \gg r$, this field is very nearly uniform over the whole area of the small coil, which is $A = \pi r^2$.

The flux linked with the small coil is

$$\Phi = B A = \frac{\mu_0 I}{2R} \times \pi r^2$$

$$\Phi = \frac{\mu_0 \pi r^2 I}{2R}$$

By definition the mutual inductance is the flux linkage per unit current,

$$M = \frac{\Phi}{I} = \frac{\mu_0 \pi r^2}{2R}$$

Hence the **mutual inductance of the arrangement is** $M = \dfrac{\mu_0 \pi r^2}{2R}$.

TIP

Always pass the current through the LARGER coil. Its central field is uniform across the tiny inner coil, so the flux integral is trivial; doing it the other way round needs the field of the small coil over the large area, which is far harder.

Final answer. The mutual inductance is $M = \dfrac{\mu_0 \pi r^2}{2R}$.

← *Back to Q94*

Q95

(i) Define mutual inductance of a pair of coils. Write its SI unit.
(ii) A long solenoid of radius R and length L has n turns per unit length. A circular loop of radius ***r*** **(is placed inside at the centre of the solenoid such that its axis coincides with the axis of the solenoid. Obtain the mutual inductance of the solenoid and the loop.**

[2026 • Set 55-2-1]

SOLUTION

(i) The mutual inductance M of a pair of coils is the magnetic flux linked with one coil per unit current flowing in the other coil. If a current I_1 in coil 1 produces a flux linkage Φ_{21} with coil 2, then

$$M = \frac{\Phi_{21}}{I_1}$$

Its SI unit is the **henry (H)**, where $\mathbf{1\ H = 1\ Wb\ A^{-1}}$.

(ii) Let the long solenoid of radius R and length L have n turns per unit length and carry a current I. The circular loop of radius r ($r < R$) is placed coaxially inside it.

For a long solenoid, the magnetic field is uniform inside and effectively zero outside,

$$B = \mu_0 \, n \, I$$

Since the loop of radius r lies entirely within the solenoid, the flux through it is

$$\Phi = B \times \pi \, r^2 = \mu_0 \, n \, I \, \pi \, r^2$$

Therefore the mutual inductance is

$$M = \frac{\Phi}{I} = \mu_0 \, n \, \pi \, r^2$$

Hence the **mutual inductance is $M = \mu_0 \, n \, \pi \, r^2$**, which depends on the loop radius r and not on the solenoid radius R, because the flux is confined to the loop's own area.

Final answer. Mutual inductance $M = \mu_0 \, n \, \pi \, r^2$, SI unit henry (H).

← *Back to Q95*

Q96

State Faraday's law of electromagnetic induction. Briefly describe two methods for producing induced emf.

[2026 • Set 55-3-3]

SOLUTION

Faraday's law of electromagnetic induction states that whenever the magnetic flux linked with a closed circuit changes, an emf is induced in it, and the magnitude of this induced emf is equal to the rate of change of magnetic flux through the circuit. The negative sign (Lenz's law) shows that the induced emf opposes the change causing it:

$$\varepsilon = -\frac{d\Phi}{dt}$$

For a coil of N turns, the flux links each turn, so

$$\varepsilon = -N\frac{d\Phi}{dt}$$

Two methods of producing an induced emf:

(1) By changing the magnetic field with time. The conductor is kept stationary while the magnetic field through it is varied, for example by moving a bar magnet towards or away from a stationary coil, or by switching the current on and off in a neighbouring coil. The changing $\vec{B}$ changes the flux and induces an emf.

(2) By moving a conductor in a magnetic field (motional emf). A conductor is moved through a steady magnetic field so that it cuts the field lines, for example by sliding a straight rod across a magnetic field region, or by rotating a coil in a uniform field as in an AC generator. The change in area or orientation changes the flux and induces an emf.

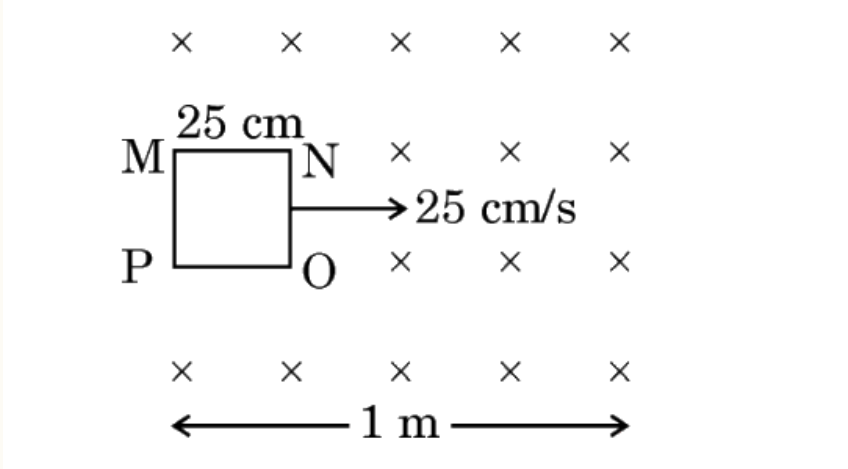
Thus an emf can be induced **either by changing $\vec{B}$ in time or by moving a conductor through a steady field**.

Final answer. Faraday's law: $\boldsymbol{\varepsilon = -\frac{d\Phi}{dt}}$; emf can be induced by changing $\vec{B}$ with time or by moving a conductor through a steady field.

← *Back to Q96*

Q97

The figure given below shows a square-shaped loop MNOP of side 25 cm placed horizontally in a uniform magnetic field $\vec{B}$ directed vertically downward. The position of the loop at $t = 0$ s is as shown in figure. The loop is pulled with a constant velocity of 25 cm/s till it goes out of the field.



(a) What will be the direction of the induced current in the loop as it goes out of the field? For how long would the current in the loop persist?
(b) Plot graphs showing the variation of magnetic flux and magnitude of induced emf as a function of time.

[2026 • Set 55-4-1]

SOLUTION

The square loop MNOP has side $l = 25\ \text{cm} = 0.25\ m$ and lies horizontally in a uniform field $\vec{B}$ directed vertically downward. It is pulled horizontally with constant speed $v = 25\ \text{cm/s} = 0.25\ m/s$ until it leaves the field.

(a) As the loop moves out of the field, the downward flux threading it decreases. By Lenz's law the induced current opposes this decrease, so it must set up a magnetic field in the same downward direction through the loop. By the right-hand rule, this requires the induced current to flow clockwise when the loop is viewed from above.

The induced current flows only while the loop is partly inside and partly outside the field, that is, while the flux is actually changing. The trailing edge has to travel the full side length of the loop to clear the field boundary, so the time for which the current persists is

$$t = \frac{l}{v} = \frac{0.25\ m}{0.25\ m/s} = 1\ s$$

Hence the induced current is **clockwise (seen from above) and lasts for 1 s**.

(b) Graphs:

Magnetic flux Φ versus time: While the loop is fully inside the field the flux stays constant at its maximum value $\Phi_{max} = B\,l^2 = B(0.25)^2 = 0.0625\,B$. Once the

loop begins to exit, Φ falls linearly with time and reaches zero after 1 *s* of exit. The graph is a horizontal line at Φ_{max} followed by a straight line sloping down to zero.

Magnitude of induced emf $|\varepsilon|$ versus time: While the flux is constant, $|\varepsilon| = 0$. During the 1 *s* exit the flux changes at a constant rate, so the emf jumps to a constant value

$$|\varepsilon| = B\,v\,l = B(0.25)(0.25) = 0.0625\,B$$

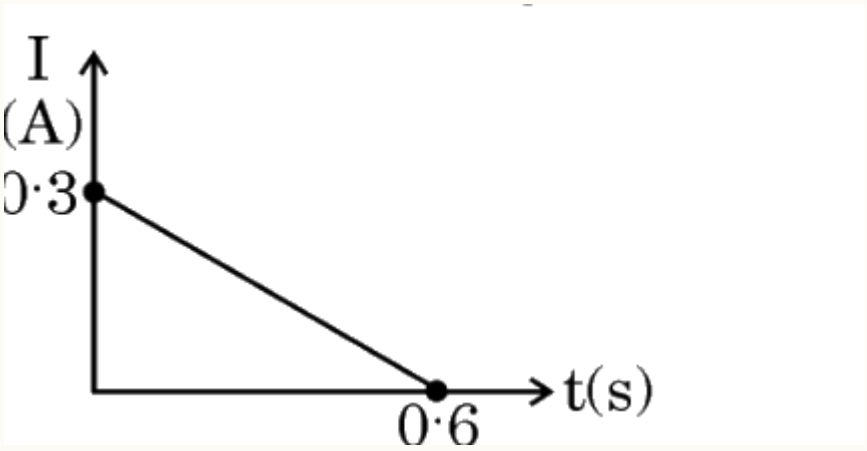
and stays flat for that 1 *s*, then drops back to zero. The graph is a rectangular pulse of height $0.0625\,B$ and width 1 *s*.

Final answer. The induced current is clockwise (viewed from above) and persists for **1 *s***; **Φ** stays constant then falls linearly to zero, and **$|\varepsilon|$** is a rectangular pulse of height **$Bvl = 0.0625\,B$**.

← *Back to Q97*

Q98

A conducting rectangular loop of area 5 cm^2 and resistance 4 Ω is removed from a region of uniform magnetic field, acting normal to the plane of the loop. The value of induced current I in the loop varies with time t, as shown in the figure.



Find:

(a) total charge that passed through the loop

(b) change in magnetic flux through the loop

(c) magnitude of magnetic field in the region

[2026 • Set 55-4-2]

SOLUTION

The rectangular loop has area $A = 5\ \text{cm}^2 = 5\times 10^{-4}\ \text{m}^2$ and resistance $R = 4\ \Omega$. From the graph, the induced current falls linearly from $I_0 = 0.3\ A$ at $t = 0$ to zero at $t = 0.6\ s$, forming a right-angled triangle.

(a) The total charge that flows is the area under the current-time graph:

$$q = \int I\,dt = \frac{1}{2} I_0\, t_{max}$$

$$q = \frac{1}{2}(0.3)(0.6)$$

$$q = 0.09\ C$$

(b) The charge is also related to the change in flux and the resistance by $q = \frac{|\Delta\Phi|}{R}$, so

$$|\Delta\Phi| = q\,R$$

$$|\Delta\Phi| = (0.09)(4)$$

$$|\Delta\Phi| = 0.36\ \text{Wb}$$

(c) Since the field acts normal to the loop, $|\Delta\Phi| = B\,A$ (the field is removed completely, so the flux falls from BA to 0). Hence

$$B = \frac{|\Delta\Phi|}{A} = \frac{q\,R}{A}$$

$$B = \frac{0.36}{5\times 10^{-4}}$$

$$B = 720\ T$$

Hence $q = 0.09\ C$, $|\Delta\Phi| = 0.36\ \text{Wb}$, and $B = 720\ T$ for the data given.

CONCEPT

The key relation is $q = \frac{|\Delta\Phi|}{R}$, found by integrating $I = \frac{1}{R}\frac{d\Phi}{dt}$. Charge depends only on total flux change and resistance, never on how fast the field is removed.

Final answer. Charge $q = 0.09\ C$, change in flux $|\Delta\Phi| = 0.36\ Wb$, and field $B = 720\ T$.

← Back to Q98

Q99

Two coils, one of radius 0·5 cm having 10 turns and the other of radius 5 cm having 50 turns are placed coaxially in air such that their centres are coincident. Calculate:
(a) the magnetic flux through the smaller coil if the larger coil carries a current of 3 A, and
(b) the mutual inductance of the two coils.

[2026 • Set 55-4-3]

SOLUTION

The smaller coil X has radius $r = 0.5\ cm = 0.005\ m$ with $N_X = 10$ turns. The larger coil Y has radius $R = 5\ cm = 0.05\ m$ with $N_Y = 50$ turns. They are coaxial and concentric, and the larger coil carries $I = 3\ A$.

The magnetic field at the common centre due to the larger coil Y is

$$B = \frac{\mu_0 N_Y I}{2R}$$

$$B = \frac{(4\pi\times 10^{-7})(50)(3)}{2\times 0.05}$$

$$B = \frac{(4\pi\times 10^{-7})(150)}{0.1}$$

$$B = 6\pi\times 10^{-4}\ T = 1.885\times 10^{-3}\ T$$

Since $r \ll R$, this field is uniform over the area of the smaller coil, $A = \pi r^2 = \pi(0.005)^2 = 7.854\times 10^{-5}\ m^2$.

(a) The total flux linked with the smaller coil (all N_X turns) is

$$\Phi_X = N_X\, B\, A = (10)(6\pi\times 10^{-4})(7.854\times 10^{-5})$$

$$\Phi_X = 1.48\times 10^{-6}\ \text{Wb}$$

So the flux through the smaller coil is about 1.48 μWb.

(b) The mutual inductance is the flux linkage per unit current,

$$M = \frac{\Phi_X}{I} = \frac{1.48\times 10^{-6}}{3}$$

$$M = 4.93\times 10^{-7}\ H \approx 0.5\ \mu\text{H}$$

Hence **the flux through the smaller coil is 1.48 μWb and the mutual inductance is about 4.93×10^{-7} H.**

Final answer. Flux through the smaller coil $\Phi_X \approx$ **1.48 μWb** and mutual inductance $M \approx 4.93\times 10^{-7}$ H.

← *Back to Q99*

Q100

A long solenoid of length L and radius r_1*, having* N_1 *turns is surrounded symmetrically by a coil of radius* r_2 *(*$>\mathbf{r_1}$*) having* N_2 *turns (*$N_2 \ll N_1$*) around its mid-point. Derive an expression for the mutual inductance of solenoid and coil. Is* $\mathbf{M_{12} = M_{21}}$ *valid in this case?*

[2026 • Set 55-5-1]

SOLUTION

Let the long solenoid have length L, radius r_1 and N_1 turns, so its number of turns per unit length is $\mathbf{n_1} = \frac{N_1}{L}$. The outer coil of radius $\mathbf{r_2}$ **(**$>\mathbf{r_1}$**)** has N_2 turns and is wound symmetrically around the midpoint of the solenoid. Pass a current I through the solenoid.

For a long solenoid the field is uniform and confined to its interior:

$B = \mu_0 n_1 I = \dfrac{\mu_0 N_1 I}{L}$ inside, and $B \approx 0$ outside.

Because the field is zero in the region between r_1 and r_2, the outer coil links flux only through the solenoid's cross-sectional area πr_1^2, not through its own larger area πr_2^2.

The flux through one turn of the outer coil is

$$\phi = B \times \pi r_1^2 = \mu_0 n_1 I \pi r_1^2$$

The total flux linkage with all N_2 turns of the outer coil is

$$\Phi_2 = N_2 \phi = \mu_0 n_1 N_2 \pi r_1^2 I = \frac{\mu_0 N_1 N_2 \pi r_1^2}{L} I$$

Therefore the mutual inductance is

$$M = \frac{\Phi_2}{I} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{L}$$

Is $M_{12} = M_{21}$ valid here? Yes. By the reciprocity theorem the mutual inductance of any pair of coils is symmetric, so $M_{12} = M_{21} = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{L}$. Here it is far easier to compute the flux of the solenoid through the outer coil, since the solenoid's field is simple and uniform; the reverse calculation (field of the outer coil at the solenoid) is harder, but the equality is guaranteed.

Hence the **mutual inductance is $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{L}$, and $M_{12} = M_{21}$ holds.**

COMMON MISTAKE

Note the area is πr_1^2, the solenoid radius, not πr_2^2. Since $B \approx 0$ outside the solenoid, the extra area of the outer coil links no flux.

Final answer. Mutual inductance $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{L}$, and yes, $M_{12} = M_{21}$ by the reciprocity theorem.

← Back to Q100

Q101

(a) *Show that the energy required to build up the current I in a coil of inductance L is* $\frac{1}{2} L I^2$.

(b) *Considering the case of magnetic field produced by air-filled current carrying solenoid, show that the magnetic energy density of a magnetic field B is* $\frac{B^2}{2\mu_0}$.

[2025 • Set 55-1-1]

SOLUTION

(a) When current is being established in a coil, it changes with time. By Faraday's law a back-emf is set up in the coil that opposes this growth of current. To keep the current increasing, the source must do work against this back-emf, and this work is stored as magnetic energy in the coil.

The magnitude of the induced (back) emf is

$$\varepsilon = L \frac{di}{dt}$$

The work done by the source in time dt to drive a charge **dq = *i* dt** against this emf is

$$dW = \varepsilon\, i\, dt = L \frac{di}{dt}\, i\, dt = L\, i\, di$$

The total work done as the current grows from 0 to I is

$$U = {}_0^I L\, i\, di$$

$$U = L[\frac{i^2}{2}]_0^I$$

$$\mathbf{U = \frac{1}{2} L I^2}$$

This is the magnetic energy stored in the inductor when it carries a current I.

(b) For an air-filled solenoid of length l, cross-sectional area A and n turns per unit length, the self-inductance is

$$L = \mu_0 n^2 A\, l$$

and the magnetic field inside is

$$B = \mu_0\, n\, I \Rightarrow I = \frac{B}{\mu_0\, n}$$

The energy stored is

$$U = \frac{1}{2}\, L\, \mathrm{I}^2 = \frac{1}{2}\, (\mu_0\, \mathrm{n}^2\, A\, l)\left(\frac{B}{\mu_0\, n}\right)^2$$

$$U = \frac{1}{2}\, \mu_0\, \mathrm{n}^2\, A\, l \cdot \frac{\mathrm{B}^2}{\mu_0{}^2\, \mathrm{n}^2}$$

$$U = \frac{\mathrm{B}^2}{2\mu_0}\, (A\, l)$$

Here $A\, l$ is the volume of the solenoid. So the magnetic energy per unit volume, called the magnetic energy density, is

$$\mathrm{u}_B = \frac{U}{A\, l}$$

$$\mathbf{u_B = \frac{B^2}{2\mu_0}}$$

This result, though derived for a solenoid, holds for any magnetic field in vacuum.

Final answer. The energy stored is $\boldsymbol{U = \frac{1}{2}LI^2}$ and the magnetic energy density is $\mathbf{u_B = \frac{B^2}{2\mu_0}}$.

← *Back to Q101*

Q102

Consider two long co-axial solenoids S_1 *and* S_2*, each of length* l ($\gg \mathrm{r}_2$) *and of radius* r_1 *and* $\mathbf{r_2}$ **($\mathbf{r_2 > r_1}$)**. *The number of turns per unit length are* n_1 *and* n_2 *respectively. Derive an expression for mutual inductance* M_{12} *of solenoid* S_1 *with respect to solenoid* S_2*. Show that* $\mathbf{M_{12} = M_{21}}$.

[2025 • Set 55-1-2]

SOLUTION

Take the two long coaxial solenoids: inner S_1 of radius r_1 with n_1 turns per unit length, and outer S_2 of radius r_2 with n_2 turns per unit length, both of length $l \gg r_2$. Mutual inductance M_{12} is the flux linked with S_1 per unit current in S_2.

Pass a current I through the outer solenoid S_2. The magnetic field inside it is uniform and directed along the axis,

$$B_2 = \mu_0 n_2 I$$

This field is uniform over the cross-section of the inner solenoid S_1, whose area is πr_1^2. So the flux through one turn of S_1 is

$$\phi = B_2 (\pi r_1^2) = \mu_0 n_2 I \pi r_1^2$$

The number of turns of S_1 is $\mathbf{N_1 = n_1 \, l}$, so the total flux linkage with S_1 is

$$\Phi_1 = N_1 \phi = (n_1 l)(\mu_0 n_2 I \pi r_1^2)$$

$$\Phi_1 = \mu_0 n_1 n_2 \pi r_1^2 l I$$

By definition $\mathbf{\Phi_1 = M_{12} \, I}$, hence

$$\mathbf{M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l}$$

Now reverse the roles and pass a current I through the inner solenoid S_1. The field inside S_1 is

$$B_1 = \mu_0 n_1 I$$

This field exists essentially only within the inner cylinder of radius r_1; the region between r_1 and r_2 has practically no field. So the flux from S_1 that links each turn of S_2 passes only through the area πr_1^2,

$$\phi' = B_1 (\pi r_1^2) = \mu_0 n_1 I \pi r_1^2$$

With $\mathbf{N_2 = n_2 \, l}$ turns of S_2, the total linkage is

$$\Phi_2 = N_2 \phi' = (n_2 l)(\mu_0 n_1 I \pi r_1^2)$$

$$\Phi_2 = \mu_0 n_1 n_2 \pi r_1^2 l I$$

Hence

$$M_{21} = \frac{\Phi_2}{I} = \mu_0 n_1 n_2 \pi r_1^2 l$$

Comparing the two results,

$$\mathbf{M_{12} = M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l}$$

COMMON MISTAKE

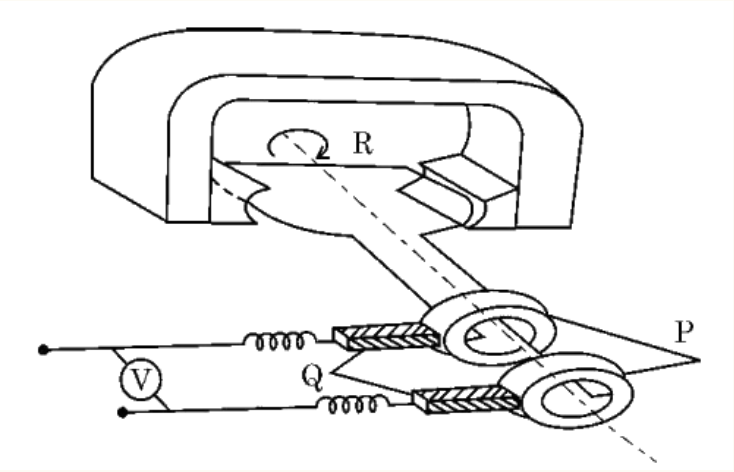
A common error is using πr_2^2 when computing M_{21}. The linking flux is set by the smaller (inner) radius r_1 in both cases, which is precisely why $\mathbf{M_{12} = M_{21}}$.

Final answer. The mutual inductance is $\mathbf{M_{12} = M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l}$.

← *Back to Q102*

Q103

(a) State Lenz's law.
(b) In the given figure:
(i) Identify the machine.
(ii) Name the parts P and Q and R of the machine.
(iii) Give the polarities of the magnetic poles.
(iv) Write the two ways of increasing the output voltage.



[2025 • Set 55-2-1]

SOLUTION

(a) Lenz's law states that the direction of the induced current (or induced emf) is always such that it opposes the change in magnetic flux that produces it. It is a direct consequence of the law of conservation of energy.

(b) The figure shows a rectangular coil mounted on an axle, free to rotate between the poles of a magnet, with its ends connected through two rings to external contacts.

(i) The machine is an AC generator (alternator). It converts mechanical energy of rotation into electrical energy using electromagnetic induction.

(ii) The parts are identified as follows. P are the brushes, which are carbon pieces pressing against the rings to carry current to the external circuit. Q are the slip rings, which rotate along with the coil and connect its ends to the brushes. R is the armature coil, a rectangular coil of many turns wound on a soft-iron core, mounted on the rotating axle.

(iii) The magnet provides a fixed field across the coil. One pole piece facing the coil is a North pole (N) and the opposite pole piece is a South pole (S), so the magnetic field lines run across the coil from the N-pole to the S-pole.

(iv) The peak (maximum) output emf of the generator is given by

$$\varepsilon_0 = N\,B\,A\,\omega$$

So the output voltage can be increased by any of the following. First, by increasing the number of turns N in the armature coil. Second, by using a stronger magnetic field B (stronger magnets) or by increasing the angular speed ω of rotation. (Increasing the area A of the coil also raises the output.)

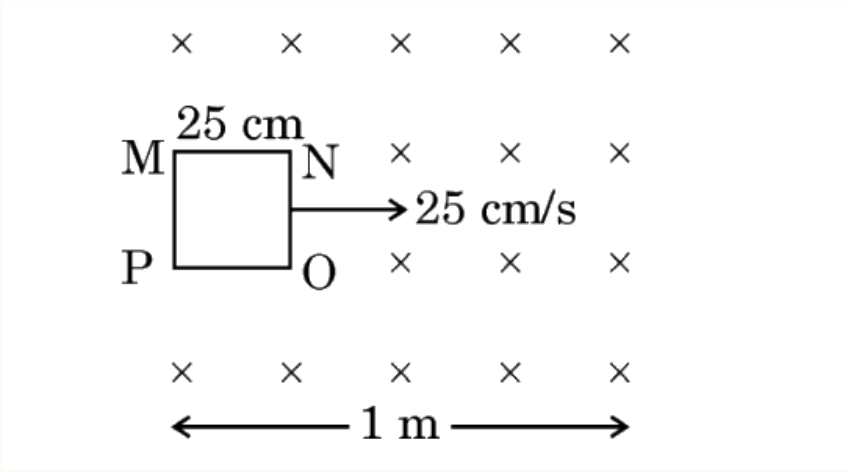
Final answer. The machine is an AC generator; $\boldsymbol{P}$ = brushes, $\boldsymbol{Q}$ = slip rings, $\boldsymbol{R}$ = armature coil; the output rises with more turns $\boldsymbol{N}$ and a stronger field $\boldsymbol{B}$ or higher speed $\boldsymbol{\omega}$.

← *Back to Q103*

Q104

Two coils '1' and '2' are placed close to each other as shown in the figure. Find the direction of induced current in coil '1' in each of the following situations, justifying

your answers :



(a) *Coil '2' is moving towards coil '1'.*

(b) *Coil '2' is moving away from coil '1'.*

(c) *The resistance connected with coil '2' is increased keeping both the coils stationary.*

[2025 • Set 55-4-1]

SOLUTION

Coil 2 carries a steady current I_2 driven by a battery through a resistor R, and coil 1 is a closed loop placed close to it. Whenever the flux through coil 1 due to coil 2 changes, an induced current appears in coil 1, with its direction fixed by Lenz's law.

(a) When coil 2 moves towards coil 1, the magnetic field of coil 2 at coil 1 becomes stronger, so the flux linked with coil 1 increases. By Lenz's law the induced current in coil 1 must oppose this increase, so it sets up a magnetic field that opposes the field of coil 2. Hence the induced current in coil 1 flows in a direction opposite to the current I_2 in coil 2 (viewed from the same side). The two coils behave like like poles facing each other and tend to repel.

(b) When coil 2 moves away from coil 1, its field at coil 1 weakens, so the flux through coil 1 decreases. By Lenz's law the induced current opposes this decrease, setting up a field in the same sense as that of coil 2. Hence the induced current in coil 1 flows in the same direction as I_2 in coil 2. The coils now attract, opposing the separation.

(c) When the resistance in coil 2's circuit is increased (both coils stationary), the current falls,

$$I_2 = \frac{V}{R} \text{ decreases}$$

so the flux through coil 1 decreases. By Lenz's law the induced current opposes this decrease, just as in case (b). Hence the induced current in coil 1 flows in the same direction as I_2 in coil 2.

Final answer. Induced current in coil 1 is opposite to $\mathbf{I_2}$ in (a), and in the same direction as $\mathbf{I_2}$ in both (b) and (c).

← *Back to Q104*

Q105

(a) *Differentiate between magnetic flux through an area and magnetic field at a point.*
(b) *A bar magnet is held with its length along the axis of a closed coil. Initially the south pole of the magnet faces the coil. If the magnet is moved towards the coil, explain how a current is induced in the coil and in what direction.*

[2025 • Set 55-4-2]

SOLUTION

(a) Magnetic field and magnetic flux differ as follows.

Magnetic field $\vec{B}$ is a vector quantity defined at each point in space; it measures the strength and direction of the magnetic influence at that point. Its SI unit is the tesla (T), and it is a local (point) property.

Magnetic flux Φ is a scalar quantity defined for a surface; it measures the total magnetic field passing through that surface and is given by

$$\Phi = \int \vec{B} \cdot d\vec{S} = B\,S\,\theta$$

where θ is the angle between $\vec{B}$ and the area vector $d\vec{S}$. Its SI unit is the weber (Wb), where $\mathbf{1\ Wb = 1\ T^2}$. Thus $\vec{B}$ is defined at a point, while Φ depends on the area, its orientation and the field, and so is a property of the whole surface.

(b) The bar magnet lies along the axis of the coil with its South pole facing the coil. As the magnet is pushed towards the coil, the magnetic flux linked with

the coil increases (the field at the coil grows stronger as the magnet comes nearer).

By Faraday's law, this changing flux induces an emf and hence a current in the closed coil. By Lenz's law, the induced current must oppose the approach of the magnet. To do so, the face of the coil nearer the magnet must become a South pole, so that it repels the approaching South pole.

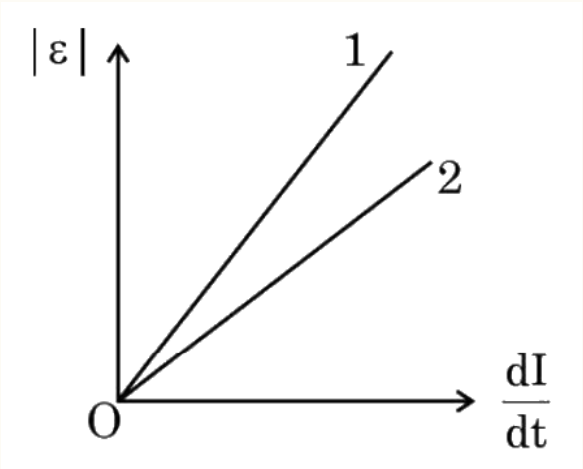
Applying the clock rule, when the near face of the coil acts as a South pole, the induced current in that face flows clockwise as seen by an observer on the magnet's side (i.e. looking at the coil from the magnet). This induced current lasts only while the magnet is moving and the flux is changing.

Final answer. The increasing flux induces a current that makes the coil's near face a South pole, flowing clockwise as seen from the approaching magnet, opposing its motion.

← *Back to Q105*

Q106

(a) *Obtain an expression for the self-inductance of a long solenoid of length 'l', cross-sectional area 'A' and having 'N' turns.*
(b) *The figure shows the plot of magnitude of induced emf (|ε|) versus the rate of change of current in two coils '1' and '2'. Which coil has greater value of self-inductance and why ?*



[2025 • Set 55-4-3]

SOLUTION

(a) Consider a long solenoid of length l, cross-sectional area A and total number of turns N, carrying a current I. The number of turns per unit length is $\boldsymbol{n = N/l}$.

The magnetic field inside a long solenoid is uniform and given by

$$B = \mu_0 \, n \, I = \mu_0 \, \frac{N}{l} \, I$$

The flux through one turn of area A is

$$\phi = B \, A = \mu_0 \, \frac{N}{l} \, I \, A$$

The total flux linkage with all N turns is

$$\Phi = N\phi = \mu_0 \, \frac{N^2 \, A \, I}{l}$$

Since $\boldsymbol{\Phi = L \, I}$ by definition of self-inductance,

$$L = \frac{\Phi}{I}$$

$$\mathbf{L = \frac{\mu_0 \, N^2 \, A}{l}}$$

(b) The induced emf is related to the rate of change of current by

$$|\varepsilon| = L \left| \frac{dI}{dt} \right|$$

So in a graph of $|\varepsilon|$ against $\frac{dI}{dt}$, the slope of the straight line equals the self-inductance L. The coil whose line is steeper (greater slope) has the larger self-inductance.

From the given plot, coil 1 has the steeper line, so coil 1 has the greater value of self-inductance, because for the same rate of change of current it produces a larger induced emf.

Final answer. Self-inductance is $\boldsymbol{L = \frac{\mu_0 \, N^2 \, A}{l}}$, and the coil with the steeper $\boldsymbol{|\varepsilon|}$ vs $\boldsymbol{dI/dt}$ line (coil 1) has the greater self-inductance.

← Back to Q106

Q107

(a) State Lenz's law. A rod MN *of length L is rotated about an axis passing through its end M perpendicular to its length, with a constant angular velocity* $\vec{\omega}$ *in a uniform magnetic field* $\vec{B}$ *parallel to the axis. Obtain an expression for emf induced between its ends.*

OR

(b) Define 'self-inductance' of a coil. Derive an expression for self-inductance of a long solenoid of cross-sectional area A and length l, having n turns per unit length.

[2025 • Set 55-6-1]

SOLUTION

(a) Lenz's law states that the direction of the induced current is always such that it opposes the change in magnetic flux that produces it. It follows from the law of conservation of energy.

The rod MN of length L rotates about an axis through its end M, perpendicular to its own length, with constant angular velocity ω in a uniform magnetic field $\vec{B}$ directed parallel to the axis (perpendicular to the plane of rotation).

Consider a small element of the rod of length dx at a distance x from the end M. Its linear (tangential) speed is

$$v = x\omega$$

The motional emf induced across this element of length dx is

$$d\varepsilon = B\, v\, dx = B(x\omega)\, dx$$

The total emf between the ends M and N is obtained by integrating from $\boldsymbol{x = 0}$ to $\boldsymbol{x = L}$,

$$\varepsilon = {}_0^L B\omega\, x\, dx$$

$$\varepsilon = B\omega[\frac{x^2}{2}]_0^L$$

$$\varepsilon = \frac{1}{2} B\omega L^2$$

(b) Self-inductance of a coil is defined as the magnetic flux linked with the coil per unit current flowing through it, $L = \Phi/I$. Equivalently, it is the emf induced in the coil per unit rate of change of current, $L = -\varepsilon/(dI/dt)$. It measures the opposition of the coil to any change in the current through it.

For a long solenoid of length l, cross-sectional area A and n turns per unit length carrying current I, the field inside is

$$B = \mu_0 n I$$

The flux through one turn is

$$\phi = B A = \mu_0 n I A$$

The total number of turns is $N = n l$, so the total flux linkage is

$$\Phi = N\phi = (n l)(\mu_0 n I A) = \mu_0 n^2 A l I$$

Hence the self-inductance is

$$L = \frac{\Phi}{I}$$

$$L = \mu_0 n^2 A l$$

Final answer. The induced emf is $\varepsilon = \frac{1}{2} B\omega L^2$; the solenoid self-inductance is $L = \mu_0 n^2 A l$.

← Back to Q107

Q108

(a) *State Faraday's law of electromagnetic induction and explain the role of negative sign in its expression.*
(b) *Explain, with an example, that Lenz's law is consistent with the law of conservation of energy.*

[2025 • Set 55-7-1]

SOLUTION

(a) Faraday's law of electromagnetic induction states that the emf induced in a closed circuit is equal to the negative of the rate of change of magnetic flux linked with the circuit. For a coil of N turns,

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -N\,\frac{d\Phi}{dt} \text{ (for } N \text{ turns)}$$

Role of the negative sign: the negative sign is the mathematical statement of Lenz's law. It shows that the induced emf, and the current it drives, always acts in such a direction as to oppose the change in flux that produces it. If the flux is increasing, the induced current opposes the increase; if it is decreasing, the induced current opposes the decrease. Thus the negative sign ensures the result is consistent with the conservation of energy.

(b) Consider moving the North pole of a bar magnet towards a closed coil. As the magnet approaches, the flux through the coil increases. By Lenz's law, the induced current makes the near face of the coil a North pole, which repels the approaching magnet.

To keep the magnet moving against this repulsion, the person must do mechanical work. This mechanical work is converted into electrical energy in the coil, which appears as heat dissipated in the resistance of the circuit ($I^2 R$ heating).

Now suppose the induced current flowed the opposite way, making the near face a South pole. It would then attract the magnet, pulling it in on its own and at the same time producing electrical energy, all without any external work. This would create energy from nothing, violating energy conservation. Hence the opposing direction demanded by Lenz's law is exactly what is required for energy to be conserved.

Final answer. Faraday's law $\varepsilon = -d\Phi/dt$; the negative sign expresses Lenz's law, and the work done against the induced repulsion (which becomes electrical energy) shows Lenz's law conserves energy.

← *Back to Q108*

Q109

(a) *Two concentric circular coils of radii* r_1 *and* r_2 *(*$r_2 \gg r_1$*) are placed coaxially with their centres coinciding. If a current I is passed through the outer coil, obtain the expression for mutual inductance of the arrangement.*

(b) *The current in a solenoid decreases steadily from* 6 mA *to* 2 mA *in* 50 ms. *If an average emf of* 0.4 *V is induced, find the self-inductance of the solenoid.*

[2025 • Set 55-7-2]

SOLUTION

(a) Two concentric coplanar circular coils have radii r_1 and r_2 with $r_2 \gg r_1$, placed coaxially with coinciding centres. A current I is passed through the larger (outer) coil of radius r_2.

The magnetic field produced at the common centre by the outer coil is

$$B = \frac{\mu_0 I}{2 r_2}$$

Since $r_2 \gg r_1$, this field is nearly uniform over the entire area of the small inner coil, πr_1^2. So the flux linked with the inner coil is

$$\Phi = B (\pi r_1^2) = \frac{\mu_0 I}{2 r_2} (\pi r_1^2)$$

$$\Phi = \frac{\mu_0 \pi r_1^2 I}{2 r_2}$$

The mutual inductance is $M = \Phi/I$, hence

$$\mathbf{M = \frac{\mu_0 \pi r_1^2}{2 r_2}}$$

(b) For the solenoid, the induced emf in terms of self-inductance is

$$\varepsilon = L \frac{\Delta I}{\Delta t}$$

The given data are

$$\Delta I = 6 - 2 = 4 \text{ mA} = 4 \times 10^{-3} A$$

$$\Delta t = 50 \text{ ms} = 0.05 s$$

$$\varepsilon = 0.4\ V$$

Solving for L,

$$L = \frac{\varepsilon\, \Delta t}{\Delta I}$$

$$L = \frac{(0.4)(0.05)}{4\times 10^{-3}}$$

$$L = \frac{0.02}{4\times 10^{-3}}$$

$$\mathbf{L = 5\ H}$$

TIP

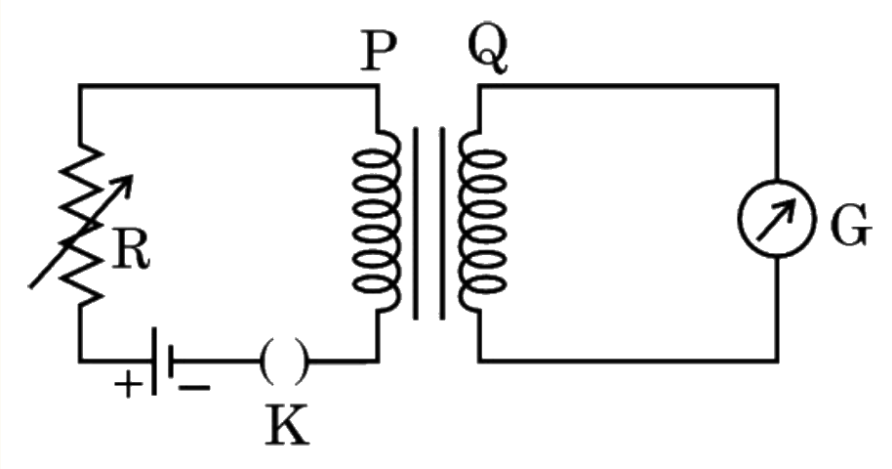
Always convert mA to A and ms to s before substituting. Forgetting the 10^{-3} factors here would give L off by a factor of 1000.

Final answer. Mutual inductance $\boldsymbol{M = \frac{\mu_0 \pi r_1^2}{2 r_2}}$ and the solenoid's self-inductance is $\boldsymbol{L = 5\ H}$.

← *Back to Q109*

Q110

Consider the arrangement of two coils P and Q shown in the figure. When current in coil P is switched on or switched off, a current flows in coil Q.
(a) Explain the phenomenon involved in it.
(b) Mention two factors on which the current produced in coil Q depends.
(c) Give the direction of current in coil Q when there is a current in the coil P and (i) R is increased, and (ii) R is decreased.



[2024 • Set 55-1-2]

SOLUTION

Coils P and Q are placed close together. Coil P carries a current set by a battery and a variable resistor R, and coil Q is connected to a galvanometer.

(a) When the current in coil P is switched on or off (or changed), the magnetic flux linked with the neighbouring coil Q changes. By Faraday's law, this changing flux induces an emf and hence a current in coil Q. This phenomenon, in which a changing current in one coil induces an emf in a nearby coil, is called mutual induction. The induced emf is

$$\varepsilon_Q = -M \frac{dI_P}{dt}$$

where M is the mutual inductance of the pair.

(b) The current induced in coil Q depends on:

(i) the mutual inductance M of the two coils, which is fixed by their number of turns, size, separation, relative orientation and the presence of a core;

(ii) the rate of change of current in coil P, namely $\frac{dI_P}{dt}$.

(c) The direction of the induced current in Q follows from Lenz's law.

(i) When R is increased, the current I_P decreases, so the flux through Q decreases. By Lenz's law, the induced current in Q opposes this decrease by trying to maintain the flux, so it flows in the same direction as I_P in coil P.

(ii) When R is decreased, the current I_P increases, so the flux through Q increases. By Lenz's law, the induced current in Q opposes this increase, so it flows in the direction opposite to I_P in coil P.

Final answer. The effect is mutual induction; the current in Q depends on M and $\mathbf{dI_P/dt}$; it flows in the same direction as $\mathbf{I_P}$ when R is increased and opposite to $\mathbf{I_P}$ when R is decreased.

← Back to Q110

Q111

A 100-turn coil of radius 1.6 cm and resistance 5.0 Ω *is co-axial with a solenoid of 250 turns/cm and radius 1.8 cm. The solenoid current drops from 1.5 A to zero in 25 ms. Calculate the current induced in the coil in this duration. (Take* π^2**=10**)

[2024 • Set 55-2-1]

SOLUTION

Here the coil sits inside the solenoid, so the field of the solenoid passes through the coil. The induced current comes from the changing solenoid current, through the mutual inductance of the pair.

The magnetic field inside the long solenoid is $\mathbf{B_{sol}} = \mu_0\, n\, I$, where n is the number of turns per unit length.

Since the coil (radius $\mathbf{r_c = 1.6\ cm}$) lies inside the solenoid (radius 1.8 cm), this uniform field threads the coil over its own area $\mathbf{A_c} = \pi\, \mathbf{r_c}^2$.

The flux linked with all N turns of the coil is $\Phi = N\, \mathbf{B_{sol}}\, \mathbf{A_c} = N\, \mu_0\, n\, I\, \pi\, \mathbf{r_c}^2$.

Hence the mutual inductance is $M = \dfrac{\Phi}{I} = N\, \mu_0\, n\, \pi\, \mathbf{r_c}^2$.

Putting $N = 100$, $\mu_0 = 4\pi \times 10^{-7}\ T\, m\, A^{-1}$, $n = 250\ \mathbf{turns/cm} = 25000\ \mathbf{turns}/m$ and $\mathbf{r_c} = 0.016\ m$ so that $\mathbf{r_c}^2 = 2.56\times 10^{-4}\ \mathbf{m^2}$:

$$M = (100)(4\pi \times 10^{-7})(25000)(\pi)(2.56\times 10^{-4})$$

$M = 4\pi^2\times (100)(25000)(2.56\times 10^{-4})\times 10^{-7}\ H.$

Using $\pi^2 = 10$, $4\pi^2 = 40$, so $M = 40 \times 6.4\times 10^2 \times 10^{-7}\ H = 2.56\times 10^{-3}\ H = 2.56\ \mathbf{mH}$.

The magnitude of the induced emf is $|\varepsilon| = M\, \dfrac{\mathbf{dI}}{\mathbf{dt}} = M\, \dfrac{\Delta I}{\Delta t}$.

$|\varepsilon| = (2.56\times 10^{-3})\times \dfrac{1.5 - 0}{25\times 10^{-3}}\ V.$

$|\varepsilon| = (2.56\times 10^{-3})(60)\ V = 0.1536\ V.$

The induced current in the coil is $I_{ind} = \frac{|\varepsilon|}{R} = \frac{0.1536}{5.0}\ A.$

$I_{ind} = 0.0307\ A \approx 30.7\ \text{mA}.$

Therefore the **current induced in the coil is about 30.7 mA** while the solenoid current falls to zero.

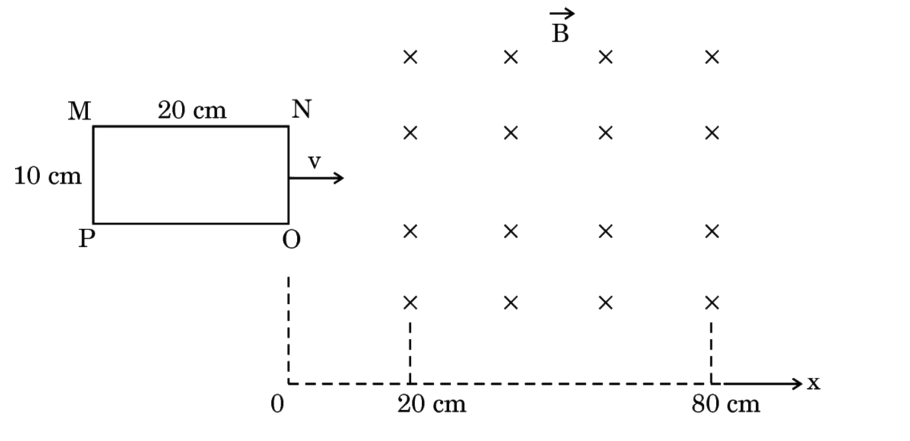
> **COMMON MISTAKE**
>
> *Use the smaller radius (the coil's own radius) for the area, since flux is only over the region where the coil actually sits, not the solenoid's full cross-section.*

Final answer. The induced current in the coil is approximately **30.7 mA** (about **0.031 *A***).

← *Back to Q111*

Q112

A rectangular loop of sides 10 cm × 20 cm is kept outside a region of uniform magnetic field **|*B*|=5 mT** *as shown in the figure. The loop is moved with the velocity of 5 cm/s till it goes completely out of the magnetic field.*



(a) Plot a graph showing variation of the magnetic flux ϕ with ***x*** **(0.**

(b) Find the maximum value of magnetic flux linked with the loop.

(c) Will an external work be required to be done to move the loop through the magnetic field?

[2024 • Set 55-2-2]

SOLUTION

The flux through the loop is $\Phi = B\,A_{in}$, where A_{in} is the part of the loop area that lies inside the field region. As the loop moves, A_{in} changes, so the flux changes.

Let the side of length 20 cm be along the direction of motion (x) and the side of length 10 cm be perpendicular to it. Take x as the position of the loop's leading edge.

(a) Variation of flux with x: while the loop is entering the field, the area inside grows steadily, so the flux rises linearly from 0 to its maximum value Φ_{max}. Once the whole loop is inside the field, the flux stays constant at Φ_{max}. As the loop leaves, the inside area shrinks, so the flux falls linearly back to 0.

The graph of Φ against x is therefore trapezoidal: a straight sloping line up (loop entering), a flat horizontal top (loop fully inside, $\Phi = \Phi_{max}$), and a straight sloping line down (loop leaving), returning to zero.

(b) The maximum flux occurs when the loop is completely inside the field, so the whole area A is threaded.

$A = (0.10\ m)(0.20\ m) = 0.02\ m^2$.

$\Phi_{max} = B\,A = (5\times 10^{-3}\ T)(0.02\ m^2)$.

$\Phi_{max} = 1.0\times 10^{-4}\ Wb = 100\ \mu Wb$.

(c) **Yes, external work is required**, but only while the loop is entering and leaving the field. During these phases the flux is changing, so an emf and an induced current are set up. By Lenz's law this current opposes the change, producing a retarding force on the loop. The external agent must do work against this force, and that work appears as heat dissipated in the loop's resistance. When the loop is fully inside, the flux is constant, so there is no emf, no induced current, no force, and hence no external work is needed during that phase (apart from overcoming friction, which is ignored here).

Final answer. The maximum flux is $\Phi_{max} = 1.0\times 10^{-4}\ Wb$, and external work is needed only while the loop enters and leaves the field.

← Back to Q112

Q113

Prove that induced charge depends on the net change in the magnetic flux and not on the time interval of the flux change.

[2024 • Set 55-3-3]

SOLUTION

By Faraday's law, the magnitude of the induced emf in a circuit equals the rate of change of magnetic flux linked with it.

$$\varepsilon = -\frac{d\phi}{dt}.$$

By Ohm's law, if the total resistance of the circuit is R, the induced current is

$$I = \frac{\varepsilon}{R} = -\frac{1}{R}\frac{d\phi}{dt}.$$

The current is the rate of flow of charge, $I = \frac{dq}{dt}$, so the small charge flowing in time dt is

$$dq = I\,dt = -\frac{1}{R}\frac{d\phi}{dt}\,dt = -\frac{1}{R}\,d\phi.$$

Integrating over the interval during which the flux changes from ϕ_1 at time t_1 to ϕ_2 at time t_2:

$$q = \int_{t_1}^{t_2} I\,dt = -\frac{1}{R}\int_{\phi_1}^{\phi_2} d\phi.$$

$$q = -\frac{1}{R}\,[\,\phi_2 - \phi_1\,] = -\frac{\Delta\phi}{R}.$$

Taking magnitudes, $q = \frac{|\Delta\phi|}{R}$.

This final expression contains only the net change in flux $\Delta\phi$ and the resistance R. The time interval $(t_2 - t_1)$ over which the change happens does not appear at all.

Hence the **induced charge depends only on the net change in magnetic flux (and the resistance), not on how fast or slow that change takes place.** Hence

proved.

CONCEPT

Note the difference: induced emf and induced current do depend on time (rate of change), but the total charge that flows does not.

Final answer. The induced charge is $q = \frac{\Delta\Phi}{R}$, which depends only on the net flux change and the resistance, not on the time taken.

← *Back to Q113*

Q114

(i) Define mutual inductance. Write its SI unit.
(ii) Derive an expression for the mutual inductance of a system of two long coaxial solenoids of same length l, having turns N_1 *and* N_2 *and of radii* r_1, r_2 $(\mathbf{r_2 > r_1})$.

[2024 • Set 55-4-1]

SOLUTION

(i) Mutual inductance of a pair of coils is the property by which a changing current in one coil induces an emf in the other due to their mutual magnetic flux linkage.

Quantitatively, if a current I_1 in coil 1 produces a flux linkage $N_2\Phi_{21}$ with coil 2, then the mutual inductance is $\mathbf{M_{21}} = \frac{N_2\Phi_{21}}{I_1}$. It is numerically the flux linked with one coil per unit current in the other.

Its SI unit is the **henry (H)**.

(ii) Consider two long coaxial solenoids of the same length l. The inner one has N_1 turns and radius r_1; the outer one has N_2 turns and radius r_2 (with $\mathbf{r_2 > r_1}$), so the inner solenoid lies completely inside the outer one.

Pass a current I_2 through the outer solenoid (turns per unit length $\mathbf{n_2 = N_2/l}$). The uniform field it produces inside is

$$\mathbf{B_2 = \mu_0\, n_2\, I_2 = \mu_0 \frac{N_2}{l}\, I_2}.$$

This field is uniform over the cross-section of the inner solenoid, area πr_1^2. So the flux through one turn of the inner solenoid is

$$\Phi_1 = B_2 \times \pi r_1^2 = \mu_0 \frac{N_2}{l} I_2 \pi r_1^2.$$

The inner solenoid has N_1 turns, so the total flux linkage with it is

$$N_1\Phi_1 = \mu_0 \frac{N_1 N_2}{l} I_2 \pi r_1^2.$$

By definition the mutual inductance is

$$M = \frac{N_1\Phi_1}{I_2} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}.$$

Therefore $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$. Note that r_1 (the smaller radius) enters, because the shared flux is confined to the inner solenoid's cross-section.

Final answer. Mutual inductance is the flux linked per unit current in the other coil (SI unit henry); for the two solenoids, $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$.

← *Back to Q114*

Q115

(i) State Lenz's Law. In a closed circuit, the induced current opposes the change in magnetic flux that produced it as per the law of conservation of energy. Justify.
(ii) A metal rod of length 2 m is rotated with a frequency 60 rev/s about an axis passing through its centre and perpendicular to its length. A uniform magnetic field of 2 T perpendicular to its plane of rotation is switched-on in the region. Calculate the e.m.f. induced between the centre and the end of the rod.

[2024 • Set 55-5-1]

SOLUTION

(i) Lenz's law states that the direction of the induced current (or induced emf) is always such that it opposes the change in magnetic flux that produces it.

Justification from conservation of energy: Suppose the induced current did not oppose the change but instead aided it. Then the motion or change would be

self-strengthening, the induced current would keep growing, and electrical energy would be produced continuously without any work being done. This would create energy out of nothing, violating the law of conservation of energy. Because the induced current opposes the change, the external agent must do work against the opposing (retarding) force; this work is exactly what is converted into the electrical energy of the induced current (and then into heat). So Lenz's law is a direct consequence of energy conservation.

(ii) The rod is rotated about an axis through its centre, perpendicular to its length. The total length is 2 *m*, so each half has length $r = 1\ m$.

The frequency is $\nu = 60\ \text{rev/}s$, so the angular frequency is

$\omega = 2\pi\nu = 2\pi \times 60 = 120\pi\ \text{rad s}^{-1}$.

The field is $B = 2\ T$, perpendicular to the plane of rotation.

The emf induced between the centre (axis) and one end of a rotating rod of length *r* is

$$\varepsilon = \frac{1}{2} B\,\omega\,r^2.$$

Substituting,

$$\varepsilon = \frac{1}{2}(2)(120\pi)(1)^2\ V.$$

$\varepsilon = 120\pi\ V \approx 377\ V$.

Therefore the **emf induced between the centre and the end of the rod is $120\pi\ V \approx 377\ V$.**

COMMON MISTAKE

Take r as the half-length (1 m), not the full 2 m, because the axis passes through the centre of the rod.

Final answer. The emf between the centre and the end is $\varepsilon = 120\pi\ V \approx 377\ V$.

← *Back to Q115*

Q116

A long solenoid of radius r consists of n turns per unit length. A current $I = I_0 \omega t$ *flows in the solenoid. A coil of N turns is wound tightly around it near its centre. What is:*
(a) the induced emf in the coil?
(b) the mutual inductance between the solenoid and the coil?

[2023 • Set 55-1-1]

SOLUTION

The coil is wound tightly over the solenoid near its centre, so the field of the solenoid passes fully through every turn of the coil.

The magnetic field inside the long solenoid is

$B = \mu_0 n I = \mu_0 n I_0 \omega t.$

The coil has the same radius r as the solenoid, so the area threaded is πr^2. The flux through one turn of the coil is

$\phi_1 = B \pi r^2 = \mu_0 n I_0 \pi r^2 \omega t.$

The coil has N turns, so the total flux linkage is

$\Phi = N\phi_1 = \mu_0 N n \pi r^2 I_0 \omega t.$

(a) The induced emf in the coil is

$$\varepsilon = -\frac{d\Phi}{dt} = -\mu_0 N n \pi r^2 I_0 \frac{d}{dt}(\omega t).$$

$\varepsilon = -\mu_0 N n \pi r^2 \omega I_0 \omega t.$

So $\varepsilon = -\mu_0 N n \pi r^2 \omega I_0 \omega t.$

(b) The mutual inductance between the solenoid and the coil is the flux linkage per unit current:

$$M = \frac{\Phi}{I} = \frac{\mu_0 N n \pi r^2 I_0 \omega t}{I_0 \omega t}.$$

$M = \mu_0 N n \pi r^2.$

So $M = \mu_0 N n \pi r^2.$

As a check, $\varepsilon = -M \frac{dI}{dt} = -\mu_0 N n \pi r^2 \omega I_0 \omega t$, which matches part (a).

Final answer. The induced emf is $\varepsilon = -\mu_0 N n \pi r^2 \omega I_0 \omega t$ and the mutual inductance is $M = \mu_0 N n \pi r^2$.

← *Back to Q116*

Q117

What is meant by the term 'mutual inductance' of a pair of coils? Obtain an expression for the mutual inductance of two long coaxial solenoids, each of length l but having different number of turns N_1 and N_2 and radii r_1 and r_2 ($r_2 > r_1$).

[2023 • Set 55-3-1]

SOLUTION

Mutual inductance of a pair of coils is the property by which a changing current in one coil sets up an induced emf in the neighbouring coil, due to the magnetic flux of the first coil linking the second.

If a current I_1 in coil 1 links a flux $N_2\Phi_{21}$ with coil 2, then the mutual inductance is $M = \dfrac{N_2\Phi_{21}}{I_1}$, i.e. the flux linked with one coil per unit current in the other. Its SI unit is the henry (H).

Now consider two long coaxial solenoids of the same length l. The inner has N_1 turns and radius r_1; the outer has N_2 turns and radius r_2 (with $r_2 > r_1$), so the inner solenoid lies entirely within the outer.

Pass a current I_2 in the outer solenoid, whose turns per unit length are $n_2 = N_2/l$. The uniform field it produces inside is

$$B_2 = \mu_0 n_2 I_2 = \mu_0 \frac{N_2}{l} I_2.$$

This field is uniform across the inner solenoid's cross-section, area πr_1^2. The flux through one turn of the inner solenoid is

$$\phi_1 = B_2 \pi r_1^2 = \mu_0 \frac{N_2}{l} I_2 \pi r_1^2.$$

With N_1 turns, the total flux linkage with the inner solenoid is

$$\Phi_1 = N_1\phi_1 = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l} I_2.$$

Therefore the mutual inductance is

$$M = \frac{\Phi_1}{I_2} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}.$$

So $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$. The same result is obtained if instead a current is passed through the inner solenoid, confirming $M_{12} = M_{21}$.

Final answer. Mutual inductance is the flux linked per unit current in the other coil (unit henry); for the two coaxial solenoids, $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$.

← *Back to Q117*

Q118

A rectangular loop of sides 25 cm *and* 20 cm *is lying in x-y plane. It is subjected to a magnetic field* $\vec{B} = (5t^2 + 2t + 10)\,\hat{k}$*, where B is in tesla and t is in seconds. If the resistance of the loop is* 4 Ω*, find the emf induced and the induced current in the loop at* $t = 5\ s$.

[2023 • Set 55-3-2]

SOLUTION

The loop lies in the *x-y* plane, so its area vector points along $\hat{k}$, parallel to the field. The field is uniform over the loop but changes with time, so the flux changes and an emf is induced (Faraday's law).

The area of the loop is

$A = (0.25\ m)(0.20\ m) = 0.05\ m^2$.

Since $\vec{B} = (5t^2 + 2t + 10)\,\hat{k}$ is along the area vector, the flux is

$\Phi = \vec{B} \cdot \vec{A} = B\,A = (5t^2 + 2t + 10)(0.05)\ Wb$.

The induced emf is the magnitude of the rate of change of flux:

$$\varepsilon = \left|\frac{d\Phi}{dt}\right| = A\,\frac{dB}{dt} = 0.05\,\frac{d}{dt}(5t^2 + 2t + 10).$$

$\varepsilon = 0.05\,(10t + 2)\ V$.

At $t = 5\ s$:

$\varepsilon = 0.05\,[\,10(5) + 2\,] = 0.05\,(52)\,V = 2.6\,V.$

The induced current is, by Ohm's law,

$$i = \frac{\varepsilon}{R} = \frac{2.6}{4}\,A = 0.65\,A.$$

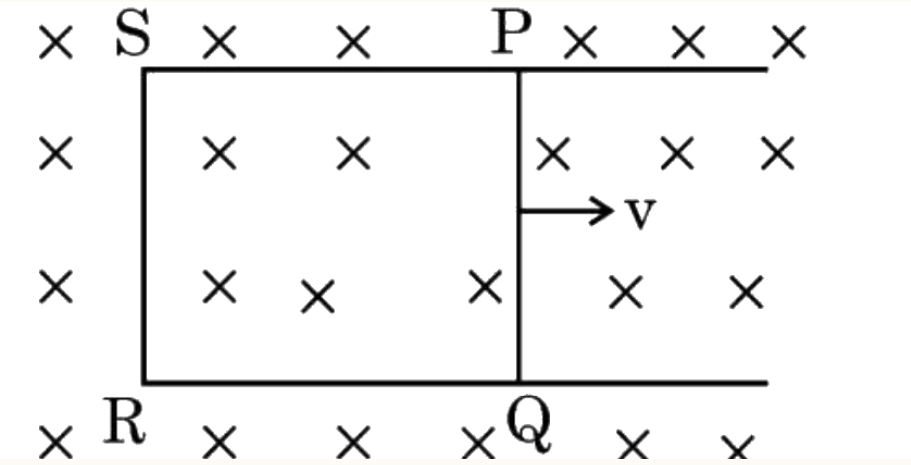
Therefore at $t = 5\,s$ the **induced emf is 2.6 V and the induced current is 0.65 A.**

Final answer. At $t = 5\,s$, the induced emf is $\varepsilon = 2.6\,V$ and the induced current is $i = 0.65\,A$.

← *Back to Q118*

Q119

The figure shows a rectangular conductor PQRS *in which the arm* PQ *of length* 10 cm *and resistance* 0.4 Ω *is free to move. It is kept in a uniform magnetic field* $B = 0.2\,T$ *acting perpendicular into the plane of* PQRS. *If arm* PQ *is moved with a velocity* v *of* 5 cm/*s as shown, find:*
(a) the current induced in the loop, and
(b) the power required to move the arm.
(Resistances of arms PS, SR *and* RQ *are negligible.)*



[2023 • Set 55-3-3]

SOLUTION

When the arm PQ slides through the field, the area of the loop PQRS changes, so the flux changes and a motional emf is set up across PQ.

The motional emf is

$\varepsilon = B\,v\,L,$

where $L = 0.10\ m$ is the length of PQ, $v = 5\ cm/s = 0.05\ m/s$, and $B = 0.2\ T$.

$\varepsilon = (0.2)(0.05)(0.10)\ V.$

$\varepsilon = 1.0\times 10^{-3}\ V = 1.0\ mV.$

(a) Since only PQ has resistance ($r = 0.4\ \Omega$; the other arms are negligible), the induced current is

$$i = \frac{\varepsilon}{r} = \frac{1.0\times 10^{-3}}{0.4}\ A.$$

$i = 2.5\times 10^{-3}\ A = 2.5\ mA.$

(b) The power required to move the arm equals the electrical power dissipated (since the rod moves at steady speed, the external force balances the magnetic retarding force):

$P = \varepsilon\, i = (1.0\times 10^{-3})(2.5\times 10^{-3})\ W.$

$P = 2.5\times 10^{-6}\ W = 2.5\ \mu W.$

Therefore the **induced current is 2.5 mA and the power required to move the arm is 2.5 μW.**

CONCEPT

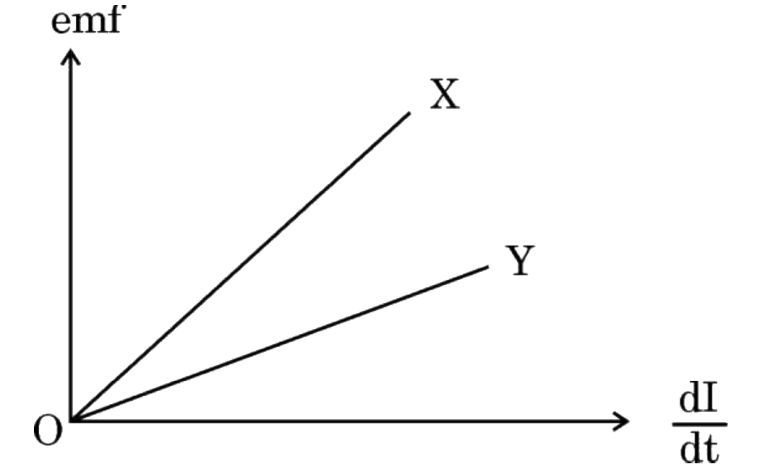
The power supplied by the agent equals $\varepsilon\, i = i^2 r = F\,v$; all three give the same answer, since at constant speed the applied work all becomes heat in the resistance.

Final answer. The induced current is $i = 2.5\ mA$ and the power required to move the arm is $P = 2.5\ \mu W$.

← *Back to Q119*

Q120

(a) The figure shows the variation of induced emf as a function of rate of change of current for two identical solenoids X and Y. One is air cored and the other is iron cored. Which one of them is iron cored? Why?



(b) Obtain an expression for self-inductance of a long solenoid of length L and cross-sectional area A having N turns.

[2023 • Set 55-4-1]

SOLUTION

(a) The graph plots induced emf ε against the rate of change of current $\frac{dI}{dt}$. For a coil, $\varepsilon = L\,\frac{dI}{dt}$, so each line is a straight line through the origin whose slope is the self-inductance L:

$$\text{slope} = \frac{\varepsilon}{dI/dt} = L.$$

An iron core has a relative permeability $\mu_r \gg 1$, which greatly increases the self-inductance compared with an air core. So the iron-cored solenoid has the larger L and therefore the **steeper line on the graph.**

Hence the solenoid whose line has the greater slope is the iron-cored one; the line with the smaller slope is air-cored.

(b) Consider a long solenoid of length L, cross-sectional area A, with N turns, carrying current I. Its turns per unit length are $n = N/L$.

The uniform field inside is

$$B = \mu_0\, n\, I = \mu_0\, \frac{N}{L}\, I.$$

The flux through one turn is

$$\Phi = B\,A = \mu_0\, \frac{N}{L}\, I\,A.$$

The total flux linkage with all N turns is

$\Phi = N\phi = \frac{\mu_0 N^2 A I}{L}$.

By definition $\Phi = L_{self} I$, so the self-inductance is

$L_{self} = \frac{\Phi}{I} = \frac{\mu_0 N^2 A}{L}$.

Therefore $L_{self} = \frac{\mu_0 N^2 A}{L}$, where L is the length of the solenoid. With an iron core of permeability $\mu = \mu_0 \mu_r$, this becomes $L_{self} = \frac{\mu_0 \mu_r N^2 A}{L}$, which explains the larger inductance in part (a).

Final answer. The iron-cored solenoid is the one with the steeper line (larger L); the self-inductance of a long solenoid is $L_{self} = \frac{\mu_0 N^2 A}{L}$.

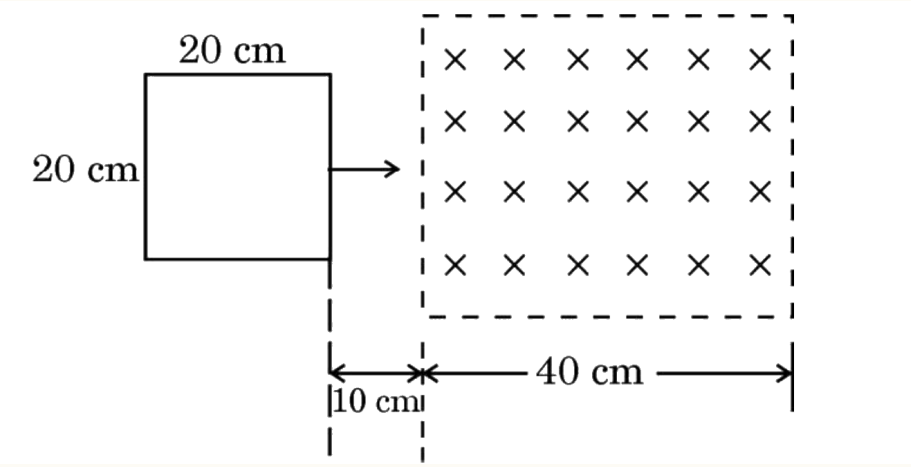
← *Back to Q120*

Q121

(a) Two coplanar concentric circular loops of radii R and r ($r \ll R$) are arranged coaxially. Obtain the expression for their mutual inductance.

OR

(b) A square loop of side 20 cm starts moving at $t = 0$ with a velocity of 5 cm/s towards a region of uniform magnetic field as shown in the figure. Specify the time interval(s) during which induced emf is produced in the loop.



[2023 • Set 55-4-2]

SOLUTION

(a) Mutual inductance of two coplanar concentric coaxial loops of radii R and r, with $r \ll R$.

Let a current I flow in the outer (larger) loop of radius R. By the formula for the magnetic field at the centre of a current loop, the field produced at the common centre is

$$B = \frac{\mu_0 I}{2R}$$

Since $r \ll R$, the small inner loop lies in a region where this field is practically uniform and directed along the common axis. The flux linked with the inner loop of area πr^2 is therefore

$$\Phi = B \times \pi r^2 = \frac{\mu_0 I}{2R} \times \pi r^2$$

$$\Phi = \frac{\mu_0 \pi r^2}{2R} I$$

By definition of mutual inductance, $\Phi = M I$, so

$$M = \frac{\Phi}{I} = \frac{\mu_0 \pi r^2}{2R}$$

Hence the **mutual inductance** of the arrangement is $\boldsymbol{M = \frac{\mu_0 \pi r^2}{2R}}$.

OR

(b) A square loop of side 20 cm moves at $\boldsymbol{v = 5\ cm/s}$ towards a uniform field region (width 10 cm from the figure). An emf is induced only while the flux through the loop is changing, that is, only while the loop is partly entering or partly leaving the field.

Entering the field: the leading edge reaches the near boundary at $\boldsymbol{t = 0}$ and the loop keeps entering until the leading edge reaches the far boundary of the field, a distance of 10 cm.

$$t = \frac{10\ \text{cm}}{5\ \text{cm/}s} = 2\ s$$

So flux increases and emf is induced during $0 \le t \le 2\ s$.

Fully inside: after $\boldsymbol{t = 2\ s}$ the field region (width 10 cm) lies wholly within the loop (side 20 cm). The trailing edge enters the field at

$$t = \frac{20\ \text{cm}}{5\ \text{cm/s}} = 4\ s$$

Between $t = 2\ s$ and $t = 4\ s$ the flux through the loop stays constant, so no emf is induced.

Leaving the field: the loop starts to exit at $t = 4\ s$ and the trailing edge clears the far boundary at

$$t = \frac{30\ \text{cm}}{5\ \text{cm/s}} = 6\ s$$

So flux decreases and emf is induced during $4\ s \le t \le 6\ s$.

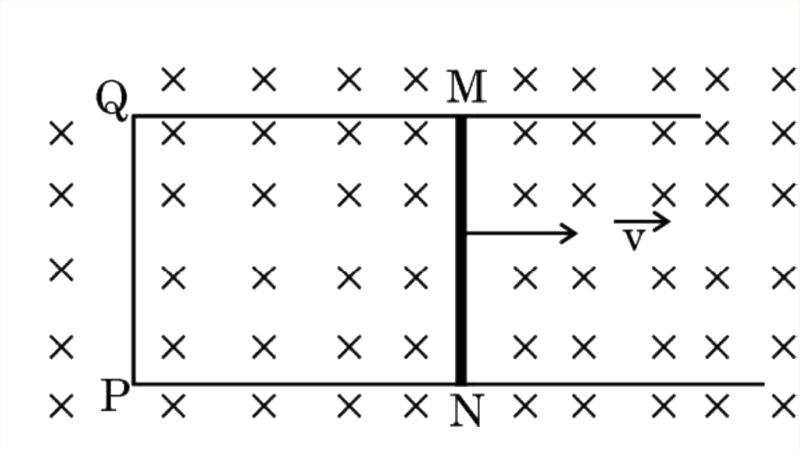
Hence **induced emf is produced during $0 \le t \le 2\ s$ and $4\ s \le t \le 6\ s$**; it is zero in between.

Final answer. (a) $M = \frac{\mu_0 \pi r^2}{2R}$. (b) emf is induced during **0** to **2** s and **4** s to **6** s.

← *Back to Q121*

Q122

A rectangular conductor MNPQ with a movable arm MN (resistance r) is kept in a uniform magnetic field as shown in the figure. Resistance of arms MQ, QP and PN are negligible. Obtain the expression for the:
(a) current induced in the loop specifying its direction, and
(b) power required to move the arm.



[2023 • Set 55-4-3]

SOLUTION

Let the arm MN have length L and move with speed v in a uniform field $\vec{B}$ directed into the plane of the loop. As MN moves, the area of the loop changes, so the flux changes and an emf is induced.

(a) Induced current and its direction.

The free charges in the moving rod experience a magnetic force, setting up a motional emf. In time dt the area swept by MN is $\mathbf{dA = L\,v\,dt}$, so the change of flux is $\mathbf{d\Phi = B\,dA = B\,L\,v\,dt}$. By Faraday's law the magnitude of the induced emf is

$$\varepsilon = \frac{d\Phi}{dt} = B\,L\,v$$

Since arm MN has resistance r and the other arms have negligible resistance, the induced current is

$$i = \frac{\varepsilon}{r} = \frac{B\,L\,v}{r}$$

Direction (Lenz's law): with $\vec{B}$ into the page and MN moving so as to increase the enclosed area, the inward flux through the loop increases. The induced current must oppose this increase, so it flows so as to set up a magnetic field out of the page inside the loop, that is, in the anticlockwise sense (from M to Q to P to N and back to M).

$i = \frac{B\,L\,v}{r}$, directed **anticlockwise** in the loop.

(b) Power required to move the arm.

The current-carrying rod in the field experiences a retarding (opposing) force $\boldsymbol{F = B\,i\,L}$. To keep the rod moving with constant speed v, an equal external force must be applied. The mechanical power supplied equals the electrical power dissipated:

$$P = \varepsilon\, i = (B\,L\,v)\left(\frac{B\,L\,v}{r}\right)$$

$$P = \frac{B^2\,L^2\,v^2}{r}$$

Hence the **power required to move the arm** is $P = \dfrac{B^2 L^2 v^2}{r}$, which appears entirely as heat in the resistance r.

TIP

Always justify the current direction with Lenz's law in words; just writing the formula loses the direction mark.

Final answer. $i = \dfrac{BLv}{r}$ (anticlockwise) and $P = \dfrac{B^2 L^2 v^2}{r}$.

← *Back to Q122*

Q123

State the basic principle behind the working of an ac generator. Briefly describe its working and obtain the expression for the instantaneous value of emf induced.

[2023 • Set 55-5-1]

SOLUTION

Principle: An a.c. generator works on the principle of electromagnetic induction. When the magnetic flux linked with a coil changes, an emf is induced in it (Faraday's law). Here the change in flux is produced by rotating the coil in a magnetic field.

Working: A rectangular coil of N turns and area A is mounted on an axle and rotated with a uniform angular speed ω about an axis perpendicular to a uniform magnetic field $\vec{B}$. The two ends of the coil are connected to two slip rings, which press against carbon brushes that lead to the external circuit. As the coil rotates, the angle between the area vector of the coil and $\vec{B}$ keeps changing, so the flux linked with the coil changes continuously and an alternating emf is induced.

Expression for instantaneous emf: At time t, let the normal to the coil make an angle $\theta = \omega t$ with $\vec{B}$. The flux linked with the coil is

$$\Phi = N B A \theta = N B A \omega t$$

By Faraday's law, the induced emf is

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d}{dt}(N\,B\,A\,\omega\,t)$$

$$\varepsilon = N\,B\,A\,\omega\,\omega\,t$$

Writing the peak value as $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$, the instantaneous emf is

$$\varepsilon = \varepsilon_0\,\omega\,t$$

Thus the output emf varies sinusoidally with time, completing one full cycle in each rotation of the coil, and its peak value is $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$.

Final answer. Instantaneous emf $\boldsymbol{\varepsilon = N\,B\,A\,\omega\,\omega\,t = \varepsilon_0\,\omega\,t}$, with $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$.

← *Back to Q123*

Q124

Two concentric circular coils X and Y of radii r_1 *and* r_2 $\boldsymbol{(r_1 > r_2)}$ *having* N_1 *and* N_2 *turns respectively are placed coaxially with centres coinciding. Obtain an expression for (i) the mutual inductance for the arrangement, and (ii) the magnetic flux linked with coil Y when current I flows through coil X.*

[2021]

SOLUTION

Coil X has radius r_1 and N_1 turns; coil Y has radius r_2 and N_2 turns, with $\boldsymbol{r_1 > r_2}$. Both are placed coaxially with coinciding centres.

(i) Mutual inductance.

Let a current I flow through the larger coil X. The magnetic field produced at its centre (which is also the centre of Y) is

$$B = \frac{\mu_0\,N_1\,I}{2\,r_1}$$

Since $\boldsymbol{r_2 < r_1}$, the inner coil Y lies in this practically uniform field. The flux linked with the N_2 turns of coil Y is

$$\Phi_Y = N_2\, B\,(\pi r_2^2) = N_2 \times \frac{\mu_0 N_1 I}{2 r_1} \times \pi r_2^2$$

$$\Phi_Y = \frac{\mu_0 N_1 N_2 \pi r_2^2}{2 r_1} I$$

By definition $\Phi_Y = M I$, so the mutual inductance is

$$M = \frac{\Phi_Y}{I} = \frac{\mu_0 N_1 N_2 \pi r_2^2}{2 r_1}$$

Hence $M = \dfrac{\mu_0 N_1 N_2 \pi r_2^2}{2 r_1}$.

(ii) Magnetic flux linked with coil Y when current I flows through X.

Using $\Phi_Y = M I$ with the value of M found above,

$$\Phi_Y = \frac{\mu_0 N_1 N_2 \pi r_2^2}{2 r_1} I$$

Thus the **flux linked with coil Y** is $\Phi_Y = \dfrac{\mu_0 N_1 N_2 \pi r_2^2 I}{2 r_1}$.

TIP

Pass the current through the LARGER coil so the field can be treated as uniform over the smaller coil; doing it the other way needs the field at an off-centre point and is far harder.

Final answer. $M = \dfrac{\mu_0 N_1 N_2 \pi r_2^2}{2 r_1}$ and $\Phi_Y = \dfrac{\mu_0 N_1 N_2 \pi r_2^2 I}{2 r_1}$.

← *Back to Q124*

Q125

What are eddy currents? Why does the pendulum plate with holes or slots reduce electromagnetic damping? How are the eddy currents minimised in the core of a transformer? Why is the electromagnetic braking effect smooth in a train?

[2021]

SOLUTION

Eddy currents: When the magnetic flux through the body of a bulk conductor changes, induced currents are set up that circulate in closed loops within the volume of the conductor. These circulating induced currents are called eddy currents. By Lenz's law, their direction is such that they oppose the change in flux that produces them.

Why slots/holes reduce damping: A solid metal plate swinging into a magnetic field develops large eddy currents over its wide cross-section. These currents dissipate energy as heat, producing strong electromagnetic damping that quickly stops the plate. When holes or slots are cut in the plate, the continuous conducting paths are broken up. The eddy currents are confined to small regions, so their magnitude falls sharply. With weaker eddy currents, far less energy is dissipated, so the opposing (damping) force is greatly reduced and the plate swings more freely.

Eddy currents in a transformer core: The core is built from thin sheets called laminations, each coated with an insulating varnish and stacked parallel to the magnetic field. The insulation breaks the core into thin strips, greatly increasing the resistance of the paths available to eddy currents and reducing the area of each eddy-current loop. This sharply lowers the eddy currents and the resulting I^2R heat loss in the core.

Smooth braking in a train: When electromagnetic brakes are switched on, strong eddy currents are induced in the metal rotating parts (wheels or rails). These currents oppose the motion and produce a retarding force that is proportional to the speed. Because the braking force builds up and dies away gradually with speed, and there is no mechanical contact or friction, the braking is **smooth and jerk-free**.

Final answer. Eddy currents are circulating induced currents in bulk conductors; slots break their paths, laminations cut them in the core, and their speed-dependence makes train braking smooth.

← Back to Q125

Q126

(a) Differentiate between self inductance and mutual inductance.
(b) The mutual inductance of two coaxial coils is 2 H. The current in one coil is changed uniformly from zero to 0.5 A in 100 ms. Find the:
(i) change in magnetic flux through the other coil.
(ii) emf induced in the other coil during the change.

[2020 • Set 55-4-1]

SOLUTION

(a) Difference between self inductance and mutual inductance.

Self inductance (L): It is the property of a single coil by virtue of which it opposes any change in the current flowing through itself. The flux linked with the coil due to its own current is $\Phi = L\,I$, and the self-induced emf is $\varepsilon = -L\frac{dI}{dt}$.

Mutual inductance (M): It is the property of a pair of coils by virtue of which a change of current in one coil induces an emf in the other. The flux linked with the second coil due to current I_1 in the first is $\Phi_2 = M\,I_1$, and the emf induced in the second coil is $\varepsilon_2 = -M\frac{dI_1}{dt}$. Self inductance involves one coil and its own current; mutual inductance involves two coils, the emf appearing in the second due to current change in the first.

(b) Given $M = 2\ H$, current changes from 0 to 0.5 A, so $\Delta I = 0.5\ A$, in $\Delta t = 100\ ms = 0.1\ s$.

(i) Change in magnetic flux through the other coil:

$$\Delta\Phi = M\,\Delta I$$

$$\Delta\Phi = 2 \times 0.5$$

$$\Delta\Phi = 1.0\ \text{Wb}$$

(ii) emf induced in the other coil:

$$|\varepsilon| = M\,\frac{\Delta I}{\Delta t} = \frac{\Delta\Phi}{\Delta t}$$

$$|\varepsilon| = \frac{1.0}{0.1}$$

$$|\varepsilon| = 10\ V$$

Hence the **change in flux is 1.0 Wb** and the **induced emf is 10 *V***.

Final answer. $\Delta\Phi = 1.0$ Wb and **$|\varepsilon| = 10$ *V***.

← Back to Q126

Q127

Draw the labelled diagram of an AC generator. Briefly explain its working and obtain the expression for the emf produced in the coil.

[2020 • Set 55-4-3]

SOLUTION

Labelled diagram (described): Draw a rectangular coil ABCD wound on an axle, placed between the curved pole pieces of a permanent magnet marked N (left) and S (right), with the field lines $\vec{B}$ pointing from N to S across the coil. The two ends of the coil are joined to two separate slip rings R_1 and R_2 mounted on the same axle. Two carbon brushes B_1 and B_2 press against the slip rings and connect through them to an external load (resistor) in the outer circuit. An arrow shows the coil rotating about the axle.

Principle: The a.c. generator works on the principle of electromagnetic induction: rotating a coil in a magnetic field changes the flux linked with it and induces an emf.

Working: The coil ABCD of N turns and area A is rotated with constant angular speed ω in the uniform field $\vec{B}$. As it turns, the angle between the coil's normal and $\vec{B}$ keeps changing, so the flux changes and an alternating emf appears across the coil. The slip rings rotate with the coil while the brushes stay fixed, delivering this alternating emf to the external circuit.

Expression for emf: At time t the normal to the coil makes angle $\theta = \omega t$ with $\vec{B}$, so the flux linked with the coil is

$$\Phi = N\,B\,A\,\omega\,t$$

By Faraday's law the induced emf is

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d}{dt}(N\,B\,A\,\omega\,t)$$

$$\varepsilon = N\,B\,A\,\omega\,\omega\,t$$

With peak emf $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$, this is

$$\varepsilon = \varepsilon_0\,\omega\,t$$

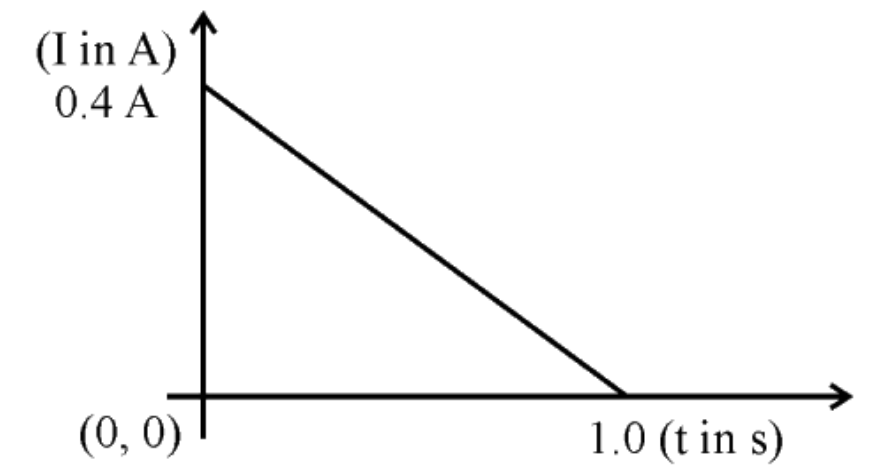
Hence the generator produces an **alternating emf $\boldsymbol{\varepsilon = N\,B\,A\,\omega\,\omega\,t}$** whose peak value is $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$.

Final answer. emf produced $\boldsymbol{\varepsilon = N\,B\,A\,\omega\,\omega\,t = \varepsilon_0\,\omega\,t}$, with $\boldsymbol{\varepsilon_0 = N\,B\,A\,\omega}$.

← *Back to Q127*

Q128

When a conducting loop of resistance 10 Ω *and area* 10 cm^2 *is removed from an external magnetic field acting normally, the variation of induced current in the loop with time is shown in the figure.*



Find the
(i) total charge passed through the loop.

(ii) change in magnetic flux through the loop.
(iii) magnitude of the magnetic field applied.

[2020 • Set 55-5-1]

SOLUTION

From the graph, the induced current falls linearly from its initial value $\mathbf{I_0 = 0.4}$ $\boldsymbol{A}$ at $\boldsymbol{t = 0}$ to 0 at $\boldsymbol{t = 1.0\ s}$. Given $\boldsymbol{R = 10\ \Omega}$ and $\boldsymbol{A = 10\ cm^2 = 10 \times 10^{-4}\ m^2}$.

(i) Total charge through the loop.

The charge equals the area under the current-time graph, which is a triangle:

$$q = \int I\, dt = \frac{1}{2} \times I_0 \times t$$

$$q = \frac{1}{2} \times 0.4 \times 1.0$$

$$q = 0.20\ C$$

(ii) Change in magnetic flux through the loop.

The charge is related to the flux change by $\boldsymbol{q = \frac{\Delta\Phi}{R}}$, so

$$\Delta\Phi = q \times R$$

$$\Delta\Phi = 0.20 \times 10$$

$$\Delta\Phi = 2.0\ \text{Wb}$$

(iii) Magnitude of the magnetic field.

Since the field acts normally and the loop is fully removed from it, $\boldsymbol{\Delta\Phi = B\,A}$, so

$$B = \frac{\Delta\Phi}{A} = \frac{q\,R}{A}$$

$$B = \frac{2.0}{10 \times 10^{-4}}$$

$$B = 2000\ T$$

Hence, with the data given, $\boldsymbol{q = 0.20\ C}$, $\boldsymbol{\Delta\Phi = 2.0\ Wb}$, and $\boldsymbol{B = 2000\ T}$ (a very large value, indicating a unit mismatch in the printed data; the method and relations $\boldsymbol{q = \Delta\Phi/R}$ and $\boldsymbol{\Delta\Phi = BA}$ are the marked steps).

CONCEPT

The key relation $q = \Delta\Phi/R$ does not contain time, so the total charge depends only on the net flux change, not on how fast the field is removed.

Final answer. $q = 0.20\ C$, $\Delta\Phi = 2.0\ \text{Wb}$, and $B = \dfrac{\Delta\Phi}{A} = 2000\ T$ (per the given data).

← Back to Q128

Q129

Define mutual inductance and write its S.I. unit.

[2019 • Set 55-1-1]

SOLUTION

Mutual inductance: The mutual inductance of a pair of coils is the property by which a change of current in one coil induces an emf in the neighbouring coil.

If a current I_1 flowing in coil 1 links a magnetic flux Φ_2 with coil 2, then Φ_2 is proportional to I_1:

$$\Phi_2 = M\, I_1$$

where the constant M is the mutual inductance of the pair. Equivalently, the emf induced in coil 2 due to a changing current in coil 1 is

$$\varepsilon_2 = -M\,\frac{dI_1}{dt}$$

Thus M may be defined as the flux linked with one coil per unit current in the other, or as the emf induced in one coil per unit rate of change of current in the other.

SI unit: The SI unit of mutual inductance is the **henry (H)**, where $1\ H = 1\ \text{Wb}/A = 1\ V\,s/A$.

Final answer. Mutual inductance M is the flux linked per unit current in the other coil; its SI unit is the henry (H).

← Back to Q129

Q130

(a) How are eddy currents generated in a conductor which is subjected to a magnetic field?
(b) Write two examples of their useful applications.
(c) How can the disadvantages of eddy currents be minimized?

[2019 • Set 55-3-1]

SOLUTION

(a) How eddy currents are generated.

When a bulk conductor is placed in a region where the magnetic flux through it is changing, either because the field itself varies with time or because the conductor moves through a non-uniform field, an emf is induced in the body of the conductor by Faraday's law. This induced emf drives currents that circulate in closed loops within the volume of the conductor. These circulating induced currents are the eddy currents, and by Lenz's law their direction opposes the change in flux producing them.

(b) Two useful applications.

First, electromagnetic braking: eddy currents induced in the moving metal parts of trains and machines produce a retarding force that brings them smoothly to rest without mechanical contact.

Second, induction heating: eddy currents induced in a metal object heat it strongly, as used in induction furnaces for melting metals and in induction cooktops. (Other examples include induction motors, speedometers, and dead-beat galvanometers.)

(c) Minimising the disadvantages of eddy currents.

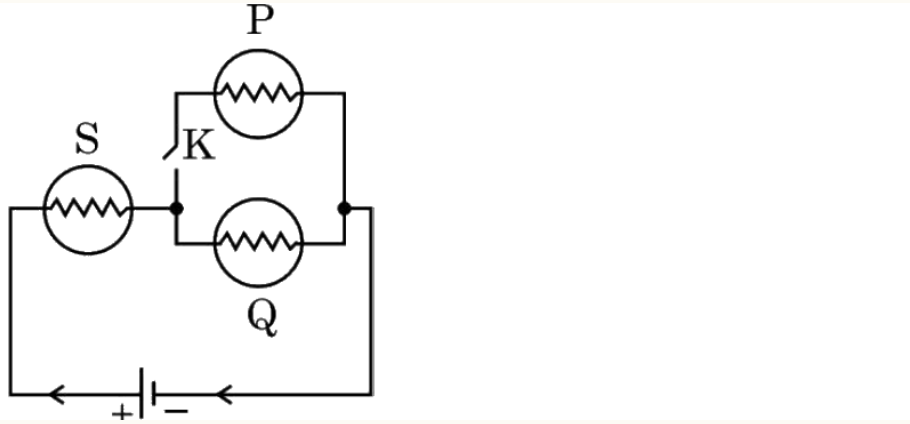
In the cores of transformers, motors and other devices, the metal is used in the form of thin laminations (sheets) coated with an insulating layer and stacked together, instead of a single solid block. The insulation increases the resistance of the eddy-current paths and breaks them into small loops, so the eddy currents and the resulting I^2R heat loss are **greatly reduced**.

Final answer. Eddy currents arise from changing flux in a bulk conductor; used in braking and induction heating; minimised by using laminated cores.

← *Back to Q130*

Q131

The figure shows a rectangular conducting frame MNOP of resistance R placed partly in a perpendicular magnetic field $\vec{B}$ and moved with velocity $\vec{v}$ as shown in the figure. Obtain the expressions for the
(a) force acting on the arm 'ON' and its direction, and
(b) power required to move the frame to get a steady emf induced between the arms MN and PO.



[2019 • Set 55-4-1]

SOLUTION

The frame MNOP has the arm ON (length L) inside the magnetic field $\vec{B}$, which points perpendicular to the plane of the frame. The frame is pulled with velocity $\vec{v}$, so the area of the frame inside the field changes and an emf is induced.

As the arm ON moves with velocity v, the free charges in it experience a motional emf. The magnitude of this emf is:

$$\varepsilon = BvL$$

Since the frame has resistance R, the induced current is:

$$i = \frac{\varepsilon}{R} = \frac{BvL}{R}$$

(a) The current-carrying arm ON lies in the magnetic field $\vec{B}$, so it experiences a magnetic force:

$$F = BiL$$

$$F = B \cdot \frac{BvL}{R} \cdot L$$

$$F = \frac{B^2L^2v}{R}$$

By Lenz's law, this force opposes the cause producing it, namely the motion of the frame. Hence the force on arm ON acts in a direction opposite to the velocity $\vec{v}$, i.e. it tries to drag the frame back into (or out of) the field so as to oppose the change in flux.

(b) To keep the frame moving with a steady velocity v, an external agent must apply an equal and opposite force, doing work at the rate (power):

$$P = F \cdot v$$

$$P = \frac{B^2L^2v}{R} \cdot v$$

$$P = \frac{B^2L^2v^2}{R}$$

This power equals $\frac{\varepsilon^2}{R}$, and it is entirely dissipated as Joule heat in the resistance R of the frame, in agreement with conservation of energy.

Thus the **force on arm ON is $F = \frac{B^2L^2v}{R}$ directed opposite to $\vec{v}$**, and the **power required is $P = \frac{B^2L^2v^2}{R}$**.

Final answer. The force on arm ON is $\boldsymbol{\frac{B^2L^2v}{R}}$ opposite to $\vec{\boldsymbol{v}}$, and the power required is $\boldsymbol{\frac{B^2L^2v^2}{R}}$.

← *Back to Q131*

Q132

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.

[2017]

SOLUTION

Mutual inductance of a pair of coils is defined as the magnetic flux linked with one coil per unit current flowing in the other coil. If a current I_1 in coil 1 links flux Φ_{21} with coil 2, then:

$$\Phi_{21} = M\, I_1 \Rightarrow M = \frac{\Phi_{21}}{I_1}$$

Its SI unit is the **henry (H)**.

Consider two long coaxial solenoids of the same length l, the inner one of radius r_1 with N_1 turns and the outer one of radius r_2 ($\mathbf{r_2 > r_1}$) with N_2 turns, wound one over the other.

Let a current I_2 flow through the outer solenoid. The magnetic field inside it is uniform and given by:

$$B_2 = \mu_0\, n_2\, I_2 = \frac{\mu_0\, N_2\, I_2}{l}$$

The inner solenoid (radius r_1) lies completely within this field, so the flux through one turn of the inner solenoid is:

$$\phi = B_2 \cdot \pi\, r_1^{\,2} = \frac{\mu_0\, N_2\, I_2}{l}\, \pi\, r_1^{\,2}$$

The total flux linkage with all N_1 turns of the inner solenoid is:

$$\Phi_1 = N_1\, \phi = \frac{\mu_0\, N_1\, N_2\, \pi\, r_1^{\,2}\, I_2}{l}$$

By definition, the mutual inductance is:

$$M = \frac{\Phi_1}{I_2}$$

$$M = \frac{\mu_0\, N_1\, N_2\, \pi\, r_1^{\,2}}{l}$$

Thus the **mutual inductance of the two coaxial solenoids is** $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{l}$.

COMMON MISTAKE

Always take the field of the OUTER solenoid and the area of the INNER solenoid (πr_1^2). Using r_2 here is the most common mistake and costs a mark.

Final answer. The mutual inductance of the two coaxial solenoids is $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{l}$.

← Back to Q132

Q133

Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

[2017]

SOLUTION

Self-inductance of a coil is defined as the magnetic flux linked with the coil per unit current flowing through it:

$$L = \frac{\Phi}{I}$$

Equivalently, it is the magnitude of the back-emf induced in the coil per unit rate of change of current, since:

$$\varepsilon = -L\,\frac{dI}{dt}$$

Its SI unit is the henry (H).

When an inductor L is connected to a source of emf, the current grows from 0 to its final value I. While the current is increasing, a back-emf $\varepsilon = L\,\dfrac{di}{dt}$ opposes it. The source must do work against this back-emf.

The work done in time dt, when current is i and charge $dq = i\,dt$ flows, is:

$$dW = \varepsilon\, i\, dt = L\, \frac{di}{dt}\, i\, dt = L\, i\, di$$

The total work done in building the current from 0 to I is obtained by integration:

$$W = {}_0^I L\, i\, di$$

$$W = L[\frac{i^2}{2}]_0^I$$

$$W = \frac{1}{2} LI^2$$

This work is stored as magnetic potential energy in the inductor:

$$U = \frac{1}{2} LI^2$$

Hence the **energy stored in an inductor carrying current I is $U = \frac{1}{2} LI^2$.**

Final answer. The energy stored in the inductor is $\boldsymbol{U = \frac{1}{2} LI^2}$.

← *Back to Q133*

Q134

Define mutual inductance between a pair of coils. Derive an expression for the mutual inductance of two long coaxial solenoids of same length wound one over the other.
OR
Define self-inductance of a coil. Obtain the expression for the energy stored in an inductor L connected across a source of emf.

[2017]

SOLUTION

This is an either/or question; both alternatives are answered fully.

Mutual inductance of a pair of coils is the magnetic flux linked with one coil per unit current in the other coil:

$$M = \frac{\Phi_{21}}{I_1}$$

Its SI unit is the henry (H).

For two long coaxial solenoids of the same length l, inner radius r_1 with N_1 turns and outer radius r_2 with N_2 turns, pass a current I_2 through the outer solenoid. The field inside it is:

$$B_2 = \frac{\mu_0 N_2 I_2}{l}$$

The flux through one turn of the inner solenoid (radius r_1):

$$\phi = B_2 \cdot \pi r_1^2 = \frac{\mu_0 N_2 I_2 \pi r_1^2}{l}$$

Total flux linkage with the inner solenoid's N_1 turns:

$$\Phi_1 = N_1 \phi = \frac{\mu_0 N_1 N_2 \pi r_1^2 I_2}{l}$$

$$M = \frac{\Phi_1}{I_2} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$

OR

Self-inductance of a coil is the magnetic flux linked with it per unit current:

$L = \frac{\Phi}{I}$, with $\varepsilon = -L \frac{dI}{dt}$

When current grows from 0 to I in an inductor L, the source works against the back-emf. Work done in time dt:

$$dW = \varepsilon i \, dt = L i \, di$$

Integrating from 0 to I:

$$W = \int_0^I L i \, di = \frac{1}{2} LI^2$$

This is stored as magnetic field energy:

$$U = \frac{1}{2} LI^2$$

Thus **either** $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$ **or** $U = \frac{1}{2}LI^2$, as required.

Final answer. Mutual inductance $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$; OR energy stored $U = \frac{1}{2}LI^2$.

← *Back to Q134*

Q135

State Lenz's law. Illustrate, by giving an example, how this law helps in predicting the direction of the current in a loop in the presence of a changing magnetic flux.
In a given coil of self-inductance of 5 mH, current changes from 4 A to 1 A in 30 ms. Calculate the emf induced in the coil.

[2015]

SOLUTION

Lenz's law states that the direction of the induced current (or induced emf) is always such that it opposes the change in magnetic flux that produces it. This law is a consequence of the conservation of energy.

Example: When the north pole of a bar magnet is moved towards a coil, the magnetic flux through the coil increases. By Lenz's law, the induced current flows in a direction that makes the near face of the coil a north pole, so that it repels the approaching magnet and opposes the increase of flux. To keep the magnet moving, the external agent must do work against this repulsion, and this work appears as electrical energy in the coil. If instead the magnet is pulled away, the flux decreases and the induced current reverses to make the near face a south pole, attracting the magnet and again opposing the change.

Numerical: For the coil,

$$L = 5\ \text{mH} = 5 \times 10^{-3}\ H$$

$$\Delta I = 4\ A - 1\ A = 3\ A$$

$$\Delta t = 30\ \text{ms} = 0.03\ s$$

The magnitude of the induced emf is:

$$|\varepsilon| = L\left|\frac{\Delta I}{\Delta t}\right|$$

$$|\varepsilon| = (5 \times 10^{-3}) \times \frac{3}{0.03}$$

$$|\varepsilon| = (5 \times 10^{-3}) \times 100$$

$$|\varepsilon| = 0.5\ V$$

Thus the **emf induced in the coil is** 0.5 V.

Final answer. The emf induced in the coil is **0.5 V.**

← *Back to Q135*

Q136

(a) A rod of length l is moved horizontally with a uniform velocity 'v' in a direction perpendicular to its length through a region in which a uniform magnetic field is acting vertically downward. Derive the expression for the emf induced across the ends of the rod.
(b) How does one understand this motional emf by invoking the Lorentz force acting on the free charge carriers of the conductor ? Explain.

[2014]

SOLUTION

(a) Consider a rod of length l moving horizontally with uniform velocity v perpendicular to its own length, through a region where a uniform magnetic field $\vec{B}$ acts vertically downward (perpendicular to both the rod and its velocity).

In a small time dt, the rod sweeps out an area:

$$dA = l\,(v\,dt)$$

The change in magnetic flux linked with the circuit in this time is:

$$d\Phi = B\,dA = B\,l\,v\,dt$$

By Faraday's law, the magnitude of the induced (motional) emf is:

$$\varepsilon = \frac{d\Phi}{dt} = \frac{B\,l\,v\,dt}{dt}$$

$$\varepsilon = Bvl$$

(b) Lorentz-force interpretation: Each free charge q inside the rod moves along with the rod at velocity $\vec{v}$. In the magnetic field $\vec{B}$ it experiences a magnetic (Lorentz) force:

$$\vec{F}_m = q\,(\vec{v} \times \vec{B})$$

This force is directed along the length of the rod, so positive charges accumulate at one end and negative charges at the other. This charge separation sets up an electric field $\vec{E}$ inside the rod. The separation continues until the electric force balances the magnetic force, i.e. **qE = qvB**, giving ***E* = vB**.

The potential difference (motional emf) between the ends of the rod equals the work done by the magnetic force per unit charge:

$$\varepsilon = \int_0^l (\vec{v} \times \vec{B}) \cdot d\vec{\ell} = \int_0^l vB\,d\ell = Bvl$$

Hence the **motional emf ε = Bvl** arises because the magnetic force on the free charges, moving with the rod, drives them along the rod, doing work per unit charge equal to Bvl.

Final answer. The induced emf across the rod is **ε = Bvl**, arising from the Lorentz force on its free charges.

← Back to Q136

Q137

A metallic rod of length 'l' is rotated with a frequency ν with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius r, about an axis passing through the centre and perpendicular to the plane of the ring. A constant uniform magnetic field B parallel to the axis is present every where. Using Lorentz force, explain how emf is induced between the centre and the metallic ring and hence obtain the expression for it.

[2013]

SOLUTION

The rod of length l is hinged at the centre of the ring and rotated with frequency ν, so its angular speed is $\omega = 2\pi\nu$. The uniform magnetic field $\vec{B}$ is parallel to the axis of rotation, hence perpendicular to the plane in which the rod sweeps.

Consider a small element of the rod of length $d\xi$ at a distance ξ from the hinge (centre). As the rod rotates, this element moves with a tangential speed:

$$v = \xi\,\omega$$

A free positive charge q in this element experiences a Lorentz force:

$$\vec{F} = q\,(\vec{v} \times \vec{B})$$

Since $\vec{v}$ is tangential and $\vec{B}$ is axial, the force $\vec{v} \times \vec{B}$ is directed radially along the rod, pushing the charges along its length. The force per unit charge (the motional electric field) on the element is:

$$\frac{F}{q} = vB = \xi\,\omega\,B$$

The emf contributed by the element is:

$$d\varepsilon = vB\,d\xi = B\,\omega\,\xi\,d\xi$$

The total emf induced between the centre and the rim is obtained by integrating over the whole rod from $\xi = 0$ to $\xi = l$:

$$\varepsilon = \int_0^l B\,\omega\,\xi\,d\xi$$

$$\varepsilon = B\,\omega\left[\frac{\xi^2}{2}\right]_0^l$$

$$\varepsilon = \frac{1}{2}B\,\omega\,l^2$$

Writing $\omega = 2\pi\nu$:

$$\varepsilon = \frac{1}{2}B(2\pi\nu)l^2 = \pi\nu\,B\,l^2$$

Thus the **emf induced between the centre and the ring is** $\varepsilon = \frac{1}{2}B\omega\,l^2 = \pi\nu\,B\,l^2$.

Final answer. The induced emf is $\varepsilon = \frac{1}{2} B\omega l^2 = \pi \nu B l^2$.

← Back to Q137

Q138

(a) Define self inductance. Write its S.I. units.
(b) Derive an expression for self inductance of a long solenoid of length l, cross-sectional area A having N number of turns.

[2009]

SOLUTION

(a) **Self-inductance** of a coil is defined as the magnetic flux linked with the coil per unit current flowing through it:

$$L = \frac{\Phi}{I}$$

Equivalently, it is the back-emf induced per unit rate of change of current, since $\varepsilon = -L\,\frac{dI}{dt}$. Its SI unit is the **henry (H)**.

(b) Consider a long solenoid of length l, cross-sectional area A and total number of turns N, so the number of turns per unit length is $n = \frac{N}{l}$. Let a current I flow through it.

The magnetic field inside a long solenoid is uniform:

$$B = \mu_0 n I = \frac{\mu_0 N I}{l}$$

The magnetic flux through one turn of the solenoid is:

$$\phi = B A = \frac{\mu_0 N I}{l} A$$

The total flux linkage with all N turns is:

$$\Phi = N\phi = \frac{\mu_0 N^2 A I}{l}$$

By the definition of self-inductance:

$$L = \frac{\Phi}{I}$$

$$L = \frac{\mu_0 N^2 A}{l}$$

In terms of $n = N/l$, this can also be written as $L = \mu_0 n^2 A l$.

Thus the **self-inductance of the long solenoid is** $L = \frac{\mu_0 N^2 A}{l}$.

Final answer. The self-inductance of the long solenoid is $L = \frac{\mu_0 N^2 A}{l} = \mu_0 n^2 A l$.

← *Back to Q138*

Q139

(i) State Faraday's law of electromagnetic induction.
(ii) A jet plane is travelling towards west at a speed of 1800 km/h. *What is the voltage difference developed between the ends of the wing having a span of* 25 m, *if the Earth's magnetic field at the location has a magnitude of* 5×10^{-4} *T and the dip angle is* 30°*?*

[2009]

SOLUTION

(i) **Faraday's law of electromagnetic induction** states that the emf induced in a closed circuit is equal to the negative rate of change of magnetic flux linked with it:

$$\varepsilon = -\frac{d\Phi}{dt}$$

For a coil of N turns this becomes:

$$\varepsilon = -N\frac{d\phi}{dt}$$

The negative sign represents Lenz's law, indicating that the induced emf opposes the change in flux.

(ii) The wing of the jet acts like a rod moving through the Earth's magnetic field. The emf is produced only by the component of $\vec{B}$ perpendicular to the horizontal plane of motion, which is the vertical component B_V.

Given:

$$v = 1800 \text{ km/h} = 1800 \times \frac{5}{18} \ m/s = 500 \ m/s$$

$$L = 25 \ m, \ B = 5 \times 10^{-4} \ T, \ \delta = 30^\circ$$

The vertical component of the Earth's field is:

$$B_V = B\delta = (5 \times 10^{-4}) \times \sin 30^\circ$$

$$B_V = (5 \times 10^{-4}) \times 0.5 = 2.5 \times 10^{-4} \ T$$

The voltage developed across the wings is:

$$\varepsilon = B_V \, v \, L$$

$$\varepsilon = (2.5 \times 10^{-4}) \times 500 \times 25$$

$$\varepsilon = 3.125 \ V \approx 3.1 \ V$$

Thus the **voltage difference developed between the ends of the wing is about** 3.1 V.

> **COMMON MISTAKE**
>
> *Use the VERTICAL component Bδ, not B itself. The plane flies horizontally, so only the vertical part of the field is perpendicular to its motion.*

Final answer. The voltage developed across the wings is about **3.1 V**.

← *Back to Q139*

Q140

A metallic rod of length l is rotated at a constant angular speed ω, normal to a uniform magnetic field B. Derive an expression for the current induced in the rod, if

the resistance of the rod is R.

[2008]

SOLUTION

A metallic rod of length l is rotated with constant angular speed ω about one end, in a plane normal to a uniform magnetic field $\vec{B}$. The field is perpendicular to the plane of rotation.

Consider a small element of the rod of length dx at a distance x from the fixed end. Its tangential speed is $v = x\omega$. A free charge in this element experiences a motional electric field of magnitude $vB = x\omega B$ along the rod, so the emf contributed by the element is:

$$d\varepsilon = B\, v\, dx = B\omega\, x\, dx$$

Integrating over the full length of the rod from $x = 0$ to $x = l$:

$$\varepsilon = {}_0^{l} B\omega\, x\, dx$$

$$\varepsilon = B\omega[\frac{x^2}{2}]_0^{l}$$

$$\varepsilon = \frac{1}{2} B\omega\, l^2$$

This emf drives a current through the rod of resistance R. By Ohm's law, the induced current is:

$$I = \frac{\varepsilon}{R}$$

$$I = \frac{B\omega\, l^2}{2R}$$

Thus the **induced current in the rotating rod is** $I = \dfrac{B\omega\, l^2}{2R}$.

Final answer. The induced current in the rod is $I = \dfrac{B\omega\, l^2}{2R}$.

← *Back to Q140*

Q141

Explain with the help of a labelled diagram the underlying principle and working of a step-up transformer. Why cannot such a device be used to step up d.c. voltage?

[2007]

SOLUTION

Labelled diagram: Draw two coils wound on a common closed laminated soft-iron core (a rectangular ring). On the left limb is the primary coil with N_p turns, connected to an a.c. source. On the right limb is the secondary coil with N_s turns ($\mathbf{N_s > N_p}$), connected to the load. Mark the alternating supply on the primary side and the output terminals V_s on the secondary side.

Principle: A step-up transformer works on the principle of mutual induction. The alternating current in the primary sets up a continuously changing magnetic flux in the soft-iron core, and this same flux links the secondary, inducing an emf in it.

Working: The core confines almost all the flux, so every turn of both coils is linked by the same flux Φ and experiences the same rate of change of flux $\frac{d\Phi}{dt}$.

The emf induced in the primary and secondary are:

$$\varepsilon_p = -N_p \frac{d\Phi}{dt}, \varepsilon_s = -N_s \frac{d\Phi}{dt}$$

Dividing the two:

$$\frac{\varepsilon_s}{\varepsilon_p} = \frac{N_s}{N_p}$$

For an ideal transformer the terminal voltages equal the emfs, so $\frac{\mathbf{V_s}}{\mathbf{V_p}} = \frac{\mathbf{N_s}}{\mathbf{N_p}}$.

Since $\mathbf{N_s > N_p}$, we get $\mathbf{V_s > V_p}$, hence the voltage is stepped up.

For an ideal (loss-free) transformer, input power equals output power, $\mathbf{V_p I_p = V_s I_s}$, so the larger output voltage comes with a correspondingly smaller output current.

Why d.c. cannot be stepped up: A transformer needs a changing flux to induce an emf in the secondary (Faraday's law). A steady d.c. current gives a constant

flux, so $\frac{d\Phi}{dt} = 0$ and no emf is induced in the secondary. Therefore **a transformer cannot step up a steady d.c. voltage; it works only with alternating current.**

Final answer. A step-up transformer works on mutual induction with $\frac{V_s}{V_p} = \frac{N_s}{N_p} > 1$; it fails for d.c. because steady current gives no flux change and hence no induced emf.

← *Back to Q141*

Q142

Draw a labelled diagram of an a.c. generator. Explain briefly its principle and working.

[2007]

SOLUTION

Labelled diagram: Draw a rectangular coil (armature) of N turns placed between the curved N and S pole pieces of a magnet that supply a uniform field $\vec{B}$. The two ends of the coil are connected to two separate slip rings R_1 and R_2 that rotate with the coil. Two carbon brushes B_1 and B_2 press against the slip rings and lead to the external load. Label the coil, magnet poles, slip rings, brushes and load.

Principle: An a.c. generator works on the principle of electromagnetic induction. When the magnetic flux linked with a rotating coil changes, an induced emf is set up in the coil (Faraday's law).

Working: The coil is rotated with a constant angular speed ω about an axis perpendicular to $\vec{B}$. At time t, the area vector $\vec{A}$ of the coil makes an angle ωt with $\vec{B}$, so the flux linked with the coil is:

$$\Phi = NBA\omega t$$

By Faraday's law, the induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -\mathrm{NBA}\frac{d}{\mathrm{dt}}(\omega\, t)$$

$$\varepsilon = \mathrm{NBA}\ \omega\omega\ t$$

The emf reverses its polarity every half rotation, so the current delivered to the load alternates in direction. The slip rings and brushes carry this **alternating emf** to the external circuit.

$\varepsilon = \varepsilon_0 \omega\, t$, where the peak emf is $\varepsilon_0 = \mathrm{NBA}\omega$.

Final answer. The a.c. generator works on electromagnetic induction and produces $\varepsilon = \mathrm{NBA}\omega\omega\, t$.

← *Back to Q142*

Q143

How is the mutual inductance of a pair of coils affected when:
(i) separation between the coils is increased?
(ii) the number of turns of each coil is increased?
(iii) a thin iron sheet is placed between the two coils, other factors remaining the same?
Explain your answer in each case.

[2006]

SOLUTION

The mutual inductance M of a pair of coils depends on their geometry, the number of turns, and the medium between them, through $M \propto N_1 N_2$ and the flux linkage.

(i) Separation increased: As the coils are moved apart, the magnetic field of one coil at the location of the other gets weaker, so less flux of the first coil links the second. The flux linkage falls.

Hence **the mutual inductance decreases.**

(ii) Number of turns of each coil increased: Mutual inductance is given by $M \propto N_1 N_2$, where N_1 and N_2 are the numbers of turns of the two coils. Increasing

the turns of either coil increases both the flux produced and the linkage of that flux.

Hence **the mutual inductance increases.**

(iii) Thin iron sheet placed between the coils: Iron has a very high relative permeability ($\mu_r \gg 1$). When it is placed in the path of the flux between the coils, it strongly concentrates and channels the magnetic flux from one coil into the other, greatly increasing the flux linkage.

Hence **the mutual inductance increases.**

Final answer. (i) ***M*** decreases, (ii) ***M*** increases, (iii) ***M*** increases.

← *Back to Q143*

Q144

Explain, with the help of a diagram, the principle and working of an a.c. generator. Write the expression for the e.m.f. generated in the coil in terms of its speed of rotation.

[2005]

SOLUTION

Principle: An a.c. generator works on the principle of electromagnetic induction. A coil rotating in a uniform magnetic field experiences a continuously changing magnetic flux, which induces an alternating emf in the coil (Faraday's law).

Diagram and construction: Draw a rectangular coil of N turns and area A mounted on an axle and rotating between the N and S poles of a strong magnet that supplies the field $\vec{B}$. The ends of the coil are joined to two slip rings, which press against two carbon brushes that connect to the external circuit. Label the coil, magnet, slip rings and brushes.

Working: The coil is rotated with angular velocity ω. At time t, the normal to the coil makes an angle $\theta = \omega t$ with $\vec{B}$, so the flux through the coil is:

$$\Phi = NBA\omega\, t$$

By Faraday's law, the induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -NBA\frac{d}{dt}(\omega\, t)$$

$$\varepsilon = NBA\ \omega\omega\ t$$

Thus the emf varies sinusoidally with time. Its peak value is $\boldsymbol{\varepsilon_0 = NBA\omega}$, which depends directly on the speed of rotation ω.

$$\varepsilon = NBA\omega\omega\ t,\ \varepsilon_0 = NBA\omega$$

COMMON MISTAKE

Marks are often lost for not writing the area vector angle as ω t; always state $\boldsymbol{\Phi = NBA\omega\ t}$ *before differentiating.*

Final answer. The emf induced in the coil is $\boldsymbol{\varepsilon = NBA\omega\omega\ t}$ with peak value $\boldsymbol{\varepsilon_0 = NBA\omega}$.

← *Back to Q144*

Q145

A circular coil of N turns and radius R, is kept normal to a magnetic field, given by $\boldsymbol{B = B_0 \cos \omega\, t}$. *Deduce an expression for e.m.f. induced in this coil. State the rule which helps to detect the direction of induced current.*

[2004]

SOLUTION

The coil has N turns and radius R, so its area is $\boldsymbol{A = \pi R^2}$. It is kept normal to the field, so the flux is maximum and the field at time t is $\boldsymbol{B = B_0\omega\, t}$.

The magnetic flux linked with the coil is:

$$\Phi = NBA = N(B_0\omega\ t)(\pi\ R^2)$$

$$\Phi = N\pi\ R^2\ B_0\omega\ t$$

By Faraday's law of electromagnetic induction, the induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -N\pi\, R^2\, B_0 \frac{d}{dt}(\omega\, t)$$

$$\varepsilon = N\pi\, R^2\, B_0\, \omega\omega\, t$$

The induced emf varies sinusoidally with time, with peak value $\boldsymbol{\varepsilon_0 = N\pi\, R^2\, B_0\omega}$.

Rule for direction of induced current: The direction is given by **Lenz's law**, which states that the induced current always flows in such a direction as to oppose the change in magnetic flux that produces it.

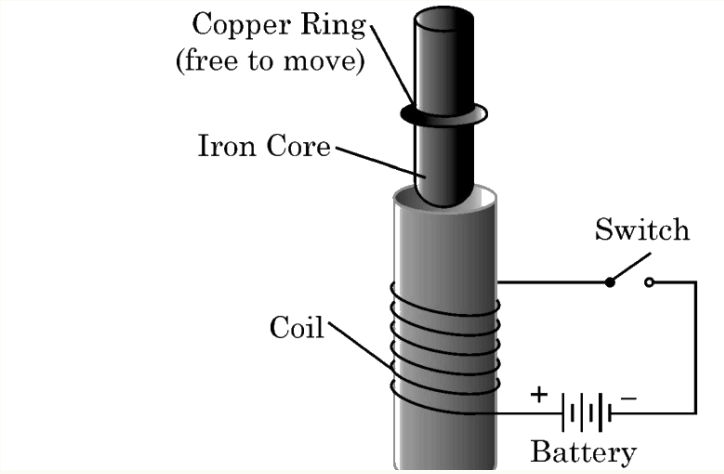
$$\varepsilon = N\pi\, R^2\, B_0\, \omega\omega\, t$$

Final answer. The induced emf is $\boldsymbol{\varepsilon = N\pi\, R^2\, B_0\omega\omega\, t}$; the direction of induced current is found from Lenz's law.

← *Back to Q145*

Q146

Read the following paragraph and answer the questions.
Consider the experimental set up shown in the figure. This jumping ring experiment is an outstanding demonstration of some simple laws of Physics. A conducting non-magnetic ring is placed over the vertical core of a solenoid. When current is passed through the solenoid, the ring is thrown off.



Answer the following questions:
(i) Explain the reason of jumping of the ring when the switch is closed in the circuit.
(ii) What will happen if the terminals of the battery are reversed and the switch is closed? Explain.
(iii) Explain the two laws that help us understand this phenomenon.

[2023 • Set 55-2-1]

SOLUTION

(i) Why the ring jumps when the switch is closed: When the switch is closed, the current in the solenoid rises rapidly from zero to its steady value. This produces a rapidly increasing magnetic flux through the conducting ring sitting above the core. By Faraday's law, this changing flux induces an emf and hence an induced current in the ring.

By Lenz's law, the induced current flows so as to oppose the increase in flux, so the face of the ring nearer the solenoid becomes the same magnetic pole as the upper face of the solenoid. Two like poles repel each other, so **the ring is thrown upward.**

(ii) If the battery terminals are reversed: Reversing the battery reverses the direction of the solenoid current and hence of its magnetic field. However, when the switch is closed, the current still rises from zero to its steady value, so the flux through the ring is still increasing.

By Lenz's law the induced current still opposes this increase, so the ring's near face again develops a like pole facing the solenoid. The two like poles repel, so **the ring still jumps off, exactly as before.**

(iii) The two laws involved:

Faraday's law of electromagnetic induction: A changing magnetic flux induces an emf, $\varepsilon = -\frac{d\Phi}{dt}$. The growing solenoid field gives a changing flux through the ring, inducing a current in it.

Lenz's law: The direction of the induced current is always such that it opposes the change in flux producing it. This opposition makes the ring's face a like pole, and the resulting repulsion throws the ring off.

COMMON MISTAKE

A common error is to say the ring drops or behaves differently when polarity is reversed; it still jumps, because Lenz's law opposes the rising current either way.

Final answer. The ring jumps because the induced current (Faraday's law) opposes the rising flux (Lenz's law), making like poles that repel; reversing the battery gives the same jump.

← *Back to Q146*

Q147

Ram is a student of class X in a village school. His uncle gifted him a bicycle with a dynamo fitted in it. He was very excited to get it. While cycling during night, he could light the bulb and see the objects on the road. He, however, did not know how this device works. He asked this question to his teacher. The teacher considered it an opportunity to explain the working to the whole class.
Answer the following questions:
(a) State the principle and working of a dynamo.
(b) Write two values each displayed by Ram and his school teacher.

[2016]

SOLUTION

(a) Principle and working of a dynamo:

A dynamo is a small generator that converts mechanical energy into electrical energy.

Principle: It works on the principle of electromagnetic induction. When a coil rotates in a magnetic field (or a magnet rotates near a coil), the magnetic flux linked with the coil changes continuously and an emf is induced in it, in accordance with Faraday's law.

Working: As the bicycle wheel turns, it rotates a small permanent magnet placed near a fixed coil inside the dynamo. The rotation produces a continuously changing magnetic flux through the coil:

$$\Phi = NBA\omega\, t$$

By Faraday's law, the induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt} = NBA\ \omega\omega\ t$$

This induced emf drives a current through the bulb, making it glow. The faster the wheel turns, the larger the value of ω and hence the brighter the bulb.

(b) Two values displayed by each:

Ram: (1) Curiosity and eagerness to learn how things work. (2) Willingness to ask questions instead of staying confused.

Teacher: (1) Encouraging students' curiosity by turning a question into a class lesson. (2) Sharing knowledge and promoting scientific awareness for the benefit of all.

Final answer. A dynamo works on electromagnetic induction, giving $\varepsilon = NBA\omega\omega\, t$; Ram showed curiosity and eagerness to learn, the teacher showed responsiveness and the spirit of sharing knowledge.

← *Back to Q147*

Q148

A group of students while coming from the school noticed a box marked "Danger H.T. 2200 V" at a substation in the main street. They did not understand the utility of such a high voltage, while they argued, the supply was only 220 V. They asked their teacher this question the next day. The teacher thought it to be an important question and therefore explained to the whole class.
Answer the following questions:
(i) What device is used to bring the high voltage down to low voltage of a.c. current and what is the principle of its working?
(ii) Is it possible to use this device for bringing down the high dc voltage to the low voltage? Explain.
(iii) Write the values displayed by the students and the teacher.

[2015]

SOLUTION

(i) Device and its principle: The device used to bring high a.c. voltage down to a low voltage is a **step-down transformer.**

Principle: It works on the principle of mutual induction. An alternating current in the primary coil sets up a continuously changing magnetic flux in the common soft-iron core, which links the secondary coil and induces an emf in it.

For the two coils linked by the same flux:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

In a step-down transformer $N_s < N_p$, so $V_s < V_p$ and the voltage is reduced.

(ii) Can it be used for d.c.? No, a transformer cannot be used to bring down a high d.c. voltage. A transformer works only with alternating current, because mutual induction requires a continuously changing flux. A steady d.c. current produces a constant flux, so $\frac{d\Phi}{dt} = 0$ and no emf is induced in the secondary. (A d.c. supply would induce only a momentary pulse at the instant of switching on or off.)

(iii) Values displayed:

Students: scientific curiosity, willingness to question and reason, and awareness of public safety.

Teacher: responsiveness to students' doubts, recognising a teachable moment, and encouraging scientific thinking in the whole class.

Final answer. A step-down transformer (working on mutual induction) lowers the a.c. voltage; it cannot be used for d.c. as steady current gives no flux change; the students showed curiosity and the teacher showed a willingness to nurture it.

← *Back to Q148*

Q149

(a) *State Faraday's law of electromagnetic induction.*
(b) *Derive an expression for the self-inductance of an air-filled long solenoid of length l and cross-sectional area A having N turns.*
(c) *A conducting rod of length 50 cm, with one end pivoted, is rotated with angular speed of 60 rpm in a uniform magnetic field of 4.0 mT directed perpendicular to the plane of rotation of rod. Find the emf induced in the rod.*

OR

(a) *Draw a labelled diagram of a step-up transformer. State the principle on which it works and obtain the ratio of secondary voltage to primary voltage in terms of number of turns and currents in the two coils.*
(b) *The ratio of the number of turns in the primary to the secondary of an ideal transformer is* 1:5. *If 5 kW power at 200 V is supplied to the primary, find (i) current in the primary, and (ii) output voltage.*

[2026 • Set 55-1-1]

SOLUTION

(a) Faraday's law of electromagnetic induction: The magnitude of the induced emf in a circuit is equal to the rate of change of magnetic flux linked with the circuit, $\boldsymbol{\varepsilon = -\frac{d\Phi}{dt}}$. The negative sign (Lenz's law) shows that the induced emf opposes the change in flux.

(b) Self-inductance of a long air-filled solenoid: For a long solenoid of length l, area A and N turns, the number of turns per unit length is $\boldsymbol{n = \frac{N}{l}}$.

The magnetic field inside the solenoid carrying current I is:

$$B = \mu_0 \, n \, I = \mu_0 \frac{N}{l} \mathrm{I}$$

The flux through one turn is $\boldsymbol{\Phi_1 = BA = \mu_0 \frac{N}{l} IA}$.

The total flux linkage through all N turns is:

$$\mathrm{N}\Phi_1 = \mu_0 \frac{\mathrm{N}^2}{l} \mathrm{IA}$$

By definition $\boldsymbol{N\Phi_1 = LI}$, so:

$$LI = \mu_0 \frac{N^2 A}{l} I$$

$$L = \frac{\mu_0 N^2 A}{l}$$

(c) EMF in the rotating rod: Given $l = 0.50\ m$, $B = 4.0 \times 10^{-3}\ T$, and angular speed = **60 rpm**.

$$\omega = \frac{2\pi \times 60}{60} = 2\pi \text{ rad/}s$$

The emf induced between the centre and the end of a rod rotating about one end is:

$$\varepsilon = \frac{1}{2} B\omega l^2$$

$$\varepsilon = \frac{1}{2} \times (4.0 \times 10^{-3}) \times (2\pi) \times (0.50)^2$$

$$\varepsilon = \frac{1}{2} \times (4.0 \times 10^{-3}) \times (2\pi) \times 0.25$$

$$\varepsilon = \pi \times 10^{-3}\ V \approx 3.14 \times 10^{-3}\ V$$

So $\varepsilon \approx 3.14$ **mV**.

OR

(a) Step-up transformer: Draw two coils on a common closed laminated soft-iron core: a primary of N_p turns fed by a.c. and a secondary of N_s turns ($N_s > N_p$) feeding the load.

Principle: It works on mutual induction; the changing flux due to the a.c. primary current links the secondary and induces an emf in it.

Since the same flux links every turn, $\frac{V_s}{V_p} = \frac{N_s}{N_p}$, and for an ideal transformer $V_p I_p = V_s I_s$, giving:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(b) Given $\frac{N_p}{N_s} = \frac{1}{5}$, input power $P = 5$ **kW** $= 5000\ W$ at $V_p = 200\ V$.

(i) Primary current:

$$I_p = \frac{P}{V_p} = \frac{5000}{200} = 25\ A$$

(ii) Output voltage:

$$V_s = V_p \times \frac{N_s}{N_p} = 200 \times 5 = 1000\ V$$

So $\mathbf{I_p = 25\ A}$ and $\mathbf{V_s = 1000\ V}$.

COMMON MISTAKE

For a rod rotating about one end the emf is $\frac{1}{2} B\omega l^2$, *not* $B\omega l^2$; *the factor* $\frac{1}{2}$ *comes from the average speed of the rod and is a frequent slip.*

Final answer. $L = \dfrac{\mu_0 N^2 A}{l}$ and the rod's emf $\approx$ **3.14 mV**; (OR) $\mathbf{I_p = 25\ A}$ and output $\mathbf{V_s = 1000\ V}$.

← *Back to Q149*

Q150

(a) (i) With the help of a labelled diagram, explain the principle, construction and working of an a.c. generator.
(ii) Deduce an expression for the induced emf in the coil of the generator.

[2026 • Set 55-2-1]

SOLUTION

(i) Principle, construction and working of an a.c. generator:

Principle: An a.c. generator works on the principle of electromagnetic induction. A coil rotating in a uniform magnetic field has a continuously changing flux linked with it, which induces an alternating emf (Faraday's law).

Construction (labelled diagram): Draw a rectangular coil (armature) of N turns, area A, wound on a soft-iron core and mounted on an axle. It rotates between the N and S pole pieces of a strong magnet that supplies the uniform field $\vec{B}$. The two ends of the coil are connected to two slip rings R_1 and R_2, which rotate

with the coil and press against two carbon brushes B_1 and B_2 that lead to the external circuit. Label coil, magnet, slip rings, brushes and load.

Working: The coil is rotated with a constant angular speed ω about an axis perpendicular to $\vec{B}$. As it rotates, the angle between the area vector $\vec{A}$ and $\vec{B}$ changes continuously, so the flux through the coil changes and an alternating emf is induced. The slip rings and brushes carry this alternating emf to the external load, where it drives an alternating current.

(ii) Expression for the induced emf:

At time t, the area vector makes an angle ωt with $\vec{B}$, so the flux linked with the coil is:

$$\Phi = NBA\omega\, t$$

By Faraday's law:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -NBA\frac{d}{dt}(\omega\, t)$$

$$\varepsilon = NBA\ \omega\omega\, t$$

Writing $\varepsilon_0 = \mathbf{NBA\omega}$ as the peak emf:

$$\varepsilon = \varepsilon_0 \omega\, t$$

The output is therefore an alternating emf that completes one full sine cycle in each rotation of the coil.

Final answer. The a.c. generator works on electromagnetic induction and the induced emf is $\boldsymbol{\varepsilon = NBA\omega\omega\, t = \varepsilon_0 \omega\, t}$ with $\boldsymbol{\varepsilon_0 = NBA\omega}$.

← *Back to Q150*

Q151

(b) If T is the time period of the rotation of the coil, at what values of T in a cycle, the

emf generated is maximum?

[2026 • Set 55-2-1]

SOLUTION

When a coil rotates with uniform angular speed ω in a uniform magnetic field, the instantaneous induced emf is sinusoidal and is given by:

$$\varepsilon(t) = \varepsilon_0 \omega\, t$$

where $\boldsymbol{\varepsilon_0 = NBA\omega}$ is the peak emf and $\boldsymbol{\omega = \frac{2\pi}{T}}$ in terms of the time period T of rotation.

Writing the emf in terms of T:

$$\varepsilon(t) = \varepsilon_0 \sin\left(\frac{2\pi\, t}{T}\right)$$

The magnitude of the emf is maximum (equal to ε_0) when the sine term reaches its extreme value ± 1:

$$\sin\left(\frac{2\pi\, t}{T}\right) = \pm 1$$

This happens when the angle equals an odd multiple of $\frac{\pi}{2}$:

$$\frac{2\pi\, t}{T} = (2n+1)\frac{\pi}{2},\ n = 0, 1, 2, \ldots$$

Solving for t:

$$t = (2n+1)\frac{T}{4}$$

$$t = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \ldots$$

Within one complete cycle (from $\boldsymbol{t = 0}$ to $\boldsymbol{t = T}$), only two of these values lie in the range. Hence the induced emf is maximum at $\boldsymbol{t = \frac{T}{4}}$ and $\boldsymbol{t = \frac{3T}{4}}$.

Physically, these instants correspond to the moment when the plane of the coil is parallel to $\vec{B}$ (the coil's normal is perpendicular to the field), so the rate of

change of flux is greatest. Therefore, in one cycle the emf generated is **maximum at $t = \frac{T}{4}$ and $t = \frac{3T}{4}$**, i.e. at odd multiples of $\frac{T}{4}$.

Final answer. The emf is maximum at $t = T/4$ and $t = 3T/4$ in one cycle (odd multiples of $T/4$).

← *Back to Q151*

Q152

(i) State Lenz's law and explain that it follows the law of conservation of energy.

[2026 • Set 55-3-1]

SOLUTION

Statement of Lenz's law: The direction of the induced current (or induced emf) is always such that it opposes the change in magnetic flux that produces it.

Mathematically, the negative sign in Faraday's law expresses Lenz's law:

$$\varepsilon = -\frac{d\Phi}{dt}$$

The minus sign shows that the induced emf acts to oppose the change in flux Φ linked with the circuit.

Lenz's law as a consequence of conservation of energy: Consider a bar magnet with its north pole being pushed towards a closed coil. As the magnet approaches, the magnetic flux linked with the coil increases.

By Lenz's law, the induced current flows in a direction that makes the near face of the coil behave as a north pole, so that it repels the approaching magnet.

Because of this opposing (repulsive) force, the external agent must do work to push the magnet towards the coil against the repulsion.

This mechanical work done by the agent is converted into electrical energy in the coil, which finally appears as Joule heat (I^2R) in its resistance. Thus the energy balance is exact:

Work done by agent = Electrical (heat) energy produced in the coil

Now suppose, against Lenz's law, the induced current supported the change, i.e. the near face became a south pole and attracted the magnet. Then the magnet would accelerate on its own, the agent would do no work, yet electrical energy would still be generated in the coil and dissipated as heat.

This would mean electrical energy is created out of nothing, which violates the law of conservation of energy. Such a situation is impossible.

Hence the induced current must oppose the change in flux, exactly as stated by Lenz's law. Therefore **Lenz's law is a direct consequence of the principle of conservation of energy.**

Final answer. Lenz's law states the induced current opposes the change in flux; the work done against this opposition equals the electrical energy generated, so the law upholds conservation of energy.

← *Back to Q152*

Q153

(ii) Write the dimensional formula for self-inductance.
The current in a coil changes from 8·0 *A to* 2·0 *A in* 0·6 *s. If an average emf induced in the coil is* 50 *V, calculate the self-inductance of the coil.*

[2026 • Set 55-3-1]

SOLUTION

Dimensional formula of self-inductance: The induced emf in a coil is related to self-inductance L by:

$$\varepsilon = -L\frac{dI}{dt} \Rightarrow L = \frac{\varepsilon\, dt}{dI}$$

The dimensional formula of emf (potential difference) is $[\varepsilon] = [ML^2T^{-3}A^{-1}]$, time is $[T]$, and current is $[A]$. Substituting:

$$[L] = \frac{[ML^2T^{-3}A^{-1}][T]}{[A]}$$

$$[L] = \mathrm{ML^2T^{-2}A^{-2}}$$

So the dimensional formula of self-inductance is $[\mathrm{ML^2T^{-2}A^{-2}}]$ (its SI unit is the henry, H).

Numerical: The average induced emf is related to the rate of change of current by:

$$\varepsilon = L\,\frac{\Delta I}{\Delta t}$$

Given: initial current = **8.0 *A***, final current = **2.0 *A***, so

$$\Delta I = 8.0 - 2.0 = 6.0\ A$$

$$\Delta t = 0.6\ s,\ \varepsilon = 50\ V$$

Rearranging for L:

$$L = \frac{\varepsilon\,\Delta t}{\Delta I}$$

$$L = \frac{50 \times 0.6}{6.0}$$

$$L = \frac{30}{6.0}$$

$$L = 5\ H$$

Therefore the self-inductance of the coil is ***L* = 5 *H***.

COMMON MISTAKE

Use the magnitude of Δ I here; the average emf already accounts for the direction, so do not carry the minus sign into the arithmetic.

Final answer. Dimensional formula **$[L]=\mathrm{ML^2T^{-2}A^{-2}}$**; the self-inductance of the coil is **5 *H***.

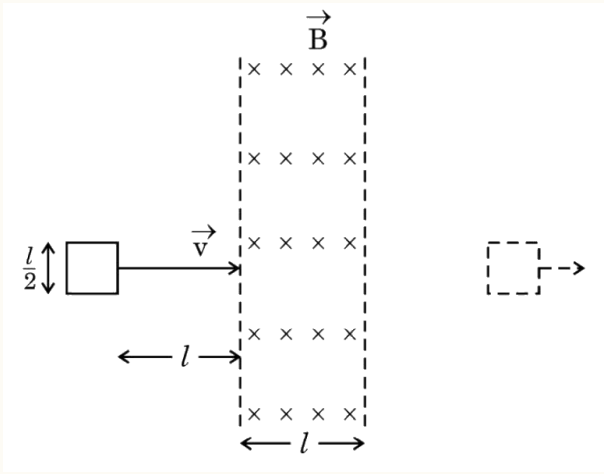
← *Back to Q153*

Q154

(a)

(i) State Lenz's law and explain how this law is a consequence of conservation of energy principle.

(ii) A square shaped loop of side is initially lying outside a region of uniform magnetic field $\vec{B}$ as shown in the figure. The loop is moved towards right with a constant velocity v till it goes out of the region of magnetic field.



(I) What will be the directions of induced current when the loop enters the field and when it leaves the field ?

(II) Draw the plots showing the variation of magnetic flux ϕ linked with the loop with time t and variation of induced emf E with time t. Mark the relevant values of E, ϕ and t on the graphs.

OR

(b)

(i) Differentiate between peak and rms values of alternating current. How are they related ?

(ii) A current element X is connected across an ac source of emf $V = V_0 \sin 2\pi \nu t$. *It is found that the voltage leads the current in phase by* $\frac{\pi}{2}$ *radian. If element X was replaced by element Y, the voltage lags behind the current in phase by* $\frac{\pi}{2}$ *radian.*

(I) Identify elements X and Y by drawing phasor diagrams.

(II) Obtain the condition of resonance when both elements X and Y are connected in series to the source and obtain expression for resonant frequency. What is the impedance value in this case ?

[2025 • Set 55-4-1]

SOLUTION

(a)(i) Lenz's law and conservation of energy: Lenz's law states that the direction of the induced current is always such that it opposes the change in magnetic flux producing it. This is shown by the negative sign in Faraday's law, $\varepsilon = -\frac{d\Phi}{dt}$.

When a magnet is pushed towards a coil, the flux increases and the induced current opposes this by making the near face a like pole, repelling the magnet. The external agent must do work against this repulsion, and this work converts into electrical energy (Joule heat) in the coil.

If instead the induced current aided the motion, the magnet would be pulled in on its own and electrical energy would appear without any work being done, creating energy from nothing. This violates conservation of energy. Hence Lenz's law is a consequence of the conservation of energy principle.

(a)(ii)(I) Directions of induced current: Take $\vec{B}$ as directed into the plane of the page in the field region. As the square loop enters from the left, the inward flux through it increases. By Lenz's law the induced current opposes this increase by setting up flux out of the page inside the loop, so the induced current flows in the **anticlockwise** direction while entering.

While the loop is fully inside the field, the flux is constant, so there is no induced current.

As the loop leaves the field on the right, the inward flux decreases. The induced current now opposes this decrease by setting up flux into the page, so it flows in the **clockwise** direction while leaving.

(a)(ii)(II) Graphs: Let the loop start at $t = 0$ and move with constant speed v. The time to fully enter is $t_1 = \overline{v}$.

Flux versus time (Φ vs t): From $t = 0$ to $t = \overline{v}$, the area inside the field grows uniformly, so Φ rises linearly from 0 to the maximum value B^2. While the loop is fully inside, Φ stays constant at B^2. As the loop exits, Φ falls linearly back to 0. The plot is a trapezoid: a straight line up, a flat top at $\Phi = B^2$, then a straight line down to zero.

EMF versus time (E vs t): Since $E = -\frac{d\Phi}{dt}$, the emf equals the (negative) slope of the flux graph. During entry, $|E| = Bv$ (constant positive rectangular pulse).

While fully inside, $E = 0$. During exit, $E = -Bv$ (constant rectangular pulse of equal magnitude but opposite sign). Mark $+Bv$ for the entry pulse and $-Bv$ for the exit pulse, with time markers at $\overline{v}$ (entry ends) and at the start and end of exit.

Therefore the induced current is **anticlockwise while entering and clockwise while leaving**, the flux is a trapezoidal pulse peaking at B^2, and the emf is two equal rectangular pulses of magnitude Bv with opposite signs.

Final answer. Induced current is anticlockwise on entry and clockwise on exit; flux is a trapezoidal pulse peaking at B^2 and emf is two opposite rectangular pulses of magnitude Bv.

← *Back to Q154*

Q155

(a) (i) *Define self-inductance of a coil. Derive the expression for the energy required to build up a current I in a coil of self-inductance L.*
(ii) *The currents passing through two inductors of self-inductances* 10 mH *and* 20 mH *increase with time at the same rate. Draw graphs showing the variation of:*
(I) the magnitude of emf induced with the rate of change of current in each inductor.
(II) the energy stored in each inductor with the current flowing through it.
OR
(b) (i) *Define the term mutual inductance. Deduce the expression for the mutual inductance of two long coaxial solenoids of the same length having different radii and different number of turns.*
(ii) *The current through an inductor is uniformly increased from zero to* 2 *A in* 40 *s. An emf of* 5 mV *is induced during this period. Find the flux linked with the inductor at* $t = 10$ *s.*

[2025 • Set 55-5-1]

SOLUTION

(a)(i) Self-inductance: The self-inductance L of a coil is numerically equal to the magnetic flux linkage produced per unit current flowing through it:

$$L = \frac{\Phi}{I} \Rightarrow \Phi = LI$$

Equivalently, it equals the emf induced per unit rate of change of current, $\boldsymbol{\varepsilon = -L\frac{dI}{dt}}$.

Energy stored in the coil: When the current is being built up, the induced (back) emf opposes the growth of current. To maintain the current against this back emf, the source does work. At an instant when the current is i and is increasing at rate $\frac{di}{dt}$, the back emf is:

$$\varepsilon = L\frac{di}{dt}$$

The work done by the source in time dt to drive charge $\boldsymbol{dq = i\,dt}$ against this emf is:

$$dW = \varepsilon\, i\, dt = L\frac{di}{dt}\, i\, dt = Li\, di$$

The total work done in raising the current from 0 to I is obtained by integration:

$$W = {}_0^I L\, i\, di$$

$$W = L[\frac{i^2}{2}]_0^I$$

$$W = \frac{1}{2}LI^2$$

This work is stored as magnetic potential energy in the coil:

$$U = \frac{1}{2}LI^2$$

(a)(ii)(I) Graph of $|\varepsilon|$ vs $\frac{dI}{dt}$: Since $|\varepsilon| = L\frac{dI}{dt}$, the graph is a straight line through the origin with slope equal to L. The two inductors give two straight lines: one of slope 10 mH and a steeper one of slope 20 mH. The 20 mH inductor's line lies above (steeper than) the 10 mH line.

(a)(ii)(II) Graph of U vs I: Since $U = \frac{1}{2}LI^2$, each graph is an upward parabola passing through the origin. For a given current, the 20 mH inductor stores more energy, so its parabola rises faster (it lies above the 10 mH parabola).

Therefore the energy required to build current I is $U = \frac{1}{2}LI^2$, the emf is linear in $\frac{dI}{dt}$ (steeper for 20 mH), and the stored energy is parabolic in I (higher for 20 mH).

Final answer. Self-inductance is flux linkage per unit current and the energy stored is $U = \frac{1}{2}LI^2$; emf vs **dI/dt** is a straight line (slope L) and U vs I is a parabola, both steeper for the **20 mH** coil.

← *Back to Q155*

Q156

(i) A rectangular coil of N turns and area of cross-section A is rotated at a steady angular speed ω in a uniform magnetic field. Obtain an expression for the emf induced in the coil at any instant of time.
(ii) Two coplanar and concentric circular loops L_1 *and* L_2 *are placed coaxially with their centres coinciding. The radii of* L_1 *and* L_2 *are* 1 *cm and* 100 *cm respectively. Calculate the mutual inductance of the loops. (Take* $\pi^2 = 10$*)*

[2024 • Set 55-3-1]

SOLUTION

(i) EMF induced in a rotating coil: Consider a rectangular coil of N turns and area A rotating with uniform angular speed ω in a uniform magnetic field B, about an axis perpendicular to $\vec{B}$.

At an instant t, let the normal to the coil make an angle $\theta = \omega t$ with the field. The magnetic flux linked with the coil is:

$$\Phi = NBA\theta = NBA\omega\, t$$

By Faraday's law, the induced emf is the negative rate of change of flux:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -\frac{d}{dt}(NBA\omega\, t)$$

$$\varepsilon = NBA\omega\omega\, t$$

Writing $\varepsilon_0 = NBA\omega$ as the peak emf:

$$\varepsilon = \varepsilon_0 \omega\, t$$

This is the required expression for the instantaneous induced emf.

(ii) Mutual inductance of the two loops: Here the inner loop L_1 has radius $r_1 = 1\ cm = 0.01\ m$ and the outer loop L_2 has radius $r_2 = 100\ cm = 1\ m$. Since $r_1 \ll r_2$, when a current I_2 flows in the larger loop L_2, the field at its centre may be treated as uniform over the small inner loop L_1:

$$B = \frac{\mu_0 I_2}{2 r_2}$$

The flux through the small inner loop L_1 (area πr_1^2) is:

$$\Phi_1 = B \times \pi r_1^2 = \frac{\mu_0 I_2}{2 r_2} \pi r_1^2$$

The mutual inductance is the flux linkage per unit current:

$$M = \frac{\Phi_1}{I_2} = \frac{\mu_0 \pi r_1^2}{2 r_2}$$

Substituting $\mu_0 = 4\pi \times 10^{-7}\ T\ m\ A^{-1}$, $r_1 = 0.01\ m$, $r_2 = 1\ m$:

$$M = \frac{(4\pi \times 10^{-7})\, \pi\, (0.01)^2}{2 \times 1}$$

$$M = \frac{4\pi^2 \times 10^{-7} \times 10^{-4}}{2}$$

$$M = 2\pi^2 \times 10^{-11}\ H$$

Using $\pi^2 = 10$:

$$M = 2 \times 10 \times 10^{-11}\ H = 2 \times 10^{-10}\ H$$

Therefore the mutual inductance of the loops is $M = 2 \times 10^{-10}\ H = 0.2\ nH$.

TIP

Always compute the field of the larger loop and the flux through the smaller loop. Mutual inductance M is the same either way, but using the small loop's area makes the uniform-field approximation valid.

Final answer. Instantaneous emf is $\varepsilon = NBA\omega\omega\, t$; the mutual inductance is $M = 2\times 10^{-10}\, H$.

← *Back to Q156*

Q157

(i) With the help of a diagram, briefly explain the construction and working of ac generator.
(ii) An electron is revolving around a proton in an orbit of radius r with a speed v. Obtain expression for magnetic moment associated with the electron.

[2024 • Set 55-4-1]

SOLUTION

(i) AC generator: An ac generator works on the principle of electromagnetic induction: when the magnetic flux linked with a coil changes, an emf is induced in it.

Construction (diagram description): A rectangular armature coil ABCD of N turns and area A is mounted on a rotor shaft and placed between the curved pole pieces N and S of a strong permanent magnet (or field magnet), which produce a uniform magnetic field $\vec{B}$. The two ends of the coil are connected to two separate slip rings R_1 and R_2 fixed on the shaft, which rotate with the coil. Two carbon brushes B_1 and B_2 press against the slip rings and lead the current to the external load resistance R. The axis of rotation of the coil is perpendicular to $\vec{B}$.

Working: As the coil is rotated with a constant angular speed ω about the axis perpendicular to $\vec{B}$, the angle between the coil's normal and the field changes continuously, so the flux linked with the coil changes. If the normal makes angle $\theta = \omega\, t$ with $\vec{B}$, the flux is:

$$\Phi = NBA\omega\, t$$

By Faraday's law the induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt} = NBA\omega\omega\, t = \varepsilon_0 \omega\, t$$

where $\boldsymbol{\varepsilon_0 = NBA\omega}$ is the peak emf. Because of the sine term, the emf and hence the current reverse direction every half rotation, giving alternating current. The slip rings and brushes deliver this alternating emf to the external circuit.

(ii) Magnetic moment of the orbiting electron: An electron of charge e revolving with speed v in a circular orbit of radius r constitutes a current loop. The time period of one revolution is:

$$T = \frac{2\pi r}{v}$$

The equivalent current (charge passing per unit time) is:

$$I = \frac{e}{T} = \frac{ev}{2\pi r}$$

The magnetic moment is the product of this current and the area of the orbit, $\boldsymbol{A = \pi r^2}$:

$$\mu = I A = \frac{ev}{2\pi r} \times \pi r^2$$

$$\mu = \frac{evr}{2}$$

Direction: Since the electron carries negative charge, the magnetic moment points opposite to the direction of its orbital angular momentum $\vec{L}$. In vector form $\vec{\boldsymbol{\mu}} = -\frac{e}{2m_e}\vec{L}$.

Therefore the magnetic moment associated with the electron is $\boldsymbol{\mu = \frac{evr}{2}}$, directed opposite to the orbital angular momentum.

Final answer. The ac generator induces $\boldsymbol{\varepsilon = \varepsilon_0 \omega\, t}$ with $\boldsymbol{\varepsilon_0 = NBA\omega}$; the electron's magnetic moment is $\boldsymbol{\mu = \frac{evr}{2}}$.

← Back to Q157

Q158

(i) Define coefficient of self-induction. Obtain an expression for self-inductance of a long solenoid of length l, area of cross-section A having N turns.
(ii) Calculate the self-inductance of a coil using the following data obtained when an AC source of frequency $(\frac{200}{\pi})$ Hz *and a DC source is applied across the coil.*

AC Source: *V (V), I (A) = (3.0, 0.5), (6.0, 1.0), (9.0, 1.5).*
DC Source: *V (V), I (A) = (4.0, 1.0), (6.0, 1.5), (8.0, 2.0).*

[2023 • Set 55-2-1]

SOLUTION

(i) Coefficient of self-induction: The coefficient of self-induction (self-inductance) L of a coil is the magnetic flux linkage set up in the coil per unit current flowing through it:

$$L = \frac{\Phi}{I}$$

Equivalently, it is the emf induced in the coil per unit rate of change of current, since $\varepsilon = -L\frac{dI}{dt}$.

Self-inductance of a long solenoid: Consider a long solenoid of length l, cross-sectional area A, with N turns carrying current I. The number of turns per unit length is $n = \frac{N}{l}$. The magnetic field inside a long solenoid is uniform:

$$B = \mu_0 n I = \mu_0 \frac{N}{l} I$$

The flux through one turn is $B A$, so the total flux linkage through all N turns is:

$$\Phi = N B A = N(\mu_0 \frac{N}{l} I)A$$

$$\Phi = \frac{\mu_0 N^2 A I}{l}$$

By definition $L = \frac{\Phi}{I}$, therefore:

$$L = \frac{\mu_0 N^2 A}{l}$$

(ii) Self-inductance from the data: From the DC source, the steady current is limited only by the coil's resistance R (inductance plays no role at steady DC). For each pair, $\frac{V}{I}$ gives:

$$R = \frac{4.0}{1.0} = \frac{6.0}{1.5} = \frac{8.0}{2.0} = 4\ \Omega$$

From the AC source, the current is limited by the impedance Z. For each pair, $\frac{V}{I}$ gives:

$$Z = \frac{3.0}{0.5} = \frac{6.0}{1.0} = \frac{9.0}{1.5} = 6\ \Omega$$

For an R–L coil, the impedance is related to R and the inductive reactance $X_L = \omega L$ by:

$$Z^2 = R^2 + (\omega L)^2$$

$$(\omega L)^2 = Z^2 - R^2 = 6^2 - 4^2 = 36 - 16 = 20$$

$$\omega L = \sqrt{20} = 2\sqrt{5}\ \Omega$$

The angular frequency is:

$$\omega = 2\pi f = 2\pi\left(\frac{200}{\pi}\right) = 400\ \text{rad s}^{-1}$$

Therefore:

$$L = \frac{2\sqrt{5}}{400} = \frac{\sqrt{5}}{200}$$

$$L = \frac{2.236}{200} \approx 1.12 \times 10^{-2}\ H$$

Therefore the self-inductance of the coil is $L \approx 11.2$ **mH**.

CONCEPT

Remember that a coil offers only its resistance R to DC but an impedance $Z=\sqrt{R^2+(\omega L)^2}$ to AC. Using DC to get R and AC to get Z is the standard trick to separate L.

Final answer. Self-inductance of a long solenoid is $L = \dfrac{\mu_0 N^2 A}{l}$, and from the data $L \approx 11.2$ mH.

← *Back to Q158*

Q159

(i) With the help of a labelled diagram, describe the principle and working of an ac generator. Hence, obtain an expression for the instantaneous value of the emf generated.
(ii) The coil of an ac generator consists of 100 *turns of wire, each of area* 0.5 m^2. *The resistance of the wire is* 100 Ω. *The coil is rotating in a magnetic field of* 0.8 *T perpendicular to its axis of rotation, at a constant angular speed of* 60 *radian per second. Calculate the maximum emf generated and power dissipated in the coil.*

[2023 • Set 55-2-1]

SOLUTION

(i) AC generator (principle, working, expression): An ac generator is based on the principle of electromagnetic induction: a changing magnetic flux through a coil induces an emf in it.

Labelled diagram description: A rectangular armature coil ABCD of N turns and area A is placed between the concave poles N and S of a field magnet producing a uniform field $\vec{B}$. The coil ends are joined to two slip rings R_1 and R_2 that rotate with the coil. Two stationary carbon brushes B_1 and B_2 touch the slip rings and connect to the external load. The coil rotates about an axis perpendicular to $\vec{B}$.

Working: When the coil is rotated with uniform angular speed ω, the angle between the coil's normal and $\vec{B}$ becomes $\theta = \omega t$ at time t. The flux linked with the coil is:

$$\Phi = NBA\omega\, t$$

By Faraday's law, the instantaneous induced emf is:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -\frac{d}{dt}(NBA\omega\, t)$$

$$\varepsilon = NBA\omega\omega\, t$$

$$\varepsilon = \varepsilon_0 \omega\, t$$

where $\boldsymbol{\varepsilon_0 = NBA\omega}$ is the peak emf. The emf reverses every half cycle, producing alternating current in the external circuit.

(ii) Numerical: Given $N = 100$, $A = 0.5\ \text{m}^2$, $R = 100\ \Omega$, $B = 0.8\ T$, $\omega = 60\ \text{rad s}^{-1}$.

Maximum (peak) emf:

$$\varepsilon_0 = NBA\omega$$

$$\varepsilon_0 = (100)(0.8)(0.5)(60)$$

$$\varepsilon_0 = 2400\ V$$

Power dissipated in the coil (using rms emf, since average power depends on rms values):

$$P = \frac{\varepsilon_{rms}^2}{R} = \frac{(\varepsilon_0/\sqrt{2})^2}{R} = \frac{\varepsilon_0^2}{2R}$$

$$P = \frac{(2400)^2}{2\times 100}$$

$$P = \frac{5.76\times 10^6}{200}$$

$$P = 2.88\times 10^4\ W$$

Therefore the maximum emf is $\boldsymbol{\varepsilon_0 = 2400\ V}$ and the power dissipated is $\boldsymbol{P = 2.88\times 10^4\ W = 28.8\ \text{kW}}$.

COMMON MISTAKE

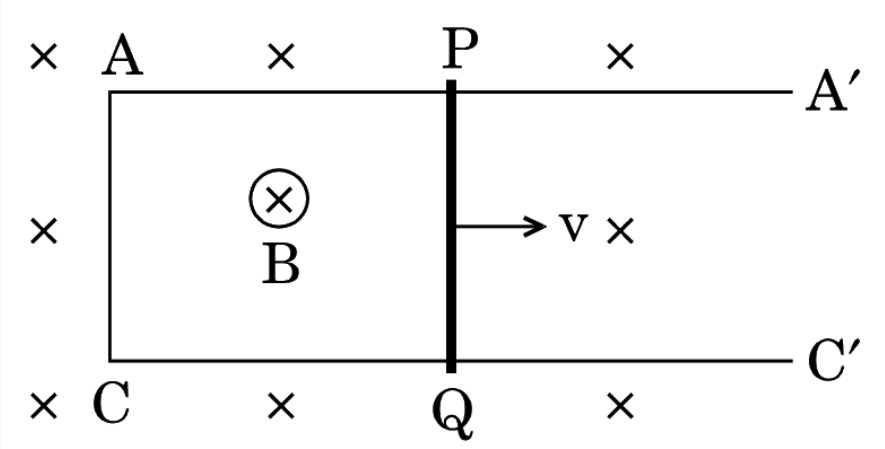
Average power must use the rms emf, not the peak emf. Forgetting the factor of 12 *(from* $\varepsilon_{rms}^2 = \varepsilon_0^2/2$*) doubles your answer and loses marks.*

Final answer. The peak emf is $\varepsilon_0 = NBA\omega = 2400\ V$ and the power dissipated is $P = 28.8\ kW$.

← *Back to Q159*

Q160

A conducting rod PQ *of length* 20 cm *and resistance* 0·1 Ω *rests on two smooth parallel rails of negligible resistance* AA' *and* CC'. *It can slide on the rails and the arrangement is positioned between the poles of a permanent magnet producing uniform magnetic field* $B = 0{\cdot}4\ T$. *The rails, the rod and the magnetic field are in three mutually perpendicular directions as shown in the figure. If the ends A and C of the rails are short circuited, find the*
(i) external force required to move the rod with uniform velocity $v = 10\ cm/s$, *and*
(ii) power required to do so.



[2020 • Set 55-1-1]

SOLUTION

Given: length of rod $L = 20\ cm = 0.20\ m$, resistance $r = 0.1\ \Omega$, magnetic field $B = 0.4\ T$, and velocity $v = 10\ cm/s = 0.10\ m/s$. The rod, rails and field are mutually perpendicular, and the ends *A* and *C* are short-circuited so the rod's resistance is the only resistance in the loop.

As the rod PQ slides, it sweeps across the field and a motional emf is induced:

$$\varepsilon = B\, v\, L$$

$$\varepsilon = (0.4)(0.10)(0.20)$$

$$\varepsilon = 8\times 10^{-3}\ V$$

The induced current in the circuit (resistance = r) is:

$$i = \frac{\varepsilon}{r}$$

$$i = \frac{8\times 10^{-3}}{0.1}$$

$$i = 0.08\ A$$

(i) External force required: The current-carrying rod in the field experiences a magnetic (retarding) force $F = \mathbf{BiL}$ that opposes its motion (Lenz's law). To move the rod with uniform velocity, the external force must exactly balance this retarding force:

$$\mathrm{F_{ext}} = B\, i\, L$$

$$\mathrm{F_{ext}} = (0.4)(0.08)(0.20)$$

$$\mathrm{F_{ext}} = 6.4\times 10^{-3}\ N$$

(ii) Power required: The power supplied by the external force to keep the rod moving at constant speed is:

$$P = \mathrm{F_{ext}}\times v$$

$$P = (6.4\times 10^{-3})(0.10)$$

$$P = 6.4\times 10^{-4}\ W$$

As a check, this equals the electrical power dissipated, $P = \mathbf{i^2\, r = (0.08)^2(0.1) = 6.4\times 10^{-4}\ W}$, confirming energy conservation.

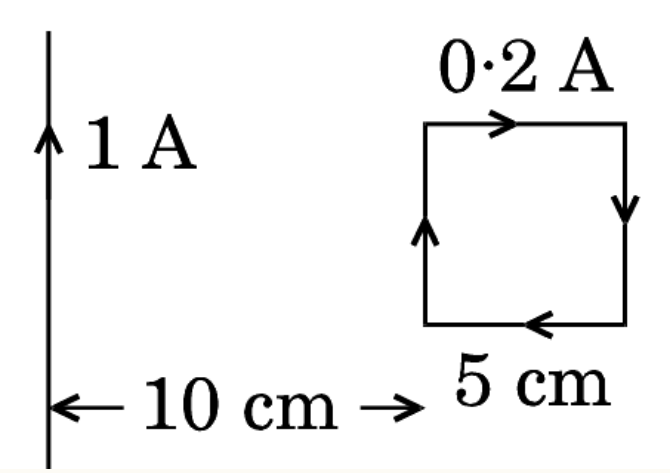
Therefore the external force required is $\mathbf{F_{ext} = 6.4\times 10^{-3}\ N}$ and the power required is $\mathbf{P = 6.4\times 10^{-4}\ W = 640\ \mu W}$.

Final answer. The external force needed is $6.4\times 10^{-3}\ N$ and the power required is $6.4\times 10^{-4}\ W$ (640 μW).

← Back to Q160

Q161

(a) Derive an expression for the induced emf developed when a coil of N turns, and area of cross-section A, is rotated at a constant angular speed ω in a uniform magnetic field B.
(b) A wheel with 100 *metallic spokes each* 0.5 *m long is rotated with a speed of* 120 rev/min *in a plane normal to the horizontal component of the Earth's magnetic field. If the resultant magnetic field at that place is* 4×10^{-4} *T and the angle of dip at the place is* 30°*, find the emf induced between the axle and the rim of the wheel.*
OR
(a) Derive the expression for the magnetic energy stored in an inductor when a current I develops in it. Hence, obtain the expression for the magnetic energy density.
(b) A square loop of sides 5 cm *carrying a current of* 0.2 *A in the clockwise direction is placed at a distance of* 10 cm *from an infinitely long wire carrying a current of* 1 *A as shown. Calculate (i) the resultant magnetic force, and (ii) the torque, if any, acting on the loop.*



[2019 • Set 55-2-1]

SOLUTION

(a) Consider a coil of N turns, each of area A, rotating with constant angular speed ω in a uniform magnetic field $\vec{B}$, with the axis of rotation perpendicular

to $\vec{B}$.

At any instant t, let the normal to the coil make an angle $\theta = \omega t$ with $\vec{B}$. The magnetic flux linked with the coil is

$$\Phi = NBA\theta = NBA\omega t$$

By Faraday's law, the induced emf is the negative rate of change of flux:

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -\frac{d}{dt}(NBA\omega t)$$

$$\varepsilon = NBA\omega\omega t$$

The emf is maximum when $\omega t = 1$, giving the **peak emf** $\varepsilon_0 = NBA\omega$. Hence

$$\varepsilon = \varepsilon_0 \omega t$$

This is a sinusoidally varying (alternating) emf, which is the principle of the AC generator.

(b) The wheel rotates in a plane normal to the horizontal component of the Earth's field, so each spoke cuts the vertical component B_V.

$$B_V = B\delta = (4\times 10^{-4})\sin 30^\circ$$

$$B_V = (4\times 10^{-4})(0.5) = 2\times 10^{-4}\ T$$

The angular speed is

$$\omega = 2\pi\nu = 2\pi\left(\frac{120}{60}\right) = 4\pi \text{ rad s}^{-1}$$

Each spoke is a rod of length $L = 0.5\ m$ rotating about one end, so the emf between the axle and the rim is

$$\varepsilon = \frac{1}{2} B_V \omega L^2$$

$$\varepsilon = \frac{1}{2}(2\times 10^{-4})(4\pi)(0.5)^2$$

$$\varepsilon = \frac{1}{2}(2\times 10^{-4})(4\pi)(0.25)$$

$$\varepsilon = \pi \times 10^{-4}\ V \approx 3.14 \times 10^{-4}\ V$$

All 100 spokes are connected in parallel between the axle and the rim, so they all have the same emf and the number of spokes does not change the result.

The emf induced between the axle and the rim is $\boldsymbol{\varepsilon \approx 3.14 \times 10^{-4}\ V = 0.314\ mV}$.

OR

(a) When a current grows in an inductor of self-inductance L, a back-emf $\boldsymbol{\varepsilon = -L\frac{di}{dt}}$ opposes the growth. The work done against this back-emf is stored as magnetic energy.

The small work done in time dt to drive current i against the back-emf is

$$dW = -\varepsilon\, i\, dt = L\frac{di}{dt}\, i\, dt = L\, i\, di$$

Total energy stored as the current rises from 0 to I:

$$U = \int_0^I L\, i\, di = L\left[\frac{i^2}{2}\right]_0^I$$

$$U = \frac{1}{2}LI^2$$

For a long solenoid, $\boldsymbol{L = \mu_0 n^2 Al}$ and $\boldsymbol{B = \mu_0 nI}$, so $\boldsymbol{I = \frac{B}{\mu_0 n}}$. Substituting,

$$U = \frac{1}{2}(\mu_0 n^2 Al)\left(\frac{B}{\mu_0 n}\right)^2$$

$$U = \frac{B^2}{2\mu_0}(Al)$$

Since Al is the volume of the solenoid, the **magnetic energy density** is

$$u = \frac{U}{Al} = \frac{B^2}{2\mu_0}$$

(b) The long wire carries $\mathbf{I_1 = 1\ A}$. The near and far sides of the square loop (parallel to the wire) are at distances $\mathbf{r_1 = 10\ cm = 0.10\ m}$ and $\mathbf{r_2 = 0.15\ m}$ and carry $\mathbf{I_2 = 0.2\ A}$.

The two sides perpendicular to the wire carry equal and opposite forces that cancel. The net force is between the two parallel sides:

$$F = \frac{\mu_0 I_1 I_2 L}{2\pi}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

with $L = 0.05\ m$.

$$F = \frac{(4\pi\times 10^{-7})(1)(0.2)(0.05)}{2\pi}\left(\frac{1}{0.10} - \frac{1}{0.15}\right)$$

$$F = (2\times 10^{-7})(0.2)(0.05)(10 - 6.67)$$

$$F = (2\times 10^{-9})(3.33) \approx 6.7\times 10^{-9}\ N$$

The force is directed away from the wire (the nearer side, carrying current opposite to the wire, is repelled more strongly).

(ii) Since both the wire and the loop lie in the same plane and the net force passes through the plane of the loop with no perpendicular separation, the **torque on the loop is zero**.

So the resultant force is about $6.7\times 10^{-9}\ N$ (away from the wire) and the torque is zero.

> **COMMON MISTAKE**
>
> *In the spokes problem use the VERTICAL component $B\delta$ (axis along B_H), and remember all spokes are in parallel so the count of spokes does not enter the emf.*

Final answer. (b) Wheel emf ≈ $\mathbf{3.14\times 10^{-4}\ V}$; OR net force ≈ $\mathbf{6.7\times 10^{-9}\ N}$ away from wire, torque = 0.

← *Back to Q161*

Q162

(a) A metallic rod of length 'l' and resistance 'R' is rotated with a frequency 'ν' with one end hinged at the centre and the other end at the circumference of a circular metallic ring of radius 'l', about an axis passing through the centre and perpendicular to the plane of the ring. A constant and uniform magnetic field 'B' parallel to the axis is present everywhere.
(i) Derive the expression for the induced emf and the current in the rod.
(ii) Due to the presence of current in the rod and of the magnetic field, find the

expression for the magnitude and direction of the force acting on this rod.
(iii) Hence, obtain an expression for the power required to rotate the rod.
(b) A copper coil is taken out of a magnetic field with a fixed velocity. Will it be easy to remove it from the same field if its ohmic resistance is increased?

OR

(a) A rectangular coil rotates in a uniform magnetic field. Obtain an expression for induced emf and current at any instant. Also find their peak values. Show the variation of induced emf versus angle of rotation (ω t) on a graph.
(b) An iron bar falling through the hollow region of a thick cylindrical shell made of copper experiences a retarding force. What can you conclude about the nature of the iron bar? Explain.

[2019 • Set 55-5-1]

SOLUTION

(a)(i) The rod of length l rotates at frequency ν about one end, so $\boldsymbol{\omega = 2\pi\nu}$. A small element at distance x from the hinge moves with speed $\boldsymbol{v = \omega x}$. The motional emf across the element is $dx \cdot B\omega x$ (since $\boldsymbol{d\varepsilon = Bv\,dx = B\omega x\,dx}$). Integrating from the centre to the rim:

$$\varepsilon = \int_0^l B\omega x\,dx = B\omega\left[\frac{x^2}{2}\right]_0^l$$

$$\varepsilon = \frac{1}{2}B\omega l^2 = \pi\nu B l^2$$

The current in the rod is

$$i = \frac{\varepsilon}{R} = \frac{\pi\nu B l^2}{R}$$

(a)(ii) Each element of the current-carrying rod in field B experiences a force $\boldsymbol{dF = Bi\,dx}$. The total force on the rod is

$$F = \int_0^l Bi\,dx = Bil$$

Substituting $\boldsymbol{i = \frac{\pi\nu B l^2}{R}}$:

$$F = B\left(\frac{\pi\nu B l^2}{R}\right)l = \frac{\pi\nu B^2 l^3}{R}$$

By Lenz's law, this force is **tangential and directed so as to oppose the rotation** of the rod.

(a)(iii) To keep the rod rotating at constant speed, the external agent must supply power equal to that dissipated. Power is emf times current:

$$P = \varepsilon\, i = \frac{\varepsilon^2}{R}$$

$$P = \frac{1}{R}(\pi \nu\, B\, l^2)^2$$

$$P = \frac{\pi^2 \nu^2 B^2 l^4}{R}$$

(b) The induced current is $\boldsymbol{i = \varepsilon/R}$, so increasing R reduces the induced current and hence reduces the opposing (retarding) force $\boldsymbol{F \propto i}$. Therefore, **yes, it becomes easier to remove the coil** when its ohmic resistance is increased.

OR

(a) Consider a rectangular coil of N turns, area A, rotating with angular speed ω in a uniform field $\vec{B}$. At time t the normal makes angle $\omega\, t$ with $\vec{B}$, so the flux is

$$\Phi = NBA\omega\, t$$

The induced emf is

$$\varepsilon = -\frac{d\Phi}{dt} = NBA\omega\omega\, t$$

The current at any instant (coil resistance R) is

$$i = \frac{\varepsilon}{R} = \frac{NBA\omega}{R}\omega\, t$$

The **peak values** occur when $\boldsymbol{\omega\, t = 1}$:

$$\varepsilon_0 = NBA\omega\ i_0 = \frac{NBA\omega}{R}$$

Graph: Plot ε on the vertical axis against the angle $\omega\, t$ on the horizontal axis. The curve is a sine wave starting from the origin, rising to $+\varepsilon_0$ at $\boldsymbol{\omega\, t = 90°}$, returning to zero at 180°, falling to $-\varepsilon_0$ at 270°, and back to zero at 360°.

(b) The retarding force on the falling iron bar arises from eddy currents induced in it. Such eddy currents (and the associated electromagnetic braking) are produced strongly only in a magnetic material that becomes magnetised. Hence the **iron bar must be a magnet (or magnetised)**, so that as it falls through the copper shell it produces a changing flux, inducing eddy currents in the copper that oppose its motion (Lenz's law) and retard it.

Final answer. (a) $\varepsilon = \pi\nu\, Bl^2$, $i = \dfrac{\pi\nu\, Bl^2}{R}$, $F = \dfrac{\pi\nu\, B^2l^3}{R}$, $P = \dfrac{\pi^2\nu^2B^2l^4}{R}$; OR the iron bar is a magnet.

← *Back to Q162*

Q163

(a) State the principle of an ac generator and explain its working with the help of a labelled diagram. Obtain the expression for the emf induced in a coil having N turns each of cross-sectional area A, rotating with a constant angular speed 'ω' in a magnetic field $\vec{B}$*, directed perpendicular to the axis of rotation.*
(b) An aeroplane is flying horizontally from west to east with a velocity of 900 km/hour. *Calculate the potential difference developed between the ends of its wings having a span of* 20 *m. The horizontal component of the Earth's magnetic field is* 5×10^{-4} *T and the angle of dip is* 30°.

[2018]

SOLUTION

(a) Principle: An AC generator works on the principle of **electromagnetic induction**, that is, when the magnetic flux linked with a coil changes, an emf is induced in it.

Labelled diagram (described): A rectangular coil ABCD (armature) of *N* turns is mounted on a shaft and rotated between the poles N and S of a strong permanent magnet (field $\vec{B}$ horizontal, perpendicular to the rotation axis). The two ends of the coil are connected to two separate slip rings R_1 and R_2, which press against two carbon brushes B_1 and B_2. The brushes are connected to the external load. Main elements: field magnet, armature coil, slip rings, brushes.

Working: As the coil rotates, the flux through it changes continuously, inducing an alternating emf. The slip rings rotate with the coil and feed this alternating emf to the external circuit through the fixed brushes.

Derivation: Let the coil of N turns, area A, rotate with angular speed ω so that at time t the normal makes angle $\omega\, t$ with $\vec{B}$. The flux is

$$\Phi = \mathrm{NBA}\omega\, t$$

By Faraday's law,

$$\varepsilon = -\frac{\mathrm{d}\Phi}{\mathrm{dt}} = -\frac{d}{\mathrm{dt}}(\mathrm{NBA}\omega\, t)$$

$$\varepsilon = \mathrm{NBA}\omega\omega\, t = \varepsilon_0 \omega\, t$$

where the peak emf is $\boldsymbol{\varepsilon_0 = \mathrm{NBA}\omega}$.

(b) Convert the speed:

$$v = 900\ \mathrm{km/h} = \frac{900\times 1000}{3600} = 250\ m\ \mathrm{s}^{-1}$$

The wings move horizontally and cut the vertical component of the Earth's field. Since $\boldsymbol{\delta = \frac{B_V}{B_H}}$,

$$\mathrm{B}_V = \mathrm{B}_H\delta = (5\times 10^{-4})\tan 30^\circ$$

$$\mathrm{B}_V = \frac{5\times 10^{-4}}{\sqrt{3}} \approx 2.89\times 10^{-4}\ T$$

The potential difference (motional emf) between the wing tips, with span $\boldsymbol{L = 20\ m}$, is

$$\varepsilon = \mathrm{B}_V\, v\, L$$

$$\varepsilon = (2.89\times 10^{-4})(250)(20)$$

$$\varepsilon \approx 1.44\ V$$

The potential difference developed across the wings is about **1.44 *V***.

COMMON MISTAKE

A horizontally flying plane cuts the VERTICAL field component, so use $\mathbf{B_V = B_H\delta}$*, not* $B_H\delta$.

Final answer. Induced emf $\boldsymbol{\varepsilon = NBA\omega\omega\, t}$; potential difference across wings $\approx$ **1.44 *V*.**

← *Back to Q163*

Q164

(a) Draw a labelled diagram of AC generator. Derive the expression for the instantaneous value of the emf induced in the coil.
(b) A circular coil of cross-sectional area 200 cm^2 *and 20 turns is rotated about the vertical diameter with angular speed of* 50 rad s^{-1} *in a uniform magnetic field of magnitude* 3.0×10^{-2} *T. Calculate the maximum value of the current in the coil.*
OR
(a) Draw a labelled diagram of a step-up transformer. Obtain the ratio of secondary to primary voltage in terms of number of turns and currents in the two coils.
(b) A power transmission line feeds input power at 2200 *V to a step-down transformer with its primary windings having 3000 turns. Find the number of turns in the secondary to get the power output at* 220 *V.*

[2017]

SOLUTION

(a) Labelled diagram (described): A rectangular armature coil ABCD of N turns is rotated on a shaft between the poles N and S of a permanent magnet, with field $\vec{B}$ perpendicular to the rotation axis. The coil ends connect to two slip rings R_1, R_2, which press against carbon brushes B_1, B_2 leading to the external load.

Derivation: Let the coil rotate with angular speed ω; at time t the normal makes angle $\omega\, t$ with $\vec{B}$. The flux linked is

$$\Phi = \text{NBA}\omega\, t$$

By Faraday's law, the instantaneous emf is

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d}{dt}(NBA\omega\, t)$$

$$\varepsilon = NBA\omega\omega\, t$$

with peak value $\boldsymbol{\varepsilon_0 = NBA\omega}$.

(b) Given $\boldsymbol{N = 20}$, $\boldsymbol{A = 200\ cm^2 = 200\times 10^{-4}\ m^2 = 0.02\ m^2}$, $\boldsymbol{\omega = 50\ rad\ s^{-1}}$, $\boldsymbol{B = 3.0\times 10^{-2}\ T}$.

The peak (maximum) emf is

$$\varepsilon_0 = NBA\omega$$

$$\varepsilon_0 = (20)(3.0\times 10^{-2})(0.02)(50)$$

$$\varepsilon_0 = 0.6\ V$$

The maximum current is $\boldsymbol{I_{max} = \frac{\varepsilon_0}{R}}$, where R is the total resistance of the closed circuit. No resistance value is given in the problem, so the maximum current cannot be evaluated to a single number; it is

$$I_{max} = \frac{\varepsilon_0}{R} = \frac{0.6}{R}\ A$$

Hence the maximum emf is $\boldsymbol{\varepsilon_0 = 0.6\ V}$, and the maximum current is $\boldsymbol{I_{max} = 0.6/R\ A}$ (for example, if the circuit resistance were 10 Ω, then $\boldsymbol{I_{max} = 0.06\ A}$).

OR

(a) Labelled diagram (described): A step-up transformer has a primary coil of N_p turns and a secondary coil of N_s turns (with $\boldsymbol{N_s > N_p}$) wound on a common closed laminated soft-iron core. The AC input is applied across the primary; the output is taken across the secondary.

Since the same changing flux Φ links every turn, the emf per turn is identical in both coils:

$$V_p = N_p\frac{d\Phi}{dt}\quad V_s = N_s\frac{d\Phi}{dt}$$

Dividing,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

For an ideal (loss-free) transformer, input power = output power, so $V_p I_p = V_s I_s$, giving

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(b) Here it is a step-down transformer: $V_p = 2200\ V$, $N_p = 3000$, $V_s = 220\ V$.

$$\frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$N_s = N_p \frac{V_s}{V_p} = 3000 \times \frac{220}{2200}$$

$$N_s = 300 \text{ turns}$$

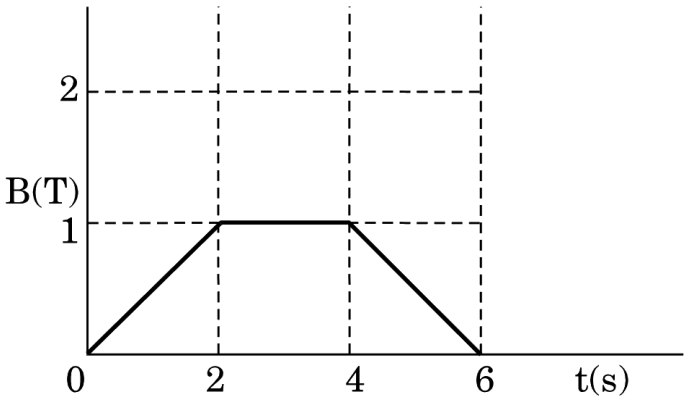
The secondary must have 300 turns.

Final answer. (b) $\varepsilon_0 = 0.6\ V$, $I_{max} = \frac{0.6}{R}\ A$ (no resistance given, so current is indeterminate in magnitude); OR $N_s = 300$ turns.

← *Back to Q164*

Q165

(a) State Faraday's law of electromagnetic induction.
(b) The magnetic field through a circular loop of wire 12 cm *in radius and* 8·5 Ω *resistance, changes with time as shown in the figure. The magnetic field is perpendicular to the plane of the loop. Calculate the induced current in the loop and plot it as a function of time.*



(c) Show that Lenz's law is a consequence of conservation of energy.

OR

(a) Describe, with the help of a suitable diagram, the working principle of a step-up transformer. Obtain the relation between input and output voltages in terms of the number of turns of primary and secondary windings and the currents in the input and output circuits.

(b) Given the input current 15 *A and the input voltage of* 100 *V for a step-up transformer having* 90% *efficiency, find the output power and the voltage in the secondary if the output current is* 3 *A.*

[2017]

SOLUTION

(a) Faraday's law of electromagnetic induction: Whenever the magnetic flux linked with a closed circuit changes, an emf is induced in it, and this induced emf is equal to the negative rate of change of magnetic flux:

$$\varepsilon = -\frac{d\Phi}{dt}$$

For a coil of N turns, $\boldsymbol{\varepsilon = -N\frac{d\Phi}{dt}}$. The negative sign represents Lenz's law (the induced emf opposes the change in flux).

(b) The area of the loop is

$$A = \pi r^2 = \pi(0.12)^2$$

$$A = 0.0452\ m^2$$

The induced emf for a region where the field changes at rate $\frac{dB}{dt}$ is

$$\varepsilon = A\left|\frac{dB}{dt}\right|$$

and the induced current is

$$i = \frac{\varepsilon}{R} = \frac{A}{R}\left|\frac{dB}{dt}\right|$$

The constant factor is

$$\frac{A}{R} = \frac{0.0452}{8.5} \approx 5.32\times 10^{-3}\ \mathrm{m^2\,\Omega^{-1}}$$

For each straight-line segment of the given B versus t graph, $\frac{dB}{dt}$ equals the slope of that segment. The induced current in each interval is the slope multiplied by $\frac{A}{R}$. For example, if the field rises by 0.5 T in 2 s then $\mathbf{\frac{dB}{dt} = 0.25\ T\ s^{-1}}$ and

$$i = (5.32\times 10^{-3})(0.25) \approx 1.33\times 10^{-3}\ A$$

Where the graph is flat (constant B), $\mathbf{\frac{dB}{dt} = 0}$ and $\boldsymbol{i = 0}$; where the slope reverses, i reverses sign. The plot of i versus t is therefore **piecewise constant**, with the height of each step proportional to the slope of $B(t)$ in that interval, and zero wherever B is constant.

(c) Lenz's law from energy conservation: By Lenz's law the induced current opposes the change in flux. So when a magnet is pushed toward a coil, the induced current repels it, and the external agent must do positive work to push it in. This work is converted into the electrical energy of the induced current (dissipated as heat in the resistance). Thus energy is conserved.

If instead the induced current aided the motion (attracted the approaching magnet), the magnet would accelerate on its own, producing current and energy without any external work being done, creating energy from nothing. This is impossible. Hence the induced current must oppose the change, which is exactly **Lenz's law**, a direct consequence of conservation of energy.

OR

(a) Step-up transformer: A primary coil (N_p turns) and a secondary coil (N_s turns, $\mathbf{N_s > N_p}$) are wound on a common closed laminated soft-iron core. AC applied to the primary sets up a changing flux in the core. By mutual induction this changing flux links the secondary and induces an emf in it.

Since the same flux Φ threads every turn,

$$V_p = N_p\frac{d\Phi}{dt}\quad V_s = N_s\frac{d\Phi}{dt}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

For an ideal transformer, $\mathbf{V_p I_p = V_s I_s}$, so

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$$

(b) Input power $\mathbf{P_{in} = V_p I_p = (100)(15) = 1500\ W}$. With 90% efficiency,

$$P_{out} = 0.90\times P_{in} = 0.90\times 1500 = 1350\ W$$

Output (secondary) voltage with output current $\mathbf{I_s = 3\ A}$:

$$V_s = \frac{P_{out}}{I_s} = \frac{1350}{3} = 450\ V$$

The output power is **1350 *W*** and the secondary voltage is **450 *V***.

Final answer. (b) $\boldsymbol{i = \frac{A}{R}\left|\frac{dB}{dt}\right|}$ (step graph); OR $\mathbf{P_{out} = 1350\ W}$, $\mathbf{V_s = 450\ V}$.

← *Back to Q165*

Q166

(a) Draw a labelled diagram of an ac generator. Obtain the expression for the emf induced in the rotating coil of N turns each of cross-sectional area A, in the presence of a magnetic field $\vec{B}$.
(b) A horizontal conducting rod 10 m long extending from east to west is falling with a speed 5.0 m s^{-1} at right angles to the horizontal component of the Earth's magnetic field, 0.3×10^{-4} Wb m^{-2}. Find the instantaneous value of the emf induced in the rod.

[2017]

SOLUTION

(a) Labelled diagram (described): A rectangular armature coil ABCD of *N* turns, each of area *A*, is mounted on a shaft and rotated between the poles N and S of a permanent magnet (field $\vec{B}$ perpendicular to the rotation axis). The coil ends are joined to two slip rings R_1, R_2 that press against carbon brushes B_1, B_2 connected to the external load.

Derivation: As the coil rotates with angular speed ω, at time *t* its normal makes angle ω *t* with $\vec{B}$, so the flux linked is

$$\Phi = \mathrm{NBA}\omega\, t$$

By Faraday's law,

$$\varepsilon = -\frac{\mathrm{d}\Phi}{\mathrm{dt}} = -\frac{d}{\mathrm{dt}}(\mathrm{NBA}\omega\, t)$$

$$\varepsilon = \mathrm{NBA}\omega\omega\, t$$

with peak emf $\boldsymbol{\varepsilon_0 = \mathrm{NBA}\omega}$, giving the alternating emf $\boldsymbol{\varepsilon = \varepsilon_0 \omega\, t}$.

(b) The horizontal rod lies east to west and falls vertically with speed v, cutting the horizontal component B_H of the Earth's field at right angles.

Given $\boldsymbol{L = 10\ m}$, $\boldsymbol{v = 5.0\ m\ s^{-1}}$, $\boldsymbol{\mathrm{B}_H = 0.3\times 10^{-4}\ \mathrm{Wb\ m^{-2}}}$.

The motional emf is

$$\varepsilon = \mathrm{B}_H\, v\, L$$

$$\varepsilon = (0.3\times 10^{-4})(5.0)(10)$$

$$\varepsilon = 1.5\times 10^{-3}\ V$$

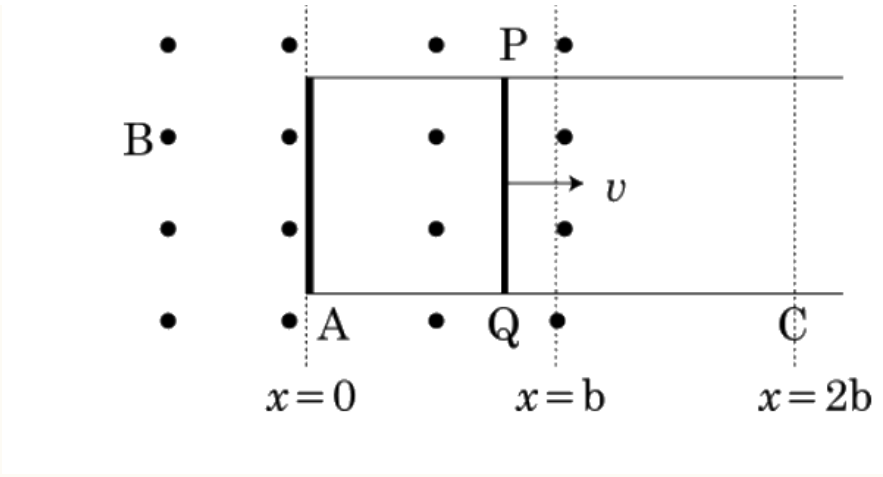
The instantaneous induced emf in the rod is $\boldsymbol{1.5\times 10^{-3}\ V = 1.5\ \mathrm{mV}}$.

Final answer. $\boldsymbol{\varepsilon = \mathrm{NBA}\omega\omega\, t}$; induced emf in falling rod $\boldsymbol{= 1.5\times 10^{-3}\ V = 1.5\ \mathrm{mV}}$.

← *Back to Q166*

Q167

(a) When a bar magnet is pushed towards (or away) from the coil connected to a galvanometer, the pointer in the galvanometer deflects. Identify the phenomenon causing this deflection and write the factors on which the amount and direction of the deflection depends. State the laws describing this phenomenon.
(b) Sketch the change in flux, emf and force when a conducting rod PQ *of resistance R and length l moves freely to and fro between A and C with speed v on a rectangular conductor placed in uniform magnetic field as shown in the figure.*



[2016]

SOLUTION

(a) Phenomenon: The deflection of the galvanometer is caused by **electromagnetic induction**. When the magnet moves toward or away from the coil, the magnetic flux linked with the coil changes, inducing an emf and hence a current that deflects the galvanometer.

Factors on which the amount of deflection depends: (i) the speed of the relative motion of the magnet and coil (faster motion gives a larger emf), (ii) the strength of the bar magnet, (iii) the number of turns in the coil, and the area of the coil.

Factor on which the direction of deflection depends: the direction of relative motion and which pole faces the coil (i.e. whether flux is increasing or decreasing), as fixed by Lenz's law.

Laws describing the phenomenon:

Faraday's law: the induced emf equals the negative rate of change of flux,

$$\varepsilon = -\frac{d\Phi}{dt}$$

Lenz's law: the direction of the induced current is always such as to oppose the change in flux that produces it.

(b) Let the rod PQ oscillate with speed v between A and C.

Flux Φ versus time: As PQ moves, the area of the loop in the field changes linearly with the rod's position. As PQ goes from A to C the enclosed flux

increases linearly from a minimum to a maximum, and decreases linearly on the return. So the Φ versus t graph is a **triangular wave**.

EMF ε versus time: $\boldsymbol{\varepsilon = -\frac{d\Phi}{dt}}$, which equals the slope of the flux graph. During steady motion the magnitude $\boldsymbol{\varepsilon = Blv}$ is constant, but its sign reverses each time PQ changes direction at A and C. So the ε versus t graph is a **square wave** of magnitude Blv, flipping sign at each turning point.

Force F versus time: The induced current is $\boldsymbol{i = \frac{Blv}{R}}$, so the force on PQ is

$$F = \mathrm{Bil} = \frac{\mathrm{B^2l^2v}}{R}$$

directed opposite to the velocity (Lenz's law). Its magnitude is constant during steady motion and its sign reverses at each turning point, so the F versus t graph is also a **square wave** of magnitude $\frac{\mathrm{B^2l^2v}}{R}$, always opposing the motion.

Final answer. Phenomenon = electromagnetic induction; $\boldsymbol{\Phi}$ is a triangular wave, while $\boldsymbol{\varepsilon}$ (=$\boldsymbol{Blv}$) and $\boldsymbol{F}$ (=$\boldsymbol{B^2l^2v/R}$) are square waves opposing the motion.

← *Back to Q167*

Q168

Draw a necessary arrangement for winding of primary and secondary coils in a step-up transformer. State its underlying principle and derive the relation between the primary and secondary voltages in terms of number of primary and secondary turns. Mention the two basic assumptions used in obtaining the above relation.
State any two causes of energy loss in actual transformers.

[2015]

SOLUTION

Arrangement (described): A step-up transformer has a primary coil of N_p turns and a secondary coil of N_s turns (with $\boldsymbol{N_s > N_p}$) wound on a common closed laminated soft-iron core. Often the two coils are wound one over the other on the same limb of the core (or on separate limbs of a closed rectangular core) so that nearly all the flux produced by the primary links the secondary. The AC

input is applied across the primary; the stepped-up output is taken across the secondary.

Underlying principle: **Mutual induction**. An alternating current in the primary produces a continuously changing magnetic flux in the iron core; this changing flux links the secondary and induces an alternating emf in it.

Derivation: Let the instantaneous flux through each turn of the core be Φ. Since (ideally) the same flux links every turn of both coils, the induced emfs are

$$V_p = N_p \frac{d\Phi}{dt}$$

$$V_s = N_s \frac{d\Phi}{dt}$$

Dividing the second equation by the first,

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

Since $N_s > N_p$, we get $V_s > V_p$, so the voltage is stepped up.

Two basic assumptions: (i) The primary and secondary have no resistance (no I^2R loss in the windings). (ii) There is no flux leakage, so the same flux links each turn of both the primary and the secondary (the core is ideal).

Two causes of energy loss in actual transformers: (i) **Copper loss**, the I^2R heating in the resistance of the primary and secondary windings. (ii) **Iron loss**, comprising eddy-current loss in the core (reduced by laminating the core) and hysteresis loss due to repeated magnetisation and demagnetisation of the core material. (Flux leakage is a further cause.)

Final answer. $\mathbf{\frac{V_s}{V_p} = \frac{N_s}{N_p}}$, by mutual induction; assumptions = no winding resistance and no flux leakage; losses = copper loss and iron (eddy + hysteresis) loss.

← Back to Q168

Q169

(a) Define mutual inductance and write its S.I. units.
(b) Derive an expression for the mutual inductance of two long co-axial solenoids of same length wound one over the other.
(c) In an experiment, two coils c_1 *and* c_2 *are placed close to each other. Find out the expression for the emf induced in the coil* c_1 *due to a change in the current through the coil* c_2.

[2015]

SOLUTION

(a) Mutual inductance: The mutual inductance M of a pair of coils is the magnetic flux linked with one coil per unit current in the other coil. If a current I_1 in coil 1 links flux Φ_{21} with coil 2, then

$$\Phi_{21} = M\, I_1 \Rightarrow M = \frac{\Phi_{21}}{I_1}$$

Its SI unit is the **henry (H)**, where $\mathbf{1\ H = 1\ Wb\ A^{-1} = 1\ V\,s\,A^{-1}}$.

(b) Consider two long coaxial solenoids of the same length l, the inner one S_1 of radius r_1 with N_1 turns and the outer S_2 with N_2 turns. Let $\mathbf{n_1 = N_1/l}$ and $\mathbf{n_2 = N_2/l}$ be their turns per unit length.

Pass a current I_2 through the outer solenoid S_2. The (nearly) uniform field it produces inside is

$$B_2 = \mu_0\, n_2\, I_2 = \frac{\mu_0\, N_2\, I_2}{l}$$

This field passes through the inner solenoid S_1, whose cross-sectional area is $\pi\, r_1^2$. The total flux linked with the N_1 turns of S_1 is

$$\Phi_1 = N_1\, B_2\, (\pi\, r_1^2) = N_1 \left(\frac{\mu_0\, N_2\, I_2}{l}\right) \pi\, r_1^2$$

By definition $\mathbf{\Phi_1 = M\, I_2}$, so

$$M = \frac{\Phi_1}{I_2} = \frac{\mu_0\, N_1\, N_2\, \pi\, r_1^2}{l}$$

This is the mutual inductance of the two coaxial solenoids.

(c) When the current I_2 through coil c_2 changes, the flux it links with coil c_1 is $\Phi_1 = M\, I_2$. By Faraday's law the emf induced in c_1 is

$$\varepsilon_1 = -\frac{d\Phi_1}{dt} = -\frac{d}{dt}(M\, I_2)$$

$$\varepsilon_1 = -M\frac{dI_2}{dt}$$

Thus the emf induced in c_1 is proportional to the rate of change of current in c_2, the constant of proportionality being the mutual inductance M (the negative sign expressing Lenz's law).

Final answer. $M = \dfrac{\Phi_{21}}{I_1}$ (unit henry); for coaxial solenoids $M = \dfrac{\mu_0 N_1 N_2 \pi r_1^2}{l}$; and $\varepsilon_1 = -M\dfrac{dI_2}{dt}$.

← *Back to Q169*

Q170

(a) Describe a simple experiment (or activity) to show that the polarity of emf induced in a coil is always such that it tends to produce a current which opposes the change of magnetic flux that produces it.
(b) The current flowing through an inductor of self inductance L is continuously increasing. Plot a graph showing the variation of
(i) Magnetic flux versus the current
(ii) Induced emf versus dI/dt
(iii) Magnetic potential energy stored versus the current.
OR
(a) Draw a schematic sketch of an ac generator describing its basic elements. State briefly its working principle. Show a plot of variation of
(i) Magnetic flux and
(ii) Alternating emf versus time generated by a loop of wire rotating in a magnetic field.
(b) Why is choke coil needed in the use of fluorescent tubes with ac mains ?

[2014]

SOLUTION

(a) Experiment to demonstrate Lenz's law: Connect a coil to a sensitive galvanometer. Take a bar magnet and push its north pole toward the coil.

As the magnet approaches, the flux through the coil increases, and the galvanometer deflects, showing an induced current. The direction of this induced current is such that the face of the coil toward the magnet becomes a north pole, which repels the approaching magnet, opposing the increase in flux. Work has to be done against this repulsion to push the magnet in.

Now pull the magnet away. The flux now decreases, and the galvanometer deflects in the opposite direction. The induced current reverses so that the near face becomes a south pole, which attracts the receding magnet, opposing the decrease in flux.

In both cases the induced current opposes the change that produces it, demonstrating that the polarity of the induced emf always tends to oppose the change in flux (Lenz's law).

(b) Graphs for an inductor of self-inductance L:

(i) Magnetic flux versus current: Since $\mathbf{\Phi = LI}$, the graph of Φ against I is a **straight line through the origin** with slope L.

(ii) Induced emf versus $\frac{dI}{dt}$: Since $\mathbf{|\varepsilon| = L\frac{dI}{dt}}$, the graph of $|\varepsilon|$ against $\frac{dI}{dt}$ is a **straight line through the origin** with slope L.

(iii) Magnetic potential energy versus current: Since $\mathbf{U = \frac{1}{2}LI^2}$, the graph of U against I is an **upward-opening parabola** passing through the origin.

OR

(a) Basic elements of an AC generator: a field magnet (provides $\vec{B}$), an armature coil of N turns rotated about an axis perpendicular to $\vec{B}$, two slip rings, and two brushes that connect the rotating coil to the external circuit.

Working principle: It works on **electromagnetic induction**. As the coil rotates in the magnetic field, the flux linked with it changes continuously, inducing an alternating emf, which is fed to the external circuit through the slip rings and brushes.

Plots (described): With $\Phi = NBA\omega\, t$ and $\varepsilon = NBA\omega\omega\, t$, plot both against time t. The flux curve is a cosine starting at its maximum NBA at $t = 0$. The emf curve is a sine starting at zero at $t = 0$, reaching peak NBAω. The two curves are 90° (a quarter cycle) out of phase, the emf being maximum where the flux is zero and vice versa.

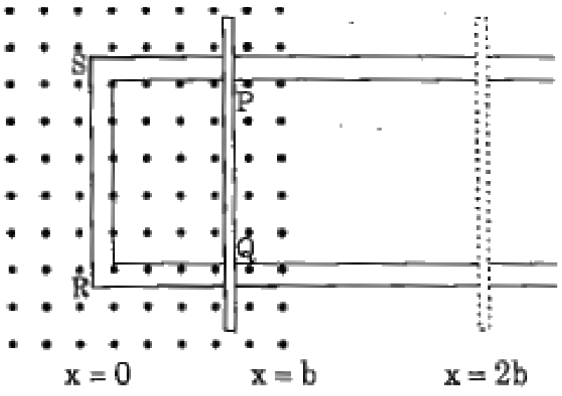
(b) A choke coil (an inductor) is needed in series with a fluorescent tube to **limit the current** flowing through the tube. It provides a large inductive reactance $X_L = \omega L$, which controls the current without dissipating power (an ideal inductor consumes no average power, since voltage and current are 90° out of phase). A resistor would limit the current too but would waste energy as heat, so a choke coil is preferred.

Final answer. Part (a) experiment confirms Lenz's law; (b) graphs are: Φ-I a line (slope L), ε-dI/dt a line (slope L), U-I a parabola; OR the choke coil limits current with negligible power loss.

← *Back to Q170*

Q171

State Faraday's law of electromagnetic induction.
Figure shows a rectangular conductor PQRS in which the conductor PQ is free to move in a uniform magnetic field B perpendicular to the plane of the paper. The field extends from $x = 0$ *to* $x = b$ *and is zero for* $x > b$*. Assume that only the arm PQ possesses resistance r. When the arm PQ is pulled outward from* $x = 0$ *to* $x = 2b$ *and is then moved backward to* $x = 0$ *with constant speed v, obtain the expressions for the flux and the induced emf. Sketch the variations of these quantities with distance* $0 \le x \le 2b$.



OR

Draw a schematic diagram of a step-up transformer. Explain its working principle. Deduce the expression for the secondary to primary voltage in terms of the number of turns in the two coils. In an ideal transformer, how is this ratio related to the currents in the two coils?

How is the transformer used in large scale transmission and distribution of electrical energy over long distances?

[2012]

SOLUTION

Faraday's law of electromagnetic induction states that whenever the magnetic flux linked with a closed circuit changes, an emf is induced in the circuit. The magnitude of the induced emf equals the negative rate of change of magnetic flux linked with the circuit.

$$\varepsilon = -\frac{d\Phi}{dt}$$

For a coil of N turns, $\boldsymbol{\varepsilon = -N\frac{d\Phi}{dt}}$. The negative sign represents Lenz's law, that the induced emf opposes the change in flux.

In the figure, the rectangular conductor PQRS lies in the plane of the paper, the arm PQ has length l and resistance r, and the uniform field B points into the page over the strip from $\boldsymbol{x = 0}$ to $\boldsymbol{x = b}$ only. PQ is pulled out from $\boldsymbol{x = 0}$ to $\boldsymbol{x = 2b}$ at constant speed v and then pushed back to $\boldsymbol{x = 0}$.

Outward journey, while PQ is inside the field, $0 \le x \le b$. The area of the loop lying in the field is $l\,x$, so the flux is

$$\Phi = B\,l\,x$$

This increases linearly from 0 at $x = 0$ to $B\,l\,b$ at $x = b$.

Outward journey, while PQ is beyond the field, $b \le x \le 2b$. The whole field strip of width b is now enclosed, so the flux stays constant at

$$\Phi = B\,l\,b$$

The induced emf is the rate of change of flux. Since $x = v\,t$, we have $\frac{dx}{dt} = v$.

For $0 \le x \le b$ (outward), $\varepsilon = -\frac{d\Phi}{dt} = -B\,l\,\frac{dx}{dt} = -B\,l\,v$, of magnitude Blv.

For $b \le x \le 2b$ (outward), $\frac{d\Phi}{dt} = 0$, so $\varepsilon = 0$.

On the return journey from $x = 2b$ to $x = b$, the flux is again constant at Blb, so $\varepsilon = 0$.

On the return from $x = b$ to $x = 0$, the flux falls from Blb back to 0, so $\varepsilon = +B\,l\,v$, of magnitude Blv but opposite in sign to the outward emf, since the flux is now decreasing.

Sketch of flux versus x: For the outward trip, Φ rises as a straight line from 0 at $x=0$ to Blb at $x=b$, then runs flat at Blb from $x=b$ to $x=2b$. The graph is a triangle-ramp followed by a flat top.

Sketch of emf versus x: A negative rectangular block of height Blv over $0 \le x \le b$ (outward), zero over $b \le x \le 2b$, and on the return a positive rectangular block of height Blv over the b to 0 stretch. The emf appears only while PQ is crossing the field region; it is two equal-magnitude rectangular pulses of opposite sign.

COMMON MISTAKE

A very common slip is to forget that emf is induced only when the flux is changing. Once PQ leaves the field region the enclosed flux is fixed, so the emf is exactly zero even though the rod is still moving.

Final answer. Flux: $\Phi = Blx$ (rising), then $\Phi = Blb$ (constant); emf: magnitude **Blv** only while PQ crosses the field, of opposite sign on the way out and back, zero otherwise.

← *Back to Q171*

Q172

(a) State the principle on which AC generator works. Draw a labelled diagram and explain its working.
(b) A conducting rod held horizontally along East-West direction is dropped from rest from a certain height near the Earth's surface. Why should there be an induced emf across the ends of the rod? Draw a plot showing the instantaneous variation of emf as a function of time from the instant it begins to fall.
OR
(a) State the principle of a step-up transformer. Explain, with the help of a labelled diagram, its working.
(b) Describe briefly any two energy losses, giving the reasons for their occurrence in actual transformers.

[2012]

SOLUTION

(a) Principle of an AC generator. It works on the principle of electromagnetic induction. When a coil is rotated in a uniform magnetic field, the magnetic flux linked with it changes continuously, and so an alternating emf is induced in the coil.

Labelled diagram (described): Draw a rectangular coil (armature) of N turns placed between the curved pole pieces of a magnet marked N on the left and S on the right, giving a uniform field $\vec{B}$ across the gap. The two ends of the coil are connected to two separate slip rings R_1 and R_2 mounted on the same axle. Two carbon brushes B_1 and B_2 press against these rings and lead to the external load resistance R. An arrow shows the coil rotating about the central axis.

Working: As the coil rotates with angular speed ω, the angle between the area vector $\vec{A}$ of the coil and $\vec{B}$ at time t is $\theta = \omega t$. The flux linked is

$$\Phi = NBA\omega\, t$$

By Faraday's law the induced emf is

$$\varepsilon = -\frac{d\Phi}{dt} = -NBA\frac{d}{dt}(\omega\, t)$$

$$\varepsilon = NBA\ \omega\omega\ t = \varepsilon_0 \omega\ t$$

where $\varepsilon_0 = NBA\omega$ is the peak emf. The slip rings reverse the connection to the external circuit every half rotation, giving an alternating output.

(b) Falling rod along East-West. A rod held horizontally along the East-West direction is dropped from rest and falls vertically downward. The Earth's magnetic field has a horizontal component B_H pointing along the South-North direction. As the rod falls, it cuts these horizontal field lines, so a motional emf is induced across its ends.

The instantaneous speed under free fall is

$$v = g\,t$$

The induced motional emf across a rod of length L is

$$\varepsilon = B_H\,v\,L = B_H\,L\,g\,t$$

So the emf grows linearly with time from zero.

Plot (described): A straight line starting from the origin and rising steadily with positive slope $B_H\,L\,g$, that is, emf on the vertical axis increasing in direct proportion to time t on the horizontal axis.

An emf is induced because the falling rod cuts the horizontal component of the Earth's field, and that emf increases linearly with time as the rod speeds up.

Final answer. AC generator gives $\varepsilon = NBA\omega\omega\,t$; the falling E-W rod cuts B_H, giving $\varepsilon = B_H\,L\,g\,t$, a straight line rising linearly from the origin.

← *Back to Q172*

Q173

State the working of a.c. generator with the help of a labelled diagram.
The coil of an a.c. generator having N turns, each of area A, is rotated with a constant angular velocity ω. Deduce the expression for the alternating e.m.f. generated in the coil.
What is the source of energy generation in this device?

[2011]

SOLUTION

An AC generator works on the principle of electromagnetic induction. When a coil rotates in a uniform magnetic field, the flux through it changes continuously and an alternating emf is induced in it.

Labelled diagram (described): Draw a rectangular coil ABCD of N turns placed between the poles of a magnet, N-pole on the left and S-pole on the right, so that a uniform field $\vec{B}$ runs across the gap. The ends of the coil are joined to two slip rings R_1 and R_2 fixed on the axle; carbon brushes B_1 and B_2 touch these rings and connect to the external load R. A curved arrow shows the coil rotating about its axis.

Working: The coil of N turns, each of area A, is rotated with constant angular velocity ω in the field $\vec{B}$. At time t the area vector makes an angle $\theta = \omega t$ with $\vec{B}$. The flux linked with the coil is

$$\Phi = N B A\theta = N B A\omega t$$

By Faraday's law the induced emf is

$$\varepsilon = -\frac{d\Phi}{dt}$$

$$\varepsilon = -N B A \frac{d}{dt}(\omega t)$$

$$\varepsilon = -N B A (-\omega\omega t)$$

$$\varepsilon = N B A \omega\omega t$$

Writing the peak value as $\varepsilon_0 = NBA\omega$, the alternating emf is

$$\varepsilon = \varepsilon_0 \omega t$$

The emf is maximum when the coil is parallel to the field (plane along $\vec{B}$) and zero when the coil is perpendicular to the field.

Source of energy: The source of electrical energy is the mechanical energy of the agent that rotates the coil, for example the turbine driven by falling water, steam, or wind. This mechanical work done against the opposing torque on the current-carrying coil is converted into electrical energy by electromagnetic induction.

Final answer. The alternating emf is $\varepsilon = NBA\omega\omega\, t = \varepsilon_0 \omega\, t$, and the energy comes from the mechanical work done in rotating the coil.

← *Back to Q173*

Q174

(a) Derive the expression for the mutual inductance of two long coaxial solenoids of same length l having radii r_1 and r_2 ($\mathbf{r_2 > r_1}$ and $l \gg r_2$).
(b) Show that mutual inductance of solenoid 1 due to solenoid 2, M_{12}, is the same as that of 2 due to 1 i.e. M_{21}.
(c) A power transmission line feeds power at 2200 V with a current of 5 A to a step down transformer with its primary winding having 4000 turns. Calculate the number of turns and the current in the secondary in order to get output power at 220 V.

[2011 • Set 55-2-1]

SOLUTION

(a) Mutual inductance of two long coaxial solenoids. Let the inner solenoid have N_1 turns and radius r_1, and the outer solenoid have N_2 turns and radius r_2, both of length l with $l \gg r_2$. Let $\mathbf{n_2 = N_2/l}$ be the number of turns per unit length of the outer solenoid.

Pass a current I_2 through the outer solenoid. The uniform field it produces inside is

$$B_2 = \mu_0 n_2 I_2 = \frac{\mu_0 N_2 I_2}{l}$$

The inner solenoid lies wholly within this field. The flux through one turn of the inner solenoid (area πr_1^2) is

$$\phi = B_2 \pi r_1^2 = \frac{\mu_0 N_2 I_2}{l} \pi r_1^2$$

The total flux linkage with all N_1 turns of the inner solenoid is

$$N_1\phi = \frac{\mu_0 N_1 N_2 \pi r_1^2 I_2}{l}$$

The mutual inductance M_{12} is defined by $\mathbf{N_1\phi = M_{12} I_2}$, so

$$M_{12} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$

(b) Showing $\mathbf{M_{12} = M_{21}}$. Now pass a current I_1 through the inner solenoid. The field it produces inside is $\mathbf{B_1} = \frac{\mu_0 N_1 I_1}{l}$, confined to the inner cross-section of area πr_1^2 (the field outside a long solenoid is negligible).

So the flux through each turn of the outer solenoid is only the part over the inner area, $\boldsymbol{\phi' = B_1 \pi r_1^2 = \frac{\mu_0 N_1 I_1}{l} \pi r_1^2}$.

Total linkage with the N_2 turns of the outer solenoid is

$$N_2\phi' = \frac{\mu_0 N_1 N_2 \pi r_1^2 I_1}{l} = M_{21} I_1$$

$$M_{21} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$

Comparing the two results,

$$M_{12} = M_{21} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$

(c) Step-down transformer. Given primary voltage $\mathbf{V_p = 2200\ V}$, primary current $\mathbf{I_p = 5\ A}$, primary turns $\mathbf{N_p = 4000}$, secondary voltage $\mathbf{V_s = 220\ V}$.

Using the turns relation $\frac{\mathbf{V_s}}{\mathbf{V_p}} = \frac{\mathbf{N_s}}{\mathbf{N_p}}$,

$$N_s = N_p \frac{V_s}{V_p} = 4000 \times \frac{220}{2200}$$

$$N_s = 400 \text{ turns}$$

For an ideal transformer, input power equals output power, $\mathbf{V_p I_p = V_s I_s}$, so

$$I_s = I_p \frac{V_p}{V_s} = 5 \times \frac{2200}{220}$$

$$I_s = 50\ A$$

The secondary has 400 turns and carries a current of 50 A.

TIP

In part (b), the trick is that the flux through the outer solenoid only spans the inner area πr_1^2, not πr_2^2, because a long solenoid's field is confined inside it. Using πr_2^2 is the most common error here.

Final answer. $\mathbf{M_{12} = M_{21} = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}}$; the secondary has $\mathbf{N_s = 400}$ turns and $\mathbf{I_s = 50\ A}$.

← *Back to Q174*

Q175

Describe briefly, with the help of a labelled diagram, the basic elements of an A.C. generator. State its underlying principle. Show diagrammatically how an alternating emf is generated by a loop of wire rotating in a magnetic field. Write the expression for the instantaneous value of the emf induced in the rotating loop.

[2010]

SOLUTION

Basic elements of an AC generator:

Armature: A rectangular coil of N turns, each of area A, wound on a soft iron core and mounted on a rotatable axle. This is where the emf is induced.

Field magnet: A strong permanent magnet (or electromagnet) with curved pole pieces N and S providing a uniform magnetic field $\vec{B}$ in the gap where the coil rotates.

Slip rings: Two metallic rings R_1 and R_2 fixed to the two ends of the coil; they rotate along with the coil.

Brushes: Two carbon brushes B_1 and B_2 that press lightly against the slip rings and connect the rotating coil to the stationary external circuit (load R).

Principle: It works on the principle of electromagnetic induction, that a changing magnetic flux through a coil induces an emf in it.

Diagram (described): Draw the coil between the N (left) and S (right) pole faces with the uniform field $\vec{B}$ pointing from N to S. Show the coil ends going to slip rings R_1, R_2 on the axle, brushes B_1, B_2 touching them, and wires from the brushes to the external load R. A curved arrow shows the coil rotating with angular speed ω.

Generation of alternating emf: As the coil rotates with angular speed ω, the angle between its area vector and $\vec{B}$ is $\theta = \omega t$, so the flux linked is

$$\Phi = \mathrm{NBA}\omega\, t$$

By Faraday's law,

$$\varepsilon = -\frac{d\Phi}{dt} = -\mathrm{NBA}\frac{d}{dt}(\omega\, t)$$

$$\varepsilon = \mathrm{NBA}\ \omega\omega\ t$$

When the coil plane is parallel to $\vec{B}$ the rate of cutting of field lines is maximum and the emf is maximum; when the coil plane is perpendicular to $\vec{B}$ the emf is zero. As the coil keeps rotating, the emf reverses direction every half turn, giving an alternating output that varies sinusoidally with time.

Instantaneous value of the induced emf:

$$\varepsilon = \mathrm{NBA}\ \omega\omega\ t$$

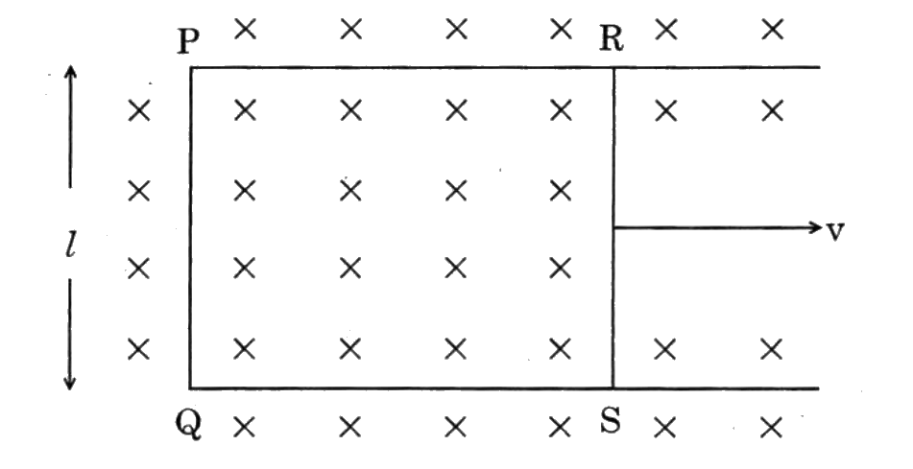
where the peak emf is $\varepsilon_0 = \mathrm{NBA}\omega$.

Final answer. The instantaneous induced emf is $\varepsilon = \mathrm{NBA}\omega\omega\, t$, with peak value $\varepsilon_0 = \mathrm{NBA}\omega$.

← *Back to Q175*

Q176

(a) What are eddy currents? Write their two applications.
(b) Figure shows a rectangular conducting loop PQSR *in which arm* RS *of length l is movable. The loop is kept in a uniform magnetic field B directed downward perpendicular to the plane of the loop. The arm* RS *is moved with a uniform speed v.*



Deduce an expression for
(i) the emf induced across the arm RS,
(ii) the external force required to move the arm, and
(iii) the power dissipated as heat.

[2009]

SOLUTION

(a) Eddy currents. When the magnetic flux linked with a bulk piece of conductor changes, induced currents are set up in closed loops within the body of the conductor. These circulating induced currents are called eddy currents. By Lenz's law they flow so as to oppose the change in flux that produces them.

Two applications:

Electromagnetic braking, used in electric trains and moving machinery, where eddy currents in a metal drum or rail produce a retarding force.

Induction furnace or induction heating, used to melt metals and in induction cooktops, where strong eddy currents heat the conductor.

(b) Rectangular loop PQSR with movable arm RS of length l moving at uniform speed v in a uniform field B perpendicular to the plane of the loop, with total circuit resistance R.

(i) EMF induced across RS. As RS moves a distance $\mathbf{dx = v\,dt}$, the area enclosed changes by $l\,dx$, so the flux change is $\mathbf{d\Phi = B\,l\,dx}$. The induced emf is

$$\varepsilon = \frac{d\Phi}{dt} = B\,l\,\frac{dx}{dt} = B\,l\,v$$

$$\varepsilon = B\,l\,v$$

(ii) External force required. The induced current in the loop is

$$i = \frac{\varepsilon}{R} = \frac{\mathrm{Blv}}{R}$$

The current-carrying arm RS in the field B experiences a magnetic force

$$\mathrm{F_{mag}} = B\,i\,l = B(\frac{\mathrm{Blv}}{R})l = \frac{\mathrm{B^2l^2v}}{R}$$

By Lenz's law this force opposes the motion, so to keep RS moving at uniform speed the external applied force must just balance it,

$$\mathrm{F_{ext}} = \frac{\mathrm{B^2l^2v}}{R}$$

(iii) Power dissipated as heat. The external agent does work against the opposing force at the rate

$$P = \mathrm{F_{ext}} \times v = \frac{\mathrm{B^2l^2v}}{R} \times v$$

$$P = \frac{\mathrm{B^2l^2v^2}}{R}$$

This equals the joule heating $\frac{\varepsilon^2}{R}$ in the circuit, confirming that all the mechanical work done in moving the arm appears as heat, in accordance with conservation of energy.

Final answer. $\varepsilon = \mathrm{Blv}$, $\mathbf{F_{ext}} = \frac{\mathrm{B^2l^2v}}{R}$, and $P = \frac{\mathrm{B^2l^2v^2}}{R}$.

← *Back to Q176*

Q177

(a) State Lenz's law. Give one example to illustrate this law. "The Lenz's law is a consequence of the principle of conservation of energy." Justify this statement.

(b) Deduce an expression for the mutual inductance of two long coaxial solenoids but having different radii and different number of turns.

[2009]

SOLUTION

(a) Lenz's law. The direction of the induced current (and induced emf) is always such that it opposes the very change in magnetic flux that produces it.

Example: When the north pole of a bar magnet is pushed towards a coil, the flux linked with the coil increases. By Lenz's law the induced current flows in a direction that makes the near face of the coil a north pole, which repels the approaching magnet and opposes its motion. (If the magnet is pulled away, the induced current reverses to attract it and oppose the withdrawal.)

Lenz's law as a consequence of conservation of energy: Because the induced current opposes the motion of the magnet, the person moving the magnet has to do work against this opposing force. This mechanical work done is exactly equal to the electrical energy that appears as heat in the resistance of the coil. Thus energy is merely converted from mechanical to electrical form and none is created. If, instead, the induced current aided the motion, the magnet would accelerate on its own and produce electrical energy without any work being done, creating energy from nothing. That would violate the principle of conservation of energy. Hence Lenz's law is a direct consequence of conservation of energy.

(b) Mutual inductance of two long coaxial solenoids of different radii and different numbers of turns. Let the inner solenoid have radius r_1 and N_1 turns, and the outer solenoid have radius r_2 and N_2 turns, both of the same length l.

Pass a current I_2 through the outer solenoid. The uniform magnetic field produced inside it is

$$B_2 = \mu_0 n_2 I_2 = \frac{\mu_0 N_2 I_2}{l}$$

The inner solenoid (cross-sectional area πr_1^2) lies entirely inside this field, so the flux through one of its turns is

$$\phi = B_2 \pi r_1^2 = \frac{\mu_0 N_2 I_2}{l} \pi r_1^2$$

The total flux linkage with all N_1 turns of the inner solenoid is

$$\Phi_1 = N_1 \phi = \frac{\mu_0 N_1 N_2 \pi r_1^2 I_2}{l}$$

The mutual inductance is given by $\Phi_1 = M I_2$, hence

$$M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$$

Note that r_1 (the smaller, inner radius) appears in the result, because the flux is linked only over the inner cross-sectional area.

Final answer. Lenz's law follows from energy conservation, and the mutual inductance is $M = \frac{\mu_0 N_1 N_2 \pi r_1^2}{l}$.

← *Back to Q177*